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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the Introduction and/or on the ISO list of patent declarations received (see www.iso.org/patents).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation on the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT) see www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 184, *Automation systems and integration*, Subcommittee SC 4, *Industrial data*.

This sixth edition cancels and replaces the fifth edition (ISO 10303-41:2018-11), which has been technically revised.

A detailed description of the changes is provided in Annex G. A list of all parts in the ISO 10303 series can be found on the ISO website.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html.

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Introduction

ISO 10303 is an International Standard for the computer-interpretable representation of product information and for the exchange of product data. The objective is to provide a neutral mechanism capable of describing products throughout their life cycle. This mechanism is suitable not only for neutral file exchange, but also as a basis for implementing and sharing product databases, and as a basis for retention and archiving.

Major subdivisions of this part of ISO 10303 are:

— action_schema;

— application_context_schema;

— approval_schema;

— basic_attribute_schema;

— certification_schema;

— contract_schema;

— date_time_schema;

— document_schema;

— effectivity_schema;

— experience_schema;

— external_reference_schema;

— group_schema;

— language_schema;

— location_schema;

— management_resources_schema;

— measure_schema;

— person_organization_schema;

— product_definition_schema;

— product_property_definition_schema;

— product_property_representation_schema;

— qualifications_schema;

— security_classification_schema;

— support_resource_schema.

The generic product description resources provide an overall organization for the integrated resources that are documented in other parts of ISO 10303. They specify resource constructs that provide consistent representation of facts about products in different application-specific views.

The generic management resources support the description of information that is used to manage and control product data. Together, the generic product description resources and the generic management resources are the foundations upon which application interpreted models, the normative conceptual schemas of application protocols, are built. Application interpreted models specialize selected generic

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management resources to elements of the integrated product description resources to satisfy the requirements that are specified in the application reference model.

The support resources are a set of shared resource constructs that are used by the ISO 10303 integrated resources. They provide an underlying consistency across the resources of ISO 10303.

In this International Standard, the same English language words may be used to refer to an object in the real world or concept, and as the name of an EXPRESS data type that represents this object or concept. The following typographical convention is used to distinguish between these. If a word or phrase occurs in the same typeface as narrative text, the referent is the object or concept. If the word or phrase occurs in a bold typeface, the referent is the EXPRESS data type. Names of EXPRESS schemas also occur in a bold typeface.

The name of an EXPRESS data type may be used to refer to the data type itself, or to an instance of the data type. The distinction between these uses is normally clear from the context. If there is a likelihood of ambiguity, either the phrase “entity data type” or “instance(s) of” is included in the text.

Double quotation marks (“”) denote quoted text. Single quotation marks (‘’) denote particular text string values.

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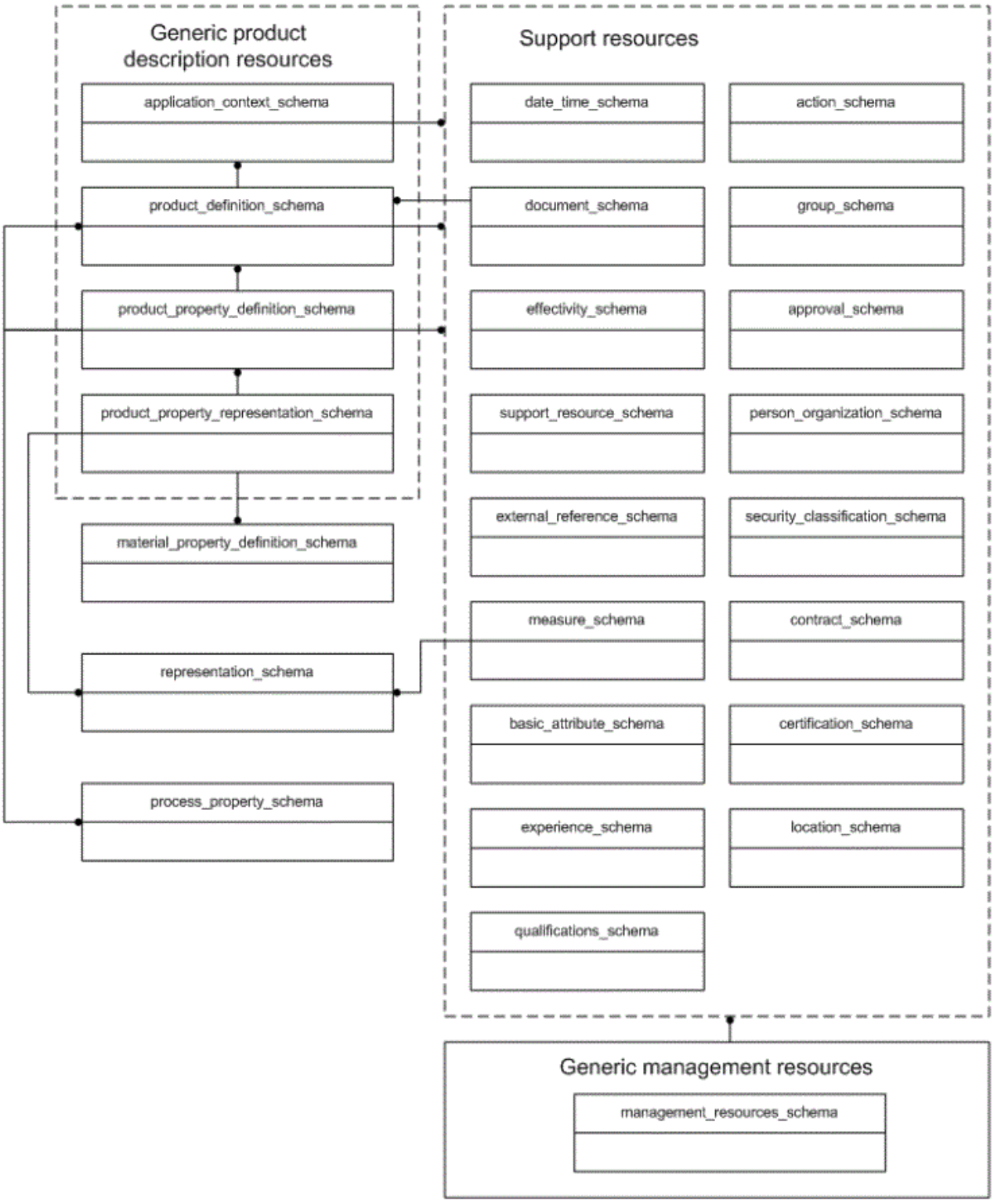


Figure 1 — The schemas of this part of ISO 10303

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Industrial automation systems and integration — Product data representation and exchange —

Part 41: Integrated generic resource: Fundamentals of product description and support

1 Scope

1.1 General

{% assign docname = sdi | replace: "_", " " %} {% case **term `sdt` not resolved via ID `sdt`** %} {% when "aic" %} This part of ISO 10303 specifies the interpretation of the integrated resources to satisfy requirements for the representation of **term `docname` not resolved via ID `docname`.** {% when "igr" %} This part of ISO 10303 specifies the integrated generic resource constructs for **term `docname` not resolved via ID `docname`.** {% when "iar" %} This part of ISO 10303 specifies the integrated application resource constructs for **term `docname` not resolved via ID `docname`.** {% endcase %} —-

The schemas that are specified in this part of ISO 10303 are organized as follows.

1.2 Generic product description resources

This part of ISO 10303 specifies the resource constructs for the high level structure for the representation of products and their properties. It also specifies ISO 10303 integrated resources for the description of generic aspects of product usage, categorization of products and associations between products.

The following are within scope in this part of ISO 10303:

— the identification of a product;

— the categorization of a product;

— the characterization of a product in one or more application contexts;

— the definition of relationships among products;

— the identification of the properties of a product or of a portion of a product;

— the identification of a representation of a property;

EXAMPLE A 3D geometric model or a table of numerical values are different representations of properties of a product.

— the description of the application context for which product data is defined.

1.3 Generic management resources

This part of ISO 1030 specifies the resource constructs to associate administrative data with product data.

The following are within scope in this part of ISO 10303:

— the structure for associating product data with related administrative data.

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1.4 Support resources

This part of ISO 10303 specifies the resource constructs for administrative data, physical quantities and their units, and basic data types.

The identification and description of following are within scope in this part of ISO 10303:

— the identification of documents;

— the actions, action requests, and the status of actions;

— the certification, approvals, security classifications, and effectivities;

— the contracts;

— the people and organizations;

— the specification of dates and times;

— the provision of mechanisms for grouping items and referring to information that is defined outside an exchange;

— the physical quantities and their units;

— the provision of mechanisms to define values for id, name, description and role attributes;

— the location of a person, organization, or product;

— the experience, qualifications, and position in an organization of a person.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 8601:2000[1)] , *Data elements and interchange formats — Information interchange — Representation of dates and times*

ISO 10303-1[2)] , *Industrial automation systems and integration — Product data representation and exchange — Part 1: Overview and fundamental principles*

ISO 10303-11:2004, *Industrial automation systems and integration — Product data representation and exchange — Part 11: Description methods: The EXPRESS language reference manual*

ISO 10303-43, *Industrial automation systems and integration — Product data representation and exchange — Part 43: Integrated generic resource: Representation structures*

ISO 10303-44, *Industrial automation systems and integration — Product data representation and exchange — Part 44: Integrated generic resource: Product structure configuration*

ISO 10303-49, *Industrial automation systems and integration — Product data representation and exchange — Part 49: Integrated generic resources: Process structure and properties*

ISO 80000 (all parts)[3)] , *Quantities and units*

1) Withdrawn.

2) Withdrawn.

3) Withdrawn.

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IEC 80000 (all parts), *Quantities and units*

3 Terms, definitions and abbreviations

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at https://www.iso.org/obp

— IEC Electropedia: available at https://www.electropedia.org

3.1 Terms defined in ISO 8601

For the purposes of this document, the following terms defined in ISO 8601 apply:

— calendar date;

— calendar week;

— calendar year;

— common year;

— coordinated universal time;

— day;

— Gregorian calendar;

— hour;

— leap year;

— local time;

— minute;

— month;

— ordinal day;

— second;

— week;

— year.

3.2 Terms defined in ISO 10303-1

For the purposes of this document, the following terms defined in ISO 10303-1 apply:

— integrated resource;

— application;

— application context;

— application interpreted model;

— application protocol;

— application reference model;

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— data;

— information;

— product;

— product data;

— structure.

3.3 Terms defined in ISO 10303-44

For the purposes of this document, the following terms defined in ISO 10303-44 apply:

— child node;

— descendent node;

— directed acyclic graph;

— leaf node;

— link;

— node;

— parent node;

— root node;

— tree.

3.4 Terms and definitions

3.4.1
agreement of common understanding
result of discussions between the partners of product data exchange or sharing that ensures that all of them have the same understanding of the transferred or shared information

Note 1 to entry: The agreement may be formalized in a document.

3.4.2
annotated EXPRESS schema
EXPRESS schema for which a natural language definition is given with each CONSTANT, TYPE, ENTITY, RULE, FUNCTION or PROCEDURE construct

EXAMPLE Clause 5.2 of an application protocol contains an annotated EXPRESS schema.

3.4.3
chain
path in which all nodes and links are distinct

3.4.4
class of products
set of all products satisfying a type of product; the elements of the set are referred to as members of the class

Note 1 to entry: A class need not have any members.

Note 2 to entry: The definition of the type will determine whether the size of the set varies with time.

Note 3 to entry: This definition was adapted from ISO/IEC 10746-2:1996 [1].

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3.4.5
contract
a binding agreement

3.4.6
degree of a node
number of direct connections to other nodes

3.4.7
management information
Information required for proper control and handling of a specific product or item.

EXAMPLE An inspection certification indicates that the product has passed inspection and is ready to move to the next step in the product life cycle.

Note 1 to entry: Management data may include information related to scheduling, roles, certifications, approvals, effectivity and life-cycle stages.

3.4.8
path
a sequence of nodes such that from each of itse nodes thre is an edge to the next node in the sequence; it has two terminal nodes (nodes that have degree 1), while all others (if any) have degree 2

Note 1 to entry: The first and last nodes in the path are different. No nodes are referenced more than once by a link.

3.4.9
real object
object that exists or did exist in the real world and that can or could be uniquely identified

EXAMPLE The Moon is an individual object that is not the result of a human process.

Note 1 to entry: An individual object may result from a manufacturing process or it may exist independently of any human process.

3.4.10
type_of_product
predicate characterizing a collection of products; a product is of the type, or satisfies the type, if the predicate holds for that product

EXAMPLE This definition was adapted from ISO/IEC 10746-2:1996 [1].

3.4.11
version
a unique revision of an item

3.5 Abbreviated terms

DAG	Directed Acyclic Graph
pos	person organization schema
SI	International System of Units

4 Action schema

4.1 General

The subject of the **action_schema** is the description of actions, the reasons for these actions, and the status of these actions.

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This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **action_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA action_schema '{iso standard 10303 part(41) version(8) object(1) action_
schema(1)}';

REFERENCE FROM basic_attribute_schema
  (description_attribute,
   description_attribute_select,
   get_description_value,
   get_id_value,
   get_name_value,
   get_role,
   id_attribute,
   id_attribute_select,
   name_attribute,
   name_attribute_select,
   object_role,
   role_select,
   role_association);
REFERENCE FROM support_resource_schema
  (bag_to_set,
   identifier,
   label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

basic_attribute_schema	ISO 10303-41
support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 2 for a graphical representation of this schema.

4.2 Fundamental concepts and assumptions

Action information can be attached to any aspect of product data.

4.3 Action type definitions

4.3.1 as_description_attribute_select

The **as_description_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE as_description_attribute_select = SELECT BASED_ON description_attribute_select WITH
  (action_request_solution);
END_TYPE;
```


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4.3.2 as_id_attribute_select

The **as_id_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE as_id_attribute_select = SELECT BASED_ON id_attribute_select WITH
  (action);
END_TYPE;
```

4.3.3 as_name_attribute_select

The **as_name_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE as_name_attribute_select = SELECT BASED_ON name_attribute_select WITH
  (action_request_solution);
END_TYPE;
```

4.3.4 as_role_select

The **as_role_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE as_role_select = SELECT BASED_ON role_select WITH
  (directed_action_assignment);
END_TYPE;
```

4.3.5 supported_item

The **supported_item** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

The **supported_item** allows for the designation of an action_directive, an action, or an action_method.

EXPRESS specification:

```
TYPE supported_item = SELECT
  (action_directive,
   action,
   action_method);
END_TYPE;
```

4.4 Action entity definitions

4.4.1 action

An **action** is the identification of the occurrence of an activity and a description of its result.

An **action** identifies an activity that has taken place, is taking place, or is expected to take place in the future.

An action has a definition that is specified by an action_method.

EXPRESS specification:

```
ENTITY action;
```

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```
  name : label;
  description : OPTIONAL text;
  chosen_method : action_method;
DERIVE
  id : identifier := get_id_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ID_ATTRIBUTE.IDENTIFIED_ITEM'))
<= 1;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **action** is known.

description: the text that characterizes the **action**. The value of the attribute need not be specified.

chosen_method: the action_method that specifies the procedure selected to carry out the **action** and a description of the results of that **action**.

id: the identifier that distinguishes the **action**.

Formal propositions:

WR1: Each **action** shall be the identified_item in at most one id_attribute.

4.4.2 action_directive 

An **action_directive** is an authoritative instrument that provides directions to achieve the specified results.

EXPRESS specification:

```
ENTITY action_directive;
  name : label;
  description : OPTIONAL text;
  analysis : text;
  comment : text;
  requests : SET [1:?] OF versioned_action_request;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **action_directive** is known.

description: the text that characterizes the **action_directive**. The value of the attribute need not be specified.

analysis: an informal description of the results of the analysis that was carried out on the elements of the requests set. The request sets are the versioned_actions referenced in the requests attribute.

comment: an informal description of any other pertinent information.

requests: a set of versioned_action_request entity data types that defines the expected results.

4.4.3 action_method 

An **action_method** is the definition of an activity. This definition includes the activity's objectives and effects.

EXPRESS specification:

```
ENTITY action_method;
  name : label;
```

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```
  description : OPTIONAL text;
  consequence : text;
  purpose : text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **action_method** is known.

description: the text that characterizes the **action_method**. The value of the attribute need not be specified.

consequence: an informal description of the effects of the **action_method**.

purpose: an informal description of the objectives of the **action_method**.

4.4.4 action_method_relationship

An **action_method_relationship** is a relationship between two instances of the entity data type action_method and provides an identification and description of this relationship.

EXPRESS specification:

```
ENTITY action_method_relationship;
  name : label;
  description : OPTIONAL text;
  relating_method : action_method;
  related_method : action_method;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **action_method_relationship** is known.

description: the text that characterizes the **action_method_relationship**. The value of the attribute need not be specified.

relating_method: one of the instances of action_method that is a part of the relationship.

related_method: the other instance of action_method that is a part of the relationship. If one element of the relationship is dependent upon the other this attribute shall be the dependent one.

4.4.5 action_relationship

An **action_relationship** is a relationship between two instances of the entity data type action and provides an identification and description of this relationship.

EXPRESS specification:

```
ENTITY action_relationship;
  name : label;
  description : OPTIONAL text;
  relating_action : action;
  related_action : action;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **action_relationship** is known.

description: the text that characterizes the **action_relationship**. The value of the attribute need not be specified.

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relating_action: one of the instances of action that is a part of the relationship.

related_action: the other instance of action that is a part of the relationship. If one element of the relationship is dependent upon the other this attribute shall be the dependent one.

4.4.6 action_request_solution



An **action_request_solution** is an association between a versioned_action_request and an action_method that is a potential solution for the request.

EXPRESS specification:

```
ENTITY action_request_solution;
  method : action_method;
  request : versioned_action_request;
DERIVE
  description : text := get_description_value(SELF);
  name : label := get_name_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
  WR2: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'NAME_ATTRIBUTE.NAMED_ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

method: the action_method that is a potential solution.

request: the versioned_action_request for which a solution is specified.

description: the text that characterizes the **action_request_solution**.

name: the label by which the **action_request_solution** is known.

Formal propositions:

WR1: Each **action_request_solution** shall be the described_item in at most one description_attribute.

WR2: Each **action_request_solution** shall be the named_item in at most one name_attribute.

4.4.7 action_request_status



An **action_request_status** is the association of a status with a versioned_action_request.

EXPRESS specification:

```
ENTITY action_request_status;
  status : label;
  assigned_request : versioned_action_request;
END_ENTITY;
```

Attribute definitions:

status: the label that provides a user interpretable designation for the level of completion of the action.

assigned_request: the versioned_action_request to which the **status** applies.

4.4.8 action_resource



An **action_resource** is a thing that is identified as being needed to carry out an action.

EXPRESS specification:

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```
ENTITY action_resource;
  name : label;
  description : OPTIONAL text;
  usage : SET [1:?] OF supported_item;
  kind : action_resource_type;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **action_resource** is known.

description: the text that characterizes the **action_resource**. The value of the attribute need not be specified.

usage: a set of supported_items for which the **action_resource** is used.

kind: the action_resource_type that specifies the sort of **action_resource** that is being used.

4.4.9 action_resource_relationship

An **action_resource_relationship** is a relationship between two instances of the entity data type action_resource and provides an identification and description of this relationship.

EXPRESS specification:

```
ENTITY action_resource_relationship;
  name : label;
  description : OPTIONAL text;
  relating_resource : action_resource;
  related_resource : action_resource;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **action_resource_relationship** is known.

description: the text that characterizes the **action_resource_relationship**. The value of the attribute need not be specified.

relating_resource: one of the instances of action_resource that is a part of the relationship.

related_resource: the other instance of action_resource that is a part of the relationship. If one element of the relationship is dependent upon the other this attribute shall be the dependent one.

4.4.10 action_resource_type

An **action_resource_type** is the identification of the kind of action_resource needed to carry out an action.

EXPRESS specification:

```
ENTITY action_resource_type;
  name : label;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **action_resource_type** is known.

4.4.11 action_status

An **action_status** is the association of a status with an executed_action.

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EXPRESS specification:

```
ENTITY action_status;
  status : label;
  assigned_action : executed_action;
END_ENTITY;
```

Attribute definitions:

status: the label that provides a user interpretable designation for the level of completion of the action.

assigned_action: the executed_action to which the status applies.

4.4.12 directed_action

A **directed_action** is a type of executed_action that is governed by an action_directive.

EXPRESS specification:

```
ENTITY directed_action
  SUBTYPE OF (executed_action);
  directive : action_directive;
END_ENTITY;
```

Attribute definitions:

directive: the action_directive that governs the **directed_action**.

4.4.13 directed_action_assignment

A **directed_action_assignment** is an association of a directed_action with product data.

EXPRESS specification:

```
ENTITY directed_action_assignment
  ABSTRACT SUPERTYPE;
  assigned_directed_action : directed_action;
DERIVE
  role : object_role := get_role(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

assigned_directed_action: the instance of the **directed_action** entity data that is to be associated with product data.

role: the **object_role** that specifies the purpose of the association of the **directed_action_assignment** with product data.

Formal propositions:

WR1: Each **directed_action_assignment** shall be **item_with_role** in at most one role_association.

4.4.14 executed_action

An **executed_action** is a type of action that is completed, partially completed, or just identified. It may but need not have status information associated with it.

EXPRESS specification:

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```
ENTITY executed_action
  SUBTYPE OF (action);
END_ENTITY;
```

4.4.15 versioned_action_request

A **versioned_action_request** is a specification of a desired result.

EXPRESS specification:

```
ENTITY versioned_action_request;
  id : identifier;
  version : OPTIONAL label;
  purpose : text;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **versioned_action_request**.

version: the identification of the version of the **versioned_action_request**. The value of the attribute need not be specified.

purpose: an informal description of the reason for the **versioned_action_request**.

description: the text that characterizes the **versioned_action_request**. The value of the attribute need not be specified.

4.4.16 versioned_action_request_relationship

A **versioned_action_request_relationship** is a relationship between two versioned_action_request objects.

EXPRESS specification:

```
ENTITY versioned_action_request_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating_versioned_action_request : versioned_action_request;
  related_versioned_action_request : versioned_action_request;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **versioned_action_request_relationship**.

name: the label by which the **versioned_action_request_relationship** is known.

description: the text that characterizes the **versioned_action_request_relationship**. The value of this attribute need not be specified.

relating_versioned_action_request: one of the instances of versioned_action_request that is a part of the relationship.

related_versioned_action_request: the other instance of versioned_action_request that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

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4.4.17 action_directive_relationship

An **action_directive_relationship** is a relationship between two action_directive objects.

EXPRESS specification:

```
ENTITY action_directive_relationship;
  name : label;
  description : OPTIONAL STRING;
  relating : action_directive;
  related : action_directive;
  relation_type : STRING;
WHERE
  WR1: acyclic_action_directive_relationship(SELF, [related], 'ACTION_SCHEMA.ACTION_
DIRECTIVE_RELATIONSHIP');
END_ENTITY;
```

Attribute definitions:

name: shall be unique within a population of **action_directive_relationship**.

description: the text or the set of texts that provide further information about the **action_directive_relationship**. The value of this attribute need not be specified.

relating: one object of action_directive that is part of the relationship.

related: the other object of action_directive that is part of the relationship.

relation_type: the meaning of the relationship.

Formal propositions:

WR1: A **action_directive_relationship** shall not participate in its own definition.

4.5 Action function definitions

4.5.1 acyclic_action_method_relationship

The **acyclic_action_method_relationship** function determines whether the graph of instances of the entity data type action_method that contains relation as one of its links contains a cycle. This function may be used to evaluate either an action_method_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

EXPRESS specification:

```
FUNCTION acyclic_action_method_relationship(relation : action_method_relationship;
                                            relatives : SET [1:?] OF action_method;
                                            specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF action_method_relationship;
  END_LOCAL;
  IF relation.relating_method IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(am <* bag_to_set(USEDIN(relation.relating_method, ('ACTION_SCHEMA.' +
'ACTION_METHOD_RELATIONSHIP.') + 'RELATED')) | specific_relation IN TYPEOF(am));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_action_method_relationship(x[i], relatives + relation.relating_method,
specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
```

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```
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate action_method_relationship to be checked.

relatives: (input) the set of instances of the entity data type action_method that the function is searching for in the relating_method parameter of the relation argument.

specific_relation: (input) the fully qualified name of a subtype of the action_method_relationship entity.

4.5.2 acyclic_action_relationship

The **acyclic_action_relationship** function determines whether the graph of instances of the entity data type action that contains relation as one of its links contains a cycle. This function may be used to evaluate either an action_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

EXPRESS specification:

```
FUNCTION acyclic_action_relationship(relation : action_relationship;
                                     relatives : SET [1:?] OF action;
                                     specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF action_relationship;
  END_LOCAL;
  IF relation.relating_action IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(actn <* bag_to_set(USEDIN(relation.relating_action, ('ACTION_SCHEMA.' +
'ACTION_RELATIONSHIP.') + 'RELATED_ACTION')) | specific_relation IN TYPEOF(actn));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_action_relationship(x[i], relatives + relation.relating_action,
specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate action_relationship to be checked.

relatives: (input) the set of instances of the entity data type action for which the function is searching in the relating_action parameter of the relation argument.

specific_relation: (input) the fully qualified name of a subtype of the action_relationship entity.

4.5.3 acyclic_action_resource_relationship

The **acyclic_action_resource_relationship** function determines whether the graph of instances of the entity data type action_resource that contains relation as one of its links contains a cycle. This function may be used to evaluate either an action_resource_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

EXPRESS specification:

```
FUNCTION acyclic_action_resource_relationship(relation : action_resource_relationship;
                                              relatives : SET [1:?] OF action_resource;
                                              specific_relation : STRING) : BOOLEAN;
  LOCAL
```

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```
    x : SET OF action_resource_relationship;
  END_LOCAL;
  IF relation.relating_resource IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(ar <* bag_to_set(USEDIN(relation.relating_resource, ('ACTION_SCHEMA.
' + 'ACTION_RESOURCE_RELATIONSHIP.') + 'RELATED_RESOURCE')) | specific_relation IN
TYPEOF(ar));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_action_resource_relationship(x[i], relatives + relation.relating_
resource, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate action_resource_relationship to be checked.

relatives: (input) the set of instances of the entity data type action_resource that the function is searching for in the relating_resource parameter of the relation argument.

specific_relation: (input) the fully qualified name of a subtype of the action_resource_relationship entity.

4.5.4 acyclic_versioned_action_request_relationship

The **acyclic_versioned_action_request_relationship** function determines whether the graph of instances of the entity data type versioned_action_request that contains relation as one of its links contains a cycle. This function may be used to evaluate either an action_resource_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

EXPRESS specification:

```
FUNCTION acyclic_versioned_action_request_relationship(relation : versioned_action_
request_relationship;
                                                       relatives : SET OF versioned_
action_request;
                                                       specific_relation : STRING) :
BOOLEAN;
  LOCAL
    x : SET OF versioned_action_request_relationship;
  END_LOCAL;
  IF relation.relating_versioned_action_request IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(varr <* bag_to_set(USEDIN(relation.relating_versioned_action_request,
('ACTION_SCHEMA.' + 'VERSIONED_ACTION_REQUEST_RELATIONSHIP.') + 'RELATED_VERSIONED_ACTION_
REQUEST')) | specific_relation IN TYPEOF(varr));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_versioned_action_request_relationship(x[i], relatives + relation.
relating_versioned_action_request, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate versioned_action_request_relationship to be checked.

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relatives: (input) the set of instances of the entity data type versioned_action_request that the function is searching for in the relating_versioned_action_request parameter of the relation argument.

specific_relation: (input) the fully qualified name of a subtype of the versioned_action_request_relationship entity.

4.5.5 acyclic_action_directive_relationship

The **acyclic_action_directive_relationship** function determines whether the graph of instance of the entity data type action_directive that contains relation as one of its links contains a cycle. this function may be used to evaluate either a action_directive_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

EXPRESS specification:

```
FUNCTION acyclic_action_directive_relationship(relation : action_directive_relationship;
                                               relatives : SET [1:?] OF action_directive;
                                               specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF action_directive_relationship;
  END_LOCAL;
  IF relation.relating IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(adr <* bag_to_set(USEDIN(relation.relating, ('ACTION_SCHEMA.' + 'ACTION_
DIRECTIVE_RELATIONSHIP.') + 'RELATED')) | specific_relation IN TYPEOF(adr));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_action_directive_relationship(x[i], relatives + relation.relating,
specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate action_directive_relationship to be checked.

relatives: (input) the set of instances of the entity data type action_directive for which the function is searching in the relating parameter of the relation argument.

specific_relation: (input) the fully qualified name of a subtype of the action_directive_relationship entity.

```
END_SCHEMA; -- action_schema
```

5 Application context schema

5.1 General

The subject of the **application_context_schema** is the applicable usages of product data. This schema provides a mechanism for application protocols to define a frame of reference or context that applies to particular sets of product data.

This schema also provides a means to identify application protocols.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **application_context_schema** and identifies the necessary external references.

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Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA application_context_schema '{iso standard 10303 part(41) version(8) object(1)
application_context_schema(2)}';

REFERENCE FROM basic_attribute_schema
  (description_attribute,
   description_attribute_select,
   get_description_value,
   get_id_value,
   id_attribute,
   id_attribute_select);
REFERENCE FROM date_time_schema
  (year_number);
REFERENCE FROM language_schema
  (language);
REFERENCE FROM management_resources_schema
  (identification_assignment);
REFERENCE FROM person_organization_schema
  (organization);
REFERENCE FROM support_resource_schema
  (identifier,
   label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

basic_attribute_schema	ISO 10303-41
date_time_schema	ISO 10303-41
language_schema	ISO 10303-41
management_resources_schema	ISO 10303-41
person_organization_schema	ISO 10303-41
support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 3 for a graphical representation of this schema.

5.2 Fundamental concepts and assumptions

The meaningful exchange of product data requires the identification of the application context in which that product data is defined. An application context may have several context elements. Each context element may be referenced by a different set of product data.

5.3 Application context type definitions

5.3.1 acs_description_attribute_select

The **acs_description_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE acs_description_attribute_select = SELECT BASED_ON description_attribute_select WITH
  (application_context);
```

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```
END_TYPE;
```

5.3.2 acs_id_attribute_select

The **acs_id_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE acs_id_attribute_select = SELECT BASED_ON id_attribute_select WITH
  (application_context);
END_TYPE;
```

5.3.3 exchange_identification_context_select

The **exchange_identification_context_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

The **exchange_identification_context_select** type allows for the designation of the data types identification_assignment and organization .

EXPRESS specification:

```
TYPE exchange_identification_context_select = SELECT
  (identification_assignment,
   organization);
END_TYPE;
```

5.4 Application context entity definitions

5.4.1 application_context

An **application_context**, as defined in ISO 10303-1, is a context in which product data is defined and has meaning. An **application_context** represents various types of information that relate to product data and may affect the meaning and usage of that data.

EXPRESS specification:

```
ENTITY application_context;
  application : label;
DERIVE
  description : text := get_description_value(SELF);
  id : identifier := get_id_value(SELF);
INVERSE
  context_elements : SET [1:?] OF application_context_element FOR frame_of_reference;
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
  WR2: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ID_ATTRIBUTE.IDENTIFIED_ITEM'))
<= 1;
END_ENTITY;
```

Attribute definitions:

application: the label by which the **application_context** is known.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **application_context**.

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NOTE 2 The description attribute aids in defining the usage of product data in an **application_context**.

NOTE 3 This attribute is an upwardly compatible addition to **application_context** as specified in ISO 10303-41:1994.

id: the identifier that distinguishes the **application_context**.

NOTE 4 This attribute is an upwardly compatible addition to **application_context** as specified in ISO 10303-41:1994.

NOTE 5 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

context_elements: the set of instances of the entity data type application_context_element that defines the various aspects of the **application_context**.

Formal propositions:

WR1: Each **application_context** shall be the described_item in at most one description_attribute.

NOTE 6 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

WR2: Each **application_context** shall be the identified_item in at most one id_attribute.

NOTE 7 The id_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 8 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

5.4.2 application_context_element 

An **application_context_element** is an aspect of an application context that contains information about the context in which product data is defined. This aspect contains information about the conditions in which the product data exists.

An **application_context_element** may be either a library_context or a product_concept_context, or a product_context, or a product_definition_context.

EXPRESS specification:

```
ENTITY application_context_element
  SUPERTYPE OF (ONEOF(library_context, product_concept_context, product_context, product_
definition_context));
  name : label;
  frame_of_reference : application_context;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **application_context_element** is known.

EXAMPLE 'Functional definition', 'physical definition', 'usage occurrence', and 'manufacturable definition' are examples of names.

frame_of_reference: the application_context of which this **application_context_element** is a member.

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5.4.3 application_context_relationship

An **application_context_relationship** is a relationship between two instances of application_context and provides a description of their relationship.

NOTE 1 The role of **application_context_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 Without any further specialization, this relationship does not imply that the product data associated with **relating_context** are also associated with the **related_context**.

NOTE 3 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 4 This entity, together with the application_context entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY application_context_relationship;
  name : label;
  description : OPTIONAL text;
  relating_context : application_context;
  related_context : application_context;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **application_context_relationship** is known.

description: the text that characterizes the **application_context_relationship**. The value of the attribute need not be specified.

relating_context: one of the instances of application_context that is a part of the relationship.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_context: the other instance of application_context that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 6 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

5.4.4 application_protocol_definition

An **application_protocol_definition** is the identification of an application protocol.

EXPRESS specification:

```
ENTITY application_protocol_definition;
  status : label;
  application_interpreted_model_schema_name : label;
  application_protocol_year : year_number;
  application : application_context;
END_ENTITY;
```

Attribute definitions:

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status: the label that provides a user interpretable designation for the level of completion of the application protocol.

EXAMPLE "IS", "DIS", "TS" or "CD" are examples of application protocol states.

application_interpreted_model_schema_name: the EXPRESS schema name of the application interpreted model.

application_protocol_year: the year when the application protocol attained the status given by the **status** attribute.

application: the application context of the application protocol.

5.4.5 library_context



A **library_context** is a type of application_context_element that defines a context for elements of a library.

The inherited attribute name identifies or characterizes the type of library.

NOTE The name attribute can represent the nature of the source of the library, the nature of the contents of the library, or both.

EXAMPLE 1 'Dewey Decimal', 'ISO 13584 library', and 'supplier catalogue' are examples of type of library.

EXPRESS specification:

```
ENTITY library_context
  SUBTYPE OF (application_context_element);
  library_reference : label;
END_ENTITY;
```

Attribute definitions:

library_reference: the label that identifies the library.

EXAMPLE 2 'ACME self-propelled rocket sleds catalogue' is an example of a library reference.

5.4.6 product_concept_context



A **product_concept_context** is a type of application_context_element that defines a context for a product_concept (see ISO 10303-44).

A **product_concept_context** represents information that relates to the characterization of potential purchasers of a product. Such information may affect the meaning and usage of the product data.

EXPRESS specification:

```
ENTITY product_concept_context
  SUBTYPE OF (application_context_element);
  market_segment_type : label;
END_ENTITY;
```

Attribute definitions:

market_segment_type: the label that identifies the kind of consumer preferences associated with a product.

EXAMPLE 'Luxury automobiles', 'laptop personal computers', and 'budget personal stereos' are examples of *market_segment_type*s.

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5.4.7 product_context

A **product_context** is a type of application_context_element that represents life cycle independent information about a product. This information describes the discipline in which data about the product are created.

NOTE 1 Different annotated EXPRESS schemas that use or specialize this entity data type can make use of common values for the attributes of the **product_context** entity data type in order to establish consistency among schemas that relate to the same types or kinds of products.

EXAMPLE 1 Two annotated EXPRESS schemas that specify product data structures for design and construction of buildings respectively can each constrain the values of the name attribute to be 'building', and the **discipline_type** attribute to be 'architecture'.

The inherited attribute name identifies a particular industry, subject area, or kind of product for which product data is created or used.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

EXAMPLE 2 'ship', 'aircraft' and 'building' are examples of names of product contexts.

EXPRESS specification:

```
ENTITY product_context
  SUBTYPE OF (application_context_element);
  discipline_type : label;
END_ENTITY;
```

Attribute definitions:

discipline_type: the label that identifies a particular field of practice involved in creating or using product data.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

EXAMPLE 3 'electronics', 'engineering', or 'architecture' are examples of discipline types.

5.4.8 product_definition_context

A **product_definition_context** is a type of application_context_element that represents information about the stage in the product life cycle for which a product_definition is created or used.

NOTE 1 Different annotated EXPRESS schemas that use or specialize this entity data type can make use of common values for the attributes of the **product_definition_context** entity data type in order to maintain consistency between schemas that relate to the same or related life-cycle phases.

EXAMPLE 1 Two annotated EXPRESS schemas that specify product data structures for conceptual design and detailed design of process plants respectively can each constrain the values of the name attribute to be 'conceptual design' and 'detailed design', and the **life_cycle_stage** attribute to be 'design'.

The inherited attribute name identifies a subdivision or characterization of a life cycle stage for which product data is created or used.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

EXAMPLE 2 'required' and 'planned' are examples of names of product definition contexts, used in conjunction with a life cycle stage of 'thermal analysis'.

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EXPRESS specification:

```
ENTITY product_definition_context
  SUBTYPE OF (application_context_element);
  life_cycle_stage : label;
END_ENTITY;
```

Attribute definitions:

life_cycle_stage: the label that identifies a stage in the life cycle of a product.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

EXAMPLE 3 A 'preliminary design' and 'manufacturing planning' are examples of life cycle stages.

5.4.9 dependent_product_definition_exchange_context

A **dependent_product_definition_exchange_context** is a type of product_definition_context and is an abstract entity that describes a specific context for exchange.

EXPRESS specification:

```
ENTITY dependent_product_definition_exchange_context
  ABSTRACT SUPERTYPE
  SUBTYPE OF (product_definition_context);
WHERE
  WR1: SIZEOF(TYPEOF(SELF) * ['APPLICATION_CONTEXT_SCHEMA.INDENTIFICATION_DEPENDENT_
PRODUCT_DEFINITION_EXCHANGE_CONTEXT', 'APPLICATION_CONTEXT_SCHEMA.LANGUAGE_DEPENDENT_
PRODUCT_DEFINITION_EXCHANGE_CONTEXT']) > 0;
END_ENTITY;
```

Formal propositions:

WR1: at least one of the entities identification_dependent_product_definition_exchange_context or language_dependent_product_definition_exchange_context shall be specified.

5.4.10 identification_dependent_product_definition_exchange_context

An **identification_dependent_product_definition_exchange_context** is a type of dependent_product_definition_exchange_context that establishes the context of the product_definition exchange.

EXPRESS specification:

```
ENTITY identification_dependent_product_definition_exchange_context
  SUBTYPE OF (dependent_product_definition_exchange_context);
  identification_context : exchange_identification_context_select;
END_ENTITY;
```

Attribute definitions:

identification_context: specifies the context of the exchange to be either identification_assignment or organization .

5.4.11 language_dependent_product_definition_exchange_context

A **language_dependent_product_definition_exchange_context** is a type of dependent_product_definition_exchange_context that establishes the specified language of the product_definition exchange context.

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EXPRESS specification:

```
ENTITY language_dependent_product_definition_exchange_context
  SUBTYPE OF (dependent_product_definition_exchange_context);
  language : language;
END_ENTITY;
```

Attribute definitions:

language: specifies the default language of the exchange context.

```
END_SCHEMA; -- application_context_schema
```

6 Approval schema

6.1 General

The subject of the **approval_schema** is the description of authorization data. Approval information concerns the acceptance of product data.

EXAMPLE One version of a product may be approved for manufacture whilst another may still be undergoing design.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **approval_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA approval_schema '{iso standard 10303 part(41) version(7) object(1) approval_
schema(3)}';

REFERENCE FROM basic_attribute_schema
  (description_attribute,
   description_attribute_select,
   get_description_value,
   get_role,
   object_role,
   role_association,
   role_select);
REFERENCE FROM date_time_schema
  (date_time_select);
REFERENCE FROM person_organization_schema
  (person_organization_select);
REFERENCE FROM support_resource_schema
  (bag_to_set,
   label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

basic_attribute_schema	ISO 10303-41
date_time_schema	ISO 10303-41
person_organization_schema	ISO 10303-41
support_resource_schema	ISO 10303-41

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NOTE 2 See Annex D, Paragraph 4 for a graphical representation of this schema.

6.2 Fundamental concepts and assumptions

Approval information can be attached to any aspect of product data.

6.3 Approval type definitions

6.3.1 apps_description_attribute_select

The **apps_description_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE apps_description_attribute_select = SELECT BASED_ON description_attribute_select WITH
  (approval_role);
END_TYPE;
```

6.3.2 apps_role_select

The **apps_role_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE apps_role_select = SELECT BASED_ON role_select WITH
  (approval_date_time);
END_TYPE;
```

6.4 Approval entity definitions

6.4.1 approval

An **approval** is a confirmation of the product data for its intended use. An **approval** is an incompletely modelled concept and requires population of related entity data types to form a complete approval.

NOTE The role of **approval** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

EXPRESS specification:

```
ENTITY approval;
  status : approval_status;
  level : label;
END_ENTITY;
```

Attribute definitions:

status: the type or level of **approval** in terms of its usage.

EXAMPLE One possible status is 'released for production'; this explicitly identifies the approved usage. Another possible status is 'preliminary design completed'; this only implies the approved usage that will depend upon company-specific procedures.

level: the label that provides a user interpretable designation for the level of completion of the approval.

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6.4.2 approval_date_time

An **approval_date_time** is an association between an approval and a date, time, or date and time.

EXPRESS specification:

```
ENTITY approval_date_time;
  date_time : date_time_select;
  dated_approval : approval;
DERIVE
  role : object_role := get_role(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

date_time: the moment when the **dated_approval** is given.

dated_approval: the approval with which the date or time is associated.

role: the object_role that specifies the purpose of the association of the **approval_date_time** with product data.

NOTE 1 This attribute is an upwardly compatible addition to **approval_date_time** as specified in ISO 10303-41:1994.

Formal propositions:

WR1: Each **approval_date_time** shall be the item_with_role in at most one role_association.

NOTE 2 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

6.4.3 approval_person_organization

An **approval_person_organization** is an association between an approval and a given person, organization, or person in the context of an organization.

EXPRESS specification:

```
ENTITY approval_person_organization;
  person_organization : person_organization_select;
  authorized_approval : approval;
  role : approval_role;
END_ENTITY;
```

Attribute definitions:

person_organization: the person or organization playing the given role.

authorized_approval: the approval that is affected by the person or organization.

role: the approval_role that specifies the purpose of the association of the approval with a person or organization.

6.4.4 approval_relationship

An **approval_relationship** is a relationship between two instances of the entity data type approval with a description of their relationship.

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NOTE 1 The role of **approval_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 This entity, together with the approval entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY approval_relationship;
  name : label;
  description : OPTIONAL text;
  relating_approval : approval;
  related_approval : approval;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **approval_relationship** is known.

description: the text that characterizes the **approval_relationship**. The value of the attribute need not be specified.

relating_approval: one of the instances of approval that is a part of the relationship.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_approval: the other instance of approval that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

6.4.5 approval_role 

An **approval_role** is a definition of a function. An **approval_role** is an incompletely modelled concept and requires population of related entity data types to form a complete confirmation.

EXPRESS specification:

```
ENTITY approval_role;
  role : label;
DERIVE
  description : text := get_description_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

role: the label by which the **approval_role** is known.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **approval_role**.

NOTE 2 This attribute is an upwardly compatible addition to **approval_role** as specified in ISO 10303-41:1994.

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Formal propositions:

WR1: Each **approval_role** shall be the described_item in at most one description_attribute.

NOTE 3 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 4 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

6.4.6 approval_status 

An **approval_status** specifies the state of an approval.

EXAMPLE 'Approved' and 'disapproved' are examples of **approval_status**.

EXPRESS specification:

```
ENTITY approval_status;
  name : label;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **approval_status** is known.

NOTE The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

6.5 Approval function definitions

6.5.1 acyclic_approval_relationship

The **acyclic_approval_relationship** function determines whether the graph of instances of the entity data type approval that contains relation as one of its links contains a cycle. This function may be used to evaluate either a approval_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the approval_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_approval_relationship(relation : approval_relationship;
                                       relatives : SET [1:?] OF approval;
                                       specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF approval_relationship;
  END_LOCAL;
  IF relation.relating_approval IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(app <* bag_to_set(USEDIN(relation.relating_approval, ('APPROVAL_SCHEMA.' +
'APPROVAL_RELATIONSHIP.') + 'RELATED_APPROVAL')) | specific_relation IN TYPEOF(app));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_approval_relationship(x[i], relatives + relation.relating_approval,
specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
```

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```
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate approval_relationship to be checked.

relatives: (input) the set of instances of the entity data type approval for which the function is searching in the relating_approval parameter of the relation argument.

specific_relation: (input) the fully qualified name of a subtype of the approval_relationship entity.

```
END_SCHEMA; -- approval_schema
```

7 Basic attribute schema

7.1 General

The subject of the **basic_attribute_schema** is the assignment of id, name, description and role attributes to aspects of product data. This schema defines the mechanism to define a value for these attributes and associate it with product data.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **basic_attribute_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA basic_attribute_schema '{iso standard 10303 part(41) version(9) object(1) basic_
attribute_schema(4)}';

REFERENCE FROM support_resource_schema
  (identifier,
   label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 5 for a graphical representation of this schema.

7.2 Fundamental concepts and assumptions

The following concepts are supported by this schema:

— the assignment of an identifier to product data;

— the assignment of a label to product data;

— the assignment of a descriptive text to product data;

— the assignment of a role to an association of management type data with other aspects of product data.

The assignment of an identifier corresponds to the **id_attribute** entity data type. The entities to which an identifier may be assigned are specified in the **id_attribute_select** data type. These entities have a derived attribute **id**. The corresponding identifier value is provided by the **get_id_value** function. A constraint

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ensures that there is at most one instance of **id_attribute** assigned to an instance of an entity in the **id_attribute_select** type.

The assignment of a name corresponds to the **name_attribute** entity data type. The entities to which a name may be assigned are specified in the **name_attribute_select** data type. These entities have a derived attribute name. The corresponding label value is provided by the **get_name_value** function. A constraint ensures that there is at most one instance of **name_attribute** assigned to an instance of an entity in the **name_attribute_select** type.

The assignment of a descriptive text corresponds to the **description_attribute** entity data type. The entities to which a description may be assigned are specified in the **description_attribute_select** data type. These entities have a derived attribute description. The corresponding text value is provided by the **get_description_value** function. A constraint ensures that there is at most one instance of **description_attribute** assigned to an instance of an entity in the **description_attribute_select** type.

The assignment of a role corresponds to the **role_association** entity data type. The entities to which a role may be assigned are specified in the **role_select** data type. These entities have a derived attribute role. The corresponding **object_role** is provided by the **get_role** function. A constraint ensures that there is at most one instance of **role_association** assigned to an instance of an entity in the **role_select** type.

7.3 Basic attribute type definitions

7.3.1 description_attribute_select

The **description_attribute_select** type is an extensible list of alternate data types. It provides a mechanism to refer to instances of the data types included in the types that extend the **description_attribute_select** type.

NOTE This empty extensible select requires extension in a further schema to ensure that entities that refer to it have at least one valid instantiation.

A **description_attribute_select** specifies those objects that may be assigned a descriptive text.

EXPRESS specification:

```
TYPE description_attribute_select = EXTENSIBLE GENERIC_ENTITY SELECT;
END_TYPE;
```

7.3.2 generic_attribute_list_select

The **generic_attribute_list_select** type is an extensible list of alternate data types. It provides a mechanism to refer to instances of the data types included in the types that extend the **generic_attribute_list_select** type.

NOTE This empty extensible select requires extension in a further schema to ensure that entities that refer to it have at least one valid instantiation.

A **generic_attribute_list_select** specifies those objects that may be assigned a generic attribute list.

EXPRESS specification:

```
TYPE generic_attribute_list_select = EXTENSIBLE GENERIC_ENTITY SELECT;
END_TYPE;
```

7.3.3 generic_attribute_select

The **generic_attribute_select** type is an extensible list of alternate data types. It provides a mechanism to refer to instances of the data types included in the types that extend the **generic_attribute_select** type.

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NOTE This empty extensible select requires extension in a further schema to ensure that entities that refer to it have at least one valid instantiation.

A **generic_attribute_select** specifies those objects that may be assigned a generic attribute.

EXPRESS specification:

```
TYPE generic_attribute_select = EXTENSIBLE GENERIC_ENTITY SELECT;
END_TYPE;
```

7.3.4 id_attribute_select 

The **id_attribute_select** type is an extensible list of alternate data types. It provides a mechanism to refer to instances of the data types included in the types that extend the **id_attribute_select** type.

NOTE This empty extensible select requires extension in a further schema to ensure that entities that refer to it have at least one valid instantiation.

An **id_attribute_select** specifies those objects that may be assigned an identifier.

EXPRESS specification:

```
TYPE id_attribute_select = EXTENSIBLE GENERIC_ENTITY SELECT;
END_TYPE;
```

7.3.5 name_attribute_select 

The **name_attribute_select** type is an extensible list of alternate data types. It provides a mechanism to refer to instances of the data types included in the types that extend the **name_attribute_select** type.

NOTE This empty extensible select requires extension in a further schema to ensure that entities that refer to it have at least one valid instantiation.

A **name_attribute_select** specifies those objects that may be assigned a name.

EXPRESS specification:

```
TYPE name_attribute_select = EXTENSIBLE GENERIC_ENTITY SELECT;
END_TYPE;
```

7.3.6 role_select 

The **role_select** type is an extensible list of alternate data types. It provides a mechanism to refer to instances of the data types included in the types that extend the **role_select** type.

NOTE This empty extensible select requires extension in a further schema to ensure that entities that refer to it have at least one valid instantiation.

A **role_select** specifies those objects that may be assigned a role.

EXPRESS specification:

```
TYPE role_select = EXTENSIBLE GENERIC_ENTITY SELECT;
END_TYPE;
```

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7.4 Basic attribute entity definitions

7.4.1 aggregate_id_attribute

An **aggregate_id_attribute** is used for the case where there is more than one product data element for a specific identifier.

EXPRESS specification:

```
ENTITY aggregate_id_attribute;
  attribute_value : identifier;
  identified_item : SET [1:?] OF id_attribute_select;
END_ENTITY;
```

Attribute definitions:

attribute_value: the identifier that distinguishes the **identified_item**.

identified_item: the collection of items that are identified.

7.4.2 description_attribute

A **description_attribute** is the assignment of a descriptive text string to product data.

EXPRESS specification:

```
ENTITY description_attribute;
  attribute_value : text;
  described_item : description_attribute_select;
END_ENTITY;
```

Attribute definitions:

attribute_value: the text that characterizes the **described_item**.

described_item: the item that is described.

7.4.3 generic_attribute

A **generic_attribute** is the declaration of a named attribute as a text string to enable it to be associated with product data. This capability shall not be used to replace more specific assignment capabilities if both are available to a specified target.

EXPRESS specification:

```
ENTITY generic_attribute;
  attribute_name : label;
  attribute_value : text;
END_ENTITY;
```

Attribute definitions:

attribute_name: the label used to name the **generic_attribute**.

attribute_value: the text that characterizes the **generic_attribute**.

7.4.4 generic_attribute_assignment

A **generic_attribute_assignment** is the assignment of a generic attribute to element of product data.

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EXPRESS specification:

```
ENTITY generic_attribute_assignment;
  attribute : generic_attribute;
  item_with_attribute : generic_attribute_select;
END_ENTITY;
```

Attribute definitions:

attribute: the generic_attribute to be associated with the product data.

item_with_attribute: the item to which the attribute is assigned.

7.4.5 generic_attribute_list_assignment

A **generic_attribute_list_assignment** is the assignment of a named list of generic attributes to element of product data.

EXPRESS specification:

```
ENTITY generic_attribute_list_assignment;
  attribute_name : label;
  attributes : LIST [1:?] OF generic_attribute;
  item_with_attribute : generic_attribute_list_select;
END_ENTITY;
```

Attribute definitions:

attribute_name: the label used to name the list of generic attributes.

attributes: the list of generic_attributes that are assigned to the element of product data.

item_with_attribute: the item to which the attributes are assigned.

7.4.6 id_attribute

An **id_attribute** is the assignment of an identifier to product data.

EXPRESS specification:

```
ENTITY id_attribute;
  attribute_value : identifier;
  identified_item : id_attribute_select;
END_ENTITY;
```

Attribute definitions:

attribute_value: the identifier that distinguishes the **identified_item**.

identified_item: the item that is identified.

7.4.7 name_attribute

A **name_attribute** is the assignment of a label by which the product data is known.

EXPRESS specification:

```
ENTITY name_attribute;
  attribute_value : label;
  named_item : name_attribute_select;
END_ENTITY;
```


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Attribute definitions:

attribute_value: the label by which the **named_item** is known.

named_item: the item to which the name is applied.

7.4.8 object_role

An **object_role** is a specification of a role for the association of management type data with other aspects of product data and a description of that role.

EXPRESS specification:

```
ENTITY object_role;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **object_role** is known.

description: the text that characterizes the **object_role**. The value of the attribute need not be specified.

7.4.9 role_association

A **role_association** is the assignment of an object_role to an association of management type data with other aspects of product data.

EXPRESS specification:

```
ENTITY role_association;
  role : object_role;
  item_with_role : role_select;
END_ENTITY;
```

Attribute definitions:

role: the object_role that specifies the purpose of the association of the **role_association** with product data.

item_with_role: the item to which a role is assigned.

7.5 Basic attribute function definitions

7.5.1 get_description_value

The **get_description_value** function accepts an object to which a descriptive text string may be assigned and returns this description if present.

If a description_attribute references the object, the function will return the value of the attribute_value attribute of this description_attribute. Otherwise, the indeterminate value will be returned.

EXPRESS specification:

```
FUNCTION get_description_value(obj : description_attribute_select) : text;
  LOCAL
    description_bag : BAG OF description_attribute := USEDIN(obj, ('BASIC_ATTRIBUTE_
SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.') + 'DESCRIBED_ITEM');
  END_LOCAL;
  IF SIZEOF(description_bag) = 1 THEN
    RETURN (description_bag[1].attribute_value);
```

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```
  ELSE
    RETURN (?);
  END_IF;
END_FUNCTION;
```

Argument definitions:

obj: (input) the object for which a text is determined. This is an input to the function.

7.5.2 get_id_value

The **get_id_value** function accepts an object to which an identifier may be assigned and returns this identifier if present.

If an id_attribute references the object, the function will return the value of the attribute_value attribute of this id_attribute. Otherwise, the indeterminate value will be returned.

EXPRESS specification:

```
FUNCTION get_id_value(obj : id_attribute_select) : identifier;
  LOCAL
    id_bag : BAG OF id_attribute := USEDIN(obj, ('BASIC_ATTRIBUTE_SCHEMA.' + 'ID_
ATTRIBUTE.') + 'IDENTIFIED_ITEM');
  END_LOCAL;
  IF SIZEOF(id_bag) = 1 THEN
    RETURN (id_bag[1].attribute_value);
  ELSE
    RETURN (?);
  END_IF;
END_FUNCTION;
```

Argument definitions:

obj: (input) the object for which an identifier is determined. This is an input to the function.

7.5.3 get_aggregate_id_value

The **get_aggregate_id_value** function accepts an object to which an identifier may be assigned and returns this identifier if present.

If an aggregate_id_attribute references the object, the function will return the value of the attribute_value attribute of this aggregate_id_attribute. Otherwise, the indeterminate value will be returned.

EXPRESS specification:

```
FUNCTION get_aggregate_id_value(obj : id_attribute_select) : identifier;
  LOCAL
    aggregate_id_bag : BAG OF aggregate_id_attribute := USEDIN(obj, ('BASIC_ATTRIBUTE_
SCHEMA.' + 'AGGREGATE_ID_ATTRIBUTE.') + 'IDENTIFIED_ITEM');
  END_LOCAL;
  IF SIZEOF(aggregate_id_bag) = 1 THEN
    RETURN (aggregate_id_bag[1].attribute_value);
  ELSE
    RETURN (?);
  END_IF;
END_FUNCTION;
```

Argument definitions:

obj: (input) the object for which an identifier is determined. This is an input to the function.

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7.5.4 get_name_value

The **get_name_value** function accepts an object to which a label may be assigned and returns this label if present.

If a name_attribute references the object, the function will return the value of the attribute_value attribute of this name_attribute. Otherwise, the indeterminate value will be returned.

EXPRESS specification:

```
FUNCTION get_name_value(obj : name_attribute_select) : label;
  LOCAL
    name_bag : BAG OF name_attribute := USEDIN(obj, ('BASIC_ATTRIBUTE_SCHEMA.' + 'NAME_
ATTRIBUTE.') + 'NAMED_ITEM');
  END_LOCAL;
  IF SIZEOF(name_bag) = 1 THEN
    RETURN (name_bag[1].attribute_value);
  ELSE
    RETURN (?);
  END_IF;
END_FUNCTION;
```

Argument definitions:

obj: the object for which a label is determined. This is an input to the function.

7.5.5 get_role

The **get_role** function accepts an object to which a role may be assigned and returns this role if present.

If a role_association references the object, the function will return the object_role of the role attribute of this role_association. Otherwise, the indeterminate value will be returned.

EXPRESS specification:

```
FUNCTION get_role(obj : role_select) : object_role;
  LOCAL
    role_bag : BAG OF role_association := USEDIN(obj, ('BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_
ASSOCIATION.') + 'ITEM_WITH_ROLE');
  END_LOCAL;
  IF SIZEOF(role_bag) = 1 THEN
    RETURN (role_bag[1].role);
  ELSE
    RETURN (?);
  END_IF;
END_FUNCTION;
```

Argument definitions:

obj: (input) the object for which an object_role is determined. This is an input to the function.

```
END_SCHEMA; -- basic_attribute_schema
```

8 Certification schema

8.1 General

The subject of the **certification_schema** is the definition of certifications. A certification assures and validates product data.

EXAMPLE 1 A material certificate states the chemical composition of one or more physical pieces of material, for example, purchased raw material. The presence of the material certificate removes the need to test the

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composition of the material; it allows the specified material composition to be accepted as fact without further investigation.

EXAMPLE 2 A certified supplier can supply goods that do not require checking.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **certification_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA certification_schema '{iso standard 10303 part(41) version(3) object(1)
certification_schema(5)}';

REFERENCE FROM support_resource_schema
  (label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 6 for a graphical representation of this schema.

8.2 Fundamental concepts and assumptions

Certification information can be attached to any aspect of product data.

8.3 Certification entity definitions

8.3.1 certification 

A **certification** is an assertion of facts. A **certification** is an incompletely modeled concept and requires population of related entity data types to form a complete certification.

NOTE The role of **certification** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

EXPRESS specification:

```
ENTITY certification;
  name : label;
  purpose : text;
  kind : certification_type;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **certification** is known.

purpose: an informal description of the reason the **certification** is applied.

EXAMPLE A purpose for material certification could be the fact that certain customers demand material certification for the products that are supplied to them.

kind: the certification_type that defines the class of **certification** that is applied.

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8.3.2 certification_type 

A **certification_type** is the kind of certification granted.

EXAMPLE 1 Suppliers and manufacturers can be certified.

EXPRESS specification:

```
ENTITY certification_type;
  description : label;
END_ENTITY;
```

Attribute definitions:

description: the text that characterizes the **certification_type**.

NOTE The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

EXAMPLE 2 A certification type in an example supply chain is 'supplier'.

```
END_SCHEMA; -- certification_schema
```

9 Contract schema

9.1 General

The subject of the **contract_schema** is the description of contract agreements.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **contract_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA contract_schema '{iso standard 10303 part(41) version(3) object(1) contract_
schema(6)}';

REFERENCE FROM support_resource_schema
  (bag_to_set,
   identifier,
   label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 7 for a graphical representation of this schema.

9.2 Fundamental concepts and assumptions

Contract information may be attached to any aspect of a product data.

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9.3 Contract entity definitions

9.3.1 contract

A **contract** is a binding agreement. A **contract** is an incompletely modeled concept and requires population of related entity data types to form a complete contract.

NOTE Contracts may be enforceable by law.

EXPRESS specification:

```
ENTITY contract;
  name : label;
  purpose : text;
  kind : contract_type;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **contract** is known.

purpose: an informal description of the reason for the **contract**.

kind: the **contract**'s type.

9.3.2 contract_relationship

A **contract_relationship** is a relationship between two instances of the entity data type contract.

NOTE 1 The role of **contract_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 3 This entity, together with the contract entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY contract_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating_contract : contract;
  related_contract : contract;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **contract_relationship**.

NOTE 4 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **contract_relationship** is known.

description: the text that characterizes the **contract_relationship**. The value of the attribute need not be specified.

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relating_contract: one of the instances of contract that is a part of the relationship.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_contract: the other instance of contract that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 6 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

9.3.3 contract_type 

A **contract_type** is the kind of information that an instance of contract conveys.

EXAMPLE A contract for an organization that is supplying goods to another organization could be 'fixed' or 'cost plus'.

EXPRESS specification:

```
ENTITY contract_type;
  description : label;
END_ENTITY;
```

Attribute definitions:

description: the text that characterizes the **contract_type**. The value of the attribute need not be specified.

NOTE The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

9.4 Contract function definitions

9.4.1 acyclic_contract_relationship

The **acyclic_contract_relationship** function determines whether the graph of instances of the entity data type contract that contains relation as one of its links contains a cycle. This function may be used to evaluate either a contract_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the contract_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_contract_relationship(relation : contract_relationship;
                                       relatives : SET [1:?] OF contract;
                                       specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF contract_relationship;
  END_LOCAL;
  IF relation.relating_contract IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(cont <* bag_to_set(USEDIN(relation.relating_contract, ('CONTRACT_SCHEMA.' +
'CONTRACT_RELATIONSHIP.') + 'RELATED_CONTRACT')) | specific_relation IN TYPEOF(cont));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_contract_relationship(x[i], relatives + relation.relating_contract,
specific_relation) THEN
      RETURN (FALSE);
```

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```
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate contract_relationship to be checked.

relatives: (input) the set of instances of the entity data type contract for which the function is searching in the relating_contract parameter of the relation argument.

specific_relation: (input) the fully qualified entity name of a type of contract_relationship entity.

```
END_SCHEMA; -- contract_schema
```

10 Date time schema

10.1 General

The subject of the **date_time_schema** is the definition of dates, both calendar and ordinal, time of day, combinations of date and time of day, and periods of time.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **date_time_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA date_time_schema '{iso standard 10303 part(41) version(7) object(1) date_time_
schema(7)}';

REFERENCE FROM basic_attribute_schema
  (description_attribute,
   description_attribute_select,
   get_description_value);
REFERENCE FROM measure_schema
  (measure_with_unit,
   time_measure_with_unit);
REFERENCE FROM support_resource_schema
  (bag_to_set,
   identifier,
   label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

basic_attribute_schema	ISO 10303-41
measure_schema	ISO 10303-41
support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 8 for a graphical representation of this schema.

10.2 Fundamental concepts and assumptions

Any kind of product data may have a date, time, or date and time associated with it.

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10.3 Date time type definitions

10.3.1 dts_description_attribute_select

The **dts_description_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE dts_description_attribute_select = SELECT BASED_ON description_attribute_select WITH
  (date_role,
   date_time_role,
   time_role);
END_TYPE;
```

10.3.2 ahead_or_behind

The **ahead_or_behind** type is used to specify whether a given time is ahead of coordinated universal time, behind coordinated universal time or is coordinated universal time.

NOTE Coordinated Universal Time (UTC) is the international time standard. It is the current term for what was commonly referred to as Greenwich Meridian Time (GMT). Zero (0) hours UTC is midnight in Greenwich England, which lies on the zero longitudinal meridian. Coordinated universal time is based on a 24-hour clock; therefore, afternoon hours such as 4 pm are expressed as 16:00 UTC (sixteen hours, zero minutes).

EXPRESS specification:

```
TYPE ahead_or_behind = ENUMERATION OF
  (ahead,
   exact,
   behind);
END_TYPE;
```

Enumerated item definitions:

ahead a given time is ahead of coordinated universal time;

exact a given time is coordinated universal time;

behind a given time is behind coordinated universal time.

10.3.3 date_time_or_event_occurrence

The **date_time_or_event_occurrence** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

The **date_time_or_event_occurrence** type allows for the designation of a date_time_select or an event_occurrence.

EXPRESS specification:

```
TYPE date_time_or_event_occurrence = SELECT
  (date_time_select,
   event_occurrence);
END_TYPE;
```

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10.3.4 date_time_select

The **date_time_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

A **date_time_select** type allows for the designation of a date, date_and_time, or a local_time.

EXPRESS specification:

```
TYPE date_time_select = SELECT
  (date,
   date_and_time,
   local_time);
END_TYPE;
```

10.3.5 day_in_month_number

A **day_in_month_number** is the position of the specified day in a month.

EXPRESS specification:

```
TYPE day_in_month_number = INTEGER;
WHERE
  WR1: {1 <= SELF <= 31};
END_TYPE;
```

Formal propositions:

WR1: The value of the integer shall be between 1 and 31 inclusive.

10.3.6 day_in_week_number

A **day_in_week_number** is the position of the specified day in a week as defined in ISO 8601.

NOTE Monday is day number 1, Tuesday is day number 2, Wednesday is day number 3, Thursday is day number 4, Friday is day number 5, Saturday is day number 6, and Sunday is day number 7.

EXPRESS specification:

```
TYPE day_in_week_number = INTEGER;
WHERE
  WR1: {1 <= SELF <= 7};
END_TYPE;
```

Formal propositions:

WR1: The value of the integer shall be between 1 and 7 inclusive.

10.3.7 day_in_year_number

A **day_in_year_number** is the position of the specified day in a year.

EXAMPLE The 27th day of March is day 86 in years that are not leap years and day 87 in leap years.

EXPRESS specification:

```
TYPE day_in_year_number = INTEGER;
WHERE
  WR1: {1 <= SELF <= 366};
END_TYPE;
```


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Formal propositions:

WR1: The value of the integer shall be between 1 and 366 inclusive.

10.3.8 hour_in_day

An **hour_in_day** is the hour element of a specified time on a 24-hour clock. Midnight shall be represented by the value zero.

EXAMPLE The **hour_in_day** corresponding to 3 o'clock in the afternoon is 15.

NOTE Although ISO 8601 allows two representations for midnight, 0000 and 2400, this part of ISO 10303 restricts the representation to the first value.

EXPRESS specification:

```
TYPE hour_in_day = INTEGER;
WHERE
  WR1: {0 <= SELF < 24};
END_TYPE;
```

Formal propositions:

WR1: The value of the integer shall be between 0 and 23 inclusive.

10.3.9 minute_in_hour

A **minute_in_hour** is the minute element of a specified time.

EXPRESS specification:

```
TYPE minute_in_hour = INTEGER;
WHERE
  WR1: {0 <= SELF <= 59};
END_TYPE;
```

Formal propositions:

WR1: The value of the integer shall be between 0 and 59 inclusive.

10.3.10 month_in_year_number

A **month_in_year_number** is the position of the specified month in a year as defined in ISO 8601.

NOTE January is month number 1, February is month number 2, March is month number 3, April is month number 4, May is month number 5, June is month number 6, July is month number 7, August is month number 8, September is month number 9, October is month number 10, November is month number 11, and December is month number 12.

EXPRESS specification:

```
TYPE month_in_year_number = INTEGER;
WHERE
  WR1: {1 <= SELF <= 12};
END_TYPE;
```

Formal propositions:

WR1: The value of the integer shall be between 1 and 12 inclusive.

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10.3.11 second_in_minute

A **second_in_minute** is the position of the specified second in a minute.

EXPRESS specification:

```
TYPE second_in_minute = REAL;
WHERE
  WR1: {0 <= SELF < 60.0};
END_TYPE;
```

Formal propositions:

WR1: The value of the real number shall be between 0 to 60.0 inclusive.

NOTE 1 A value of 60 allows for leap seconds.

NOTE 2 The mean solar time is determined by the rotation of the earth. Leap seconds are added as required, usually in the middle or at the end of a year, and ensure that the legal time does not differ from the non-uniform mean solar time by more than one second, in spite of the variations of the earth rotation.

10.3.12 week_in_year_number

A **week_in_year_number** is the position of the specified calendar week as defined in ISO 8601.

NOTE Week number 1 is the week containing the first Thursday of the year. This is equivalent to saying that week number 1 contains the date 4th January.

EXPRESS specification:

```
TYPE week_in_year_number = INTEGER;
WHERE
  WR1: {1 <= SELF <= 53};
END_TYPE;
```

Formal propositions:

WR1: The value of the integer shall be between 1 and 53.

10.3.13 year_number

A **year_number** is the year as defined in the Gregorian calendar. The **year_number** shall be completely and explicitly specified using as many digits as necessary to unambiguously convey the century and year within the century. Truncated year numbers shall not be used.

NOTE ISO 8601:1988 defines the Gregorian calendar.

EXAMPLE The **year_number** corresponding to the first manned landing on the moon is 1969 (not 69).

EXPRESS specification:

```
TYPE year_number = INTEGER;
WHERE
  WR1: SELF > 1581;
END_TYPE;
```

Formal propositions:

WR1: The value of the integer shall be greater than 1581.

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10.4 Date time entity definitions

10.4.1 calendar_date

A **calendar_date** is a type of date defined as a day in a month of a year as defined in ISO 8601.

EXPRESS specification:

```
ENTITY calendar_date
  SUBTYPE OF (date);
  day_component : day_in_month_number;
  month_component : month_in_year_number;
WHERE
  WR1: valid_calendar_date(SELF);
END_ENTITY;
```

Attribute definitions:

day_component: the day element of the date.

month_component: the month element of the date.

Formal propositions:

WR1: The entity shall define a valid calendar date.

NOTE If the **month_component** is 'April', 'June', 'August', or 'November' the **day_component** will be between 1 and 30; when the **month_component** is 'February' and the year_component is a leap year the **day_component** will be between 1 and 29; if the **month_component** is 'February' and the year_component is not a leap year the **day_component** will be between 1 and 28. Otherwise, the **day_component** will be between 1 and 31.

10.4.2 coordinated_universal_time_offset

A **coordinated_universal_time_offset** is the oriented offset (specified in hours and possibly minutes) from the coordinated universal time. The offset is defined in ISO 8601 The offset value shall be positive.

NOTE 1 Coordinated Universal Time (UTC) is the international time standard. It is the current term for what was commonly referred to as Greenwich Meridian Time (GMT). Zero (0) hours UTC is midnight in Greenwich England, which lies on the zero longitudinal meridian. Coordinated universal time is based on a 24-hour clock; therefore, afternoon hours such as 4 pm are expressed as 16:00 UTC (sixteen hours, zero minutes).

NOTE 2 A **coordinated_universal_time_offset** is used to relate a time to coordinated universal time by an offset (specified in hours and minutes) and a direction.

EXPRESS specification:

```
ENTITY coordinated_universal_time_offset;
  hour_offset : INTEGER;
  minute_offset : OPTIONAL INTEGER;
  sense : ahead_or_behind;
DERIVE
  actual_minute_offset : INTEGER := NVL(minute_offset, 0);
WHERE
  WR1: {0 <= hour_offset < 24};
  WR2: {0 <= actual_minute_offset <= 59};
  WR3: NOT (((hour_offset <> 0) OR (actual_minute_offset <> 0)) AND (sense = exact));
END_ENTITY;
```

Attribute definitions:

hour_offset: the number of hours by which a time is offset from coordinated universal time.

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minute_offset: the number of minutes by which a time is offset from coordinated universal time.

sense: the direction of the offset.

actual_minute_offset: the value of the number of minutes offset used to compute the **coordinated_universal_time_offset**, either the value of **minute_offset** or 0.

Formal propositions:

WR1: The **hour_offset** shall be a positive number, less than 24.

WR2: The **minute_offset** shall be a positive number, less than or equal to 59.

WR3: If the value of **sense** specifies that there is no offset from the Coordinated Universal time, **hour_offset** and **actual_minute_offset** shall both be equal to zero. If either **hour_offset** or **actual_minute_offset** is different from zero, the value of sense shall specify that there is an offset, either ahead or behind, from the Coordinated Universal time.

10.4.3 date

A **date** is the identification of a day or week or month in a year. A **date** may be a calendar_date or an ordinal_date, or a week_of_year_and_day_date, or a year_month. The date is defined in ISO 8601.

EXPRESS specification:

```
ENTITY date
  SUPERTYPE OF (ONEOF(calendar_date, ordinal_date, week_of_year_and_day_date, year_
month));
  year_component : year_number;
END_ENTITY;
```

Attribute definitions:

year_component: the year in which the **date** occurs.

10.4.4 date_and_time

A **date_and_time** is a moment of time on a particular day.

EXPRESS specification:

```
ENTITY date_and_time;
  date_component : date;
  time_component : local_time;
END_ENTITY;
```

Attribute definitions:

date_component: the date element of the date time combination.

time_component: the time element of the date time combination.

10.4.5 date_role

A **date_role** is a role for a date A **date_role** is an incompletely modeled concept and requires population related entity data types to form a complete **date_role**.

NOTE 1 A **date_role** is a role for a date_assignment and a description of that role.

EXPRESS specification:

```
ENTITY date_role;
```

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```
  name : label;
DERIVE
  description : text := get_description_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **date_role** is known.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **date_role**. The value of the attribute need not be specified.

NOTE 3 This attribute is an upwardly compatible addition to **date_role** as specified in ISO 10303-41:1994.

Formal propositions:

WR1: Each **date_role** shall be the described_item in at most one description_attribute.

NOTE 4 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 5 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

10.4.6 date_time_role 

A **date_time_role** is a role for a moment of time on a particular day. A **date_time_role** is an incompletely modeled concept that requires population of related entity data types to form a complete **date_time_role**

NOTE 1 A **date_time_role** is a role for a date_and_time_assignment and a description of that role.

EXPRESS specification:

```
ENTITY date_time_role;
  name : label;
DERIVE
  description : text := get_description_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **date_time_role** is known.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **date_time_role**.

NOTE 3 This attribute is an upwardly compatible addition to **date_time_role** as specified in ISO 10303-41:1994.

Formal propositions:

WR1: Each **date_time_role** shall be the described_item in at most one description_attribute.

NOTE 4 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

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NOTE 5 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

10.4.7 event_occurrence

An **event_occurrence** is the fact of a transition between states at some point in time. The description of the **event_occurrrnece** can provide an implicit representation of either the state preceeding the **event_occurrence** or the state after the **event_occurrence**.

NOTE 1 The point in time of the existence may not be known in terms of a calendar date before the **event_occurrence** actually happens. One reason, why the **event_occurrence** cannot always be expressed as a calendar date, is that the **event_occurrence** is not possible to plan, for example a break down of a machine.

EXAMPLE The **event_occurrence** 'start of production' implicitly declares that the before the **event_occurrence** a product was not in production.

EXPRESS specification:

```
ENTITY event_occurrence;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **event_occurrence**.

NOTE 2 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **event_occurrence** is known.

description: the text that characterizes the **event_occurrence**. The value of the attribute need not be specified.

10.4.8 event_occurrence_context_role

An **event_occurrence_context_role** establishes a context for an event_occurrence_assignment. An **event_occurrence** is an incompletely modeled concept and requires population of related entity data types to form a complete **event_occurrence_context_role**

EXAMPLE For the event_occurrence 'start of production' the product, for which production starts, is the context for that event_occurrence.

EXPRESS specification:

```
ENTITY event_occurrence_context_role;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **event_occurrence_context_role** is known.

description: the text that characterizes the **event_occurrence_context_role**. The value of the attribute need not be specified.

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10.4.9 event_occurrence_relationship

An **event_occurrence_relationship** is a relationship between two event_occurrences with a description of their relationship.

NOTE 1 The role of **event_occurrence_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 This entity, together with the event_occurrence entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY event_occurrence_relationship;
  name : label;
  description : OPTIONAL text;
  relating_event : event_occurrence;
  related_event : event_occurrence;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **event_occurrence_relationship** is known.

description: the text that characterizes the **event_occurrence_relationship**. The value of the attribute need not be specified.

relating_event: one of the instances of event_occurrences that is a part of the relationship.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_event: the other instance of event_occurrence. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

10.4.10 event_occurrence_role

An **event_occurrence_role** establishes a role for an **event_occurrence**. An **event_occurrence_role** is an incompletely modeled concept and requires population of related entity data types to form a complete **event_occurrence_role**

EXPRESS specification:

```
ENTITY event_occurrence_role;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **event_occurrence_role** is known.

description: the text that characterizes the **event_occurrence_role**. The value of the attribute need not be specified.

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10.4.11 local_time

A **local_time** is an instance of time represented on a 24-hour clock by hour, minute, and second. The instance is expressed in the local time zone and the offset with the coordinate universal time shall be specified. This concept is defined in ISO 8601.

NOTE This construct is used to represent a moment of time whereas time_measure represents amounts of time.

EXPRESS specification:

```
ENTITY local_time;
  hour_component : hour_in_day;
  minute_component : OPTIONAL minute_in_hour;
  second_component : OPTIONAL second_in_minute;
  zone : coordinated_universal_time_offset;
WHERE
  WR1: valid_time(SELF);
END_ENTITY;
```

Attribute definitions:

hour_component: the number of hours.

minute_component: the number of minutes. The value of the attribute need not be specified.

second_component: the number of seconds. The value of the attribute need not be specified.

zone: the offset of the local time zone to the coordinated universal time.

Formal propositions:

WR1: The entity shall define a valid time. The **second_component** attribute shall only exist if the **minute_component** attribute exists.

10.4.12 ordinal_date

An **ordinal_date** is a type of date defined as a day of a year. This concept is defined in ISO 8601.

EXPRESS specification:

```
ENTITY ordinal_date
  SUBTYPE OF (date);
  day_component : day_in_year_number;
WHERE
  WR1: NOT leap_year(SELF.year_component) AND {1 <= day_component <= 365} OR leap_
year(SELF.year_component) AND {1 <= day_component <= 366};
END_ENTITY;
```

Attribute definitions:

day_component: the day element of the date.

Formal propositions:

WR1: The **day_component** shall be between 1 and 365 if the year_component is not a leap year. Otherwise the **day_component** shall be between 1 and 366.

10.4.13 relative_event_occurrence

A **relative_event_occurrence** is a type of event_occurrence that takes place with an offset to an event_occurrence.

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EXAMPLE 'Five days after start of production' is an example for a **relative_event_occurrence**, where 'five days' is an offset relative to an event_occurrence 'start of production'.

EXPRESS specification:

```
ENTITY relative_event_occurrence
  SUBTYPE OF (event_occurrence);
  base_event : event_occurrence;
  offset : time_measure_with_unit;
END_ENTITY;
```

Attribute definitions:

base_event: the **relative_event_occurrence** is defined with respect to the **base_event**.

offset: the amount of time that passes between the occurrence of the **base_event** and the **relative_event_occurrence**. A negative offset specifies that the **relative_event_occurrence** occurs before the **base_event**.

10.4.14 time_interval

A **time_interval** is the identification of an interval of time. This concept is defined in ISO 8601.

EXAMPLE 'Strike duration', 'delay of production', and 'Christmas holidays' are examples for *time_interval*s.

EXPRESS specification:

```
ENTITY time_interval;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **time_interval**.

NOTE The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **time_interval** is known.

description: the text that characterizes the **time_interval**. The value of the attribute need not be specified.

10.4.15 time_interval_relationship

A **time_interval_relationship** is a relationship between two time_intervals with a description of their relationship.

NOTE 1 The role of **time_interval_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 This entity, together with the time_interval entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY time_interval_relationship;
  name : label;
  description : OPTIONAL text;
```

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```
  relating_time_interval : time_interval;
  related_time_interval : time_interval;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **time_interval_relationship** is known.

description: the text that characterizes the **time_interval_relationship**. The value of the attribute need not be specified.

relating_time_interval: one of the instances of time_interval that is a part of the relationship.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_time_interval: the other instance of time_interval. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

10.4.16 time_interval_role

A **time_interval_role** is a role that may be assigned to a **time_interval** and includes a description of that role.

NOTE 1 The assignment of a role to a **time_interval** is accomplished by the **time_interval_assignment** entity data type.

EXPRESS specification:

```
ENTITY time_interval_role;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **time_interval_role** is known.

description: the text that characterizes the **time_interval_role**. The value of the attribute need not be specified.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

10.4.17 time_interval_with_bounds

A **time_interval_with_bounds** is a type of time_interval. The **time_interval_with_bounds** is bounded either on one side or both sides. If neither **secondary_bound** nor **duration** are specified, the time interval begins at the point in time identified by **primary_bound** and has no specified end point.

EXAMPLE 'From 01-01-1999 to 31-12-1999' is an example for a both side bounded **time_interval_with_bounds**, 'from 01-01-2000 on', and 'until start of production' are examples for a one side bounded **time_interval_with_bounds**.

EXPRESS specification:

```
ENTITY time_interval_with_bounds
  SUBTYPE OF (time_interval);
  primary_bound : OPTIONAL date_time_or_event_occurrence;
```

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```
  secondary_bound : OPTIONAL date_time_or_event_occurrence;
  duration : OPTIONAL time_measure_with_unit;
WHERE
  WR1: EXISTS(primary_bound) OR EXISTS(secondary_bound);
  WR2: NOT ((EXISTS(primary_bound) AND EXISTS(secondary_bound)) AND EXISTS(duration));
  WR3: (EXISTS(primary_bound) AND NOT EXISTS(secondary_bound)) AND (NOT EXISTS(duration)
OR duration_is_positive(duration));
  WR4: ((EXISTS(secondary_bound) AND NOT EXISTS(primary_bound)) AND EXISTS(duration)) AND
duration_is_negative(duration);
END_ENTITY;
```

Attribute definitions:

primary_bound: one of the bounds of the **time_interval_with_bounds**. The value of this attribute need not be specified.

secondary_bound: the other bound of the **time_interval_with_bounds**. The value of this attribute need not be specified.

duration: the time_measure_with_unit that specifies the length of a **time_interval_with_bounds**. If positive, the **time_interval_with_bounds** is a lower bounded **time_interval_with_bounds**, else an upper bounded **time_interval_with_bounds**. The value of this attribute need not be specified.

Formal propositions:

WR1: Either the **primary_bound** or the **secondary_bound** or both shall be specified.

WR2: The combination of **primary_bound**, **secondary_bound** and **duration** shall not be specified.

WR3: If **primary_bound** is specified and **secondary_bound** is not specified and if **duration** is specified, the **duration** shall be positive.

WR4: If **primary_bound** is not specified and **secondary_bound** is specified, **duration** shall be specified and the **duration** shall be negative.

10.4.18 time_role 

A **time_role** is a definition of a role for a **local_time**.

NOTE 1 The assignment of the role to a **local_time** is accomplished by the **time_assignment** entity data type.

EXPRESS specification:

```
ENTITY time_role;
  name : label;
DERIVE
  description : text := get_description_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **time_role** is known.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **time_role**.

NOTE 3 This attribute is an upwardly compatible addition to **time_role** as specified in ISO 10303-41:1994

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Formal propositions:

WR1: Each **time_role** shall be the described_item in at most one description_attribute.

NOTE 4 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 5 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

10.4.19 week_of_year_and_day_date

A **week_of_year_and_day_date** is a type of date defined by identifying the week within a year and the day within that particular week.

EXPRESS specification:

```
ENTITY week_of_year_and_day_date
  SUBTYPE OF (date);
  week_component : week_in_year_number;
  day_component : OPTIONAL day_in_week_number;
WHERE
  WR1: NOT leap_year(SELF\date.year_component) OR {1 <= day_component + 7 * (week_
component - 1) <= 366};
  WR2: leap_year(SELF\date.year_component) OR {1 <= day_component + 7 * (week_component -
1) <= 365};
END_ENTITY;
```

Attribute definitions:

week_component: the week element of the date.

day_component: the day element of the date.

Formal propositions:

WR1: The result of the expression **day_component** + (7 * week_component) shall be between 1 and 366 if the year_component is a leap year.

WR2: The result of the expression **day_component** + (7 * week_component) shall be between 1 and 365 if the year_component is not a leap year.

10.4.20 year_month

A **year_month** is a type of date defined as a month of a year.

EXPRESS specification:

```
ENTITY year_month
  SUBTYPE OF (date);
  month_component : month_in_year_number;
END_ENTITY;
```

Attribute definitions:

month_component: the month element of the date.

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10.5 Date time function definitions

10.5.1 acyclic_event_occurrence_relationship

The **acyclic_event_occurrence_relationship** function determines whether the graph of instances of the entity data type event_occurrence that contains relation as one of its links contains a cycle. This function may be used to evaluate either an event_occurrence_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the event_occurrence_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_event_occurrence_relationship(relation : event_occurrence_relationship;
                                               relatives : SET [1:?] OF event_occurrence;
                                               specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF event_occurrence_relationship;
  END_LOCAL;
  IF relation.relating_event IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(evnt <* bag_to_set(USEDIN(relation.relating_event, ('DATE_TIME_SCHEMA.
' + 'EVENT_OCCURRENCE_RELATIONSHIP.') + 'RELATED_EVENT')) | specific_relation IN
TYPEOF(evnt));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_event_occurrence_relationship(x[i], relatives + relation.relating_
event, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate event_occurrence_relationship to be checked.

relatives: (input) the set of instances of the entity data type event_occurrence for which the function is searching in the relating_event parameter of the relation argument.

specific_relation: (input) the fully qualified entity name of a type of event_occurrence_relationship entity.

10.5.2 acyclic_time_interval_relationship

The **acyclic_time_interval_relationship** function determines whether the graph of instances of the entity data type time_interval that contains relation as one of its links contains a cycle. This function may be used to evaluate either a time_interval_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the time_interval_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_time_interval_relationship(relation : time_interval_relationship;
```

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```
                                                relatives : SET [1:?] OF time_interval;
                                                specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF time_interval_relationship;
  END_LOCAL;
  IF relation.relating_time_interval IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(ti <* bag_to_set(USEDIN(relation.relating_time_interval, ('DATE_TIME_SCHEMA.
' + 'TIME_INTERVAL_RELATIONSHIP.') + 'RELATED_TIME_INTERVAL')) | specific_relation IN
TYPEOF(ti));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_time_interval_relationship(x[i], relatives + relation.relating_time_
interval, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate time_interval_relationship to be checked.

relatives: (input) the set of instances of the entity data type time_interval for which the function is searching in the relating_time_interval parameter of the relation argument.

specific_relation: (input) the fully qualified entity name of a type of time_interval_relationship entity.

10.5.3 duration_is_positive

The **duration_is_positive** function determines whether a given duration is positive. It returns TRUE if the duration is a positive. Otherwise, it returns FALSE.

EXPRESS specification:

```
FUNCTION duration_is_positive(duration : time_measure_with_unit) : BOOLEAN;
  IF duration\measure_with_unit.value_component > 0.0 THEN
    RETURN (TRUE);
  ELSE
    RETURN (FALSE);
  END_IF;
END_FUNCTION;
```

Argument definitions:

duration: (input) the candidate duration that is being checked.

10.5.4 duration_is_negative

The **duration_is_negative** function determines whether a given duration is negative. It returns TRUE if the duration is negative. Otherwise, it returns FALSE.

EXPRESS specification:

```
FUNCTION duration_is_negative(duration : time_measure_with_unit) : BOOLEAN;
  IF duration\measure_with_unit.value_component < 0.0 THEN
    RETURN (TRUE);
  ELSE
    RETURN (FALSE);
  END_IF;
END_FUNCTION;
```

Argument definitions:

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duration: (input) the candidate duration that is being checked.

10.5.5 leap_year

The **leap_year** function checks whether or not a particular year contains 366 days instead of 365 days, known as a leap year.

NOTE Leap years occur every for years.

EXPRESS specification:

```
FUNCTION leap_year(year : year_number) : BOOLEAN;
  IF (year MOD 4 = 0) AND (year MOD 100 <> 0) OR (year MOD 400 = 0) THEN
    RETURN (TRUE);
  ELSE
    RETURN (FALSE);
  END_IF;
END_FUNCTION;
```

Argument definitions:

year: an integer representing a particular year in the Gregorian calendar.

10.5.6 valid_calendar_date

The **valid_calendar_date** function determines whether the components of a calendar_date indicate a valid date. If the calendar_date is valid, the function returns TRUE. Otherwise it returns FALSE.

EXPRESS specification:

```
FUNCTION valid_calendar_date(date : calendar_date) : LOGICAL;
  CASE date.month_component OF
  1 :
    RETURN ({1 <= date.day_component <= 31});
  2 :
    BEGIN
      IF leap_year(date.year_component) THEN
        RETURN ({1 <= date.day_component <= 29});
      ELSE
        RETURN ({1 <= date.day_component <= 28});
      END_IF;
    END;
  3 :
    RETURN ({1 <= date.day_component <= 31});
  4 :
    RETURN ({1 <= date.day_component <= 30});
  5 :
    RETURN ({1 <= date.day_component <= 31});
  6 :
    RETURN ({1 <= date.day_component <= 30});
  7 :
    RETURN ({1 <= date.day_component <= 31});
  8 :
    RETURN ({1 <= date.day_component <= 31});
  9 :
    RETURN ({1 <= date.day_component <= 30});
  10 :
    RETURN ({1 <= date.day_component <= 31});
  11 :
    RETURN ({1 <= date.day_component <= 30});
  12 :
    RETURN ({1 <= date.day_component <= 31});
  END_CASE;
  RETURN (FALSE);
```

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```
END_FUNCTION;
```

Argument definitions:

date: (input) the candidate calendar_date that is to be checked.

10.5.7 valid_time

The **valid_time** function determines whether a candidate local_time has a minute_component if it has a second_component. It returns FALSE if the condition is not met. Otherwise it returns TRUE.

EXPRESS specification:

```
FUNCTION valid_time(time : local_time) : BOOLEAN;
  IF EXISTS(time.second_component) THEN
    RETURN (EXISTS(time.minute_component));
  ELSE
    RETURN (TRUE);
  END_IF;
END_FUNCTION;
```

Argument definitions:

time: (input) the candidate local_time that is to be checked.

```
END_SCHEMA; -- date_time_schema
```

11 Document schema

11.1 General

The subject of the **document_schema** is the citations of formal standards or documents that are outside the domain of this International Standard. These resource constructs are used to reference additional information that is relevant to the description of the product.

EXAMPLE International, national, and organizational standards, catalogues, and tables of engineering data are examples of formal standards or documents.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **document_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA document_schema '{iso standard 10303 part(41) version(4) object(1) document_
schema(8)}';

REFERENCE FROM product_definition_schema
  (product,
   product_definition,
   product_definition_formation);
REFERENCE FROM support_resource_schema
  (bag_to_set,
   identifier,
   label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

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product_definition_schema	ISO 10303-41
support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 9 for a graphical representation of this schema.

11.2 Fundamental concepts and assumptions

Product data can include citations of other information sources.

EXAMPLE Heat treatment processes documentation can be specified using the resource constructs that are defined in this schema.

11.3 Document type definition

11.3.1 product_or_formation_or_definition

The **product_or_formation_or_definition** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

The **product_or_formation_or_definition** type allows for the designation of a product, a product_definition, or a product_definition_formation.

EXPRESS specification:

```
TYPE product_or_formation_or_definition = SELECT
  (product,
   product_definition_formation,
   product_definition);
END_TYPE;
```

11.4 Document entity definitions

11.4.1 document

A **document** is an identification of a collection of information.

NOTE 1 The information collected need not be represented in a format conforming to any EXPRESS schema.

EXAMPLE 1 JPEG and HTML files are examples of document.

NOTE 2 This part of ISO 10303 provides several resources to describe documents and to associate them with product data. Annex F.2 provides an explanation about how to describe documents if they are to be considered products.

EXPRESS specification:

```
ENTITY document;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  kind : document_type;
INVERSE
  representation_types : SET [0:?] OF document_representation_type FOR represented_
document;
END_ENTITY;
```

Attribute definitions:

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id: the identifier that distinguishes the **document**.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 4 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **document** is known.

NOTE 5 The name may include the source of the **document**.

EXAMPLE 2 An example of the source of a **document** is "ISO".

description: the text that characterizes the **document**. The value of the attribute need not be specified.

kind: the document_type that specifies the sort of data that the **document** describes.

representation_types: the set of document_representation_type entities that specifies the way a **document** is represented.

11.4.2 document_product_association 

A **document_product_association** is an association between a document and a product, between a document and a product_definition_formation, or between a document and a product_definition.

NOTE 1 The role of **document_product_association** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 The **document_product_association** may establish an equivalence between the document and the product_definition_formation or between the document and the product of the document and the product_definition. This equivalence will provide a document with the versioning structure found in the product_definition_schema.

EXPRESS specification:

```
ENTITY document_product_association;
  name : label;
  description : OPTIONAL text;
  relating_document : document;
  related_product : product_or_formation_or_definition;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **document_product_association** is known.

description: the text that characterizes the **document_product_association**. The value of the attribute need not be specified.

relating_document: the instance of document that is part of the association.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_product: the instance of product, product_definition_formation, or product_definition entity data type associated with an instance of document.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

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11.4.3 document_relationship

A **document_relationship** is a relationship between two instances of the entity data type document and provides a description of their relationship.

NOTE 1 The role of **document_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 3 This entity, together with the document entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY document_relationship;
  name : label;
  description : OPTIONAL text;
  relating_document : document;
  related_document : document;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **document_relationship** is known.

description: the text that characterizes the **document_relationship**. The value of the attribute need not be specified.

relating_document: one of the instances of document that is a part of the relationship.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_document: the other instance of document that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

11.4.4 document_representation_type

A **document_representation_type** is a specification of the way a document is represented.

EXPRESS specification:

```
ENTITY document_representation_type;
  name : label;
  represented_document : document;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **document_representation_type** is known.

EXAMPLE 'Digital' and 'physical' are examples for the name.

represented_document: the document for which the kinds of representation are specified.

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11.4.5 document_type

A **document_type** is the kind of data that a formal standard or document may provide information about.

NOTE The applicable values of **document_type** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

EXAMPLE 1 'Material', 'surface finish', and 'heat treatment process' are all pieces of data that can be described implicitly, by reference to other documents (such as standards documents), rather than explicitly every time they are used.

EXPRESS specification:

```
ENTITY document_type;
  product_data_type : label;
END_ENTITY;
```

Attribute definitions:

product_data_type: the name of the kind of data contained in a document.

EXAMPLE 2 'Material description' is an example of **product_data_type** that characterizes the information provided by a standards document for material.

11.4.6 document_usage_constraint

A **document_usage_constraint** is an identification of a specific subject area or aspect from within a document and gives the relevant information or text that applies. The semantics of the reference can be found in the document itself.

NOTE 1 This entity could be used to identify the clause of a document and the content or portions of the content in that clause that are relevant.

EXPRESS specification:

```
ENTITY document_usage_constraint;
  source : document;
  subject_element : label;
  subject_element_value : text;
END_ENTITY;
```

Attribute definitions:

source: the document in which the **subject_element** originates.

NOTE 2 The inverse of this relationship is used to define multiple elements of a document.

subject_element: the name of an element of the **source**.

subject_element_value: the label that conveys a specific value of the **subject_element** present in the source

EXAMPLE For references to a 'surface finish' source the **subject_element** may be a 'Clause 2.1 surface imperfection' and its associated **subject_element_value** may be 'no visible imperfections', 'no more than two imperfections that are more than 0.06 inch diameter within any given square inch of surface area', or 'no imperfections under 10 times magnification'.

11.4.7 document_with_class

A **document_with_class** is a type of document for which a classification is specified.

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EXAMPLE 1 A surface finish document may be identified by various classes, class A, class B, and class C. Each class of surface finish specifies various allowances for imperfections in the surface finish. Class A may require no visible imperfections, class B may require no more than two imperfections that are more than 0.06 inch diameter within any given square inch of surface area, and class C may require no imperfections under 10 times magnification.

EXPRESS specification:

```
ENTITY document_with_class
  SUBTYPE OF (document);
  class : identifier;
END_ENTITY;
```

Attribute definitions:

class: the identification of the data classification to which the document belongs.

NOTE The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

EXAMPLE 2 The value of this attribute would be 'A', 'B' or 'C' in the previous example.

11.5 Document function definition

11.5.1 acyclic_document_relationship

The **acyclic_document_relationship** function determines whether the graph of instances of the entity data type document that contains relation as one of its links contains a cycle. This function may be used to evaluate either a document_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the document_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_document_relationship(relation : document_relationship;
                                       relatives : SET [1:?] OF document;
                                       specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF document_relationship;
  END_LOCAL;
  IF relation.relating_document IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(doc <* bag_to_set(USEDIN(relation.relating_document, ('DOCUMENT_SCHEMA.' +
'DOCUMENT_RELATIONSHIP.') + 'RELATED_DOCUMENT')) | specific_relation IN TYPEOF(doc));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_document_relationship(x[i], relatives + relation.relating_document,
specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate document_relationship to be checked.

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relatives: (input) the set of instances of the entity data type document for which the function is searching in the relating_document parameter of the relation argument.

specific_relation: (input) the fully qualified name of a subtype of the document_relationship entity.

```
END_SCHEMA; -- document_schema
```

12 Effectivity schema

12.1 General

The subject of the **effectivity_schema** is the validity of aspects of product data or product life cycle activity data specified against certain criteria.

This schema supports the representation of effectivity according to the following criteria:

— selection of a given batch of product data;

— selection of product data by serial number;

— selection of product data by a date.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **effectivity_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA effectivity_schema '{iso standard 10303 part(41) version(8) object(1) effectivity_
schema(9)}';

REFERENCE FROM action_schema
  (action_method);
REFERENCE FROM basic_attribute_schema
  (description_attribute,
   description_attribute_select,
   get_description_value,
   get_name_value,
   name_attribute,
   name_attribute_select);
REFERENCE FROM date_time_schema
  (date_time_or_event_occurrence,
   time_interval);
REFERENCE FROM measure_schema
  (measure_with_unit);
REFERENCE FROM support_resource_schema
  (bag_to_set,
   identifier,
   label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

action_schema	ISO 10303-41
basic_attribute_schema	ISO 10303-41
date_time_schema	ISO 10303-41

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measure_schema	ISO 10303-41
support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 10 for a graphical representation of this schema.

12.2 Fundamental concepts and assumptions

Effectivity information can be attached to any aspect of product data or product life cycle activity data.

Only effectivities based on date, serial number, and lot number are considered in this schema.

12.3 Effectivity type definitions

12.3.1 effs_description_attribute_select

The **effs_description_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE effs_description_attribute_select = SELECT BASED_ON description_attribute_select WITH
  (effectivity);
END_TYPE;
```

12.3.2 effs_name_attribute_select

The **effs_name_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE effs_name_attribute_select = SELECT BASED_ON name_attribute_select WITH
  (effectivity);
END_TYPE;
```

12.4 Effectivity entity definitions

12.4.1 condition

A **condition** is a type of action_method that must be fulfilled before a statement or relationship becomes valid.

EXPRESS specification:

```
ENTITY condition
  SUBTYPE OF (action_method);
END_ENTITY;
```

12.4.2 conditional_effectivity

A **conditional_effectivity** is a type of condition or effectivity for which the domain of applicability is defined by an explicit condition.

EXPRESS specification:

```
ENTITY conditional_effectivity;
END_ENTITY;
```

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12.4.3 conditional_configuration

A **conditional_configuration** is a type of conditional_effectivity and represents an association of a product_concept_feature_association object with a design or process in order to define a valid usage of it in the context of a certain product_class.

EXPRESS specification:

```
ENTITY conditional_configuration
  SUBTYPE OF (conditional_effectivity);
  configuration_type : STRING;
  inheritance_type : STRING;
END_ENTITY;
```

Attribute definitions:

configuration_type: the **configuration_type** defines the valid usage of a **conditional_configuration**.

inheritance_type: the **inheritance_type** specifies the inheritance scheme for the configuration information as either **exception**, **inherited**, or **local**.

— An **exception** inheritance_type is when no inheritance scheme is applicable and all required configuration information must be attached locally at the business object.

— An **inherited** inheritance_type is when a scheme for inheritance of configuration information applies. The complete configuration information shall be collected from the different levels in the structure by evaluation of results.

— A **local** inheritance_type is when no inheritance scheme is applicable and all required configuration information must be attached locally at the business object. Nevertheless, any potentially inherited configuration information of a higher level shall be consistent, i.e., it shall be a subset of the locally defined configuration information.

12.4.4 dated_effectivity

A **dated_effectivity** is a type of effectivity for which the domain of applicability is defined as a possibly open-ended interval of time.

EXPRESS specification:

```
ENTITY dated_effectivity
  SUBTYPE OF (effectivity);
  effectivity_end_date : OPTIONAL date_time_or_event_occurrence;
  effectivity_start_date : OPTIONAL date_time_or_event_occurrence;
WHERE
  WR1: EXISTS(effectivity_end_date) OR EXISTS(effectivity_start_date);
END_ENTITY;
```

Attribute definitions:

effectivity_end_date: the date_time_or_event_occurrence that defines the upper bound of the interval of applicability. If a value for this attribute is not defined, the interval of applicability has no upper limit.

effectivity_start_date: the date_time_or_event_occurrence that defines the lower bound of the interval of applicability. If a value for this attribute is not defined, the interval of applicability has no lower limit.

Formal propositions:

WR1: One or both of the **effectivity_start_date** or **effectivity_end_date** shall be specified.

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12.4.5 effectivity

An **effectivity** is the identification of a domain of applicability for product data.

EXAMPLE A process is effective from 1998 March 1 through 1998 December 31.

An **effectivity** may be either a serial_numbered_effectivity or a dated_effectivity, or a lot_effectivity, or a time_interval_based_effectivity.

EXPRESS specification:

```
ENTITY effectivity
  SUPERTYPE OF (ONEOF(serial_numbered_effectivity, dated_effectivity, lot_effectivity,
time_interval_based_effectivity));
  id : identifier;
DERIVE
  name : label := get_name_value(SELF);
  description : text := get_description_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'NAME_ATTRIBUTE.NAMED_ITEM')) <= 1;
  WR2: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **effectivity**.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 2 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **effectivity** is known.

NOTE 3 This attribute is an upwardly compatible addition to **effectivity** as specified in ISO 10303-41:1994.

description: the text that characterizes the **effectivity**.

NOTE 4 This attribute is an upwardly compatible addition to **effectivity** as specified in ISO 10303-41:1994.

Formal propositions:

WR1: Each **effectivity** shall be the described_item in at most one description_attribute.

NOTE 5 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

WR2: Each **effectivity** shall be the named_item in at most one name_attribute.

NOTE 6 The name_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 7 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

12.4.6 effectivity_relationship

An **effectivity_relationship** is a relationship between two instances of the entity data type effectivity with a description of their relationship.

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NOTE 1 The role of **effectivity_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 3 This entity, together with the effectivity entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY effectivity_relationship;
  name : label;
  description : OPTIONAL text;
  related_effectivity : effectivity;
  relating_effectivity : effectivity;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **effectivity_relationship** is known.

description: the text that characterizes the **effectivity_relationship**. The value of the attribute need not be specified.

related_effectivity: the other instance of effectivity that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

relating_effectivity: one of the instances of effectivity that is a part of the relationship.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

12.4.7 lot_effectivity

A **lot_effectivity** is a type of effectivity for which the domain of applicability is defined as a given batch of items.

EXPRESS specification:

```
ENTITY lot_effectivity
  SUBTYPE OF (effectivity);
  effectivity_lot_id : identifier;
  effectivity_lot_size : measure_with_unit;
END_ENTITY;
```

Attribute definitions:

effectivity_lot_id: the identification of the batch of items.

effectivity_lot_size: the size of the batch of items.

12.4.8 serial_numbered_effectivity

A **serial_numbered_effectivity** is a type of effectivity for which the domain of applicability is defined as a possibly open-ended interval of serial numbers.

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NOTE A serial number is represented in this International Standard as an identifier. It is supposed here that, regardless of their format, identifiers used to represent serial numbers are ordered.

EXPRESS specification:

```
ENTITY serial_numbered_effectivity
  SUBTYPE OF (effectivity);
  effectivity_start_id : identifier;
  effectivity_end_id : OPTIONAL identifier;
END_ENTITY;
```

Attribute definitions:

effectivity_start_id: the first valid serial number.

effectivity_end_id: the last valid serial number. If a value for this attribute is not defined, the interval of applicability has no upper bound.

12.4.9 time_interval_based_effectivity 

A **time_interval_based_effectivity** is a type of effectivity for which the domain of applicability is defined as a time_interval.

NOTE This entity data type provides greater functionalities than the entity data type dated_effectivity. In particular, it allows the specification of an effectivity period in which the period of time is defined by a date and a duration.

EXPRESS specification:

```
ENTITY time_interval_based_effectivity
  SUBTYPE OF (effectivity);
  effectivity_period : time_interval;
END_ENTITY;
```

Attribute definitions:

effectivity_period: the time_interval defining the period of time when the associated product data are effective.

12.5 Effectivity function definitions

12.5.1 acyclic_effectivity_relationship

The **acyclic_effectivity_relationship** function determines whether the graph of instances of the entity data type effectivity that contains relation as one of its links contains a cycle. This function may be used to evaluate either a effectivity_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the effectivity_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_effectivity_relationship(relation : effectivity_relationship;
                                          relatives : SET [1:?] OF effectivity;
                                          specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF effectivity_relationship;
```

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```
  END_LOCAL;
  IF relation.relating_effectivity IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(eff <* bag_to_set(USEDIN(relation.relating_effectivity, ('EFFECTIVITY_
SCHEMA.' + 'EFFECTIVITY_RELATIONSHIP.') + 'RELATED_EFFECTIVITY')) | specific_relation IN
TYPEOF(eff));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_effectivity_relationship(x[i], relatives + relation.relating_
effectivity, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate effectivity_relationship to be checked.

relatives: (input) the set of instances of the entity data type effectivity for which the function is searching in the relating_effectivity parameter of the **relation** argument.

specific_relation: (input) the fully qualified entity name of a type of effectivity_relationship entity.

```
END_SCHEMA; -- effectivity_schema
```

13 Experience schema

13.1 General

The subject of the **experience_schema** is the identification and characterization of types of experience or particular episodes of experience gained by persons. This schema provides for the definition of generic data relating to particular episodes of experience.

EXAMPLE 1 The identification of a particular episode of experience is an example of generic data relating to experience.

EXAMPLE 2 The identification of a particular type of experience is an example of generic data relating to experience.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **experience_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA experience_schema '{iso standard 10303 part(41) version(4) object(1) experience_
schema(10)}';

REFERENCE FROM support_resource_schema
  (identifier,
   label,
   text,
   bag_to_set);
```

NOTE 1
The schemas referenced above are specified in the following parts:

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support_resource_schema ISO 10303-41

NOTE 2 See Annex D, Paragraph 11 for a graphical representation of this schema.

13.2 Fundamental concepts and assumptions

This schema provides resources for the identification, naming, and description of:

— experience;

— types of experience;

— relationships between specific experiences;

— relationships between types of experience.

13.3 Experience entity definitions

13.3.1 experience 

An **experience** is an episode of practice, undertaken by a person or organization, that contributes to the accumulation of knowledge or a skill.

EXAMPLE 100 flying hours in a Tornado jet undertaken by a particular pilot in a particular year.

EXPRESS specification:

```
ENTITY experience;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **experience**.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 2 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **experience** is known.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **experience**. The value of this attribute need not be specified.

13.3.2 experience_relationship 

An **experience_relationship** is a relationship between two instances of the experience entity data type and provides for the identification, naming, and description of that relationship.

EXPRESS specification:

```
ENTITY experience_relationship;
  id : identifier;
```

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```
  name : label;
  description : OPTIONAL text;
  relating_experience : experience;
  related_experience : experience;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **experience_relationship**.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 2 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **experience_relationship** is known.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **experience_relationship**. The value of this attribute need not be specified.

relating_experience: one of the instances of experience that is a part of the relationship.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_experience: the other instance of experience that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

13.3.3 experience_type 

An **experience_type** is a category of experience.

EXAMPLE Flying experience.

NOTE 1 A relationship between an **experience_type** object and one or more experience objects is established by the declaration of an experience_type_assignment subtype containing a single attribute that references a set of a SELECT type that contains an experience.

EXPRESS specification:

```
ENTITY experience_type;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **experience_type**.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

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NOTE 3 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **experience_type** is known.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **experience_type**. The value of this attribute need not be specified.

13.3.4 experience_type_relationship 

An **experience_type_relationship** is a relationship between two instances of the experience_type entity data type and provides for the identification, naming, and description of that relationship.

EXPRESS specification:

```
ENTITY experience_type_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating_experience_type : experience_type;
  related_experience_type : experience_type;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **experience_type_relationship**.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 2 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **experience_type_relationship** is known.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **experience_type_relationship**. The value of this attribute need not be specified.

relating_experience_type: one of the instances of experience_type that is a part of the relationship.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_experience_type: the other instance of experience_type that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

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13.4 Experience function definitions

13.4.1 acyclic_experience_relationship

The **acyclic_experience_relationship** function determines whether the graph of instances of the entity data type experience that contains relation as one of its links contains a cycle. This function may be used to evaluate either an experience_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the experience_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_experience_relationship(relation : experience_relationship;
                                         relatives : SET OF experience;
                                         specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF experience_relationship;
  END_LOCAL;
  IF relation.relating_experience IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(expr <* bag_to_set(USEDIN(relation.relating_experience, ('EXPERIENCE_
SCHEMA.' + 'EXPERIENCE_RELATIONSHIP.') + 'RELATED_EXPERIENCE')) | specific_relation IN
TYPEOF(expr));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_experience_relationship(x[i], relatives + relation.relating_
experience, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate experience_relationship to be checked.

relatives: (input) the set of instances of the entity data type experience that the function is searching for in the relating_experience parameter of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the experience_relationship entity.

13.4.2 acyclic_experience_type_relationship

The **acyclic_experience_type_relationship** function determines whether the graph of instances of the entity data type experience_type that contains relation as one of its links contains a cycle. This function may be used to evaluate either an experience_type_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the experience_type_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_experience_type_relationship(relation : experience_type_relationship;
```


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```
                                                relatives : SET OF experience_type;
                                                specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF experience_type_relationship;
  END_LOCAL;
  IF relation.relating_experience_type IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(exptyp <* bag_to_set(USEDIN(relation.relating_experience_type, ('EXPERIENCE_
SCHEMA.' + 'EXPERIENCE_TYPE_RELATIONSHIP.') + 'RELATED_EXPERIENCE_TYPE')) | specific_
relation IN TYPEOF(exptyp));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_experience_type_relationship(x[i], relatives + relation.relating_
experience_type, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate experience_type_relationship to be checked.

relatives: (input) the set of instances of the entity data type experience_type that the function is searching for in the relating_experience_type parameter of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the experience_type_relationship entity.

```
END_SCHEMA; -- experience_schema
```

14 External reference schema

14.1 General

The subject of the **external_reference_schema** is the means of identifying information that is not explicitly represented in a given exchange.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **external_reference_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA external_reference_schema '{iso standard 10303 part(41) version(7) object(1)
external_reference_schema(11)}';

REFERENCE FROM basic_attribute_schema
  (description_attribute,
   description_attribute_select,
   get_description_value);
REFERENCE FROM support_resource_schema
  (label,
   text,
   identifier,
   bag_to_set);
```

NOTE 1
The schemas referenced above are specified in the following parts:

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basic_attribute_schema	ISO 10303-41
support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 12 for a graphical representation of this schema.

14.2 Fundamental concepts and assumptions

Product data may refer to information that is not explicitly represented in a given exchange. This information is either predefined, in the annotated EXPRESS schema to which the exchange conforms, or is defined elsewhere. If the information is defined in the annotated EXPRESS schema to which the exchange conforms, this requirement is satisfied by a reference that identifies the entity data type definitions in the annotated EXPRESS schema. Otherwise, this requirement is satisfied by a reference that identifies the record type definitions and exchange format.

14.3 External reference type definitions

14.3.1 ers_description_attribute_select

The **ers_description_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE ers_description_attribute_select = SELECT BASED_ON description_attribute_select WITH
  (external_source);
END_TYPE;
```

14.3.2 message

A message is a communication that is addressed to a system in order to trigger some action. The result of such an action is an external item or identification.

NOTE The role of **message** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

EXPRESS specification:

```
TYPE message = STRING;
END_TYPE;
```

14.3.3 source_item

The **source_item** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

A **source_item** type allows for the designation of an identifier or a message.

EXPRESS specification:

```
TYPE source_item = SELECT
  (identifier,
   message);
END_TYPE;
```


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14.4 External reference entity definitions

14.4.1 external_source

An **external_source** is the identification of a source of product data that is not the EXPRESS schema to which the exchange conforms.

NOTE 1 The product data may conform to some other part of this International Standard.

EXPRESS specification:

```
ENTITY external_source;
  source_id : source_item;
DERIVE
  description : text := get_description_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

source_id: the identification of the **external_source**.

description: the text that characterizes the **external_source**.

NOTE 2 This attribute is an upwardly compatible addition to **external_source** as specified in ISO 10303-41:1994.

Formal propositions:

WR1: Each **external_source** shall be the described_item in at most one description_attribute.

NOTE 3 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 4 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

14.4.2 external_source_relationship

An **external_source_relationship** is a relationship between two instances of the entity data type external_source with a description of their relationship.

NOTE 1 The role of **external_source_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

EXAMPLE One external_source may be a subset of another external_source. This entity may be specialized to specify such a relationship.

NOTE 2 This entity, together with the external_source entity, is based on the relationship template that is described in annex E.3

EXPRESS specification:

```
ENTITY external_source_relationship;
  name : label;
  description : OPTIONAL text;
  relating_source : external_source;
  related_source : external_source;
END_ENTITY;
```

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Attribute definitions:

name: the label by which the **external_source_relationship** is known.

description: the text that characterizes the **external_source_relationship**. The value of the attribute need not be specified.

relating_source: one of the instances of external_source that is a part of the relationship.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_source: the other instance of external_source that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

14.4.3 externally_defined_item

An **externally_defined_item** is the identification of information that is not explicitly represented in a given exchange and that is not defined in the annotated EXPRESS schema to which the exchange conforms.

EXPRESS specification:

```
ENTITY externally_defined_item;
  item_id : source_item;
  source : external_source;
END_ENTITY;
```

Attribute definitions:

item_id: the string that specifies the identification of the referent item.

source: an external_source that contains the referent item.

14.4.4 externally_defined_item_relationship

An **externally_defined_item_relationship** is a relationship between two instances of the entity data type externally_defined_item with an identification and description of their relationship.

NOTE 1 The role of **externally_defined_item_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 3 This entity, together with the externally_defined_item entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY externally_defined_item_relationship;
  name : label;
  description : OPTIONAL text;
  relating_item : externally_defined_item;
  related_item : externally_defined_item;
END_ENTITY;
```

Attribute definitions:

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name: the label by which the **externally_defined_item_relationship** is known.

description: the text that characterizes the **externally_defined_item_relationship**. The value of the attribute need not be specified.

relating_item: one of the instances of externally_defined_item that is a part of the relationship.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_item: the other instance of externally_defined_item that is a part of the relationship. If one element of the relationship is dependent up on the other, this attribute shall be the dependent one.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

14.4.5 externally_defined_item_with_multiple_references 

An **externally_defined_item_with_multiple_references** is a type of externally_defined_item that allows to define additional references providing information on the context and location of the target item within the source.

EXPRESS specification:

```
ENTITY externally_defined_item_with_multiple_references
  SUBTYPE OF (externally_defined_item);
  references : LIST [1:?] OF source_item;
UNIQUE
  UR1: references;
WHERE
  WR1: NOT (SELF\externally_defined_item.item_id IN references);
END_ENTITY;
```

Attribute definitions:

references: a list of source_items defining a path to the target item_id.

Formal propositions:

WR1: An **externally_defined_item_with_multiple_references** is a type of item_id that adds a list of references to provide an unambiguous path in the external data. The item_id inherited from the externally_defined_item is the target for the reference list, but is excluded from the list itself. The list ends at a entity data type that explicitly references the target or that is explicitly referenced by the target.

14.4.6 pre_defined_item 

A **pre_defined_item** is the identification of information that is not explicitly represented in a given exchange but that is defined in the annotated EXPRESS schema to which the exchange conforms.

EXAMPLE A reference to the colour 'red' without any definition of the associated red-green-blue values would be a **pre_defined_item** if the red-green-blue values of the colour 'red' were specified in the relevant annotated EXPRESS schema.

EXPRESS specification:

```
ENTITY pre_defined_item;
  name : label;
END_ENTITY;
```

Attribute definitions:

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name: the label by which the **pre_defined_item** is known.

14.5 External reference function definitions

14.5.1 acyclic_external_source_relationship

The **acyclic_external_source_relationship** function determines whether the graph of instances of the entity data type external_source that contains relation as one of its links contains a cycle. This function may be used to evaluate either an external_source_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the external_source_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_external_source_relationship(relation : external_source_relationship;
                                              relatives : SET [1:?] OF external_source;
                                              specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF external_source_relationship;
  END_LOCAL;
  IF relation.relating_source IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(es <* bag_to_set(USEDIN(relation.relating_source, ('EXTERNAL_REFERENCE_
SCHEMA.' + 'EXTERNAL_SOURCE_RELATIONSHIP.') + 'RELATED_SOURCE')) | specific_relation IN
TYPEOF(es));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_external_source_relationship(x[i], relatives + relation.relating_
source, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate external_source_relationship to be checked.

relatives: (input) the set of instances of the entity data type external_source for which the function is searching in the relating_source parameter of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the external_source_relationship entity.

14.5.2 acyclic_externally_defined_item_relationship

The **acyclic_externally_defined_item_relationship** function determines whether the graph of instances of the entity data type externally_defined_item that contains relation as one of its links contains a cycle. This function may be used to evaluate either an externally_defined_item_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the externally_defined_item_relationship entity include rules that use this function.

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EXPRESS specification:

```
FUNCTION acyclic_externally_defined_item_relationship(relation : externally_defined_item_
relationship;
                                                      relatives : SET [1:?] OF externally_
defined_item;
                                                      specific_relation : STRING) :
BOOLEAN;
  LOCAL
    x : SET OF externally_defined_item_relationship;
  END_LOCAL;
  IF relation.relating_item IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(edi <* bag_to_set(USEDIN(relation.relating_item, ('EXTERNAL_REFERENCE_SCHEMA.
' + 'EXTERNALLY_DEFINED_ITEM_RELATIONSHIP.') + 'RELATED_ITEM')) | specific_relation IN
TYPEOF(edi));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_externally_defined_item_relationship(x[i], relatives + relation.
relating_item, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate externally_defined_item_relationship to be checked.

relatives: (input) the set of instances of the entity data type externally_defined_item for which the function is searching in the relating_item parameter of the **relation** argument.

specific_relation: (input) the fully qualified entity name of a type of externally_defined_item_relationship entity.

```
END_SCHEMA; -- external_reference_schema
```

15 Group schema

15.1 General

The subject of the **group_schema** is the definition of groups of items.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **group_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA group_schema '{iso standard 10303 part(41) version(7) object(1) group_schema(12)}';

REFERENCE FROM support_resource_schema
  (label,
   bag_to_set,
   identifier,
   text);
REFERENCE FROM basic_attribute_schema
  (get_id_value,
   id_attribute,
```

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```
  id_attribute_select);
```

NOTE 1
The schemas referenced above are specified in the following parts:

support_resource_schema	ISO 10303-41
basic_attribute_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 13 for a graphical representation of this schema.

15.2 Fundamental concepts and assumptions

Groups are collections of product data that have a common identity. Any kinds of product data may be grouped together. The structure specified in this schema enables the common identity to be captured; it does not specify, or allow the specification of, the meaning of the common identity or the criteria for the common identity.

15.3 Group type definition

15.3.1 gs_id_attribute_select

The **gs_id_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE gs_id_attribute_select = SELECT BASED_ON id_attribute_select WITH
  (group);
END_TYPE;
```

15.4 Group entity definitions

15.4.1 group

A **group** is an identification of a collection of elements.

EXPRESS specification:

```
ENTITY group;
  name : label;
  description : OPTIONAL text;
DERIVE
  id : identifier := get_id_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ID_ATTRIBUTE.IDENTIFIED_ITEM'))
<= 1;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **group** is known.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **group**. The value of the attribute need not be specified.

id: the identifier that distinguishes the **group**. The value of this attribute need not be specified.

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NOTE 2 This attribute is an upwardly compatible addition to group as specified in ISO 10303-41:1994.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 4 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

Formal propositions:

WR1: Each **group** shall be the identified_item in at most one id_attribute.

NOTE 5 The id_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 6 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

15.4.2 group_relationship

A **group_relationship** is a relationship between two instances of the entity data type group with a description of their relationship.

NOTE 1 The role of **group_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 3 This entity, together with the group entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY group_relationship;
  name : label;
  description : OPTIONAL text;
  relating_group : group;
  related_group : group;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **group_relationship** is known.

description: the text that characterizes the **group_relationship**. The value of the attribute need not be specified.

relating_group: one of the instances of group that is a part of the relationship.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_group: the other instance of group that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

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15.5 Group function definition

15.5.1 acyclic_group_relationship

The **acyclic_group_relationship** function determines whether the graph of instances of the entity data type group that contains relation as one of its links contains a cycle. This function may be used to evaluate either a group_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the group_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_group_relationship(relation : group_relationship;
                                    relatives : SET [1:?] OF group;
                                    specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF group_relationship;
  END_LOCAL;
  IF relation.relating_group IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(grp <* bag_to_set(USEDIN(relation.relating_group, ('GROUP_SCHEMA.' + 'GROUP_
RELATIONSHIP.') + 'RELATED_GROUP')) | specific_relation IN TYPEOF(grp));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_group_relationship(x[i], relatives + relation.relating_group, specific_
relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate group_relationship to be checked.

relatives: (input) the set of instances of the entity data type group for which the function is searching in the relating_group parameter of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the group_relationship entity.

```
END_SCHEMA; -- group_schema
```

16 Language schema

16.1 General

The subject of the **language_schema** specifies the data structure for the representation of language support.

— IP1: The name attribute of the language shall be an alpha-3 bibliographic code of ISO 639-2.

— IP2: If present, the description attribute shall identify a country with an alpha-2 code specified in ISO 3166-1.

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This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **language_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA language_schema '{iso standard 10303 part(41) version(1) object(1) location_
schema(13)}';

REFERENCE FROM group_schema
  (group);
```

NOTE 1
The schemas referenced above are specified in the following parts:

group_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 14 for a graphical representation of this schema.

16.2 Fundamental concepts and assumptions

This specifies the representation of the language used in text attributes and the association of translations to these texts.

16.3 Language entity definition

16.3.1 language 

A **language** is a type of group that is a specification of the language in which an information is given. The group shall specify in its name a language code according to ISO 639-2, and, if present, in its description a country code according to ISO 3166-1.

Informal proposition

EXPRESS specification:

```
ENTITY language
  SUBTYPE OF (group);
WHERE
  WR1: SELF\group.name <> '';
END_ENTITY; -- Informal proposition
```

Formal propositions:

WR1: The name attribute of the language shall be specified.

```
END_SCHEMA; -- language_schema
```

17 Location schema

17.1 General

The subject of the **location_schema** is the identification and characterization of different representations of places or positions in space at which something can take place or exist.

EXAMPLE The identification of a geographical location is an example of generic data relating to a location.

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This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **location_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA location_schema '{iso standard 10303 part(41) version(4) object(1) location_
schema(14)}';

REFERENCE FROM support_resource_schema
  (identifier,
   label,
   text,
   bag_to_set);
```

NOTE 1
The schemas referenced above are specified in the following parts:

support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 15 for a graphical representation of this schema.

17.2 Fundamental concepts and assumptions

This schema provides resources for the identification, naming, and description of:

— specific locations;

— relationships between specific locations.

17.3 Location entity definitions

17.3.1 location 

A **location** is a place or position in space at which something can take place or exist.

NOTE The specification or representation of a location might take the form of a postal address, a set of geographical or spacial coordinates, or an association with a product.

EXAMPLE 1 A map reference specifies a location where a building exists.

EXAMPLE 2 The “bridge” of a particular ship specifies a location where navigation equipment should be installed.

EXPRESS specification:

```
ENTITY location;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **location**.

name: the label by which the **location** is known.

description: the text that characterizes the **location**. The value of this attribute need not be specified.

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17.3.2 location_relationship

A **location_relationship** is a relationship between two instances of the location entity data type and provides for the identification, naming, and description of that relationship.

EXPRESS specification:

```
ENTITY location_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating_location : location;
  related_location : location;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **location_relationship**.

name: the label by which the **location_relationship** is known.

description: the text that characterizes the **location_relationship**. The value of this attribute need not be specified.

relating_location: one of the instances of location that is a part of the relationship.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_location: the other instance of location that is a part of the relationship. If one element of the relationship is dependent upon the other, attribute shall be the dependent one.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

17.4 Location function definitions

17.4.1 acyclic_location_relationship

The **acyclic_location_relationship** function determines whether the graph of instances of the entity data type location that contains relation as one of its links contains a cycle. This function may be used to evaluate either a location_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the location_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_location_relationship(relation : location_relationship;
                                       relatives : SET OF location;
                                       specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF location_relationship;
  END_LOCAL;
  IF relation.relating_location IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(lctn <* bag_to_set(USEDIN(relation.relating_location, ('LOCATION_SCHEMA.' +
'LOCATION_RELATIONSHIP.') + 'RELATED_LOCATION')) | specific_relation IN TYPEOF(lctn));
```

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```
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_location_relationship(x[i], relatives + relation.relating_location,
specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate location_relationship to be checked.

relatives: (input) the set of instances of the entity data type location that the function is searching for in the relating_location parameter of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the location_relationship entity.

```
END_SCHEMA; -- location_schema
```

18 Management resources schema

18.1 General

The subject of the **management_resources_schema** is the association of management data with other aspects of product data in specific application contexts.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **management_resources_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA management_resources_schema '{iso standard 10303 part(41) version(8) object(1)
management_resources_schema(15)}';

REFERENCE FROM action_schema;
REFERENCE FROM application_context_schema
  (library_context);
REFERENCE FROM approval_schema;
REFERENCE FROM basic_attribute_schema
  (get_role,
   object_role,
   role_association,
   role_select);
REFERENCE FROM certification_schema;
REFERENCE FROM contract_schema;
REFERENCE FROM date_time_schema;
REFERENCE FROM document_schema;
REFERENCE FROM effectivity_schema;
REFERENCE FROM experience_schema;
REFERENCE FROM external_reference_schema
  (external_source);
REFERENCE FROM group_schema;
REFERENCE FROM location_schema;
REFERENCE FROM person_organization_schema;
REFERENCE FROM qualifications_schema;
REFERENCE FROM security_classification_schema;
REFERENCE FROM support_resource_schema;
```

NOTE 1

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The schemas referenced above are specified in the following parts:

action_schema	ISO 10303-41
application_context_schema	ISO 10303-41
approval_schema	ISO 10303-41
basic_attribute_schema	ISO 10303-41
certification_schema	ISO 10303-41
contract_schema	ISO 10303-41
date_time_schema	ISO 10303-41
document_schema	ISO 10303-41
effectivity_schema	ISO 10303-41
experience_schema	ISO 10303-41
external_reference_schema	ISO 10303-41
group_schema	ISO 10303-41
location_schema	ISO 10303-41
person_organization_schema	ISO 10303-41
qualifications_schema	ISO 10303-41
security_classification_schema	ISO 10303-41
support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 16 for a graphical representation of this schema.

18.2 Fundamental concepts and assumptions

The relationship between management type data and other aspects of product data is application-specific. Management data may be assigned to various kinds of product data. The assignment of management data can be defined in a generic manner and then specialized according to each considered application context.

NOTE In the following definitions, the expression "assigned product data" means "the product data that are referenced by subtypes of the generic abstract supertypes of this schema". Use of these generic constructs is defined in annex E.5.

18.3 Management resources type definitions

18.3.1 mrs_role_select

The **mrs_role_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE mrs_role_select = SELECT BASED_ON role_select WITH
  (action_assignment,
   action_request_assignment,
   approval_assignment,
   certification_assignment,
   contract_assignment,
   document_reference,
   effectivity_assignment,
```

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```
   external_referent_assignment,
   group_assignment,
   name_assignment,
   security_classification_assignment);
END_TYPE;
```

18.3.2 assignment_object_select

The **assignment_object_select** type is an extensible list of alternate data types. It provides a mechanism to refer to instances of the data types included in the types that extend the **assignment_object_select** type.

NOTE This empty extensible select requires extension in a further schema to ensure that entities that refer to it have at least one valid instantiation.

EXPRESS specification:

```
TYPE assignment_object_select = EXTENSIBLE SELECT;
END_TYPE;
```

18.3.3 attribute_type

The **attribute_type** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

The **attribute_type** data type allows for the designation of a label or a text that can be assigned to an attribute of an entity.

EXPRESS specification:

```
TYPE attribute_type = SELECT
  (label,
   text);
END_TYPE;
```

18.4 Management resources entity definitions

18.4.1 action_assignment

An **action_assignment** is an association of an action with product data.

EXPRESS specification:

```
ENTITY action_assignment
  ABSTRACT SUPERTYPE;
  assigned_action : action;
DERIVE
  role : object_role := get_role(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

assigned_action: the instance of the action entity data type that is to be associated with product data.

role: the object_role that specifies the purpose of the association of the **action_assignment** with product data.

NOTE 1 This attribute is an enhancement to the definition of **action_assignment** using a method that is upwardly compatible with ISO 10303-41:1994.

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Formal propositions:

WR1: Each **action_assignment** shall be the item_with_role in at most one role_association.

NOTE 2 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

18.4.2 action_method_assignment

An **action_method_assignment** is an association of an action_method with product data.

EXPRESS specification:

```
ENTITY action_method_assignment
  ABSTRACT SUPERTYPE;
  assigned_action_method : action_method;
  role : action_method_role;
END_ENTITY;
```

Attribute definitions:

assigned_action_method: the instance of the action_method entity data type that is to be associated with the product data.

role: the action_method_role that specifies the purpose of the association of the **action_method_assignment** with product data.

NOTE The values related to this attribute can be specified in an annotated EXPRESS schema that uses or specializes this entity, or in an agreement of common understanding between the partners sharing this information.

18.4.3 action_method_role

An **action_method_role** is a definition of a role for an action_method and a description of that role.

EXAMPLE 'Process XYZ' is an action_method. An action_method_assignment assigns the action_method to the definition of a specific mechanical part.

The **action_method_role** for the action_method_assignment is 'Process to mill mechanical part'.

EXPRESS specification:

```
ENTITY action_method_role;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **action_method_role** is known.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 2 The entity data type action_method_assignment implements the association of the role with a specific action_method.

description: the text that characterizes the **action_method_role**. The value of the attribute need not be specified.

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18.4.4 action_request_assignment

An **action_request_assignment** is an association of a versioned_action_request with product data.

EXPRESS specification:

```
ENTITY action_request_assignment
  ABSTRACT SUPERTYPE;
  assigned_action_request : versioned_action_request;
DERIVE
  role : object_role := get_role(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

assigned_action_request: the instance of the versioned_action_request entity data type that is to be associated with product data.

role: the object_role that specifies the purpose of the association of the **action_request_assignment** with product data.

NOTE 1 This attribute is an enhancement to the definition of **action_request_assignment** using a method that is upwardly compatible with ISO 10303-41:1994.

NOTE 2 The values related to this attribute can be specified in an annotated EXPRESS schema that uses or specializes this entity, or in an agreement of common understanding between the partners sharing this information.

Formal propositions:

WR1: Each **action_request_assignment** shall be the item_with_role in at most one role_association.

NOTE 3 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

18.4.5 approval_assignment

An **approval_assignment** is an association of an approval with product data.

EXPRESS specification:

```
ENTITY approval_assignment
  ABSTRACT SUPERTYPE;
  assigned_approval : approval;
DERIVE
  role : object_role := get_role(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

assigned_approval: the instance of the approval entity data type that is to be associated with product data.

role: the object_role that specifies the purpose of the association of the **approval_assignment** with product data.

NOTE 1 This attribute is an enhancement to the definition of **approval_assignment** using a method that is upwardly compatible with ISO 10303-41:1994.

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NOTE 2 The values related to this attribute can be specified in an annotated EXPRESS schema that uses or specializes this entity, or in an agreement of common understanding between the partners sharing this information.

Formal propositions:

WR1: Each **approval_assignment** shall be the item_with_role in at most one role_association.

NOTE 3 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

18.4.6 assignment_object_relationship

An **assignment_object_relationship** is a relationship between two objects of the assignment_object_select list representing the same object in different organizational contexts.

EXPRESS specification:

```
ENTITY assignment_object_relationship;
  id : OPTIONAL identifier;
  description : OPTIONAL text;
  relating : assignment_object_select;
  related : assignment_object_select;
  relation_type : STRING;
WHERE
  WR1: acyclic_assignment_object_relationship(SELF, [related], 'MANAGEMENT_RESOURCES_
SCHEMA.ASSIGNMENT_OBJECT_RELATIONSHIP');
END_ENTITY;
```

Attribute definitions:

id: the identifier or set of identifiers for the **assignment_object_relationship**.

description: the text or the set of texts that provide further information about the **assignment_object_relationship**. The value of this attribute need not be specified.

relating: one object of assignment_object_select that is part of the relationship.

related: the other object of assignment_object_select that is part of the relationship.

relation_type: the meaning of the relationship.

Formal propositions:

WR1: An **assignment_object_relationship** shall not participate in its own definition.

18.4.7 attribute_classification_assignment

An **attribute_classification_assignment** is an association of a class to an attribute of an existing entity in order to characterize it further. The meaning of the class of the existing attribute is determined by the role.

EXPRESS specification:

```
ENTITY attribute_classification_assignment
  ABSTRACT SUPERTYPE;
  assigned_class : group;
  attribute_name : label;
  role : classification_role;
END_ENTITY;
```

Attribute definitions:

assigned_class: the group that serves as a classification.

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attribute_name: the attribute for which the classification is defined. The **attribute_name** shall identify an attribute of the assigned entity.

role: the classification_role that specifies the purpose of the association of the **attribute_classification_assignment** with product data.

NOTE The values related to this attribute can be specified in an annotated EXPRESS schema that uses or specializes this entity, or in an agreement of common understanding between the partners sharing this information.

18.4.8 attribute_value_assignment

An **attribute_value_assignment** is an extendion of the cardinality of a string valued attribute in an entity data type by one. The relationship of the extension to the existing attribute is determined by the role. The entity data types that have the attribute for which the extension is provided are identified by a specialization of attribute_value_assignment.

EXPRESS specification:

```
ENTITY attribute_value_assignment
  ABSTRACT SUPERTYPE;
  attribute_name : label;
  attribute_value : attribute_type;
  role : attribute_value_role;
END_ENTITY;
```

Attribute definitions:

attribute_name: the attribute for which the additional value is defined. The **attribute_name** shall identify an attribute of the assigned entity.

NOTE 1 The identified attribute may be defined in the entity or inherited from a supertype.

EXAMPLE 'name' and 'description' are examples for the **attribute_name**.

attribute_value: the attribute_type that contains, as label, the additional value for the attribute of the assigned entity specified by **attribute_name**.

role: the attribute_value_role that specifies the purpose of the association of the **attribute_value_assignment** with product data.

NOTE 2 The values related to this attribute can be specified in an annotated EXPRESS schema that uses or specializes this entity, or in an agreement of common understanding between the partners sharing this information.

18.4.9 attribute_value_role

An **attribute_value_role** is a description of the purpose of an additional value assigned to an attribute of an entity.

EXPRESS specification:

```
ENTITY attribute_value_role;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **attribute_value_role** is known.

EXAMPLE 'Additional marketing name', 'primary value', or 'translated' are examples for name.

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description: the text that characterizes the **attribute_value_role**. The value of the attribute need not be specified.

18.4.10 certification_assignment

A **certification_assignment** is an association of a certification with product data.

EXPRESS specification:

```
ENTITY certification_assignment
  ABSTRACT SUPERTYPE;
  assigned_certification : certification;
DERIVE
  role : object_role := get_role(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

assigned_certification: the instance of the certification entity data type that is to be associated with product data.

role: the object_role that specifies the purpose of the association of the **certification_assignment** with product data.

NOTE 1 This attribute is an enhancement to the definition of **certification_assignment** using a method that is upwardly compatible with ISO 10303-41:1994.

Formal propositions:

WR1: Each **certification_assignment** shall be the item_with_role in at most one role_association.

NOTE 2 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

18.4.11 classification_assignment

A **classification_assignment** is an association of a class to product data.

EXPRESS specification:

```
ENTITY classification_assignment
  ABSTRACT SUPERTYPE;
  assigned_class : group;
  role : classification_role;
END_ENTITY;
```

Attribute definitions:

assigned_class: the group that serves as a class.

role: the classification_role that specifies the purpose of the association of the **classification_assignment** with product data.

NOTE The values related to this attribute can be specified in an annotated EXPRESS schema that uses or specializes this entity, or in an agreement of common understanding between the partners sharing this information.

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18.4.12 classification_assignment_relationship

A **classification_assignment_relationship** is a relationship between two classification_assignment objects.

EXPRESS specification:

```
ENTITY classification_assignment_relationship;
  id : OPTIONAL identifier;
  description : OPTIONAL text;
  relating : classification_assignment;
  related : classification_assignment;
  relation_type : STRING;
WHERE
  WR1: acyclic_classification_assignment_relationship(SELF, [related], 'MANAGEMENT_
RESOURCES_SCHEMA.CLASSIFICATION_ASSIGNMENT_RELATIONSHIP');
END_ENTITY;
```

Attribute definitions:

id: the identifier or set of identifiers for the **classification_assignment_relationship**.

description: the text or the set of texts that provide further information about the **classification_assignment_relationship**. The value of this attribute need not be specified.

relating: one object of classification_assignment that is part of the relationship.

related: the other object of classification_assignment that is part of the relationship.

relation_type: the meaning of the relationship.

Formal propositions:

WR1: A **classification_assignment_relationship** shall not participate in its own definition.

18.4.13 classification_role

A **classification_role** is a specified purpose for including product data in an exchange and a description of that purpose.

EXPRESS specification:

```
ENTITY classification_role;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **classification_role** is known.

description: the text that characterizes the **classification_role**. The value of the attribute need not be specified.

18.4.14 contract_assignment

A **contract_assignment** is an association of a contract with product data.

EXPRESS specification:

```
ENTITY contract_assignment
  ABSTRACT SUPERTYPE;
  assigned_contract : contract;
```


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```
DERIVE
  role : object_role := get_role(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

assigned_contract: the instance of the contract entity data type that is to be associated with product data.

role: the object_role that specifies the purpose of the association of the **contract_assignment** with product data.

NOTE 1 This attribute is an enhancement to the definition of **contract_assignment** using a method that is upwardly compatible with ISO 10303-41:1994.

NOTE 2 The values related to this attribute can be specified in an annotated EXPRESS schema that uses or specializes this entity, or in an agreement of common understanding between the partners sharing this information.

Formal propositions:

WR1: Each **contract_assignment** shall be the item_with_role in at most one role_association.

NOTE 3 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

18.4.15 date_and_time_assignment

A **date_and_time_assignment** is an association of a date_and_time with product data.

EXPRESS specification:

```
ENTITY date_and_time_assignment
  ABSTRACT SUPERTYPE;
  assigned_date_and_time : date_and_time;
  role : date_time_role;
END_ENTITY;
```

Attribute definitions:

assigned_date_and_time: the instance of the date_and_time entity data type that is to be associated with the product data.

role: the date_time_role that specifies the purpose of the association of the **date_and_time_assignment** with product data.

NOTE The values related to this attribute can be specified in an annotated EXPRESS schema that uses or specializes this entity, or in an agreement of common understanding between the partners sharing this information.

EXAMPLE 'Completed on' is an example of role that may be used to characterize the fact that the **assigned_date_and_time** defines the date and time of that day when some action completed.

18.4.16 date_assignment

A **date_assignment** is an association of a date with product data.

EXPRESS specification:

```
ENTITY date_assignment
  ABSTRACT SUPERTYPE;
  assigned_date : date;
  role : date_role;
```

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```
END_ENTITY;
```

Attribute definitions:

assigned_date: the date that is to be associated with the product data.

role: the date_role that specifies the purpose of the association of the **date_assignment** with product data.

NOTE The values related to this attribute can be specified in an annotated EXPRESS schema that uses or specializes this entity, or in an agreement of common understanding between the partners sharing this information.

EXAMPLE 'Creation date' is an example of role that may be used to characterize the fact that the assigned product data has been created on the **assigned_date**.

18.4.17 document_reference

A **document_reference** is an association of a document with product data.

EXPRESS specification:

```
ENTITY document_reference
  ABSTRACT SUPERTYPE;
  assigned_document : document;
  source : label;
DERIVE
  role : object_role := get_role(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

assigned_document: the instance of the document entity data type that is to be associated with product data.

source: the label stating the origination of the **assigned_document**.

EXAMPLE 'Engineering' and 'library' are examples of sources.

role: the object_role that specifies the purpose of the association of the **document_reference** with product data.

NOTE 1 This attribute is an enhancement to the definition of **document_reference** using a method that is upwardly compatible with ISO 10303-41:1994.

NOTE 2 The values related to this attribute can be specified in an annotated EXPRESS schema that uses or specializes this entity, or in an agreement of common understanding between the partners sharing this information.

Formal propositions:

WR1: Each **document_reference** shall be the item_with_role in at most one role_association.

NOTE 3 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

18.4.18 document_usage_constraint_assignment

A **document_usage_constraint_assignment** is an association of a document_usage_constraint with product data.

EXPRESS specification:

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```
ENTITY document_usage_constraint_assignment
  ABSTRACT SUPERTYPE;
  assigned_document_usage : document_usage_constraint;
  role : document_usage_role;
END_ENTITY;
```

Attribute definitions:

assigned_document_usage: the instance of the document_usage_constraint entity data type that is associated with the product data.

role: the document_usage_role that specifies the purpose of the association of the **document_usage_constraint_assignment** with product data.

NOTE 1 The values related to this attribute can be specified in an annotated EXPRESS schema that uses or specializes this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 2 Multiple constraints can be expressed by multiple instances of (a subtype of) document_usage_constraint. Such multiple instances can refer to the same instance of document_usage_role or to different instances.

18.4.19 document_usage_role

A **document_usage_role** is a specified purpose for including product data in an exchange and a description of that purpose.

EXPRESS specification:

```
ENTITY document_usage_role;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **document_usage_role** is known.

NOTE The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

EXAMPLE Examples of **document_usage_role.name** are 'applying process constraint' and 'applying material constraint'.

description: the text that characterizes the **document_usage_role**. The value of the attribute need not be specified.

18.4.20 effectivity_assignment

An **effectivity_assignment** is an association of an effectivity with product data.

EXPRESS specification:

```
ENTITY effectivity_assignment
  ABSTRACT SUPERTYPE;
  assigned_effectivity : effectivity;
DERIVE
  role : object_role := get_role(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

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assigned_effectivity: the instance of the effectivity entity data type that is to be associated with product data.

role: the object_role that specifies the purpose of the association of the **effectivity_assignment** with product data.

NOTE 1 This attribute is an enhancement to the definition of **effectivity_assignment** using a method that is upwardly compatible with ISO 10303-41:1994.

Formal propositions:

WR1: Each **effectivity_assignment** shall be the item_with_role in at most one role_association.

NOTE 2 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

18.4.21 effectivity_context_assignment

An **effectivity_context_assignment** is an assignment of a purpose to an effectivity_assignment.

EXPRESS specification:

```
ENTITY effectivity_context_assignment
  ABSTRACT SUPERTYPE;
  assigned_effectivity_assignment : effectivity_assignment;
  role : effectivity_context_role;
END_ENTITY;
```

Attribute definitions:

assigned_effectivity_assignment: the instance of the effectivity_assignment entity data type that is to be associated with the product data.

role: the effectivity_context_role that specifies the purpose of the association of the **effectivity_context_assignment** with product data.

18.4.22 effectivity_context_role

An **effectivity_context_role** is a specified purpose and the product data that is the context for an effectivity_context_assignment and a description of that purpose.

EXPRESS specification:

```
ENTITY effectivity_context_role;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **effectivity_context_role** is known.

NOTE The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

EXAMPLE 'Manufacturing location' is an example of the name of an **effectivity_context_role** that may be used to characterize the fact that if effectivity is assigned to product data in the context of a particular plant, this plant plays the role of a manufacturing location.

description: the text that characterizes the **effectivity_context_role**. The value of this attribute need not be specified.

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18.4.23 event_occurrence_assignment

An **event_occurrence_assignment** is an association of an event_occurrence with product data.

EXPRESS specification:

```
ENTITY event_occurrence_assignment
  ABSTRACT SUPERTYPE;
  assigned_event_occurrence : event_occurrence;
  role : event_occurrence_role;
END_ENTITY;
```

Attribute definitions:

assigned_event_occurrence: the instance of the event_occurrence entity data type that is to be associated with product data.

role: the event_occurrence_role that specifies the purpose of the association of the **event_occurrence_assignment** with product data.

18.4.24 event_occurrence_context_assignment

An **event_occurrence_context_assignment** is an identification of a context for the association of an event_occurrence_assignment with product data.

EXAMPLE For the event_occurrence 'start of production', the product for which production starts is the context for that event_occurrence.

EXPRESS specification:

```
ENTITY event_occurrence_context_assignment
  ABSTRACT SUPERTYPE;
  assigned_event_occurrence_assignment : event_occurrence_assignment;
  role : event_occurrence_context_role;
END_ENTITY;
```

Attribute definitions:

assigned_event_occurrence_assignment: the instance of the event_occurrence_assignment entity data type that is to be associated with product data.

role: the event_occurrence_context_role that specifies the purpose of the association of the **event_occurrence_context_assignment** with product data.

18.4.25 experience_assignment

An **experience_assignment** is a mechanism for associating an experience with product data.

EXAMPLE The association of a particular episode of flying practice with a particular pilot.

EXPRESS specification:

```
ENTITY experience_assignment
  ABSTRACT SUPERTYPE;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  assigned_experience : experience;
  role : experience_role;
END_ENTITY;
```

Attribute definitions:

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id: the identifier that distinguishes the **experience_assignment**.

name: the label by which the **experience_assignment** is known.

description: the text that characterizes the **experience_assignment**. The value of the attribute need not be specified.

assigned_experience: the instance of the experience entity data type that is to be associated with product data.

role: the experience_role that specifies the purpose of the association of the **experience_assignment** with product data.

18.4.26 experience_role

An **experience_role** is a specified purpose for an experience and the description of the purpose.

EXPRESS specification:

```
ENTITY experience_role;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **experience_role**.

name: the label by which the **experience_role** is known.

description: the text that characterizes the **experience_role**. The value of this attribute need not be specified.

18.4.27 experience_type_assignment

An **experience_type_assignment** is a mechanism for associating an experience_type with product data.

EXAMPLE The association of flying experience with a particular category of naval personnel.

EXPRESS specification:

```
ENTITY experience_type_assignment
  ABSTRACT SUPERTYPE;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  assigned_experience_type : experience_type;
  role : experience_type_role;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **experience_type_assignment**.

name: the label by which the **experience_type_assignment** is known.

description: the text that characterizes the **experience_type_assignment**. The value of this attribute need not be specified.

assigned_experience_type: the instance of the experience_type entity data type that is to be associated with product data.

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role: the experience_type_role that specifies the purpose of the association of the **experience_type_assignment** with product data.

18.4.28 experience_type_role

An **experience_type_role** is a specified purpose for an experience_type and a description of the purpose.

EXPRESS specification:

```
ENTITY experience_type_role;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **experience_type_role**.

name: the label by which the **experience_type_role** is known.

description: the text that characterizes the **experience_type_role**. The value of this attribute need not be specified.

18.4.29 external_identification_assignment

An **external_identification_assignment** is a type of identification_assignment that is defined in the context of an external_source. The source_id shall be a valid identifier in the context of the source.

EXPRESS specification:

```
ENTITY external_identification_assignment
  ABSTRACT SUPERTYPE
  SUBTYPE OF (identification_assignment);
  source : external_source;
END_ENTITY;
```

Attribute definitions:

source: the external_source that specifies the context in which in the source_id is defined.

18.4.30 external_identification_assignment_relationship

An **external_identification_assignment_relationship** is a relationship between two external_identification_assignment objects.

EXPRESS specification:

```
ENTITY external_identification_assignment_relationship;
  id : OPTIONAL identifier;
  description : OPTIONAL text;
  relating : external_identification_assignment;
  related : external_identification_assignment;
  relation_type : STRING;
WHERE
  WR1: acyclic_external_identification_assignment_relationship(SELF, [related],
'MANAGEMENT_RESOURCES_SCHEMA.EXTERNAL_IDENTIFICATION_ASSIGNMENT_RELATIONSHIP');
END_ENTITY;
```

Attribute definitions:

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id: the identifier or set of identifiers for the **external_identification_assignment_relationship**.

description: the text or the set of texts that provide further information about the **external_identification_assignment_relationship**. The value of this attribute need not be specified.

relating: one object of external_identification_assignment that is part of the relationship.

related: the other object of external_identification_assignment that is part of the relationship.

relation_type: the meaning of the relationship.

Formal propositions:

WR1: An **external_identification_assignment_relationship** shall not participate in its own definition.

18.4.31 external_referent_assignment

An **external_referent_assignment** is an identification of product data referenced in outside sources.

NOTE 1 This is an incompletely modeled concept and requires population of related entity data types to form a complete **external_referent_assignment**.

EXAMPLE 1 A URI for an externally defined tool is specified by assigning the value 'http://www.tool.com/mill.stp' to the name attribute of external_referent_assignment. The fact that the value is a URI is established by assigning a value of 'uri' to the role attribute of external_referent_assignment.

EXAMPLE 2 A URI fragment identifier for the bottom face of the externally defined tool is specified by assigning the value 'http://www.tool.com/mill.stp#bottom_face' to the name attribute of external_referent_assignment. The fact that the value is a URI

EXPRESS specification:

```
ENTITY external_referent_assignment
  ABSTRACT SUPERTYPE;
  assigned_name : label;
DERIVE
  role : object_role := get_role(SELF);
UNIQUE
  UR1: assigned_name;
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

assigned_name: the label by which the **external_referent_assignment** is known.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

role: the object_role that specifies the purpose of the association of the **external_referent_assignment** with product data.

NOTE 3 This attribute is an enhancement to the definition of **external_referent_assignment** using a method that is upwardly compatible with ISO 10303-41:1994.

Formal propositions:

WR1: Each **external_referent_assignment** shall be the item_with_role in at most one role_association.

NOTE 4 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

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18.4.32 group_assignment

A **group_assignment** is an association of a group with product data.

EXPRESS specification:

```
ENTITY group_assignment
  ABSTRACT SUPERTYPE;
  assigned_group : group;
DERIVE
  role : object_role := get_role(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

assigned_group: the instance of the group entity data type that is to be associated with product data.

role: the object_role that specifies the purpose of the association of the **group_assignment** with product data.

NOTE 1 This attribute is an enhancement to the definition of **group_assignment** using a method that is upwardly compatible with ISO 10303-41:1994.

Formal propositions:

WR1: Each **group_assignment** shall be the item_with_role in at most one role_association.

NOTE 2 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

18.4.33 identification_assignment

An **identification_assignment** is an association of an identification with product data. The assignment is made within a specified role.

EXPRESS specification:

```
ENTITY identification_assignment
  ABSTRACT SUPERTYPE;
  assigned_id : identifier;
  role : identification_role;
END_ENTITY;
```

Attribute definitions:

assigned_id: the identification associated with the product data.

NOTE The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

role: the identification_role that specifies the purpose of the association of the **identification_assignment** with product data.

EXAMPLE 'Alias identification' is an example of a role that may be used to characterize the fact that the **assigned_id** provides an alternate identification to some product data.

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18.4.34 identification_assignment_relationship

An **identification_assignment_relationship** is a relationship between two instances of the entity data type identification_assignment and provides an identification and description of their relationship.

NOTE 1 The role of **identification_assignment_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 3 This entity, together with the identification_assignment entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY identification_assignment_relationship;
  name : label;
  description : OPTIONAL text;
  relating_identification_assignment : identification_assignment;
  related_identification_assignment : identification_assignment;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **identification_assignment_relationship** is known.

description: the text that characterizes the **identification_assignment_relationship**. The value of the attribute need not be specified.

relating_identification_assignment: one of the instances of identification_assignment that is a part of the relationship.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_identification_assignment: the other instance of identification_assignment that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

18.4.35 identification_role

An **identification_role** is a purpose for identifying an item of product data and includes identification and description of the purpose.

EXPRESS specification:

```
ENTITY identification_role;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **identification_role** is known.

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NOTE The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **identification_role**. The value of the attribute need not be specified.

18.4.36 library_assignment

A **library_assignment** is a type of external_referent_assignment that assigns a library_context to product data.

EXPRESS specification:

```
ENTITY library_assignment
  ABSTRACT SUPERTYPE
  SUBTYPE OF (external_referent_assignment);
  frame_of_reference : library_context;
UNIQUE
  UR1: frame_of_reference;
END_ENTITY;
```

Attribute definitions:

frame_of_reference: the library_context in which the **library_assignment** is defined.

18.4.37 location_assignment

A **location_assignment** is a mechanism for associating a location with product data.

EXAMPLE The association of a geographical location with an item of heavy lifting equipment.

EXPRESS specification:

```
ENTITY location_assignment
  ABSTRACT SUPERTYPE;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  assigned_location : location;
  role : location_role;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **location_assignment**.

name: the label by which the **location_assignment** is known.

description: the text that characterizes the **location_assignment**. The value of the attribute need not be specified.

assigned_location: the instance of the location entity data type that is to be associated with product data.

role: the location_role that specifies the purpose of the association of the **location_assignment** with product data.

18.4.38 location_representation_assignment

A **location_representation_assignment** is a mechanism for associating a location with its representation.

EXAMPLE The association of a location with its representation as a set of geographical co-ordinates.

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EXPRESS specification:

```
ENTITY location_representation_assignment
  ABSTRACT SUPERTYPE;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  represented_location : location;
  role : location_representation_role;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **location_representation_assignment**.

name: the label by which the **location_representation_assignment** is known.

description: the text that characterizes the **location_representation_assignment**. The value of the attribute need not be specified.

represented_location: the instance of the location entity data type that is to be associated with product data.

role: the location_representation_role that specifies the purpose of the association of the **location_representation_assignment** with product data.

18.4.39 location_representation_role

A **location_representation_role** is a purpose for a representation of a location and provides identification and description of the purpose.

EXPRESS specification:

```
ENTITY location_representation_role;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **location_representation_role**.

name: the label by which the **location_representation_role** is known.

description: the text that characterizes the **location_representation_role**. The value of the attribute need not be specified.

18.4.40 location_role

A **location_role** is the purpose for a location in the context of product data an provides identification and description of the purpose.

EXPRESS specification:

```
ENTITY location_role;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

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id: the identifier that distinguishes the **location_role**.

name: the label by which the **location_role** is known.

description: the text that characterizes the **location_role**. The value of the attribute need not be specified.

18.4.41 name_assignment

A **name_assignment** is an identification of a name used to designate product data.

EXPRESS specification:

```
ENTITY name_assignment
  ABSTRACT SUPERTYPE;
  assigned_name : label;
DERIVE
  role : object_role := get_role(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

assigned_name: the label by which the product data is known.

role: the object_role that specifies the purpose of the association of the **name_assignment** with product data.

NOTE 1 This attribute is an enhancement to the definition of **name_assignment** using a method that is upwardly compatible with ISO 10303-41:1994

Formal propositions:

WR1: Each **name_assignment** shall be the item_with_role in at most one role_association.

NOTE 2 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

18.4.42 organization_assignment

An **organization_assignment** is an association of an organization with product data.

EXPRESS specification:

```
ENTITY organization_assignment
  ABSTRACT SUPERTYPE;
  assigned_organization : organization;
  role : organization_role;
END_ENTITY;
```

Attribute definitions:

assigned_organization: the instance of the organization entity data type that is to be associated with the product data.

role: the organization_role that specifies the purpose of the association of the **organization_assignment** with product data.

EXAMPLE 'Supplier' is an example of a role that may be used to characterize the fact that the assigned product data has been released by the **assigned_organization**.

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18.4.43 organization_type_assignment

An **organization_type_assignment** is a mechanism for associating an organization_type with product data.

EXAMPLE An association between national standards bodies and ISO.

EXPRESS specification:

```
ENTITY organization_type_assignment
  ABSTRACT SUPERTYPE;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  assigned_organization_type : organization_type;
  role : organization_type_role;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **organization_type_assignment**.

name: the label by which the **organization_type_assignment** is known.

description: the text that characterizes the **organization_type_assignment**. The value of the attribute need not be specified.

assigned_organization_type: the instance of the organization_type entity data type that is to be associated with product data.

role: the organization_type_role that specifies the purpose of the association of the **organization_type_assignment** with product data.

18.4.44 organization_type_role

An **organization_type_role** is a purpose for an **organization_type** and provides identification and description of the purpose.

EXPRESS specification:

```
ENTITY organization_type_role;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **organization_type_role**.

name: the label by which the **organization_type_role** is known.

description: the text that characterizes the **organization_type_role**. The value of the attribute need not be specified.

18.4.45 organizational_project_assignment

An **organizational_project_assignment** is an association of an organizational_project with product data.

EXPRESS specification:

```
ENTITY organizational_project_assignment
  ABSTRACT SUPERTYPE;
```


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```
  assigned_organizational_project : organizational_project;
  role : organizational_project_role;
END_ENTITY;
```

Attribute definitions:

assigned_organizational_project: the instance of the organizational_project entity data type that is to be associated with product data.

role: the organizational_project_role that specifies the purpose of the association of the **organizational_project_assignment** with product data.

18.4.46 organizational_project_role

An **organizational_project_role** is a purpose for an **organizational_project** and provides identification and description of the purpose.

EXPRESS specification:

```
ENTITY organizational_project_role;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **organizational_project_role** is known.

description: the text that characterizes the **organizational_project_role**. The value of the attribute need not be specified.

18.4.47 person_and_organization_assignment

A **person_and_organization_assignment** is an association of a person_and_organization with product data.

EXPRESS specification:

```
ENTITY person_and_organization_assignment
  ABSTRACT SUPERTYPE;
  assigned_person_and_organization : person_and_organization;
  role : person_and_organization_role;
END_ENTITY;
```

Attribute definitions:

assigned_person_and_organization: the instance of the person_and_organization entity data type that is to be associated with the product data.

role: the person_and_organization_role that specifies the purpose of the association of the **person_and_organization_assignment** with product data.

EXAMPLE 'Contact person' is an example of a role that may be used to characterize the fact that information about the assigned product data can be obtained with contacting the mentioned person in the specified organization.

18.4.48 person_assignment

A **person_assignment** is the association of a person with product data.

EXPRESS specification:

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```
ENTITY person_assignment
  ABSTRACT SUPERTYPE;
  assigned_person : person;
  role : person_role;
END_ENTITY;
```

Attribute definitions:

assigned_person: the instance of the person entity data type that is to be associated with product data.

role: the person_role that specifies the purpose of the association of the **person_assignment** with product data.

NOTE The function of the person with respect to the assignment is usually a consequence of the functions of the person within his organization.

EXAMPLE 'Designer' and 'creator' are examples of roles that may be used to characterize the fact that the assigned product data has been designed or created by the **assigned_person**.

18.4.49 person_type_assignment

A **person_type_assignment** is a mechanism for associating a person_type with product data.

EXAMPLE An association between a (generic) *structural engineer* and a particular construction project.

EXPRESS specification:

```
ENTITY person_type_assignment
  ABSTRACT SUPERTYPE;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  assigned_person_type : person_type;
  role : person_type_role;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **person_type_assignment**.

name: the label by which the **person_type_assignment** is known.

description: the text that characterizes the **person_type_assignment**. The value of the attribute need not be specified.

assigned_person_type: the instance of the person_type entity data type that is to be associated with product data.

role: the person_type_role that specifies the purpose of the association of the **person_type_assignment** with product data.

18.4.50 person_type_definition_assignment

A **person_type_definition_assignment** is a mechanism for associating a person_type_definition with product data.

EXAMPLE An association between the definition of *structural engineer* employed by the UK Institution of Structural Engineers and the definition of *structural engineer* employed by a particular construction project.

EXPRESS specification:

```
ENTITY person_type_definition_assignment
  ABSTRACT SUPERTYPE;
```

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```
  id : identifier;
  name : label;
  description : OPTIONAL text;
  assigned_person_type_definition : person_type_definition;
  role : person_type_definition_role;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **person_type_definition_assignment**.

name: the label by which the **person_type_definition_assignment** is known.

description: the text that characterizes the **person_type_definition_assignment**. The value of the attribute need not be specified.

assigned_person_type_definition: the instance of the person_type_definition entity data type that is to be associated with product data.

role: the person_type_definition_role that specifies the purpose of the association of the **person_type_definition_assignment** with product data.

18.4.51 person_type_definition_role

A **person_type_definition_role** is a purpose for a **person_type_definition** and provides identification and description of the purpose.

EXPRESS specification:

```
ENTITY person_type_definition_role;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **person_type_definition_role**.

name: the label by which the **person_type_definition_role** is known.

description: the text that characterizes the **person_type_definition_role**. The value of the attribute need not be specified.

18.4.52 person_type_role

A **person_type_role** is a purpose for a **person_type** and provides identification and description of the purpose.

EXPRESS specification:

```
ENTITY person_type_role;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **person_type_role**.

name: the label by which the **person_type_role** is known.

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description: the text that characterizes the **person_type_role**. The value of the attribute need not be specified.

18.4.53 position_in_organization_assignment

A **position_in_organization_assignment** is a mechanism for associating a position_in_organization with product data.

EXAMPLE An association between the managing director of a particular organization and a project.

EXPRESS specification:

```
ENTITY position_in_organization_assignment
  ABSTRACT SUPERTYPE;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  assigned_position_in_organization : position_in_organization;
  role : position_in_organization_role;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **position_in_organization_assignment**.

name: the label by which the **position_in_organization_assignment** is known.

description: the text that characterizes the **position_in_organization_assignment**. The value of the attribute need not be specified.

assigned_position_in_organization: the instance of the position_in_organization entity data type that is to be associated with product data.

role: the position_in_organization_role that specifies the purpose of the association of the **position_in_organization_assignment** with product data.

18.4.54 position_in_organization_role

A **position_in_organization_role** is a purpose for a **position_in_organization** and provides identification and description of the purpose

EXPRESS specification:

```
ENTITY position_in_organization_role;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **position_in_organization_role**.

name: the label by which the **position_in_organization_role** is known.

description: the text that characterizes the **position_in_organization_role**. The value of the attribute need not be specified.

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18.4.55 position_in_organization_type_assignment

A **position_in_organization_type_assignment** is a mechanism for associating a position_in_organization_type with product data.

EXAMPLE An association between a generic company manager and a project.

EXPRESS specification:

```
ENTITY position_in_organization_type_assignment
  ABSTRACT SUPERTYPE;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  assigned_position_in_organization_type : position_in_organization_type;
  role : position_in_organization_type_role;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **position_in_organization_type_assignment**.

name: the label by which the **position_in_organization_type_assignment** is known.

description: the text that characterizes the **position_in_organization_type_assignment**. The value of the attribute need not be specified.

assigned_position_in_organization_type: the instance of the position_in_organization_type entity data type that is to be associated with product data.

role: the position_in_organization_type_role that specifies the purpose of the association of the **position_in_organization_type_assignment** with product data.

18.4.56 position_in_organization_type_role

A **position_in_organization_type_role** is a purpose for a **position_in_organization_type** and provides identification and description of the purpose.

EXPRESS specification:

```
ENTITY position_in_organization_type_role;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **position_in_organization_type_role**.

name: the label by which the **position_in_organization_type_role** is known.

description: the text that characterizes the **position_in_organization_type_role**. The value of the attribute need not be specified.

18.4.57 qualification_assignment

A **qualification_assignment** is a mechanism for associating a qualification with product data.

EXAMPLE The association of a pilot's licence with a person indicates that the person is qualified to fulfill the role of a pilot.

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EXPRESS specification:

```
ENTITY qualification_assignment
  ABSTRACT SUPERTYPE;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  assigned_qualification : qualification;
  role : qualification_role;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **qualification_assignment**.

name: the label by which the **qualification_assignment** is known.

description: the text that characterizes the **qualification_assignment**. The value of the attribute need not be specified.

assigned_qualification: the instance of the qualification entity data type that is to be associated with product data.

role: the qualification_role that specifies the purpose of the association of the **qualification_assignment** with product data.

18.4.58 qualification_role

A **qualification_role** is a purpose for a qualification and provides identification and description of the purpose.

EXPRESS specification:

```
ENTITY qualification_role;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **qualification_role**.

name: the label by which the **qualification_role** is known.

description: the text that characterizes the **qualification_role**. The value of the attribute need not be specified.

18.4.59 qualification_type_assignment

A **qualification_type_assignment** is a mechanism for associating a qualification_type with product data.

EXAMPLE An association between a person and a Product Management Professional certification indicates that the person is qualified to take on the role of project manager.

EXPRESS specification:

```
ENTITY qualification_type_assignment
  ABSTRACT SUPERTYPE;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  assigned_qualification_type : qualification_type;
```


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```
  role : qualification_type_role;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **qualification_type_assignment**.

name: the label by which the **qualification_type_assignment** is known.

description: the text that characterizes the **qualification_type_assignment**. The value of the attribute need not be specified.

assigned_qualification_type: the instance of the qualification_type entity data type that is to be associated with product data.

role: the qualification_type_role that specifies the purpose of the association of the **qualification_type_assignment** with product data.

18.4.60 qualification_type_role

A **qualification_type_role** is a purpose for a **qualification_type** and provides identification and description of the purpose.

EXPRESS specification:

```
ENTITY qualification_type_role;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **qualification_type_role**.

name: the label by which the **qualification_type_role** is known.

description: the text that characterizes the **qualification_type_role**. The value of the attribute need not be specified.

18.4.61 security_classification_assignment

A **security_classification_assignment** is an association of a security_classification with product data.

EXPRESS specification:

```
ENTITY security_classification_assignment
  ABSTRACT SUPERTYPE;
  assigned_security_classification : security_classification;
DERIVE
  role : object_role := get_role(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ROLE_ASSOCIATION.ITEM_WITH_
ROLE')) <= 1;
END_ENTITY;
```

Attribute definitions:

assigned_security_classification: the instance of the security_classification entity data type that is to be associated with product data.

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role: the object_role that specifies the purpose of the association of the **security_classification_assignment** with product data.

NOTE 1 This attribute is an enhancement to the definition of **security_classification_assignment** using a method that is upwardly compatible with ISO 10303-41:1994

Formal propositions:

WR1: Each **security_classification_assignment** shall be the item_with_role in at most one role_association.

NOTE 2 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

18.4.62 time_assignment

A **time_assignment** is an association of a local_time with product data.

EXPRESS specification:

```
ENTITY time_assignment
  ABSTRACT SUPERTYPE;
  assigned_time : local_time;
  role : time_role;
END_ENTITY;
```

Attribute definitions:

assigned_time: the instance of the local_time entity data type that is to be associated with product data.

role: the time_role that specifies the purpose of the association of the **time_assignment** with product data.

EXAMPLE 'Start time' is an example of a role that may be used to characterize the fact that the **assigned_time** defines the time when some action was expected to start.

18.4.63 time_interval_assignment

A **time_interval_assignment** is an association of a time_interval with product data.

EXPRESS specification:

```
ENTITY time_interval_assignment
  ABSTRACT SUPERTYPE;
  assigned_time_interval : time_interval;
  role : time_interval_role;
END_ENTITY;
```

Attribute definitions:

assigned_time_interval: the instance of the time_interval entity data type that is to be associated with product data.

role: the time_interval_role that specifies the purpose of the association of the **time_interval_assignment** with product data.

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18.5 Management resources function definitions

18.5.1 acyclic_assignment_object_relationship

The **acyclic_assignment_object_relationship** function determines whether the graph of instances of the entity data type assignment_object_select that contains relation as one of its links contains a cycle. This function may be used to evaluate either an assignment_object_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE The algorithm of the function is explained in annex E.2.

EXPRESS specification:

```
FUNCTION acyclic_assignment_object_relationship(relation : assignment_object_relationship;
                                                relatives : SET [1:?] OF assignment_
object_select;
                                                specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF assignment_object_relationship;
  END_LOCAL;
  IF relation.relating IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(aor <* bag_to_set(USEDIN(relation.relating, ('MANAGEMENT_RESOURCES_SCHEMA.'
+ 'ASSIGNMENT_OBJECT_RELATIONSHIP.') + 'RELATED')) | specific_relation IN TYPEOF(aor));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_assignment_object_relationship(x[i], relatives + relation.relating,
specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate assignment_object_relationship to be checked.

relatives: (input) the set of instances of the entity data type assignment_object_select for which the function is searching in the relating parameter of the relation argument.

specific_relation: (input) the fully qualified name of a subtype of the assignment_object_relationship entity.

18.5.2 acyclic_classification_assignment_relationship

The **acyclic_classification_assignment_relationship** function determines whether the graph of instances of the entity data type classification_assignment that contains relation as one of its links contains a cycle. This function may be used to evaluate either a classification_assignment_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE The algorithm of the function is explained in annex E.2.

EXPRESS specification:

```
FUNCTION acyclic_classification_assignment_relationship(relation : classification_
assignment_relationship;
                                                        relatives : SET [1:?] OF
classification_assignment;
```

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```
                                                                    specific_relation : STRING) :
BOOLEAN;
  LOCAL
    x : SET OF classification_assignment_relationship;
  END_LOCAL;
  IF relation.relating IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(car <* bag_to_set(USEDIN(relation.relating, ('MANAGEMENT_RESOURCES_SCHEMA.
' + 'CLASSIFICATION_ASSIGNMENT_RELATIONSHIP.') + 'RELATED')) | specific_relation IN
TYPEOF(car));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_classification_assignment_relationship(x[i], relatives + relation.
relating, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate classification_assignment_relationship to be checked.

relatives: (input) the set of instances of the entity data type classification_assignment for which the function is searching in the relating parameter of the relation argument.

specific_relation: (input) the fully qualified name of a subtype of the classification_assignment_relationship entity.

18.5.3 acyclic_external_identification_assignment_relationship

The **acyclic_external_identification_assignment_relationship** function determines whether the graph of instances of the entity data type external_identification_assignment that contains relation as one of its links contains a cycle. This function may be used to evaluate either a external_identification_assignment_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE The algorithm of the function is explained in annex E.2.

EXPRESS specification:

```
FUNCTION acyclic_external_identification_assignment_relationship(relation : external_
identification_assignment_relationship;
                                                                 relatives : SET [1:?] OF
external_identification_assignment;
                                                                 specific_relation :
STRING) : BOOLEAN;
  LOCAL
    x : SET OF external_identification_assignment_relationship;
  END_LOCAL;
  IF relation.relating IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(eiar <* bag_to_set(USEDIN(relation.relating, ('MANAGEMENT_RESOURCES_SCHEMA.'
+ 'EXTERNAL_IDENTIFICATION_ASSIGNMENT_RELATIONSHIP.') + 'RELATED')) | specific_relation
IN TYPEOF(eiar));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_external_identification_assignment_relationship(x[i], relatives +
relation.relating, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
```

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```
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate external_identification_assignment_relationship to be checked.

relatives: (input) the set of instances of the entity data type external_identification_assignment for which the function is searching in the relating parameter of the relation argument.

specific_relation: (input) the fully qualified name of a subtype of the external_identification_assignment_relationship entity.

18.5.4 acyclic_identification_assignment_relationship

The **acyclic_identification_assignment_relationship** function determines whether the graph of instances of the entity data type identification_assignment that contains relation as one of its links contains a cycle. This function may be used to evaluate either an identification_assignment_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the identification_assignment_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_identification_assignment_relationship(relation : identification_
assignment_relationship;
                                                        relatives : SET [1:?] OF
identification_assignment;
                                                        specific_relation : STRING) :
BOOLEAN;
  LOCAL
    x : SET OF identification_assignment_relationship;
  END_LOCAL;
  IF relation.relating_identification_assignment IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(ia <* bag_to_set(USEDIN(relation.relating_identification_assignment,
('MANAGEMENT_RESOURCES_SCHEMA.' + 'IDENTIFICATION_ASSIGNMENT_RELATIONSHIP.') + 'RELATED_
IDENTIFICATION_ASSIGNMENT')) | specific_relation IN TYPEOF(ia));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_identification_assignment_relationship(x[i], relatives + relation.
relating_identification_assignment, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate identification_assignment_relationship to be checked.

relatives: (input) the set of instances of the entity data type identification_assignment that the function is searching for in the relating_identification_assignment parameter of the relation argument.

specific_relation: (input) the fully qualified entity name of a type of identification_assignment_relationship entity.

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```
END_SCHEMA; -- management_resources_schema
```

19 Measure schema

19.1 General

The subject of the **measure_schema** is the description of physical quantities.

EXAMPLE Density, length, force, and time are different kinds of physical quantities.

The resource constructs defined in this schema are based upon material in ISO 31 and ISO 1000.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **measure_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA measure_schema '{iso standard 10303 part(41) version(8) object(1) measure_
schema(16)}';

REFERENCE FROM basic_attribute_schema
  (get_name_value,
   name_attribute,
   name_attribute_select);
REFERENCE FROM representation_schema
  (representation_context);
REFERENCE FROM support_resource_schema;
```

NOTE 1
The schemas referenced above are specified in the following parts:

basic_attribute_schema ISO 10303-41

:: ISO -

support_resource_schema ISO 10303-41

NOTE 2 See Annex D, Paragraph 17 for a graphical representation of this schema.

19.2 Fundamental concepts and assumptions

The following requirements are supported by this schema:

— it shall be possible to specify the predefined kinds of physical quantity if the kind of a physical quantity is known at the time when the schema is being specified;

— it shall be possible to specify non-predefined kinds of physical quantity if the kind of a physical quantity is not known at the time when the schema is being specified.

EXAMPLE The elements of a list of material properties are physical quantities. Each element may be a different kind of physical quantity. The kind of each element is only decided upon at instantiation time.

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19.3 Measure type definitions

19.3.1 s_name_attribute_select

The **s_name_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE s_name_attribute_select = SELECT BASED_ON name_attribute_select WITH
  (derived_unit);
END_TYPE;
```

19.3.2 absorbed_dose_measure

An **absorbed_dose_measure** is the value of an absorbed dose of radiation.

EXPRESS specification:

```
TYPE absorbed_dose_measure = REAL;
END_TYPE;
```

19.3.3 acceleration_measure

An **acceleration_measure** is the value of the rate of change of velocity.

EXPRESS specification:

```
TYPE acceleration_measure = REAL;
END_TYPE;
```

19.3.4 radioactivity_measure

A **radioactivity_measure** is the value of the radioactive disintegration.

EXPRESS specification:

```
TYPE radioactivity_measure = REAL;
END_TYPE;
```

19.3.5 amount_of_substance_measure

An **amount_of_substance_measure** is the value for the quantity of a substance when compared with the number of atoms in 0.012 kilogram of carbon 12.

EXPRESS specification:

```
TYPE amount_of_substance_measure = REAL;
END_TYPE;
```

19.3.6 area_measure

An **area_measure** is the value of the extent of a surface.

EXPRESS specification:

```
TYPE area_measure = REAL;
END_TYPE;
```

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19.3.7 celsius_temperature_measure

A **celsius_temperature_measure** is the value for the degree of heat of a body.

EXPRESS specification:

```
TYPE celsius_temperature_measure = REAL;
END_TYPE;
```

19.3.8 context_dependent_measure

A **context_dependent_measure** is the value of a physical quantity that may be interpreted based on the context in which it is used.

EXPRESS specification:

```
TYPE context_dependent_measure = REAL;
END_TYPE;
```

19.3.9 count_measure

A **count_measure** is the value of a count.

EXPRESS specification:

```
TYPE count_measure = NUMBER;
END_TYPE;
```

19.3.10 descriptive_measure

A **descriptive_measure** is a textual value of a physical quantity.

EXPRESS specification:

```
TYPE descriptive_measure = STRING;
END_TYPE;
```

19.3.11 capacitance_measure

A **capacitance_measure** is the value of capacitance.

EXPRESS specification:

```
TYPE capacitance_measure = REAL;
END_TYPE;
```

19.3.12 dose_equivalent_measure

A **dose_equivalent_measure** is the value of the radiation dose equivalent.

EXPRESS specification:

```
TYPE dose_equivalent_measure = REAL;
END_TYPE;
```

19.3.13 electric_charge_measure

An **electric_charge_measure** is the value of an electrical charge.

EXPRESS specification:

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```
TYPE electric_charge_measure = REAL;
END_TYPE;
```

19.3.14 electric_current_measure

An **electric_current_measure** is the value for the movement of electrically charged particles.

EXPRESS specification:

```
TYPE electric_current_measure = REAL;
END_TYPE;
```

19.3.15 conductance_measure

A **conductance_measure** is the value of an electrical conductance.

EXPRESS specification:

```
TYPE conductance_measure = REAL;
END_TYPE;
```

19.3.16 electric_potential_measure

An **electric_potential_measure** is the value of an electrical potential.

EXPRESS specification:

```
TYPE electric_potential_measure = REAL;
END_TYPE;
```

19.3.17 energy_measure

An **energy_measure** is the value of energy, or work done, in a system.

EXPRESS specification:

```
TYPE energy_measure = REAL;
END_TYPE;
```

19.3.18 force_measure

A **force_measure** is the value of a force.

EXPRESS specification:

```
TYPE force_measure = REAL;
END_TYPE;
```

19.3.19 frequency_measure

A **frequency_measure** is the value of a frequency.

EXPRESS specification:

```
TYPE frequency_measure = REAL;
END_TYPE;
```

19.3.20 magnetic_flux_density_measure

A **magnetic_flux_density_measure** is the value of magnetic flux density.

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EXPRESS specification:

```
TYPE magnetic_flux_density_measure = REAL;
END_TYPE;
```

19.3.21 illuminance_measure

An **illuminance_measure** is the value of illuminance.

EXPRESS specification:

```
TYPE illuminance_measure = REAL;
END_TYPE;
```

19.3.22 inductance_measure

An **inductance_measure** is the value of inductance.

EXPRESS specification:

```
TYPE inductance_measure = REAL;
END_TYPE;
```

19.3.23 length_measure

A **length_measure** is the value of a distance.

EXPRESS specification:

```
TYPE length_measure = REAL;
END_TYPE;
```

19.3.24 luminous_intensity_measure

A **luminous_intensity_measure** is the value for the brightness of a body.

EXPRESS specification:

```
TYPE luminous_intensity_measure = REAL;
END_TYPE;
```

19.3.25 luminous_flux_measure

A **luminous_flux_measure** is the value of luminous flux.

EXPRESS specification:

```
TYPE luminous_flux_measure = REAL;
END_TYPE;
```

19.3.26 mass_measure

A **mass_measure** is the value of the amount of matter that a body contains.

EXPRESS specification:

```
TYPE mass_measure = REAL;
END_TYPE;
```


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19.3.27 measure_value

The **measure_value** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

A **measure_value** is a value as defined in ISO 31-0.

EXPRESS specification:

```
TYPE measure_value = SELECT
  (absorbed_dose_measure,
   dose_equivalent_measure,
   radioactivity_measure,
   acceleration_measure,
   amount_of_substance_measure,
   area_measure,
   celsius_temperature_measure,
   context_dependent_measure,
   count_measure,
   descriptive_measure,
   capacitance_measure,
   electric_charge_measure,
   conductance_measure,
   electric_current_measure,
   electric_potential_measure,
   energy_measure,
   magnetic_flux_density_measure,
   force_measure,
   frequency_measure,
   illuminance_measure,
   inductance_measure,
   length_measure,
   luminous_flux_measure,
   luminous_intensity_measure,
   magnetic_flux_measure,
   mass_measure,
   numeric_measure,
   non_negative_length_measure,
   parameter_value,
   plane_angle_measure,
   positive_length_measure,
   positive_plane_angle_measure,
   positive_ratio_measure,
   power_measure,
   pressure_measure,
   ratio_measure,
   resistance_measure,
   solid_angle_measure,
   thermodynamic_temperature_measure,
   time_measure,
   velocity_measure,
   volume_measure);
END_TYPE;
```

19.3.28 magnetic_flux_measure

A **magnetic_flux_measure** is the value of magnetic flux.

EXPRESS specification:

```
TYPE magnetic_flux_measure = REAL;
END_TYPE;
```

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19.3.29 non_negative_length_measure

A **non_negative_length_measure** type is a length_measure whose value is greater than or equal to zero.

EXPRESS specification:

```
TYPE non_negative_length_measure = length_measure;
WHERE
  WR1: SELF >= 0.0;
END_TYPE;
```

Formal propositions:

WR1: A **non_negative_length_measure** shall be positive or zero.

19.3.30 numeric_measure

A **numeric_measure** is the numeric value of a physical quantity.

NOTE In order to define measure quantities that are not included as specific types in this standard a **numeric_measure** may be used as the value_component of a measure_with_unit, the corresponding unit_component being an si_unit or a derived_unit.

EXPRESS specification:

```
TYPE numeric_measure = NUMBER;
END_TYPE;
```

19.3.31 parameter_value

A **parameter_value** is the value that specifies the amount of a parameter in a parameter space.

EXPRESS specification:

```
TYPE parameter_value = REAL;
END_TYPE;
```

19.3.32 plane_angle_measure

A **plane_angle_measure** is the value of an angle in a plane.

EXPRESS specification:

```
TYPE plane_angle_measure = REAL;
END_TYPE;
```

19.3.33 positive_length_measure

A **positive_length_measure** is a length_measure that is greater than zero.

EXPRESS specification:

```
TYPE positive_length_measure = non_negative_length_measure;
WHERE
  WR1: SELF > 0.0;
END_TYPE;
```

Formal propositions:

WR1: The value shall be positive.

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19.3.34 positive_plane_angle_measure

A **positive_plane_angle_measure** is a plane_angle_measure that is greater than zero.

EXPRESS specification:

```
TYPE positive_plane_angle_measure = plane_angle_measure;
WHERE
  WR1: SELF > 0.0;
END_TYPE;
```

Formal propositions:

WR1: The value shall be positive.

19.3.35 positive_ratio_measure

A **positive_ratio_measure** is a ratio_measure that is greater than zero.

EXPRESS specification:

```
TYPE positive_ratio_measure = ratio_measure;
WHERE
  WR1: SELF > 0.0;
END_TYPE;
```

Formal propositions:

WR1: The value shall be positive.

19.3.36 power_measure

A **power_measure** is the value of power, or the rate of doing work.

EXPRESS specification:

```
TYPE power_measure = REAL;
END_TYPE;
```

19.3.37 pressure_measure

A **pressure_measure** is the value of force per unit area.

EXPRESS specification:

```
TYPE pressure_measure = REAL;
END_TYPE;
```

19.3.38 ratio_measure

A **ratio_measure** is the value of the relation between two physical quantities that are of the same kind.

EXPRESS specification:

```
TYPE ratio_measure = REAL;
END_TYPE;
```

19.3.39 resistance_measure

A **resistance_measure** is the value of electrical resistance.

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EXPRESS specification:

```
TYPE resistance_measure = REAL;
END_TYPE;
```

19.3.40 si_prefix

An **si_prefix** is the name of a prefix that may be associated with an si_unit. The definitions of SI prefixes are specified in ISO 1000.

EXPRESS specification:

```
TYPE si_prefix = ENUMERATION OF
  (exa,
   peta,
   tera,
   giga,
   mega,
   kilo,
   hecto,
   deca,
   deci,
   centi,
   milli,
   micro,
   nano,
   pico,
   femto,
   atto);
END_TYPE;
```

Enumerated item definitions:

exa see ISO 1000;

peta see ISO 1000;

tera see ISO 1000;

giga see ISO 1000;

mega see ISO 1000;

kilo see ISO 1000;

hecto see ISO 1000;

deca see ISO 1000;

deci see ISO 1000;

centi see ISO 1000;

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milli see ISO 1000;

micro see ISO 1000;

nano see ISO 1000;

pico see ISO 1000;

femto see ISO 1000;

atto see ISO 1000.

19.3.41 si_unit_name

An **si_unit_name** is the name of an SI unit.

NOTE 1 The definitions of the names of SI units are specified in ISO 1000.

EXPRESS specification:

```
TYPE si_unit_name = ENUMERATION OF
  (metre,
   gram,
   second,
   ampere,
   kelvin,
   mole,
   candela,
   radian,
   steradian,
   hertz,
   newton,
   pascal,
   joule,
   watt,
   coulomb,
   volt,
   farad,
   ohm,
   siemens,
   weber,
   tesla,
   henry,
   degree_Celsius,
   lumen,
   lux,
   becquerel,
   gray,
   sievert);
END_TYPE;
```

Enumerated item definitions:

metre see ISO 1000;

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gram see ISO 1000;

NOTE 2 ISO 1000 gives “kilogram” as the SI unit name. This part of ISO 10303 uses “gram” as the SI unit name.

second see ISO 1000;

ampere see ISO 1000;

kelvin see ISO 1000;

mole see ISO 1000;

candela see ISO 1000;

radian see ISO 1000;

steradian see ISO 1000;

hertz see ISO 1000;

newton see ISO 1000;

pascal see ISO 1000;

joule see ISO 1000;

watt see ISO 1000;

coulomb see ISO 1000;

volt see ISO 1000;

farad see ISO 1000;

ohm see ISO 1000;

siemens see ISO 1000;

weber see ISO 1000;

tesla see ISO 1000;

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henry see ISO 1000;

degree_Celsius see ISO 1000;

lumen see ISO 1000;

lux see ISO 1000;

becquerel see ISO 1000;

gray see ISO 1000;

sievert see ISO 1000.

19.3.42 solid_angle_measure

A **solid_angle_measure** is the value of a solid angle.

EXPRESS specification:

```
TYPE solid_angle_measure = REAL;
END_TYPE;
```

19.3.43 thermodynamic_temperature_measure

A **thermodynamic_temperature_measure** is the value for the degree of heat of a body.

EXPRESS specification:

```
TYPE thermodynamic_temperature_measure = REAL;
END_TYPE;
```

19.3.44 time_measure

A **time_measure** is the value of the duration of a period.

EXPRESS specification:

```
TYPE time_measure = REAL;
END_TYPE;
```

19.3.45 unit

The **unit** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

A **unit** is a physical quantity, with a value of one.

NOTE A **unit** is used to express other physical quantities of the same kind.

EXPRESS specification:

```
TYPE unit = SELECT
  (derived_unit,
```

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```
   named_unit);
END_TYPE;
```

19.3.46 velocity_measure

A **velocity_measure** is the value of the rate of change of position.

EXPRESS specification:

```
TYPE velocity_measure = REAL;
END_TYPE;
```

19.3.47 volume_measure

A **volume_measure** is the value of the solid content of a body.

EXPRESS specification:

```
TYPE volume_measure = REAL;
END_TYPE;
```

19.4 Measure entity definitions

19.4.1 absorbed_dose_measure_with_unit

An **absorbed_dose_measure_with_unit** is a type of measure_with_unit in which the physical quantity is an absorbed dose of radiation as defined in ISO 31.

EXPRESS specification:

```
ENTITY absorbed_dose_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.ABSORBED_DOSE_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_
component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component be of type absorbed_dose_unit.

19.4.2 absorbed_dose_unit

An **absorbed_dose_unit** is a type of derived_unit in which the absorbed dose is expressed.

EXPRESS specification:

```
ENTITY absorbed_dose_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.gray);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the Gray as defined in ISO 31.

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19.4.3 si_absorbed_dose_unit

A **si_absorbed_dose_unit** is a type of absorbed_dose_unit and a type of si_unit in which the absorbed dose is expressed in SI units.

EXPRESS specification:

```
ENTITY si_absorbed_dose_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.gray;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be gray.

WR2: No other name shall be assigned to this entity.

19.4.4 acceleration_measure_with_unit

An **acceleration_measure_with_unit** is a type of measure_with_unit in which the physical quantity is an acceleration.

EXPRESS specification:

```
ENTITY acceleration_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.ACCELERATION_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_
component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component be of type acceleration_unit.

19.4.5 acceleration_unit

An **acceleration_unit** is a type of derived_unit in which the acceleration of an object is expressed.

EXPRESS specification:

```
ENTITY acceleration_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensional_exponents(1.0, 0.0, -2.0, 0.0,
0.0, 0.0, 0.0);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponent of length shall be equal to one, the dimensional exponent of time shall be equal to minus two, and all the other dimensional exponents shall be equal to zero.

19.4.6 radioactivity_measure_with_unit

A **radioactivity_measure_with_unit** is a type of measure_with_unit in which the physical quantity is radioactivity as defined in ISO 31.

EXPRESS specification:

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```
ENTITY radioactivity_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.RADIOACTIVITY_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_
component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type radioactivity_unit.

19.4.7 radioactivity_unit

A **radioactivity_unit** is a type of derived_unit in which the radioactivity is expressed.

EXPRESS specification:

```
ENTITY radioactivity_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.
becquerel);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the becquerel as defined in ISO 31.

19.4.8 si_radioactivity_unit

A **si_radioactivity_unit** is a type of radioactivity_unit and a type of si_unit in which the radioactivity is expressed in SI units.

EXPRESS specification:

```
ENTITY si_radioactivity_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.becquerel;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be becquerel.

WR2: No other name shall be assigned to this entity.

19.4.9 amount_of_substance_measure_with_unit

An **amount_of_substance_measure_with_unit** is a type of measure_with_unit. // in which the physical quantity is an This entity data type shall only be used to characterize a number of particles.

EXPRESS specification:

```
ENTITY amount_of_substance_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.AMOUNT_OF_SUBSTANCE_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_
component);
END_ENTITY;
```

Formal propositions:

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WR1: The unit shall be an amount_of_substance_unit.

19.4.10 amount_of_substance_unit

An **amount_of_substance_unit** is a type of named_unit in which the number of elementary entities of a substance as compared to the number of atoms in 0.012 kilograms of carbon-12 is expressed.

NOTE 1 This definition applies to the SI quantity 'mole'. When the mole is used, the elementary entities, whose quantity is expressed, must be specified. They may be atoms, molecules, ions, electrons or other particles or specified groups of such particles (see ISO 31-8).

NOTE 2 This unit corresponds to one of the seven fundamental quantities as specified in ISO 1000.

EXPRESS specification:

```
ENTITY amount_of_substance_unit
  SUBTYPE OF (named_unit);
WHERE
  WR1: ((((((SELF\named_unit.dimensions.length_exponent = 0.0) AND (SELF\named_unit.
dimensions.mass_exponent = 0.0)) AND (SELF\named_unit.dimensions.time_exponent = 0.0))
AND (SELF\named_unit.dimensions.electric_current_exponent = 0.0)) AND (SELF\named_unit.
dimensions.thermodynamic_temperature_exponent = 0.0)) AND (SELF\named_unit.dimensions.
amount_of_substance_exponent = 1.0)) AND (SELF\named_unit.dimensions.luminous_intensity_
exponent = 0.0);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponent of amount of substance shall be equal to one, and all the other dimensional exponents shall be equal to zero.

19.4.11 area_measure_with_unit

An **area_measure_with_unit** is a type of measure_with_unit in which the physical quantity is an area as defined in ISO 31.

EXPRESS specification:

```
ENTITY area_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.AREA_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit shall be an area_unit.

19.4.12 area_unit

An **area_unit** is a type of derived_unit used to measure the extent of a surface.

EXPRESS specification:

```
ENTITY area_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensional_exponents(2.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0);
END_ENTITY;
```

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Formal propositions:

WR1: The dimensional exponent of length shall be equal to two and all the other dimensional exponents shall be equal to zero.

19.4.13 celsius_temperature_measure_with_unit

A **celsius_temperature_measure_with_unit** is a type of measure_with_unit in which the physical quantity is a Celsius temperature as defined in ISO 31-4.

EXPRESS specification:

```
ENTITY celsius_temperature_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.THERMODYNAMIC_TEMPERATURE_UNIT' IN TYPEOF(SELF\measure_with_unit.
unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit shall be a thermodynamic_temperature_unit.

19.4.14 context_dependent_unit

A **context_dependent_unit** is a type of named_unit that is not related to the system of units defined in this part of ISO 10303.

EXAMPLE The number of parts in an assembly is a physical quantity that may be measured in a unit called "parts". Such a unit cannot be related to an SI unit.

EXPRESS specification:

```
ENTITY context_dependent_unit
  SUBTYPE OF (named_unit);
  name : label;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **context_dependent_unit** is known.

19.4.15 conversion_based_unit

A **conversion_based_unit** is a type of named_unit that defines a unit on the basis of a measure_with_unit.

NOTE The value_component attribute of the measure_with_unit defines the conversion factor.

EXAMPLE An inch is a **conversion_based_unit**. It is from the Imperial system, its name is "inch", and it can be related to the si_unit, millimetre, through a measure_with_unit whose value is 25.4 millimetre. A foot is also a **conversion_based_unit**. It is from the Imperial system, its name is "foot", and it can be related to an si_unit, millimetre, either directly or through the unit called "inch".

EXPRESS specification:

```
ENTITY conversion_based_unit
  SUBTYPE OF (named_unit);
  name : label;
  conversion_factor : measure_with_unit;
WHERE
  WR1: SELF\named_unit.dimensions = derive_dimensional_exponents(conversion_factor
\measure_with_unit.unit_component);
```

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```
END_ENTITY;
```

Attribute definitions:

name: the label by which the **conversion_based_unit** is known.

conversion_factor: the measure_with_unit that specifies the physical quantity from which the **conversion_based_unit** is derived.

Formal propositions:

WR1: The dimensional exponents shall be equal to those from the conversion_factor.

19.4.16 derived_unit 

A **derived_unit** is an expression of units.

EXAMPLE Newtons per square millimetre is a **derived_unit**.

A **derived_unit** may be either an absorbed_dose_unit or an acceleration_unit, or a radioactivity_unit, or an area_unit, or a capacitance_unit, or a dose_equivalent_unit, or an electric_charge_unit, or a conductance_unit, or an electric_potential_unit, or an energy_unit, or a magnetic_flux_density_unit or a force_unit, or a frequency_unit, or an illuminance_unit, or an inductance_unit, or a magnetic_flux_unit, or a power_unit, or a pressure_unit, or a resistance_unit, or a velocity_unit, or a volume_unit.

EXPRESS specification:

```
ENTITY derived_unit
  SUPERTYPE OF (ONEOF(absorbed_dose_unit, acceleration_unit, radioactivity_unit, area_
unit, capacitance_unit, dose_equivalent_unit, electric_charge_unit, conductance_unit,
electric_potential_unit, energy_unit, magnetic_flux_density_unit, force_unit, frequency_
unit, illuminance_unit, inductance_unit, magnetic_flux_unit, power_unit, pressure_unit,
resistance_unit, velocity_unit, volume_unit));
  elements : SET [1:?] OF derived_unit_element;
DERIVE
  name : label := get_name_value(SELF);
WHERE
  WR1: (SIZEOF(elements) > 1) OR (SIZEOF(elements) = 1) AND (elements[1].exponent <> 1.0);
  WR2: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'NAME_ATTRIBUTE.NAMED_ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

elements: the derived_unit_elements and their exponents that define the **derived_unit**.

name: the label by which the **derived_unit** is known.

NOTE 1 This attribute is an upwardly compatible addition to **derived_unit** as specified in ISO 10303-41:1994.

Formal propositions:

WR1: There shall be either more than one member in the elements set or the value of the exponent of the single element of the elements set shall not be equal to one.

WR2: Each **derived_unit** shall be the named_item in at most one name_attribute.

NOTE 2 The name_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 3 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

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19.4.17 derived_unit_element

A **derived_unit_element** is the association of a named_unit with an exponent.

NOTE This entity is used in this part of ISO 10303 to represent an element of the dimensional expression of a derived_unit.

EXAMPLE Newtons per square millimetre is a derived unit. It has two elements, Newton whose exponent has a value of 1, and millimetre whose exponent is -2.

EXPRESS specification:

```
ENTITY derived_unit_element;
  unit : named_unit;
  exponent : REAL;
END_ENTITY;
```

Attribute definitions:

unit: the named_unit that specifies the mathematical factor of the element.

exponent: the power that is applied to the unit attribute.

19.4.18 dimensional_exponents

A **dimensional_exponents** is a definition of the powers of the dimensions of the seven base quantities.

NOTE 1 The seven base quantities are defined in ISO 31.

NOTE 2 This entity enables the dimensional expression of any physical quantity with respect to these seven base quantities.

NOTE 3 Length, mass, time, electric current, thermodynamic temperature, amount of substance, and luminous intensity are the seven base quantities.

EXAMPLE 1 A length of 2 millimetres has a length exponent of 1. The remaining exponents are equal to 0.

EXAMPLE 2 A velocity of 2 millimetres per second has a length exponent of 1 and a time exponent of -1. The remaining exponents are 0.

EXPRESS specification:

```
ENTITY dimensional_exponents;
  length_exponent : REAL;
  mass_exponent : REAL;
  time_exponent : REAL;
  electric_current_exponent : REAL;
  thermodynamic_temperature_exponent : REAL;
  amount_of_substance_exponent : REAL;
  luminous_intensity_exponent : REAL;
END_ENTITY;
```

Attribute definitions:

length_exponent: the power of the length base quantity.

mass_exponent: the power of the mass base quantity.

time_exponent: the power of the time base quantity.

electric_current_exponent: the power of the electric current base quantity.

thermodynamic_temperature_exponent: the power of the thermodynamic temperature base quantity.

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amount_of_substance_exponent: the power of the amount of substance base quantity.

luminous_intensity_exponent: the power of the luminous intensity base quantity.

19.4.19 capacitance_unit

A **capacitance_unit** is a type of derived_unit in which the capacitance is expressed.

EXPRESS specification:

```
ENTITY capacitance_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.farad);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the farad as defined in ISO 31.

19.4.20 capacitance_measure_with_unit

A **capacitance_measure_with_unit** is a type of measure_with_unit in which the physical quantity is an electric capacitance as defined in ISO 31.

EXPRESS specification:

```
ENTITY capacitance_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.CAPACITANCE_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component be of type capacitance_unit.

19.4.21 si_capacitance_unit

A **si_capacitance_unit** is a type of capacitance_unit and a type of si_unit in which the electric capacitance is expressed in SI units.

EXPRESS specification:

```
ENTITY si_capacitance_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.farad;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be farad.

WR2: No other name shall be assigned to this entity.

19.4.22 dose_equivalent_measure_with_unit

A **dose_equivalent_measure_with_unit** is a type of measure_with_unit in which the physical quantity is a radiation dose equivalent as defined in ISO 31.

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EXPRESS specification:

```
ENTITY dose_equivalent_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.DOSE_EQUIVALENT_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_
component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type dose_equivalent_unit.

19.4.23 dose_equivalent_unit

A **dose_equivalent_unit** is a type of derived_unit in which the radiation dose equivalent is expressed.

EXPRESS specification:

```
ENTITY dose_equivalent_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.sievert);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the sievert as defined in ISO 31.

19.4.24 si_dose_equivalent_unit

A **si_dose_equivalent_unit** is a type of dose_equivalent_unit and a type of si_unit in which the radiation dose equivalent is expressed in SI units.

EXPRESS specification:

```
ENTITY si_dose_equivalent_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.sievert;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be sievert.

WR2: No other name shall be assigned to this entity.

19.4.25 electric_charge_unit

An **electric_charge_unit** is a type of derived_unit in which the electric charge is expressed.

EXPRESS specification:

```
ENTITY electric_charge_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.coulomb);
END_ENTITY;
```

Formal propositions:

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WR1: The dimensional exponents shall be equal to those of the coulomb as defined in ISO 31.

19.4.26 electric_charge_measure_with_unit

An **electric_charge_measure_with_unit** is a type of measure_with_unit in which the physical quantity is an electric charge as defined in ISO 31.

EXPRESS specification:

```
ENTITY electric_charge_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.ELECTRIC_CHARGE_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_
component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type electric_charge_unit.

19.4.27 si_electric_charge_unit

A **si_electric_charge_unit** is a type of electric_charge_unit and a type of si_unit in which the electric charge is expressed in SI units.

EXPRESS specification:

```
ENTITY si_electric_charge_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.coulomb;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be volt.

WR2: No other name shall be assigned to this entity.

19.4.28 electric_current_measure_with_unit

An **electric_current_measure_with_unit** is a type of measure_with_unit in which the physical quantity is an electric current as defined in ISO 31.

EXPRESS specification:

```
ENTITY electric_current_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.ELECTRIC_CURRENT_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_
component);
END_ENTITY;
```

Formal propositions:

WR1: The unit shall be an electric_current_unit.

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19.4.29 electric_current_unit

An **electric_current_unit** is a type of named_unit in which the movement of electrically charged particles is expressed.

NOTE This unit corresponds to one of the seven fundamental quantities as specified in ISO 1000.

EXPRESS specification:

```
ENTITY electric_current_unit
  SUBTYPE OF (named_unit);
WHERE
  WR1: ((((((SELF\named_unit.dimensions.length_exponent = 0.0) AND (SELF\named_unit.
dimensions.mass_exponent = 0.0)) AND (SELF\named_unit.dimensions.time_exponent = 0.0))
AND (SELF\named_unit.dimensions.electric_current_exponent = 1.0)) AND (SELF\named_unit.
dimensions.thermodynamic_temperature_exponent = 0.0)) AND (SELF\named_unit.dimensions.
amount_of_substance_exponent = 0.0)) AND (SELF\named_unit.dimensions.luminous_intensity_
exponent = 0.0);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponent of electric current shall be equal to one, and all the other dimensional exponents shall be equal to zero.

19.4.30 conductance_unit

A **conductance_unit** is a type of derived_unit in which the electrical conductance is expressed.

EXPRESS specification:

```
ENTITY conductance_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.siemens);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the siemens as defined in ISO 31.

19.4.31 conductance_measure_with_unit

A **conductance_measure_with_unit** is a type of measure_with_unit in which the physical quantity is a conductance as defined in ISO 31.

EXPRESS specification:

```
ENTITY conductance_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.CONDUCTANCE_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type conductance_unit.

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19.4.32 si_conductance_unit

A **si_conductance_unit** is a type of conductance_unit and a type of si_unit in which the electrical conductance is expressed in SI units.

EXPRESS specification:

```
ENTITY si_conductance_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.siemens;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be siemens.

WR2: No other name shall be assigned to this entity.

19.4.33 electric_potential_unit

An **electric_potential_unit** is a type of derived_unit in which the electric potential is expressed.

EXPRESS specification:

```
ENTITY electric_potential_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.volt);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the volt as defined in ISO 31.

19.4.34 electric_potential_measure_with_unit

An **electric_potential_measure_with_unit** is a type of measure_with_unit in which the physical quantity is an electric potential as defined in ISO 31.

EXPRESS specification:

```
ENTITY electric_potential_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.ELECTRIC_POTENTIAL_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_
component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type electric_potential_unit.

19.4.35 si_electric_potential_unit

A **si_electric_potential_unit** is a type of electric_potential_unit and a type of si_unit in which the electric potential is expressed in SI units.

EXPRESS specification:

```
ENTITY si_electric_potential_unit;
WHERE
```

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```
  WR1: SELF\si_unit.name = si_unit_name.volt;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be volt.

WR2: No other name shall be assigned to this entity.

19.4.36 energy_unit

An **energy_unit** is a type of derived_unit in which the energy is expressed.

EXPRESS specification:

```
ENTITY energy_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.joule);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the joule as defined in ISO 31.

19.4.37 energy_measure_with_unit

An **energy_measure_with_unit** is a type of measure_with_unit in which the physical quantity is energy as defined in ISO 31.

EXPRESS specification:

```
ENTITY energy_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.ENERGY_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type energy_unit.

19.4.38 si_energy_unit

A **si_energy_unit** is a type of energy_unit and a type of si_unit in which the energy is expressed in SI units.

EXPRESS specification:

```
ENTITY si_energy_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.joule;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be joule.

WR2: No other name shall be assigned to this entity.

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19.4.39 magnetic_flux_density_unit

A **magnetic_flux_density_unit** is a type of derived_unit in which the magnetic flux density is expressed.

EXPRESS specification:

```
ENTITY magnetic_flux_density_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.tesla);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the tesla as defined in ISO 31.

19.4.40 magnetic_flux_density_measure_with_unit

A **magnetic_flux_density_measure_with_unit** is a type of measure_with_unit in which the physical quantity is magnetic flux density as defined in ISO 31.

EXPRESS specification:

```
ENTITY magnetic_flux_density_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.MAGNETIC_FLUX_DENSITY_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_
component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type magnetic_flux_density_unit.

19.4.41 si_magnetic_flux_density_unit

A **si_magnetic_flux_density_unit** is a type of magnetic_flux_density_unit and a type of si_unit in which the magnetic flux density is expressed in SI units.

EXPRESS specification:

```
ENTITY si_magnetic_flux_density_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.tesla;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be tesla.

WR2: No other name shall be assigned to this entity.

19.4.42 force_unit

A **force_unit** is a type of derived_unit in which the force is expressed.

EXPRESS specification:

```
ENTITY force_unit
  SUBTYPE OF (derived_unit);
WHERE
```

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```
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.newton);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the newton as defined in ISO 31.

19.4.43 force_measure_with_unit

A **force_measure_with_unit** is a type of measure_with_unit in which the physical quantity is force as defined in ISO 31.

EXPRESS specification:

```
ENTITY force_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.FORCE_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type force_unit.

19.4.44 si_force_unit

A **si_force_unit** is a type of force_unit and a type of si_unit in which the force is expressed in SI units.

EXPRESS specification:

```
ENTITY si_force_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.newton;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be newton.

WR2: No other name shall be assigned to this entity.

19.4.45 frequency_unit

A **frequency_unit** is a type of derived_unit in which the frequency is expressed.

EXPRESS specification:

```
ENTITY frequency_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.hertz);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the hertz as defined in ISO 31.

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19.4.46 frequency_measure_with_unit

A **frequency_measure_with_unit** is a type of measure_with_unit in which the quantity measured is a frequency as defined in ISO 31.

EXPRESS specification:

```
ENTITY frequency_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.FREQUENCY_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type frequency_unit.

19.4.47 si_frequency_unit

A **si_frequency_unit** is a type of frequency_unit and a type of si_unit in which the frequency is expressed in SI units.

EXPRESS specification:

```
ENTITY si_frequency_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.hertz;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be hertz.

WR2: No other name shall be assigned to this entity.

19.4.48 global_unit_assigned_context

A **global_unit_assigned_context** is a type of representation_context in which the units apply to all measure_values of the applicable kind.

NOTE For an example of the use of this entity, see annex F.4.6.

EXPRESS specification:

```
ENTITY global_unit_assigned_context
  SUBTYPE OF (representation_context);
  units : SET [1:?] OF unit;
END_ENTITY;
```

Attribute definitions:

units: the units that apply in the representation_context.

19.4.49 illuminance_unit

An **illuminance_unit** is a type of derived_unit in which the illuminance is expressed.

EXPRESS specification:

```
ENTITY illuminance_unit
  SUBTYPE OF (derived_unit);
```

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```
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.lux);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the lux as defined in ISO 31.

19.4.50 illuminance_measure_with_unit

An **illuminance_measure_with_unit** is a type of measure_with_unit in which the quantity measured is an illuminance as defined in ISO 31.

EXPRESS specification:

```
ENTITY illuminance_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.ILLUMINANCE_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type illuminance_unit.

19.4.51 si_illuminance_unit

A **si_illuminance_unit** is a type of illuminance_unit and a type of si_unit in which the illuminance is expressed in SI units.

EXPRESS specification:

```
ENTITY si_illuminance_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.lux;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be lux.

WR2: No other name shall be assigned to this entity.

19.4.52 inductance_unit

An **inductance_unit** is a type of derived_unit in which the inductance is expressed.

EXPRESS specification:

```
ENTITY inductance_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.henry);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the henry as defined in ISO 31.

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19.4.53 inductance_measure_with_unit

An **inductance_measure_with_unit** is a type of measure_with_unit in which the quantity measured is an inductance as defined in ISO 31.

EXPRESS specification:

```
ENTITY inductance_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.INDUCTANCE_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type inductance_unit.

19.4.54 si_inductance_unit

A **si_inductance_unit** is a type of inductance_unit and a type of si_unit in which the inductance is expressed in SI units.

EXPRESS specification:

```
ENTITY si_inductance_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.henry;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be henry.

WR2: No other name shall be assigned to this entity.

19.4.55 length_measure_with_unit

A **length_measure_with_unit** is a type of measure_with_unit in which the physical quantity is a length as defined in ISO 31.

EXPRESS specification:

```
ENTITY length_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.LENGTH_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit shall be a length_unit.

19.4.56 positive_length_measure_with_unit

A **positive_length_measure_with_unit** is a type of length_measure_with_unit of which the value is positive.

EXPRESS specification:

```
ENTITY positive_length_measure_with_unit
  SUBTYPE OF (length_measure_with_unit);
```

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```
WHERE
  WR1: 'MEASURE_SCHEMA.POSITIVE_LENGTH_MEASURE' IN TYPEOF(SELF\measure_with_unit.value_
component);
END_ENTITY;
```

Formal propositions:

WR1: The value of the value_component shall be positive.

19.4.57 length_unit

A **length_unit** is a type of named_unit in which distances are expressed.

NOTE This unit corresponds to one of the seven fundamental quantities as specified in ISO 1000.

EXPRESS specification:

```
ENTITY length_unit
  SUBTYPE OF (named_unit);
WHERE
  WR1: ((((((SELF\named_unit.dimensions.length_exponent = 1.0) AND (SELF\named_unit.
dimensions.mass_exponent = 0.0)) AND (SELF\named_unit.dimensions.time_exponent = 0.0))
AND (SELF\named_unit.dimensions.electric_current_exponent = 0.0)) AND (SELF\named_unit.
dimensions.thermodynamic_temperature_exponent = 0.0)) AND (SELF\named_unit.dimensions.
amount_of_substance_exponent = 0.0)) AND (SELF\named_unit.dimensions.luminous_intensity_
exponent = 0.0);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponent of length shall be equal to one, and all the other dimensional exponents shall be equal to zero.

19.4.58 luminous_flux_unit

A **luminous_flux_unit** is a type of named_unit in which the luminous flux is expressed.

EXPRESS specification:

```
ENTITY luminous_flux_unit
  SUBTYPE OF (named_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.lumen);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the lumen as defined in ISO 31.

19.4.59 luminous_flux_measure_with_unit

A **luminous_flux_measure_with_unit** is a type of measure_with_unit in which the quantity measured is a luminous flux as defined in ISO 31.

EXPRESS specification:

```
ENTITY luminous_flux_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.LUMINOUS_FLUX_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_
component);
```


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```
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type luminous_flux_unit.

19.4.60 luminous_intensity_measure_with_unit

A **luminous_intensity_measure_with_unit** is a type of measure_with_unit in which the physical quantity is a luminous intensity as defined in ISO 31.

EXPRESS specification:

```
ENTITY luminous_intensity_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.LUMINOUS_INTENSITY_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_
component);
END_ENTITY;
```

Formal propositions:

WR1: The unit shall be a luminous_intensity_unit.

19.4.61 luminous_intensity_unit

A **luminous_intensity_unit** is a type of named_unit in which the brightness of a body is expressed.

NOTE This unit corresponds to one of the seven fundamental quantities as specified in ISO 1000.

EXPRESS specification:

```
ENTITY luminous_intensity_unit
  SUBTYPE OF (named_unit);
WHERE
  WR1: ((((((SELF\named_unit.dimensions.length_exponent = 0.0) AND (SELF\named_unit.
dimensions.mass_exponent = 0.0)) AND (SELF\named_unit.dimensions.time_exponent = 0.0))
AND (SELF\named_unit.dimensions.electric_current_exponent = 0.0)) AND (SELF\named_unit.
dimensions.thermodynamic_temperature_exponent = 0.0)) AND (SELF\named_unit.dimensions.
amount_of_substance_exponent = 0.0)) AND (SELF\named_unit.dimensions.luminous_intensity_
exponent = 1.0);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponent of luminous intensity shall be equal to one, and all the other dimensional exponents shall be equal to zero.

19.4.62 magnetic_flux_unit

A **magnetic_flux_unit** is a type of derived_unit in which the magnetic flux is expressed.

EXPRESS specification:

```
ENTITY magnetic_flux_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.weber);
END_ENTITY;
```

Formal propositions:

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WR1: The dimensional exponents shall be equal to those of the weber as defined in ISO 31.

19.4.63 magnetic_flux_measure_with_unit

A **magnetic_flux_measure_with_unit** is a type of measure_with_unit in which the quantity measured is a magnetic flux as defined in ISO 31.

EXPRESS specification:

```
ENTITY magnetic_flux_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.MAGNETIC_FLUX_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_
component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type magnetic_flux_unit.

19.4.64 si_magnetic_flux_unit

A **si_magnetic_flux_unit** is a type of magnetic_flux_unit and a type of si_unit in which the magnetic flux is expressed in SI units.

EXPRESS specification:

```
ENTITY si_magnetic_flux_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.weber;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be weber.

WR2: No other name shall be assigned to this entity.

19.4.65 mass_measure_with_unit

A **mass_measure_with_unit** is a type of measure_with_unit in which the physical quantity is a mass as defined in ISO 31.

EXPRESS specification:

```
ENTITY mass_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.MASS_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit shall be a mass_unit.

19.4.66 mass_unit

A **mass_unit** is a type of named_unit in which the amount of matter that a body contains is expressed.

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NOTE This unit corresponds to one of the seven fundamental quantities as specified in ISO 1000.

EXPRESS specification:

```
ENTITY mass_unit
  SUBTYPE OF (named_unit);
WHERE
  WR1: ((((((SELF\named_unit.dimensions.length_exponent = 0.0) AND (SELF\named_unit.
dimensions.mass_exponent = 1.0)) AND (SELF\named_unit.dimensions.time_exponent = 0.0))
AND (SELF\named_unit.dimensions.electric_current_exponent = 0.0)) AND (SELF\named_unit.
dimensions.thermodynamic_temperature_exponent = 0.0)) AND (SELF\named_unit.dimensions.
amount_of_substance_exponent = 0.0)) AND (SELF\named_unit.dimensions.luminous_intensity_
exponent = 0.0);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponent of mass shall be equal to one, and all the other dimensional exponents shall be equal to zero.

19.4.67 measure_with_unit

A **measure_with_unit** is the specification of a physical quantity as defined in ISO 31. A **measure_with_unit** may be either a length_measure_with_unit or a mass_measure_with_unit, or a time_measure_with_unit, or an electric_current_measure_with_unit, or a thermodynamic_temperature_measure_with_unit, or a celsius_temperature_measure_with_unit, or an amount_of_substance_measure_with_unit, or a luminous_intensity_measure_with_unit, or a plane_angle_measure_with_unit, or a solid_angle_measure_with_unit, or an area_measure_with_unit, or a volume_measure_with_unit, or a ratio_measure_with_unit, or an acceleration_measure_with_unit, or a capacitance_measure_with_unit, or an electric_charge_measure_with_unit, or a conductance_measure_with_unit, or an electric_potential_measure_with_unit, or an energy_measure_with_unit, or a magnetic_flux_density_measure_with_unit, or a force_measure_with_unit, or a frequency_measure_with_unit, or an illuminance_measure_with_unit, or an inductance_measure_with_unit, or a luminous_flux_measure_with_unit, or a magnetic_flux_measure_with_unit, or a power_measure_with_unit, or a pressure_measure_with_unit, or a resistance_measure_with_unit, or a velocity_measure_with_unit, or an absorbed_dose_measure_with_unit, or a radioactivity_measure_with_unit, or a dose_equivalent_measure_with_unit.

EXPRESS specification:

```
ENTITY measure_with_unit
  SUPERTYPE OF (ONEOF(length_measure_with_unit, mass_measure_with_unit, time_measure_
with_unit, electric_current_measure_with_unit, thermodynamic_temperature_measure_with_
unit, celsius_temperature_measure_with_unit, amount_of_substance_measure_with_unit,
luminous_intensity_measure_with_unit, plane_angle_measure_with_unit, solid_angle_measure_
with_unit, area_measure_with_unit, volume_measure_with_unit, ratio_measure_with_unit,
acceleration_measure_with_unit, capacitance_measure_with_unit, electric_charge_measure_
with_unit, conductance_measure_with_unit, electric_potential_measure_with_unit, energy_
measure_with_unit, magnetic_flux_density_measure_with_unit, force_measure_with_unit,
frequency_measure_with_unit, illuminance_measure_with_unit, inductance_measure_with_unit,
luminous_flux_measure_with_unit, magnetic_flux_measure_with_unit, power_measure_with_
unit, pressure_measure_with_unit, resistance_measure_with_unit, velocity_measure_with_
unit, absorbed_dose_measure_with_unit, radioactivity_measure_with_unit, dose_equivalent_
measure_with_unit));
  value_component : measure_value;
  unit_component : unit;
WHERE
  WR1: valid_units(SELF);
END_ENTITY;
```

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Attribute definitions:

value_component: the value of the physical quantity if expressed with respect to the **unit_component**.

unit_component: the unit in which the physical quantity is expressed.

Formal propositions:

WR1: The **unit_component** shall be a valid unit for the kind of measure specified by the **value_component**.

19.4.68 named_unit

A **named_unit** is a unit quantity. A **named_unit** may be either a si_unit or a conversion_based_unit, or a context_dependent_unit, and additionally may be a length_unit or a mass_unit, or a time_unit, or an electric_current_unit, or a thermodynamic_temperature_unit, or an amount_of_substance_unit, or a luminous_flux_unit, or a luminous_intensity_unit, or a plane_angle_unit, or a solid_angle_unit, or a ratio_unit.

EXPRESS specification:

```
ENTITY named_unit
  SUPERTYPE OF (ONEOF(si_unit, conversion_based_unit, context_dependent_unit) ANDOR
ONEOF(length_unit, mass_unit, time_unit, electric_current_unit, thermodynamic_temperature_
unit, amount_of_substance_unit, luminous_flux_unit, luminous_intensity_unit, plane_angle_
unit, solid_angle_unit, ratio_unit));
  dimensions : dimensional_exponents;
END_ENTITY;
```

Attribute definitions:

dimensions: the exponents of the base properties by which the **named_unit** is defined.

19.4.69 plane_angle_measure_with_unit

A **plane_angle_measure_with_unit** is a type of measure_with_unit in which the physical quantity is a plane angle as defined in ISO 31.

EXPRESS specification:

```
ENTITY plane_angle_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.PLANE_ANGLE_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit shall be a plane_angle_unit.

19.4.70 positive_plane_angle_measure_with_unit

A **positive_plane_angle_measure_with_unit** is a type of plane_angle_measure_with_unit of which the value is positive.

EXPRESS specification:

```
ENTITY positive_plane_angle_measure_with_unit
  SUBTYPE OF (plane_angle_measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.POSITIVE_PLANE_ANGLE_MEASURE' IN TYPEOF(SELF\measure_with_unit.
value_component);
```


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```
END_ENTITY;
```

Formal propositions:

WR1: The value of the value_component shall be positive.

19.4.71 plane_angle_unit

A **plane_angle_unit** is a type of named_unit in which angles in planes are expressed.

EXPRESS specification:

```
ENTITY plane_angle_unit
  SUBTYPE OF (named_unit);
WHERE
  WR1: ((((((SELF\named_unit.dimensions.length_exponent = 0.0) AND (SELF\named_unit.
dimensions.mass_exponent = 0.0)) AND (SELF\named_unit.dimensions.time_exponent = 0.0))
AND (SELF\named_unit.dimensions.electric_current_exponent = 0.0)) AND (SELF\named_unit.
dimensions.thermodynamic_temperature_exponent = 0.0)) AND (SELF\named_unit.dimensions.
amount_of_substance_exponent = 0.0)) AND (SELF\named_unit.dimensions.luminous_intensity_
exponent = 0.0);
END_ENTITY;
```

Formal propositions:

WR1: All the dimensional exponents shall be equal to zero.

19.4.72 power_unit

A **power_unit** is a type of derived_unit in which the power is expressed.

EXPRESS specification:

```
ENTITY power_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.watt);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the watt as defined in ISO 31.

19.4.73 power_measure_with_unit

A **power_measure_with_unit** is a type of measure_with_unit in which the quantity measured is a power as defined in ISO 31.

EXPRESS specification:

```
ENTITY power_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.POWER_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type power_unit.

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19.4.74 si_power_unit

A **si_power_unit** is a type of power_unit and a type of si_unit in which the power is expressed in SI units.

EXPRESS specification:

```
ENTITY si_power_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.watt;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be watt.

WR2: No other name shall be assigned to this entity.

19.4.75 pressure_unit

A **pressure_unit** is a type of derived_unit in which the pressure is expressed.

EXPRESS specification:

```
ENTITY pressure_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.pascal);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the pascal as defined in ISO 31.

19.4.76 pressure_measure_with_unit

A **pressure_measure_with_unit** is a type of measure_with_unit in which the quantity measured is a pressure as defined in ISO 31.

EXPRESS specification:

```
ENTITY pressure_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.PRESSURE_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type pressure_unit.

19.4.77 si_pressure_unit

A **si_pressure_unit** is a type of pressure_unit and a type of si_unit in which the pressure is expressed in SI units.

EXPRESS specification:

```
ENTITY si_pressure_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.pascal;
  WR2: NOT EXISTS(SELF\derived_unit.name);
```


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```
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be pascal.

WR2: No other name shall be assigned to this entity.

19.4.78 ratio_measure_with_unit

A **ratio_measure_with_unit** is a type of measure_with_unit in which the physical quantity is a ratio as defined in ISO 31.

EXPRESS specification:

```
ENTITY ratio_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.RATIO_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit shall be a ratio_unit.

19.4.79 ratio_unit

A **ratio_unit** is a type of named_unit in which the ratio between two physical quantities of the same kind is expressed.

EXPRESS specification:

```
ENTITY ratio_unit
  SUBTYPE OF (named_unit);
WHERE
  WR1: ((((((SELF\named_unit.dimensions.length_exponent = 0.0) AND (SELF\named_unit.
dimensions.mass_exponent = 0.0)) AND (SELF\named_unit.dimensions.time_exponent = 0.0))
AND (SELF\named_unit.dimensions.electric_current_exponent = 0.0)) AND (SELF\named_unit.
dimensions.thermodynamic_temperature_exponent = 0.0)) AND (SELF\named_unit.dimensions.
amount_of_substance_exponent = 0.0)) AND (SELF\named_unit.dimensions.luminous_intensity_
exponent = 0.0);
END_ENTITY;
```

Formal propositions:

WR1: All the dimensional exponents shall be equal to zero.

19.4.80 resistance_unit

A **resistance_unit** is a type of derived_unit in which the electrical resistance is expressed.

EXPRESS specification:

```
ENTITY resistance_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensions_for_si_unit(si_unit_name.ohm);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponents shall be equal to those of the ohm as defined in ISO 31.

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19.4.81 resistance_measure_with_unit

A **resistance_measure_with_unit** is a type of measure_with_unit in which the quantity measured is an electrical resistance as defined in ISO 31.

EXPRESS specification:

```
ENTITY resistance_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.RESISTANCE_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type resistance_unit.

19.4.82 si_resistance_unit

A **si_resistance_unit** is a type of resistance_unit and a type of si_unit in which the resistance is expressed in SI units.

EXPRESS specification:

```
ENTITY si_resistance_unit;
WHERE
  WR1: SELF\si_unit.name = si_unit_name.ohm;
  WR2: NOT EXISTS(SELF\derived_unit.name);
END_ENTITY;
```

Formal propositions:

WR1: The name of the si_unit shall be ohm.

WR2: No other name shall be assigned to this entity.

19.4.83 si_unit

An **si_unit** is a type of named_unit that defines a unit with respect to the system of units defined in ISO 80000.

EXPRESS specification:

```
ENTITY si_unit
  SUBTYPE OF (named_unit);
  prefix : OPTIONAL si_prefix;
  name : si_unit_name;
DERIVE
  SELF\named_unit.dimensions : dimensional_exponents := dimensions_for_si_unit(name);
WHERE
  WR1: NOT (('MEASURE_SCHEMA.MASS_UNIT' IN TYPEOF(SELF)) AND (SIZEOF(USEDIN(SELF,
'MEASURE_SCHEMA.DERIVED_UNIT_ELEMENT.UNIT')) > 0)) OR (prefix = si_prefix.kilo);
END_ENTITY;
```

Attribute definitions:

prefix: the si_prefix that specifies the ratio with the unit specified by name. The value of this attribute need not be specified.

name: the si_unit_name by which the **si_unit** is known.

dimensions: the dimensional_exponents for the **si_unit**.

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Formal propositions:

WR1: If the **si_unit** is a mass_unit it shall not be used by a derived_unit_element unless the prefix is kilo.

19.4.84 solid_angle_measure_with_unit

A **solid_angle_measure_with_unit** is a type of measure_with_unit in which the physical quantity is a solid angle as defined in ISO 31.

EXPRESS specification:

```
ENTITY solid_angle_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.SOLID_ANGLE_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit shall be a solid_angle_unit.

19.4.85 solid_angle_unit

A **solid_angle_unit** is a type of named_unit in which solid angles are expressed.

EXPRESS specification:

```
ENTITY solid_angle_unit
  SUBTYPE OF (named_unit);
WHERE
  WR1: ((((((SELF\named_unit.dimensions.length_exponent = 0.0) AND (SELF\named_unit.
dimensions.mass_exponent = 0.0)) AND (SELF\named_unit.dimensions.time_exponent = 0.0))
AND (SELF\named_unit.dimensions.electric_current_exponent = 0.0)) AND (SELF\named_unit.
dimensions.thermodynamic_temperature_exponent = 0.0)) AND (SELF\named_unit.dimensions.
amount_of_substance_exponent = 0.0)) AND (SELF\named_unit.dimensions.luminous_intensity_
exponent = 0.0);
END_ENTITY;
```

Formal propositions:

WR1: All the dimensional exponents shall be equal to zero.

19.4.86 thermodynamic_temperature_measure_with_unit

A **thermodynamic_temperature_measure_with_unit** is a type of measure_with_unit in which the physical quantity is a thermodynamic temperature as defined in ISO 31.

EXPRESS specification:

```
ENTITY thermodynamic_temperature_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.THERMODYNAMIC_TEMPERATURE_UNIT' IN TYPEOF(SELF\measure_with_unit.
unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit shall be a thermodynamic_temperature_unit.

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19.4.87 thermodynamic_temperature_unit

A **thermodynamic_temperature_unit** is a type of named_unit in which the degree of heat of a body is expressed.

NOTE This unit corresponds to one of the seven fundamental quantities as specified in ISO 1000.

EXPRESS specification:

```
ENTITY thermodynamic_temperature_unit
  SUBTYPE OF (named_unit);
WHERE
  WR1: ((((((SELF\named_unit.dimensions.length_exponent = 0.0) AND (SELF\named_unit.
dimensions.mass_exponent = 0.0)) AND (SELF\named_unit.dimensions.time_exponent = 0.0))
AND (SELF\named_unit.dimensions.electric_current_exponent = 0.0)) AND (SELF\named_unit.
dimensions.thermodynamic_temperature_exponent = 1.0)) AND (SELF\named_unit.dimensions.
amount_of_substance_exponent = 0.0)) AND (SELF\named_unit.dimensions.luminous_intensity_
exponent = 0.0);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponent of thermodynamic temperature shall be equal to one and the other dimensional exponents shall be equal to zero.

19.4.88 time_measure_with_unit

A **time_measure_with_unit** is a type of measure_with_unit in which the physical quantity is a time as defined in ISO 31.

EXPRESS specification:

```
ENTITY time_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.TIME_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit shall be a time_unit.

19.4.89 time_unit

A **time_unit** is a type of named_unit in which the duration of periods is expressed.

NOTE This unit corresponds to one of the seven fundamental quantities as specified in ISO 1000.

EXPRESS specification:

```
ENTITY time_unit
  SUBTYPE OF (named_unit);
WHERE
  WR1: ((((((SELF\named_unit.dimensions.length_exponent = 0.0) AND (SELF\named_unit.
dimensions.mass_exponent = 0.0)) AND (SELF\named_unit.dimensions.time_exponent = 1.0))
AND (SELF\named_unit.dimensions.electric_current_exponent = 0.0)) AND (SELF\named_unit.
dimensions.thermodynamic_temperature_exponent = 0.0)) AND (SELF\named_unit.dimensions.
amount_of_substance_exponent = 0.0)) AND (SELF\named_unit.dimensions.luminous_intensity_
exponent = 0.0);
END_ENTITY;
```


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Formal propositions:

WR1: The dimensional exponent of time shall be equal to one, and all the other dimensional exponents shall be equal to zero.

19.4.90 velocity_measure_with_unit

A **velocity_measure_with_unit** is a type of measure_with_unit in which the quantity measured is a velocity.

EXPRESS specification:

```
ENTITY velocity_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.VELOCITY_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit_component shall be of type velocity_unit.

19.4.91 velocity_unit

A **velocity_unit** is a type of derived_unit in which the velocity is expressed.

EXPRESS specification:

```
ENTITY velocity_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensional_exponents(1.0, 0.0, -1.0, 0.0,
0.0, 0.0, 0.0);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponent of length shall be equal to one, the dimensional exponent of time shall be equal to minus one, and all the other dimensional exponents shall be equal to zero.

19.4.92 volume_measure_with_unit

A **volume_measure_with_unit** is a type of measure_with_unit in which the physical quantity is a volume as defined in ISO 31.

EXPRESS specification:

```
ENTITY volume_measure_with_unit
  SUBTYPE OF (measure_with_unit);
WHERE
  WR1: 'MEASURE_SCHEMA.VOLUME_UNIT' IN TYPEOF(SELF\measure_with_unit.unit_component);
END_ENTITY;
```

Formal propositions:

WR1: The unit shall be a volume_unit.

19.4.93 volume_unit

A **volume_unit** is a type of derived_unit used to measure the extent of a solid.

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EXPRESS specification:

```
ENTITY volume_unit
  SUBTYPE OF (derived_unit);
WHERE
  WR1: derive_dimensional_exponents(SELF) = dimensional_exponents(3.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0);
END_ENTITY;
```

Formal propositions:

WR1: The dimensional exponent of length shall be equal to three and all the other dimensional exponents shall be equal to zero.

19.5 Measure function definitions

19.5.1 derive_dimensional_exponents

The **derive_dimensional_exponents** function determines the dimensional exponents of a unit. For named units the dimensions attribute is returned and for derived units the dimensional exponents are calculated from its elements.

EXPRESS specification:

```
FUNCTION derive_dimensional_exponents(x : unit) : dimensional_exponents;
  LOCAL
    result : dimensional_exponents := dimensional_exponents(0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0);
  END_LOCAL;
  IF 'MEASURE_SCHEMA.DERIVED_UNIT' IN TYPEOF(x) THEN
    REPEAT i := LOINDEX(x\derived_unit.elements) TO HIINDEX(x\derived_unit.elements);
      result.length_exponent := result.length_exponent + x\derived_unit.
elements[i]\derived_unit_element.exponent * x\derived_unit.elements[i]\derived_unit_
element.unit\named_unit.dimensions.length_exponent;
      result.mass_exponent := result.mass_exponent + x\derived_unit.elements[i]\derived_
unit_element.exponent * x\derived_unit.elements[i]\derived_unit_element.unit\named_unit.
dimensions.mass_exponent;
      result.time_exponent := result.time_exponent + x\derived_unit.elements[i]\derived_
unit_element.exponent * x\derived_unit.elements[i]\derived_unit_element.unit\named_unit.
dimensions.time_exponent;
      result.electric_current_exponent := result.electric_current_exponent + x\derived_
unit.elements[i]\derived_unit_element.exponent * x\derived_unit.elements[i]\derived_unit_
element.unit\named_unit.dimensions.electric_current_exponent;
      result.thermodynamic_temperature_exponent := result.thermodynamic_temperature_
exponent + x\derived_unit.elements[i]\derived_unit_element.exponent * x\derived_unit.
elements[i]\derived_unit_element.unit\named_unit.dimensions.thermodynamic_temperature_
exponent;
      result.amount_of_substance_exponent := result.amount_of_substance_exponent
+ x\derived_unit.elements[i]\derived_unit_element.exponent * x\derived_unit.
elements[i]\derived_unit_element.unit\named_unit.dimensions.amount_of_substance_exponent;
      result.luminous_intensity_exponent := result.luminous_intensity_exponent
+ x\derived_unit.elements[i]\derived_unit_element.exponent * x\derived_unit.
elements[i]\derived_unit_element.unit\named_unit.dimensions.luminous_intensity_exponent;
    END_REPEAT;
  ELSE
    result := x\named_unit.dimensions;
  END_IF;
  RETURN (result);
END_FUNCTION;
```

Argument definitions:

x: (input) the unit that the dimensional_exponents are being derived from.

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19.5.2 dimensions_for_si_unit

The **dimensions_for_si_unit** function returns the dimensional_exponents of the given si_unit.

EXPRESS specification:

```
FUNCTION dimensions_for_si_unit(n : si_unit_name) : dimensional_exponents;
  CASE n OF
  metre :
    RETURN (dimensional_exponents(1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0));
  gram :
    RETURN (dimensional_exponents(0.0, 1.0, 0.0, 0.0, 0.0, 0.0, 0.0));
  second :
    RETURN (dimensional_exponents(0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 0.0));
  ampere :
    RETURN (dimensional_exponents(0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0));
  kelvin :
    RETURN (dimensional_exponents(0.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0));
  mole :
    RETURN (dimensional_exponents(0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.0));
  candela :
    RETURN (dimensional_exponents(0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0));
  radian :
    RETURN (dimensional_exponents(0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0));
  steradian :
    RETURN (dimensional_exponents(0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0));
  hertz :
    RETURN (dimensional_exponents(0.0, 0.0, -1.0, 0.0, 0.0, 0.0, 0.0));
  newton :
    RETURN (dimensional_exponents(1.0, 1.0, -2.0, 0.0, 0.0, 0.0, 0.0));
  pascal :
    RETURN (dimensional_exponents(-1.0, 1.0, -2.0, 0.0, 0.0, 0.0, 0.0));
  joule :
    RETURN (dimensional_exponents(2.0, 1.0, -2.0, 0.0, 0.0, 0.0, 0.0));
  watt :
    RETURN (dimensional_exponents(2.0, 1.0, -3.0, 0.0, 0.0, 0.0, 0.0));
  coulomb :
    RETURN (dimensional_exponents(0.0, 0.0, 1.0, 1.0, 0.0, 0.0, 0.0));
  volt :
    RETURN (dimensional_exponents(2.0, 1.0, -3.0, -1.0, 0.0, 0.0, 0.0));
  farad :
    RETURN (dimensional_exponents(-2.0, -1.0, 4.0, 2.0, 0.0, 0.0, 0.0));
  ohm :
    RETURN (dimensional_exponents(2.0, 1.0, -3.0, -2.0, 0.0, 0.0, 0.0));
  siemens :
    RETURN (dimensional_exponents(-2.0, -1.0, 3.0, 2.0, 0.0, 0.0, 0.0));
  weber :
    RETURN (dimensional_exponents(2.0, 1.0, -2.0, -1.0, 0.0, 0.0, 0.0));
  tesla :
    RETURN (dimensional_exponents(0.0, 1.0, -2.0, -1.0, 0.0, 0.0, 0.0));
  henry :
    RETURN (dimensional_exponents(2.0, 1.0, -2.0, -2.0, 0.0, 0.0, 0.0));
  degree_Celsius :
    RETURN (dimensional_exponents(0.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0));
  lumen :
    RETURN (dimensional_exponents(0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0));
  lux :
    RETURN (dimensional_exponents(-2.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0));
  becquerel :
    RETURN (dimensional_exponents(0.0, 0.0, -1.0, 0.0, 0.0, 0.0, 0.0));
  gray :
    RETURN (dimensional_exponents(2.0, 0.0, -2.0, 0.0, 0.0, 0.0, 0.0));
  sievert :
    RETURN (dimensional_exponents(2.0, 0.0, -2.0, 0.0, 0.0, 0.0, 0.0));
  OTHERWISE :
    RETURN (?);
```

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```
  END_CASE;
END_FUNCTION;
```

Argument definitions:

n: (input) the name of the si_unit_name for which the dimensional_exponents will be returned.

19.5.3 valid_units

The **valid_units** function validates a measure_with_unit. If the unit of the measure_with_unit is valid the function returns TRUE. Otherwise, it returns FALSE.

EXPRESS specification:

```
FUNCTION valid_units(m : measure_with_unit) : BOOLEAN;
  IF 'MEASURE_SCHEMA.LENGTH_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(1.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.MASS_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 1.0,
0.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.TIME_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0,
1.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.ELECTRIC_CURRENT_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0,
0.0, 1.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.THERMODYNAMIC_TEMPERATURE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0,
0.0, 0.0, 1.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.CELSIUS_TEMPERATURE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0,
0.0, 0.0, 1.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.AMOUNT_OF_SUBSTANCE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0,
0.0, 0.0, 0.0, 1.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.LUMINOUS_INTENSITY_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 1.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.PLANE_ANGLE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0) THEN
```

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```
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.SOLID_ANGLE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.AREA_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(2.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.VOLUME_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(3.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.RATIO_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.POSITIVE_LENGTH_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(1.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.POSITIVE_PLANE_ANGLE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.ACCELERATION_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(1.0, 0.0, -
2.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.CAPACITANCE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(-2.0, -
1.0, 4.0, 2.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.ELECTRIC_CHARGE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0,
1.0, 1.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.CONDUCTANCE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(-2.0, -
1.0, 3.0, 2.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.ELECTRIC_POTENTIAL_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(2.0, 1.0, -
3.0, -1.0, 0.0, 0.0, 0.0) THEN
```

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```
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.ENERGY_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(2.0, 1.0, -
2.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.FORCE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(1.0, 1.0, -
2.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.FREQUENCY_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0, -
1.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.ILLUMINANCE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(-2.0, 0.0,
0.0, 0.0, 0.0, 0.0, 1.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.INDUCTANCE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(2.0, 1.0, -
2.0, -2.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.LUMINOUS_FLUX_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 1.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.MAGNETIC_FLUX_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(2.0, 1.0, -
2.0, -1.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.MAGNETIC_FLUX_DENSITY_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 1.0, -
2.0, -1.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.POWER_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(2.0, 1.0, -
3.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.PRESSURE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(-1.0, 1.0,
-2.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.RESISTANCE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(2.0, 1.0, -
3.0, -2.0, 0.0, 0.0, 0.0) THEN
```

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```
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.VELOCITY_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(1.0, 0.0, -
1.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.RADIOACTIVITY_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(0.0, 0.0, -
1.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.ABSORBED_DOSE_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(2.0, 0.0, -
2.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  IF 'MEASURE_SCHEMA.DOSE_EQUIVALENT_MEASURE' IN TYPEOF(m.value_component) THEN
    IF derive_dimensional_exponents(m.unit_component) <> dimensional_exponents(2.0, 0.0, -
2.0, 0.0, 0.0, 0.0, 0.0) THEN
      RETURN (FALSE);
    END_IF;
  END_IF;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

m: (input) the candidate measure_with_unit that is to be checked.

```
END_SCHEMA; -- measure_schema
```

20 Person organization schema

20.1 General

The subject of the **person_organization_schema** is the description of information that identifies people and organizations.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **person_organization_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA person_organization_schema '{iso standard 10303 part(41) version(8) object(1)
person_organization_schema(17)}';

REFERENCE FROM basic_attribute_schema
  (description_attribute,
   description_attribute_select,
   generic_attribute,
   generic_attribute_assignment,
   generic_attribute_list_assignment,
   generic_attribute_list_select,
   generic_attribute_select,
   get_description_value,
```

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```
   get_id_value,
   get_name_value,
   get_role,
   id_attribute,
   id_attribute_select,
   name_attribute,
   name_attribute_select,
   object_role,
   role_association);
REFERENCE FROM support_resource_schema
  (bag_to_set,
   identifier,
   label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

basic_attribute_schema	ISO 10303-41
support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 18 for a graphical representation of this schema.

20.2 Fundamental concepts and assumptions

People and organizations are associated with product data to provide data concerning administrative structures and individuals who serve as points of contact or have particular responsibility in those structures. People and organizations may be associated with any aspect of product data.

20.3 Person organization type definitions

20.3.1 person_organization_select

The **person_organization_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

The **person_organization_select** type allows for the designation of an organization, a person or a person_and_organization.

EXPRESS specification:

```
TYPE person_organization_select = SELECT
  (person,
   organization,
   person_and_organization);
END_TYPE;
```

20.3.2 pos_description_attribute_select

The **pos_description_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE pos_description_attribute_select = SELECT BASED_ON description_attribute_select WITH
  (person_and_organization_role,
   person_and_organization,
   person_role,
   organization_role,
   organizational_project);
```

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```
END_TYPE;
```

20.3.3 pos_generic_attribute_select

The **pos_generic_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE pos_generic_attribute_select = SELECT BASED_ON generic_attribute_select WITH
  (address);
END_TYPE;
```

20.3.4 pos_generic_attribute_list_select

The **pos_generic_attribute_list_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE pos_generic_attribute_list_select = SELECT BASED_ON generic_attribute_list_select
WITH
  (address,
   person);
END_TYPE;
```

20.3.5 pos_id_attribute_select

The **pos_id_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE pos_id_attribute_select = SELECT BASED_ON id_attribute_select WITH
  (address,
   organizational_project);
END_TYPE;
```

20.3.6 pos_name_attribute_select

The **pos_name_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE pos_name_attribute_select = SELECT BASED_ON name_attribute_select WITH
  (address,
   person_and_organization);
END_TYPE;
```

20.4 Person organization entity definitions

20.4.1 address

An **address** is the information necessary for communicating, using one or more communication methods.

NOTE 1 In addition to the attributes defined in **address** additional generic attributes may be assigned using generic_attribute_assignment.

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EXAMPLE 1 A generic attribute named “LinkedIn” with a value of “Fred.Bloggs” could be used to add a LinkedIn account to address information.

EXPRESS specification:

```
ENTITY address;
  internal_location : OPTIONAL label;
  street_number : OPTIONAL label;
  street : OPTIONAL label;
  postal_box : OPTIONAL label;
  town : OPTIONAL label;
  region : OPTIONAL label;
  postal_code : OPTIONAL label;
  country : OPTIONAL label;
  facsimile_number : OPTIONAL label;
  telephone_number : OPTIONAL label;
  electronic_mail_address : OPTIONAL label;
  telex_number : OPTIONAL label;
DERIVE
  name : label := get_name_value(SELF);
  url : identifier := get_id_value(SELF);
WHERE
  WR1: ((((((((((EXISTS(internal_location) OR EXISTS(street_number)) OR EXISTS(street))
OR EXISTS(postal_box)) OR EXISTS(town)) OR EXISTS(region)) OR EXISTS(postal_code))
OR EXISTS(country)) OR EXISTS(facsimile_number)) OR EXISTS(telephone_number)) OR
EXISTS(electronic_mail_address)) OR EXISTS(telex_number);
END_ENTITY;
```

Attribute definitions:

internal_location: an organization-defined address for internal mail delivery. The value of this attribute need not be specified.

street_number: the number of a location on a street. The value of this attribute need not be specified.

street: the name of a street. The value of this attribute need not be specified.

postal_box: the number of a postal box. The value of this attribute need not be specified.

town: the name of a town. The value of this attribute need not be specified.

region: the name of a region. The value of this attribute need not be specified.

EXAMPLE 2 The counties of Great Britain and the states of the United States of America are examples of regions.

postal_code: the code that is used by the **country**‘s postal service. The value of this attribute need not be specified.

country: the name of a country. The value of this attribute need not be specified.

facsimile_number: the number at which facsimiles may be received. The value of this attribute need not be specified.

telephone_number: the number at which telephone calls may be received. The value of this attribute need not be specified.

electronic_mail_address: the electronic address at which electronic mail may be received. The value of this attribute need not be specified.

telex_number: the number at which telex messages may be received. The value of this attribute need not be specified.

name: the label that specifies the name of the location identified by the **address**.

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EXAMPLE 3 "White House", "Buckingham Palace", "Kremlin".

url: the text that specifies the uniform resource locator associated with the **address**.

NOTE 2 The format of uniform resource locators is defined in IETF RFC 1738 [4].

EXAMPLE 4 URLs are used to identify items such as world wide web sites and file transfer protocol (FTP) servers.

Formal propositions:

WR1: At least one of the attributes shall have a value.

NOTE 3 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

20.4.2 organization

An **organization** is an administrative structure.

EXPRESS specification:

```
ENTITY organization;
  id : OPTIONAL identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **organization**. The value of this attribute need not be specified.

NOTE The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **organization** is known.

description: the text that characterizes the **organization**. The value of this attribute need not be specified.

20.4.3 organization_relationship

An **organization_relationship** is a relationship between two instances of the entity data type organization with a description of their relationship.

EXAMPLE Each department of an enterprise, and the enterprise itself, may be regarded as individual organizations. The fact that the departments are a part of the whole enterprise may be represented using this entity.

NOTE 1 The role of **organization_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 3 This entity, together with the organization entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY organization_relationship;
```

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```
  name : label;
  description : OPTIONAL text;
  relating_organization : organization;
  related_organization : organization;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **organization_relationship** is known.

description: the text that characterizes the **organization_relationship**. The value of the attribute need not be specified.

relating_organization: one of the instances of organization that is a part of the relationship.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_organization: the other instance of organization that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

20.4.4 organization_role 

An **organization_role** is a purpose for an **organization** and provides identification and description of the purpose.

EXPRESS specification:

```
ENTITY organization_role;
  name : label;
DERIVE
  description : text := get_description_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **organization_role** is known.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **organization_role**. The value of the attribute need not be specified.

NOTE 2 This attribute is an upwardly compatible addition to **organization_role** as specified in ISO 10303-41:1994.

Formal propositions:

WR1: Each **organization_role** shall be the described_item in at most one description_attribute.

NOTE 3 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 4 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

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20.4.5 organization_type

An **organization_type** is a recognized kind of organization.

EXAMPLE A national standards body is a type of organization.

NOTE A relationship between an **organization_type** object and one or more organization objects is established by the declaration of a subtype of organization_type_assignment that references an organization via its items attribute.

EXPRESS specification:

```
ENTITY organization_type;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **organization_type**.

name: the label by which the **organization_type** is known.

description: the text that characterizes the **organization_type**. The value of this attribute need not be specified.

20.4.6 organization_type_relationship

An **organization_type_relationship** is a relationship between two instances of the organization_type entity data type and provides for the identification, naming, and description of that relationship.

EXPRESS specification:

```
ENTITY organization_type_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating_organization_type : organization_type;
  related_organization_type : organization_type;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **organization_type_relationship**.

name: the label by which the **organization_type_relationship** is known.

description: the text that characterizes the **organization_type_relationship**. The value of this attribute need not be specified.

relating_organization_type: one of the instances of organization_type that is a part of the relationship.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_organization_type: the other instance of organization_type that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

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20.4.7 organizational_address

An **organizational_address** is a type of address for one or more organizations.

EXPRESS specification:

```
ENTITY organizational_address
  SUBTYPE OF (address);
  organizations : SET [1:?] OF organization;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

organizations: the organizations located at the address.

description: the text that characterizes the **organizational_address**. The value of the attribute need not be specified.

20.4.8 organizational_project

An **organizational_project** is a project for which one or more organization entity data types are responsible.

EXPRESS specification:

```
ENTITY organizational_project;
  name : label;
  description : OPTIONAL text;
  responsible_organizations : SET [1:?] OF organization;
DERIVE
  id : identifier := get_id_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ID_ATTRIBUTE.IDENTIFIED_ITEM'))
<= 1;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **organizational_project** is known.

description: the text that characterizes the **organizational_project**. The value of the attribute need not be specified.

responsible_organizations: the organization entity data types indicating who has project oversight.

id: the identifier that distinguishes the **organizational_project**.

NOTE 1 This attribute is an upwardly compatible addition to **organizational_project** as specified in ISO 10303-41:1994.

NOTE 2 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

Formal propositions:

WR1: Each **organizational_project** shall be the identified_item in at most one id_attribute.

NOTE 3 The id_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

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NOTE 4 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

20.4.9 organizational_project_relationship

An **organizational_project_relationship** is a relationship between two instances of the entity data type organizational_project with an identification and description of their relationship.

NOTE 1 The role of **organizational_project_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 3 This entity, together with the organizational_project entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY organizational_project_relationship;
  name : label;
  description : OPTIONAL text;
  relating_organizational_project : organizational_project;
  related_organizational_project : organizational_project;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **organizational_project_relationship** is known.

description: the text that characterizes the **organizational_project_relationship**. The value of the attribute need not be specified.

relating_organizational_project: one of the instances of organizational_project that is a part of the relationship.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_organizational_project: the other instance of organizational_project that is a part of the relationship. If one element of the relationship is dependent up on the other, this attribute shall be the dependent one.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

20.4.10 person

A **person** is an individual human being.

NOTE 1 In addition to the attributes defined in **person** additional lists of attributes may be assigned using generic_attribute_list_assignment.

EXAMPLE 1 For example a generic attribute list named “Previous name” with a value of [“Honourific”:”Mr”, “Given name”:”Fred”, “Family Name”:”Blogs”] may be assigned.

EXPRESS specification:

```
ENTITY person;
  id : identifier;
```

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```
  last_name : OPTIONAL label;
  first_name : OPTIONAL label;
  middle_names : OPTIONAL LIST [1:?] OF label;
  prefix_titles : OPTIONAL LIST [1:?] OF label;
  suffix_titles : OPTIONAL LIST [1:?] OF label;
WHERE
  WR1: EXISTS(last_name) OR EXISTS(first_name);
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **person**.

NOTE 2 The identification of a person is usually context dependent.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

EXAMPLE 2 In the USA a person's **id** would be his or her social security number whereas in the United Kingdom it would be his or her national insurance number.

NOTE 4 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

last_name: the **person**'s surname. The value of this attribute need not be specified.

first_name: the first element of the **person**'s list of forenames. The value of this attribute need not be specified.

middle_names: the **person**'s other forenames, if there are any. The value of this attribute need not be specified.

prefix_titles: the text that specifies the **person**'s social or professional standing and appear before his or her names. The value of this attribute need not be specified.

suffix_titles: the text that specifies the **person**'s social or professional standing and appear after his or her names. The value of this attribute need not be specified.

Formal propositions:

WR1: Either the last name, first name or the last and first name shall be defined.

20.4.11 person_and_organization 

A **person_and_organization** is a person in an organization.

EXPRESS specification:

```
ENTITY person_and_organization;
  the_person : person;
  the_organization : organization;
DERIVE
  name : label := get_name_value(SELF);
  description : text := get_description_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'NAME_ATTRIBUTE.NAMED_ITEM')) <= 1;
  WR2: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

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the_person: the instance of the person entity data type who is related to an instance of organization.

the_organization: the instance of the organization entity data type to which the instance of person is related.

name: the label by which the **person_and_organization** is known.

NOTE 1 This attribute is an upwardly compatible addition to **person_and_organization** as specified in ISO 10303-41:1994.

description: the text that characterizes the **person_and_organization**.

NOTE 2 This attribute is an upwardly compatible addition to **person_and_organization** as specified in ISO 10303-41:1994.

Formal propositions:

WR1: Each **person_and_organization** shall be the described_item in at most one description_attribute.

NOTE 3 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

WR2: Each **person_and_organization** shall be the named_item in at most one name_attribute.

NOTE 4 The name_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 5 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

20.4.12 person_and_organization_role

A **person_and_organization_role** is a purpose for a **person_and_organization** and provides identification and description of the purpose.

EXAMPLE The role of a person could be 'buyer' in the context of the organization where he/she works and it could be 'customer' in the context of the organization from which he/she purchases goods.

EXPRESS specification:

```
ENTITY person_and_organization_role;
  name : label;
DERIVE
  description : text := get_description_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **person_and_organization_role** is known.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the text that characterizes the **person_and_organization_role**.

NOTE 2 This attribute is an upwardly compatible addition to **person_and_organization_role** as specified in ISO 10303-41:1994.

Formal propositions:

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WR1: Each **person_and_organization_role** shall be the described_item in at most one description_attribute.

NOTE 3 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 4 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

20.4.13 person_role

A **person_role** is a purpose for a person and provides identification and description of the purpose.

EXPRESS specification:

```
ENTITY person_role;
  name : label;
DERIVE
  description : text := get_description_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **person_role** is known.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

description: the label that characterizes the **person_role**.

NOTE 2 This attribute is an upwardly compatible addition to **person_role** as specified in ISO 10303-41:1994.

Formal propositions:

WR1: Each **person_role** shall be the described_item in at most one description_attribute.

NOTE 3 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 4 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

20.4.14 person_type

A **person_type** is a category of people having common characteristics. // <note>

EXPRESS specification:

```
ENTITY person_type;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **person_type**.

name: the label by which the **person_type** is known.

description: the text that characterizes the **person_type**. The value of this attribute need not be specified.

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20.4.15 person_type_definition

A **person_type_definition** is a characterization of type of person.

EXPRESS specification:

```
ENTITY person_type_definition;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  formation : person_type_definition_formation;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **person_type_definition**.

name: the label by which the **person_type_definition** is known.

description: the text that characterizes the **person_type_definition**. The value of this attribute need not be specified.

formation: the person_type_definition_formation to which the **person_type_definition** relates.

20.4.16 person_type_definition_formation

A **person_type_definition_formation** is a collector of definitions of a type of person.

EXPRESS specification:

```
ENTITY person_type_definition_formation;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  of_person_type : person_type;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **person_type_definition_formation**.

name: the label by which the **person_type_definition_formation** is known.

description: the text that characterizes the **person_type_definition_formation**. The value of this attribute need not be specified.

of_person_type: the person_type to which the **person_type_definition_formation** belongs.

20.4.17 person_type_definition_relationship

A **person_type_definition_relationship** is a relationship between two instances of the person_type_definition entity data type and provides for the identification, naming, and description of that relationship.

EXPRESS specification:

```
ENTITY person_type_definition_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating_person_type_definition : person_type_definition;
  related_person_type_definition : person_type_definition;
```

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```
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **person_type_definition_relationship**.

name: the label by which the **person_type_definition_relationship** is known.

description: the text that characterizes the **person_type_definition_relationship**. The value of this attribute need not be specified.

relating_person_type_definition: one of the instances of person_type_definition that is a part of the relationship.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_person_type_definition: the other instance of person_type_definition that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

20.4.18 personal_address 

A **personal_address** is a type of address, which is an address for one or more persons.

EXPRESS specification:

```
ENTITY personal_address
  SUBTYPE OF (address);
  people : SET [1:?] OF person;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

people: the people who reside at the address.

description: the text that characterizes the **personal_address**. The value of this attribute need not be specified.

20.4.19 position_in_organization 

A **position_in_organization** is a specific function or job performed by a person in a particular organization. It is defined by responsibilities and activities. A position that is not fulfilled by a person is a vacancy.

EXAMPLE The chairperson of SC4.

EXPRESS specification:

```
ENTITY position_in_organization;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **position_in_organization**.

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name: the label by which the **position_in_organization** is known.

description: the text that characterizes the **position_in_organization**. The value of this attribute need not be specified.

20.4.20 position_in_organization_relationship

A **position_in_organization_relationship** is a relationship between two instances of the position_in_organization entity data type and provides for the identification, naming, and description of that relationship.

EXPRESS specification:

```
ENTITY position_in_organization_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating_position_in_organization : position_in_organization;
  related_position_in_organization : position_in_organization;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **position_in_organization_relationship**.

name: the label by which the **position_in_organization_relationship** is known.

description: the text that characterizes the **position_in_organization_relationship**. The value of this attribute need not be specified.

relating_position_in_organization: one of the instances of position_in_organization that is a part of the relationship.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_position_in_organization: the other instance of position_in_organization that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

20.4.21 position_in_organization_type

A **position_in_organization_type** is a generic function or job that is performed in organizations.

EXAMPLE Technical Director is a position in a company.

NOTE A relationship between a **position_in_organization_type** object and one or more position_in_organization objects is established by the declaration of a subtype of position_in_organization_type_assignment that references a position_in_organization via its items attribute.

EXPRESS specification:

```
ENTITY position_in_organization_type;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

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Attribute definitions:

id: the identifier that distinguishes the **position_in_organization_type**.

name: the label by which the **position_in_organization_type** is known.

description: the text that characterizes the **position_in_organization_type**. The value of this attribute need not be specified.

20.5 Person organization function definitions

20.5.1 acyclic_organization_relationship

The **acyclic_organization_relationship** function determines whether the graph of instances of the entity data type organization that contains relation as one of its links contains a cycle. This function may be used to evaluate either an organization_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the organization_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_organization_relationship(relation : organization_relationship;
                                           relatives : SET [1:?] OF organization;
                                           specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF organization_relationship;
  END_LOCAL;
  IF relation.relating_organization IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(org <* bag_to_set(USEDIN(relation.relating_organization, ('PERSON_
ORGANIZATION_SCHEMA.' + 'ORGANIZATION_RELATIONSHIP.') + 'RELATED_ORGANIZATION')) |
specific_relation IN TYPEOF(org));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_organization_relationship(x[i], relatives + relation.relating_
organization, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate organization_relationship to be checked.

relatives: (input) the set of instances of the entity data type organization for which the function is searching in the relating_organization attribute of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the organization_relationship entity.

20.5.2 acyclic_organization_type_relationship

The **acyclic_organization_type_relationship** function determines whether the graph of instances of the entity data type organization_type that contains relation as one of its links contains a cycle. This function may be used to evaluate either an organization_type_relationship or any of its subtypes.

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The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the organization_type_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_organization_type_relationship(relation : organization_type_relationship;
                                                relatives : SET OF organization_type;
                                                specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF organization_type_relationship;
  END_LOCAL;
  IF relation.relating_organization_type IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(orgtyp <* bag_to_set(USEDIN(relation.relating_organization_type, ('PERSON_
ORGANIZATION_SCHEMA.' + 'ORGANIZATION_TYPE_RELATIONSHIP.') + 'RELATED_ORGANIZATION_
TYPE')) | specific_relation IN TYPEOF(orgtyp));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_organization_type_relationship(x[i], relatives + relation.relating_
organization_type, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate organization_type_relationship to be checked.

relatives: (input) the set of instances of the entity data type organization_type that the function is searching for in the relating_organization_type parameter of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the organization_type_relationship entity.

20.5.3 acyclic_organizational_project_relationship

The **acyclic_organizational_project_relationship** function determines whether the graph of instances of the entity data type organizational_project that contains relation as one of its links contains a cycle. This function may be used to evaluate either an organizational_project_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the organizational_project_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_organizational_project_relationship(relation : organizational_project_
relationship;
                                                     relatives : SET [1:?] OF
organizational_project;
                                                     specific_relation : STRING) :
BOOLEAN;
  LOCAL
    x : SET OF organizational_project_relationship;
  END_LOCAL;
```

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```
  IF relation.relating_organizational_project IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(op <* bag_to_set(USEDIN(relation.relating_organizational_project, ('PERSON_
ORGANIZATION_SCHEMA.' + 'ORGANIZATIONAL_PROJECT_RELATIONSHIP.') + 'RELATED_ORGANIZATIONAL_
PROJECT')) | specific_relation IN TYPEOF(op));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_organizational_project_relationship(x[i], relatives + relation.
relating_organizational_project, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate organizational_project_relationship to be checked.

relatives: (input) the set of instances of the entity data type organizational_project for which the function is searching in the relating_organizational_project attribute of the **relation** argument.

specific_relation: (input) the fully qualified entity name of a type of organizational_project_relationship entity.

20.5.4 acyclic_person_type_definition_relationship

The **acyclic_person_type_definition_relationship** function determines whether the graph of instances of the entity data type person_type_definition that contains relation as one of its links contains a cycle. This function may be used to evaluate either a person_type_definition_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the person_type_definition_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_person_type_definition_relationship(relation : person_type_definition_
relationship;
                                                     relatives : SET OF person_type_
definition;
                                                     specific_relation : STRING) :
BOOLEAN;
  LOCAL
    x : SET OF person_type_definition_relationship;
  END_LOCAL;
  IF relation.relating_person_type_definition IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(ptdef <* bag_to_set(USEDIN(relation.relating_person_type_definition,
('PERSON_ORGANIZATION_SCHEMA.' + 'PERSON_TYPE_DEFINITION_RELATIONSHIP.') + 'RELATED_
PERSON_TYPE_DEFINITION')) | specific_relation IN TYPEOF(ptdef));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_person_type_definition_relationship(x[i], relatives + relation.
relating_person_type_definition, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

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Argument definitions:

relation: (input) the candidate person_type_definition_relationship to be checked.

relatives: (input) the set of instances of the entity data type person_type_definition that the function is searching for in the relating_person_type_definition parameter of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the person_type_definition_relationship entity.

20.5.5 acyclic_position_in_organization_relationship

The **acyclic_position_in_organization_relationship** function determines whether the graph of instances of the entity data type position_in_organization that contains relation as one of its links contains a cycle. This function may be used to evaluate either a position_in_organization_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the position_in_organization_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_position_in_organization_relationship(relation : position_in_
organization_relationship;
                                                       relatives : SET OF position_in_
organization;
                                                       specific_relation : STRING) :
BOOLEAN;
  LOCAL
    x : SET OF position_in_organization_relationship;
  END_LOCAL;
  IF relation.relating_position_in_organization IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(piorg <* bag_to_set(USEDIN(relation.relating_position_in_organization,
('PERSON_ORGANIZATION_SCHEMA.' + 'POSITION_IN_ORGANIZATION_RELATIONSHIP.') + 'RELATED_
POSITION_IN_ORGANIZATION')) | specific_relation IN TYPEOF(piorg));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_position_in_organization_relationship(x[i], relatives + relation.
relating_position_in_organization, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate position_in_organization_relationship to be checked.

relatives: (input) the set of instances of the entity data type position_in_organization that the function is searching for in the relating_position_in_organization parameter of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the position_in_organization_relationship entity.

```
END_SCHEMA; -- person_organization_schema
```

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21 Product definition schema

21.1 General

The subject of the **product_definition_schema** is the identification of products, the categorization of products, definitions of products, and the relationships among them. This schema provides for the generic aspects of product definition.

EXAMPLE The identification of products and definitions of products, the grouping of products according to classification schemes and the definition of various kinds of relationship between products are all examples of generic aspects of product definition.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **product_definition_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA product_definition_schema '{iso standard 10303 part(41) version(9) object(1)
product_definition_schema(18)}';

REFERENCE FROM application_context_schema
  (product_context,
   product_definition_context);
REFERENCE FROM basic_attribute_schema
  (get_id_value,
   get_name_value,
   id_attribute,
   id_attribute_select,
   name_attribute,
   name_attribute_select);
REFERENCE FROM document_schema
  (document);
REFERENCE FROM effectivity_schema
  (effectivity);
REFERENCE FROM external_reference_schema
  (external_source);
REFERENCE FROM product_property_definition_schema
  (characterized_object);
REFERENCE FROM support_resource_schema
  (bag_to_set,
   identifier,
   label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

application_context_schema	ISO 10303-41
basic_attribute_schema	ISO 10303-41
document_schema	ISO 10303-41
effectivity_schema	ISO 10303-41
external_reference_schema	ISO 10303-41
product_property_definition_schema	ISO 10303-41

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support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 19 for a graphical representation of this schema.

21.2 Fundamental concepts and assumptions

The use of the entity data types of this schema to represent products is specified in the viewpoint as defined by the requirements of an annotated EXPRESS schema that uses or specializes this part of ISO 10303. Every product need not be represented by an instance of the entity data type **product**. Some products may be represented by other constructs in the integrated resources.

EXAMPLE 1 The constructs **product_definition_formation** and **product_definition**, defined in the **product_definition_schema** of this part of ISO 10303, are examples of other entity data types that may represent a product.

EXAMPLE 2 The construct **action_resource**, defined in ISO 10303-49, is an example of an entity data type that may represent a product.

A single product may have multiple groups of definitions associated with it; each group is valid in a given application context.

EXAMPLE 3 An application context may be the manufacture of ball-point pens. A particular design of ball-point pen has multiple versions. Each version is described through a group of product definitions. An older version has a solid cap whereas a more recent version has a hole in the cap to prevent people from choking if they swallow it. The characteristics of the two versions would be different from each other because the later version would have a hole in the cap whereas the earlier version would not. This schema could be used to define the ball-point pen with a solid cap as one product and the one with a hole in the cap as another. On the other hand it could also be used to define both pens as two versions of a single product. The approach taken would depend upon the requirements of the application context.

The definition of a product may include the properties that characterize it.

EXAMPLE 4 An integrated circuit product will have a functional definition represented by a circuit schematic diagram and a physical definition represented by a circuit layout diagram.

Definitions of products may be related to each other in various ways. This part of ISO 10303 provides the resources needed to identify the relationships among products and to be able to characterize those relationships.

EXAMPLE 5 The ball-point pen cap would be related in an assembly relationship to the ball-point pen itself and one of the caps could be substituted for the other in some circumstances.

There are two ways to represent the concept of real object:

— a real object can be represented by an instance of **product** if the subject of an annotated Express schema is concerned with real objects or if it is needed to describe a real object at various life-cycle stages and in various configurations; in this case, if an real object is identified by a serial number, this serial number can be stored in the id attribute of the **product**;

EXAMPLE 6 An airplane, in service during 30 years, may have different configurations during its life.

— a real object can be represented by an instance of **product_definition** if the subject of an annotated Express schema is primarily concerned with the design phase of products and if the concept of real object is considered as a particular stage in the design phase of a product; in this case, if a real object is identified by a serial number, this **serial_number** can be stored in the id attribute of the **product_definition**.

EXAMPLE 7 A prototype may be considered as a particular definition of a product.

Product definition facts are independent of properties. The ways in which these facts may be combined are prescribed by the relationships that are defined in this schema. Each product definition fact can be

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interpreted in any application context; the ways in which these facts are to be interpreted are prescribed in annotated EXPRESS schemas that use or specialise the resources defined in this part of ISO 10303.

NOTE An example of the way in which this schema may be used is given in annex F.1.

21.3 Product definition type definitions

21.3.1 product_definition_or_reference

The **product_definition_or_reference** type is an extensible list of alternate data types. It provides a mechanism to refer to instances of the data types included in the types that extend the **product_definition_or_reference** type.

NOTE This empty extensible select requires extension in a further schema to ensure that entities that refer to it have at least one valid instantiation.

EXPRESS specification:

```
TYPE product_definition_or_reference = EXTENSIBLE GENERIC_ENTITY SELECT
  (product_definition,
   generic_product_definition_reference);
END_TYPE;
```

21.3.2 pds_id_attribute_select

The **pds_id_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE pds_id_attribute_select = SELECT BASED_ON id_attribute_select WITH
  (product_category);
END_TYPE;
```

21.3.3 pds_name_attribute_select

The **pds_name_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE pds_name_attribute_select = SELECT BASED_ON name_attribute_select WITH
  (product_definition,
   product_definition_substitute);
END_TYPE;
```

21.3.4 source

The **source** type is a list of alternate choices that enables one to characterize whether a product is to be manufactured within an organization, bought, or the fact that this information is not known.

EXPRESS specification:

```
TYPE source = ENUMERATION OF
  (made,
   bought,
   not_known);
END_TYPE;
```

Enumerated item definitions:

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made characterize a product is to be manufactured within an organization;

bought characterize a product is to be bought;

not_known information, whether a product is to be manufactured within an organization or bought, is not known.

21.4 Product definition entity definitions

21.4.1 alternative_solution_relationship

An **alternative_solution_relationship** is a product_definition_formation_relationship that specifies the type of the relationship.

EXPRESS specification:

```
ENTITY alternative_solution_relationship
  SUBTYPE OF (product_definition_formation_relationship);
  relation_type : STRING;
WHERE
  WR1: acyclic_alternative_solution_relationship(SELF, [SELF\product_definition_
formation_relationship.related_product_definition_formation], 'PRODUCT_DEFINITION_SCHEMA.
ALTERNATIVE_SOLUTION_RELATIONSHIP');
END_ENTITY;
```

Attribute definitions:

relation_type: the meaning of the relationship.

Formal propositions:

WR1: An **alternative_solution_relationship** shall not participate in its own definition.

21.4.2 characterized_product

A **characterized_product** is a type of characterized_object and a type of product for the purpose of characterizing a product.

NOTE Properties that are assigned using **characterized_object** are outside version control, whereas properties added via **product_definition** are under version control.

EXPRESS specification:

```
ENTITY characterized_product;
END_ENTITY;
```

21.4.3 generic_product_definition_reference

A **generic_product_definition_reference** is a reference to an externally defined product_definition that may or may not have a local representation. A **generic_product_definition_reference** is either a product_definition_reference or a product_definition_reference_with_local_representation.

EXPRESS specification:

```
ENTITY generic_product_definition_reference
  ABSTRACT SUPERTYPE OF (ONEOF(product_definition_reference, product_definition_reference_
with_local_representation));
  source : external_source;
```

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```
END_ENTITY;
```

Attribute definitions:

source: defines the location which contains the original definition of the product_definition.

21.4.4 product 

A **product** is a representation of a product or a type of product

NOTE 1 The term product is defined in ISO 10303-1.

EXAMPLE 1 The SS Titanic is a product that could be represented by **product**.

EXAMPLE 2 Lifeboat is a class of products that could be represented by the entity data type **product**. Each lifeboat on the SS Titanic is a member of this class.

A **product** depends on one or more instances of product_context specifying a frame of reference that determines the validity of the information held about the product or class of products.

NOTE 2 Products that this entity data type can represent include:

- products existing in the real world;
- products that may come into existence as a consequence of some realization process;
- products that are functions;
- products that require further detail so they may serve as the basis for other physically realizable products.

EXAMPLE 3 The computer used to edit this document, the Eiffel Tower, and coal are examples of products existing in the real world.

EXAMPLE 4 A company develops a new lawnmower, a product that will be realized through a manufacturing and assembly process.

EXAMPLE 5 Heating, lighting, and Internet connectivity are examples of functions.

EXAMPLE 6 A company may develop a generic engine that is described by a schematic drawing showing its main principles and components. This product may be used as the basis for a range of engines that share characteristics such as multi-valve and operating temperature. Then, when defining a new engine, designers create a specialization of this generic engine with specific characteristics such as cylinder volume and power.

NOTE 3 The products or classes of products that are represented by **product** are specified in annotated EXPRESS schemas that use or specialize this entity data type.

EXPRESS specification:

```
ENTITY product;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  frame_of_reference : SET [1:?] OF product_context;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **product**.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

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NOTE 5 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

EXAMPLE 7 Part numbers, stock item numbers, and serial numbers are examples of product identifiers.

name: the label by which the **product** is known.

EXAMPLE 8 'Widget' is an example of name.

description: the text that characterizes the **product**. The value of the attribute need not be specified.

frame_of_reference: a set of product_context entities that defines the contexts within which the data associated with the **product** is relevant.

21.4.5 product_category

A **product_category** is a classification that applies to products.

EXAMPLE In an application protocol whose context includes manufactured parts, 'mechanical part', 'electrical part', 'structural part', 'piping part', 'water pipe', and 'hot water pipe' are examples of types of product.

EXPRESS specification:

```
ENTITY product_category;
  name : label;
  description : OPTIONAL text;
DERIVE
  id : identifier := get_id_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ID_ATTRIBUTE.IDENTIFIED_ITEM'))
<= 1;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **product_category** is known.

description: the text that characterizes the **product_category**. The value of the attribute need not be specified.

id: the identifier that distinguishes the **product_category**.

NOTE 1 This attribute is an upwardly compatible addition to **product_category** as specified in ISO 10303-41:1994.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 3 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

Formal propositions:

WR1: Each **product_category** shall be the identified_item in at most one id_attribute.

NOTE 4 The id_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 5 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

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21.4.6 product_category_relationship

A **product_category_relationship** is a hierarchical relationship of one category with another and provides a description of their relationship. If a product belongs to a product_category, it also belongs to all of the parent categories.

NOTE 1 The role of **product_category_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

EXAMPLE 1 The two sub-level product_categorys called 'cold water pipe' and 'hot water pipe' can be related to each other through the more generic product_category called 'piping part' by using two **product_category_relationships**, one **product_category_relationship** between 'piping part' and 'cold water pipe' and the other **product_category_relationship** between 'piping part' and 'hot water pipe'.

NOTE 2 Networks of instances of product_category may be defined using this entity.

EXAMPLE 2 A product_category called 'piping part' may be the parent category of sub-categories called 'cold water pipe' and 'hot water pipe'. The product_category called 'hot water pipe' may be a sub-category of another product_category called 'boiler output pipe'. This is an example of a product_category network because the product_category called 'hot water pipe' has two parents.

NOTE 3 This entity, in conjunction with the product_category entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY product_category_relationship;
  name : label;
  description : OPTIONAL text;
  category : product_category;
  sub_category : product_category;
WHERE
  WR1: acyclic_product_category_relationship(SELF, [SELF.sub_category]);
END_ENTITY;
```

Attribute definitions:

name: the label by which the **product_category_relationship** is known.

description: the text that characterizes the **product_category_relationship**. The value of the attribute need not be specified.

category: the parent of the **sub_category**.

EXAMPLE 3 In the previous example, 'piping part' would be the category for 'cold water pipe' and 'hot water pipe'.

sub_category: a child of the **category**.

EXAMPLE 4 In the previous example 'cold water pipe' would be the **sub_category** for 'piping part' in one case and 'hot water pipe' would be the **sub_category** in the other.

Formal propositions:

WR1: A graph of product_category instances shall not be cyclic.

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21.4.7 product_definition

A **product_definition** is a representation of an aspect of a product, or of a class of products, for an identified life cycle stage. The life cycle stage for which a **product_definition** exists may be further characterized by discipline, by usage, or by both.

NOTE 1 The **product_definition** entity type supports the representation of different views of a product for different purposes. Multiple views of the same product, or class of products, are represented by different instances of **product_definition** for the same product_definition_formation.

EXAMPLE 1 The design of the SS Titanic and the as-built description of the SS Titanic can be represented as two instances of **product_definition** for the product that represents the ship itself.

The **product_definition** entity data type may represent particular products that are the members of an identified class of products.

EXAMPLE 2 Each individual lifeboat on the SS Titanic can be represented by an instance of **product_definition**, in which the associated product represents the class of products whose members are the lifeboats.

NOTE 2 A **product_definition** can identify an occurrence of a product.

The **product_definition** entity data type acts as an aggregator for information about the properties of products.

EXAMPLE 3 The designed shape of an aeroplane can be identified by an instance of the product_definition_shape entity data type that is associated with the design **product_definition** of the aeroplane.

The usage of a **product_definition** in another context is specified through its participation in a product_definition_relationship as the related_product_definition in which the using context is specified by the **frame_of_reference** of the relating_product_definition.

If a **product_definition** is considered in multiple contexts, the product_definition_context_association shall be used to specify a collection of product_definition_contexts.

EXPRESS specification:

```
ENTITY product_definition;
  id : identifier;
  description : OPTIONAL text;
  formation : product_definition_formation;
  frame_of_reference : product_definition_context;
DERIVE
  name : label := get_name_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'NAME_ATTRIBUTE.NAMED_ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **product_definition**.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 4 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

description: the text that characterizes the **product_definition**. The value of the attribute need not be specified.

formation: the product_definition_formation to which the **product_definition** relates.

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frame_of_reference: the product_definition_context in which the **product_definition** is defined.

name: the label by which the **product_definition** is known.

NOTE 5 This attribute is an upwardly compatible addition to **product_definition** as specified in ISO 10303-41:1994.

Formal propositions:

WR1: Each **product_definition** shall be the named_item in at most one name_attribute.

NOTE 6 The name_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 7 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

21.4.8 product_definition_context_association

A **product_definition_context_association** is an association of a product_definition_context with a product_definition, and specifies the meaning of this association.

NOTE 1 A **product_definition_context_association** may be used to associate a using context to a product_definition. The defining context is associated to the product_definition using frame_of_reference.

EXAMPLE A product_definition, initially defined in the context of 'mechanical design', and in which the representation of the shape of a product is defined, may also be relevant in the context of 'process-planning design'. For such a case, the **product_definition_context_association** would relate the product_definition with the product_definition_context characterizing the context 'process-planning design' and the name of the role attribute would be 'other relevant context'.

EXPRESS specification:

```
ENTITY product_definition_context_association;
  definition : product_definition;
  frame_of_reference : product_definition_context;
  role : product_definition_context_role;
END_ENTITY;
```

Attribute definitions:

definition: the reference to a product_definition associated to a context.

frame_of_reference: the product_definition_context that defines the stage in the product life cycle with which the product_definition is to be associated.

role: the product_definition_context_role that specifies the purpose of the association of the context of this **frame_of_reference** attribute with the product_definition of the **definition** attribute.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

21.4.9 product_definition_context_role

A **product_definition_context_role** is a specification of the purpose of a product_definition_context.

EXPRESS specification:

```
ENTITY product_definition_context_role;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```


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Attribute definitions:

name: the label by which the **product_definition_context_role** is known.

description: the text that characterizes the **product_definition_context_role**. The value of the attribute need not be specified.

21.4.10 product_definition_effectivity

A **product_definition_effectivity** is a type of effectivity. A **product_definition_effectivity** is the identification of a valid use of a particular product_definition in the context of its participation in a given product_definition_relationship. The referenced product_definition is the related_product_definition attribute of the product_definition_relationship.

NOTE Appropriate subtypes of product_definition_relationship and further information about the way to describe assemblies are specified in ISO 10303-44.

EXPRESS specification:

```
ENTITY product_definition_effectivity
  SUBTYPE OF (effectivity);
  usage : product_definition_relationship;
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'MANAGEMENT_RESOURCES_SCHEMA.' + 'EFFECTIVITY_ASSIGNMENT.
ASSIGNED_EFFECTIVITY')) = 0;
END_ENTITY;
```

Attribute definitions:

usage: the product_definition_relationship that defines the context of the effectivity.

Formal propositions:

WR1: A **product_definition_effectivity** shall not be referred to as assigned_effectivity.

21.4.11 product_definition_formation

A **product_definition_formation** is a collector of definitions of a product.

EXAMPLE 1 An application interpreted model could use this entity to support the identification of different versions of a single product. Each version would be described by a group of instances of the entity data type product_definition and each group, identified by a **product_definition_formation**, would be associated with the same product.

EXPRESS specification:

```
ENTITY product_definition_formation;
  id : identifier;
  description : OPTIONAL text;
  of_product : product;
UNIQUE
  UR1: id, of_product;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **product_definition_formation**.

NOTE 1 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

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EXAMPLE 2 Part version number is an example of a **product_definition_formation** identifier.

description: the text that characterizes the **product_definition_formation**. The value of the attribute need not be specified.

NOTE 2 This attribute may be used to depict differences in the purpose and function of different formations of a single product.

of_product: the product to which the **product_definition_formation** belongs.

NOTE 3 A product is associated with one or more **product_definition_formation** instances through the implicit inverse of this relationship.

21.4.12 final_solution

A **final_solution** is a type of product_definition_formation and is a set of additional sensual characteristics that can be applied to an occurrence that represents a neutral part in order to complete its definition.

EXPRESS specification:

```
ENTITY final_solution
  SUBTYPE OF (product_definition_formation);
  status : STRING;
INVERSE
  specification : SET [1:?] OF product_definition FOR formation;
END_ENTITY;
```

Attribute definitions:

status: the text string that describes the level of completion between the neutral part and the final part.

specification: the set of instances of the entity data type product_definition that defines the various aspects of the **final_solution**.

21.4.13 product_definition_formation_relationship

A **product_definition_formation_relationship** is a relationship between two instances of the entity data type product_definition_formation and provides an identification and description of this relationship.

NOTE 1 A relationship may exist between instances of the entity data type product_definition_formation that relate to different instances of the entity data type product or between different formations of the same product.

NOTE 2 The role of **product_definition_formation_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 3 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 4 This entity, together with the product_definition_formation entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY product_definition_formation_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating_product_definition_formation : product_definition_formation;
  related_product_definition_formation : product_definition_formation;
```


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```
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **product_definition_formation_relationship**.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 6 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **product_definition_formation_relationship** is known.

description: the text that characterizes the **product_definition_formation_relationship**. The value of the attribute need not be specified.

relating_product_definition_formation: one of the instances of product_definition_formation that is a part of the relationship.

NOTE 7 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_product_definition_formation: the other instance of product_definition_formation that is a part of the relationship. If one element of the relationship is dependent up on the other, this attribute shall be the dependent one.

NOTE 8 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

21.4.14 product_definition_formation_with_specified_source

A **product_definition_formation_with_specified_source** is a type of product_definition_formation whose source is identified to be either manufactured within an organization, bought, or the fact that this information is not known.

EXPRESS specification:

```
ENTITY product_definition_formation_with_specified_source
  SUBTYPE OF (product_definition_formation);
  make_or_buy : source;
END_ENTITY;
```

Attribute definitions:

make_or_buy: the characterization of the source of the product_definition_formation.

21.4.15 product_definition_relationship

A **product_definition_relationship** is a relationship between two instances of the entity data type product_definition or generic_product_definition_reference and provides an identification and description of this relationship.

NOTE 1 A relationship may exist between instances of the entity data type product_definition that relate to different instances of the entity data type product or between different definitions of the same product.

EXAMPLE 1 The relationships within a bill-of-materials structure are examples of **product_definition_relationship** entity data types that associate different products. The relationship between

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a sketch and a detailed design is an example of a **product_definition_relationship** that associates different definitions of a single product.

NOTE 2 A single product_definition may be used more than once within the description of a product.

EXAMPLE 2 The same component could be used more than once in the same assembly. Each usage of the component would be specified as an instance of the **product_definition_relationship** entity.

NOTE 3 The role of **product_definition_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 4 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 5 This entity, together with the product_definition entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY product_definition_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating_product_definition : product_definition_or_reference;
  related_product_definition : product_definition_or_reference;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **product_definition_relationship**.

NOTE 6 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 7 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **product_definition_relationship** is known.

description: the text that characterizes the **product_definition_relationship**. The value of the attribute need not be specified.

relating_product_definition: one of the instances of product_definition that is a part of the relationship.

NOTE 8 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 9 In the context of an ISO 10303 application protocol, the **product_definition_relationship** is an assembly component relationship, the **relating_product_definition** may be the assembly.

related_product_definition: the other instance of product_definition that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 10 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 11 In the context of an ISO 10303 application protocol, the **related_product_definition** may be the product_definition that is an element of the assembly.

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21.4.16 product_definition_relationship_relationship

A **product_definition_relationship_relationship** is a relationship between two instances of the entity data type product_definition_relationship and provides an identification and description of this relationship.

EXPRESS specification:

```
ENTITY product_definition_relationship_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating : product_definition_relationship;
  related : product_definition_relationship;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **product_definition_relationship_relationship**.

name: the label by which the **product_definition_relationship_relationship** is known.

description: the text that characterizes the **product_definition_relationship_relationship**. The value of the attribute need not be specified.

relating: one of the instances of product_definition that is a part of the relationship.

related: the other instance of product_definition that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

21.4.17 product_definition_reference

A **product_definition_reference** is a type of generic_product_definition_reference that identifies the externally defined product_definition by providing the key attributes necessary to unambiguously identify it and the associated product_definition_formation, product and organization that owns the product id.

EXPRESS specification:

```
ENTITY product_definition_reference
  SUBTYPE OF (generic_product_definition_reference);
  product_id : identifier;
  product_definition_formation_id : identifier;
  product_definition_id : identifier;
  id_owning_organization_name : OPTIONAL label;
END_ENTITY;
```

Attribute definitions:

product_id: specifies the identifier of the externally defined product, where the referenced product_definition is assigned to one of its product_definition_formation.

product_definition_formation_id: specifies the identifier of the externally defined product_definition_formation.

product_definition_id: specifies the identifier of the externally defined product_definition.

id_owning_organization_name: specifies the id owner of the externally defined product, realised by an applied organization_assignment with the role 'id owner'.

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21.4.18 product_definition_reference_with_local_representation

A **product_definition_reference_with_local_representation** is a type of generic_product_definition_reference that represents an externally defined product_definition by a local product_definition.

EXPRESS specification:

```
ENTITY product_definition_reference_with_local_representation;
END_ENTITY;
```

21.4.19 product_definition_substitute

A **product_definition_substitute** is an association of a product_definition with a product_definition_relationship. The association states the fact that the product_definition may replace, in the context of the relationship, the product_definition that is specified in the related_product_definition attribute of the product_definition_relationship.

NOTE 1 If a product_definition_relationship exists between a component part and its assembly, a **product_definition_substitute** could be used to capture an allowable substitute, that is, a different component part that may be used as a replacement in the same assembly.

EXAMPLE 1 Two kinds of ball-point pens may be manufactured: a standard model and a deluxe model. Each model of the pen would be specified as a separate product_definition related to a particular kind of nib: a standard nib and a deluxe nib, respectively. The fact that the deluxe nib and the standard nib are interchangeable only in the context of the standard model of ball-point pen is established with this construct.

EXPRESS specification:

```
ENTITY product_definition_substitute;
  description : OPTIONAL text;
  context_relationship : product_definition_relationship;
  substitute_definition : product_definition;
DERIVE
  name : label := get_name_value(SELF);
WHERE
  WR1: context_relationship.related_product_definition :<>: substitute_definition;
  WR2: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'NAME_ATTRIBUTE.NAMED_ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

description: the text that characterizes the **product_definition_substitute**. The value of the attribute need not be specified.

context_relationship: the product_definition_relationship that specifies the context in which the replacement may occur.

EXAMPLE 2 In the previous example, the relating_product_definition of a product_definition_relationship identified as a **context_relationship** would be the standard model of the ball-point pen and the related_product_definition would be the standard nib.

substitute_definition: the product_definition that acts as an allowed replacement for the related_product_definition of the **context_relationship**.

EXAMPLE 3 In the previous example, this attribute would identify the deluxe nib .

name: the label by which the **product_definition_substitute** is known.

NOTE 2 This attribute is an upwardly compatible addition to product_definition as specified in ISO 10303-41:1994.

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Formal propositions:

WR1: A product_definition shall not be defined as a substitute for itself.

WR2: Each **product_definition_substitute** shall be the named_item in at most one name_attribute.

NOTE 3 The name_attribute data type is defined in basic_attribute_schema of this part of ISO 10303

NOTE 4 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

21.4.20 product_definition_with_associated_documents

A **product_definition_with_associated_documents** is a type of product_definition in which the associated documents participate in the definition of the product.

EXPRESS specification:

```
ENTITY product_definition_with_associated_documents
  SUBTYPE OF (product_definition);
  documentation_ids : SET [1:?] OF document;
END_ENTITY;
```

Attribute definitions:

documentation_ids: the set of instances of document entity data type that are associated with the product_definition.

21.4.21 product_related_product_category

A **product_related_product_category** is a type of product_category that identifies products that satisfy the type identified by the category.

EXPRESS specification:

```
ENTITY product_related_product_category
  SUBTYPE OF (product_category);
  products : SET [1:?] OF product;
END_ENTITY;
```

Attribute definitions:

products: a set of products that belong to the **product_related_product_category**.

21.4.22 product_relationship

A **product_relationship** is a relationship between two instances of the entity data type product and provides an identification and description of their relationship.

NOTE 1 The role of **product_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 3 This entity, together with the product entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

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```
ENTITY product_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating_product : product;
  related_product : product;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **product_relationship**.

NOTE 4 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **product_relationship** is known.

description: the text that characterizes the **product_relationship**. The value of the attribute need not be specified.

relating_product: one of the instances of product that is a part of the relationship.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_product: the other instance of product that is a part of the relationship. If one element of the relationship is dependent up on the other, this attribute shall be the dependent one.

NOTE 6 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

21.5 Product definition function definitions

21.5.1 acyclic_alternative_solution_relationship

The **acyclic_alternative_solution_relationship** function determines whether the graph of instance of the entity data type product_definition_formation that contains relation as one of its links contains a cycle. this function may be used to evaluate either a alternative_solution_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE The algorithm of the function is explained in annex E.2.

EXPRESS specification:

```
FUNCTION acyclic_alternative_solution_relationship(relation : alternative_solution_
relationship;
                                                   relatives : SET [1:?] OF product_
definition_formation;
                                                   specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF alternative_solution_relationship;
  END_LOCAL;
  IF relation.relating_product_definition_formation IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(asr <* bag_to_set(USEDIN(relation.relating_product_definition_formation,
('PRODUCT_DEFINITION_SCHEMA.' + 'ALTERNATIVE_SOLUTION_RELATIONSHIP.') + 'RELATED_PRODUCT_
DEFINITION_FORMATION')) | specific_relation IN TYPEOF(asr));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_alternative_solution_relationship(x[i], relatives + relation.relating_
product_definition_formation, specific_relation) THEN
      RETURN (FALSE);
```

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```
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate alternative_solution_relationship to be checked.

relatives: (input) the set of instances of the entity data type product_definition_formation for which the function is searching in the relating_product_definition_formation parameter of the relation argument.

specific_relation: (input) the fully qualified name of a subtype of the alternative_solution_relationship entity.

21.5.2 acyclic_product_category_relationship

The **acyclic_product_category_relationship** function determines whether the graph of instances of the entity data type product_category that contains relation as one of its links contains a cycle. This function may be used to evaluate either a product_category_relationship or any of its subtypes. The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE The algorithm of the function is explained in annex E.2.

EXPRESS specification:

```
FUNCTION acyclic_product_category_relationship(relation : product_category_relationship;
                                               children : SET OF product_category) :
BOOLEAN;
  LOCAL
    x : SET OF product_category_relationship;
    local_children : SET OF product_category;
  END_LOCAL;
  REPEAT i := 1 TO HIINDEX(children);
    IF relation.category :=: children[i] THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  x := bag_to_set(USEDIN(relation.category, 'PRODUCT_DEFINITION_SCHEMA.' + 'PRODUCT_
CATEGORY_RELATIONSHIP.SUB_CATEGORY'));
  local_children := children + relation.category;
  IF SIZEOF(x) > 0 THEN
    REPEAT i := 1 TO HIINDEX(x);
      IF NOT acyclic_product_category_relationship(x[i], local_children) THEN
        RETURN (FALSE);
      END_IF;
    END_REPEAT;
  END_IF;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate product_category_relationship to be checked.

children: (input) the set of instances of the entity data type product_category that the function is searching for in the category field of the **relation** argument.

21.5.3 acyclic_product_definition_formation_relationship

The **acyclic_product_definition_formation_relationship** function determines whether the graph of instances of the entity data type product_definition_formation that contains relation as one of its links

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contains a cycle. This function may be used to evaluate either a product_definition_formation_relationship or any of its subtypes. The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the product_definition_formation_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_product_definition_formation_relationship(relation : product_definition_
formation_relationship;
                                                           relatives : SET [1:?] OF
product_definition_formation;
                                                           specific_relation : STRING) :
BOOLEAN;
  LOCAL
    x : SET OF product_definition_formation_relationship;
  END_LOCAL;
  IF relation.relating_product_definition_formation IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(pdf <* bag_to_set(USEDIN(relation.relating_product_definition_formation,
('PRODUCT_DEFINITION_SCHEMA.' + 'PRODUCT_DEFINITION_FORMATION_RELATIONSHIP.') + 'RELATED_
PRODUCT_DEFINITION_FORMATION')) | specific_relation IN TYPEOF(pdf));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_product_definition_formation_relationship(x[i], relatives + relation.
relating_product_definition_formation, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate product_definition_formation_relationship to be checked.

relatives: (input) the set of instances of the entity data type product_definition_formation that the function is searching for in the relating_product_definition_formation parameter of the relation argument.

specific_relation: (input) the fully qualified name of a subtype of the product_definition_formation_relationship entity.

21.5.4 acyclic_product_definition_relationship

The **acyclic_product_definition_relationship** function determines whether the graph of instances of the entity data type product_definition that contains relation as one of its links contains a cycle. This function may be used to evaluate either a product_definition_relationship or any of its subtypes. The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the product_definition_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_product_definition_relationship(relation : product_definition_
relationship;
                                                 relatives : SET [1:?] OF product_
definition_or_reference;
```

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```
                                                    specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF product_definition_relationship;
  END_LOCAL;
  IF relation.relating_product_definition IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(pd <* bag_to_set(USEDIN(relation.relating_product_definition, ('PRODUCT_
DEFINITION_SCHEMA.' + 'PRODUCT_DEFINITION_RELATIONSHIP.') + 'RELATED_PRODUCT_
DEFINITION')) | specific_relation IN TYPEOF(pd));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_product_definition_relationship(x[i], relatives + relation.relating_
product_definition, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate product_definition_relationship to be checked.

relatives: (input) the set of instances of the entity data type product_definition that the function is searching for in the relating_product_definition parameter of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the product_definition_relationship entity.

21.5.5 acyclic_product_relationship

The **acyclic_product_relationship** function determines whether the graph of instances of the entity data type product that contains relation as one of its links contains a cycle. This function may be used to evaluate either a product_relationship or any of its subtypes. The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the product_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_product_relationship(relation : product_relationship;
                                      relatives : SET [1:?] OF product;
                                      specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF product_relationship;
  END_LOCAL;
  IF relation.relating_product IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(prod <* bag_to_set(USEDIN(relation.relating_product, ('PRODUCT_DEFINITION_
SCHEMA.' + 'PRODUCT_RELATIONSHIP.') + 'RELATED_PRODUCT')) | specific_relation IN
TYPEOF(prod));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_product_relationship(x[i], relatives + relation.relating_product,
specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

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Argument definitions:

relation: (input) the candidate product_relationship to be checked.

relatives: (input) the set of instances of the entity data type product that the function is searching for in the relating_product parameter of the **relation** argument.

specific_relation: (input) the fully qualified entity name of a type of product_relationship entity.

21.5.6 categories_of_product

The **categories_of_product** function returns the set of product category names associated with the argument. The function evaluates the bag of instances of product_category that are associated with the input product through an instance of product_related_product_category. The function returns the SET of STRING values stored in the name attribute of the instances of product_category in the bag.

EXPRESS specification:

```
FUNCTION categories_of_product(obj : product) : SET OF STRING;
  LOCAL
    category_assignments : BAG OF product_category;
    categories : SET OF STRING := [];
  END_LOCAL;
  category_assignments := USEDIN(obj, 'PRODUCT_DEFINITION_SCHEMA.' + 'PRODUCT_RELATED_
PRODUCT_CATEGORY.PRODUCTS');
  REPEAT i := LOINDEX(category_assignments) TO HIINDEX(category_assignments) BY 1;
    categories := categories + category_assignments[i].name;
  END_REPEAT;
  RETURN (categories);
END_FUNCTION;
```

Argument definitions:

obj: the product whose categories are searched.

21.5.7 get_product_definitions

The **get_product_definitions** function returns for any instance of product the set of instances of product_definition that refer to the product through an instance of product_definition_formation.

NOTE This function is not used in this schema.

EXPRESS specification:

```
FUNCTION get_product_definitions(c_def_instance : product) : SET OF product_definition;
  LOCAL
    pd_set : SET OF product_definition_formation := [];
    pdr_set : SET OF product_definition := [];
  END_LOCAL;
  pd_set := bag_to_set(USEDIN(c_def_instance, 'PRODUCT_DEFINITION_SCHEMA.PRODUCT_
DEFINITION_FORMATION.OF_PRODUCT'));
  IF SIZEOF(pd_set) < 1 THEN
    RETURN (pdr_set);
  END_IF;
  REPEAT i := 1 TO HIINDEX(pd_set);
    pdr_set := pdr_set + bag_to_set(USEDIN(pd_set[i], 'PRODUCT_DEFINITION_SCHEMA.PRODUCT_
DEFINITION.FORMATION'));
  END_REPEAT;
  RETURN (pdr_set);
END_FUNCTION;
```

Argument definitions:

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c_def_instance: the candidate product to be checked.

```
END_SCHEMA; -- product_definition_schema
```

22 Product property definition schema

22.1 General

The subject of the **product_property_definition_schema** is the characteristics of an object.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **product_property_definition_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA product_property_definition_schema '{iso standard 10303 part(41) version(9)
object(1) product_property_definition_schema(19)}';

REFERENCE FROM basic_attribute_schema
  (get_id_value,
   id_attribute,
   id_attribute_select);
REFERENCE FROM process_property_schema
  (action_property,
   resource_property);
REFERENCE FROM product_definition_schema
  (product_definition,
   product_definition_relationship,
   product_definition_relationship_relationship);
REFERENCE FROM product_property_representation_schema
  (item_identified_representation_usage);
REFERENCE FROM product_structure_schema
  (assembly_component_usage,
   multi_level_reference_designator,
   next_assembly_usage_occurrence,
   product_definition_occurrence);
REFERENCE FROM shape_dimension_schema
  (dimensional_location,
   dimensional_size);
REFERENCE FROM shape_tolerance_schema
  (geometric_tolerance);
REFERENCE FROM support_resource_schema
  (bag_to_set,
   identifier,
   label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

basic_attribute_schema	ISO 10303-41

:: ISO -

product_definition_schema	ISO 10303-41
product_property_representation_schema	ISO 10303-41

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:: ISO -

:: ISO -

:: ISO -

support_resource_schema ISO 10303-41

NOTE 2 See Annex D, Paragraph 20 for a graphical representation of this schema.

22.2 Fundamental concepts and assumptions

This schema provides resources for the identification of:

— types of properties to be identified and described;

— properties assigned to a **characterized_object**, a **product_definition**, a **product_definition_relationship** or a **shape_definition**.

The meaning of each assigned property may be defined either by its association with an instance of the entity characterizing a type of property or by its name attribute.

EXAMPLE 1 The definition of the surface finish of a shape is independent of the way in which the shape or the surface finish is represented.

NOTE In the previous example, both the defined surface finish and the shape could have multiple representations.

Each characteristic may be associated with a single **characterized_object** or **product_definition** or with one **product_definition** in the context of another. Each characteristic is independent of the number or types of representations of that characteristic. Each characteristic may be associated with the shape of a product, an element of the shape of a product, or the relationship between elements of the shape of a product.

This part of ISO 10303 distinguishes between the definition of a product and its possible usages. This part of ISO 10303 establishes the following assumptions:

— the definition of a given object is characterized by a set of unique properties.

EXAMPLE 2 A product cannot have two shapes simultaneously.

— any usage of the object is characterized by a set of unique properties.

EXAMPLE 3 A product, like glue, may have different shapes depending on its usage.

— a property characterizes either the definition or one of the usages of an object.

EXAMPLE 4 The appearance of chair x is a unique property of that chair. The colour designating that the chair is white is a single item in a representation for the appearance property of chair x. This colour is shareable among many representations for the properties of many different objects.

22.3 Product property definition type definitions

22.3.1 characterized_definition 

The **characterized_definition** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

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The **characterized_definition** type allows for the designation of a characterized_object, a characterized_product_definition, a characterized_item or a shape_definition to which properties may be assigned.

EXPRESS specification:

```
TYPE characterized_definition = SELECT
  (characterized_item,
   characterized_object,
   characterized_product_definition,
   shape_definition);
END_TYPE;
```

22.3.2 characterized_item

The **characterized_item** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

The **characterized_item** type allows for the designation of an item_identified_representation_usage, a dimensional_size, or a geometric_tolerance to which properties may be assigned.

EXPRESS specification:

```
TYPE characterized_item = SELECT
  (item_identified_representation_usage,
   dimensional_size,
   geometric_tolerance);
END_TYPE;
```

22.3.3 ppds_id_attribute_select

The **ppds_id_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE ppds_id_attribute_select = SELECT BASED_ON id_attribute_select WITH
  (property_definition,
   shape_aspect,
   shape_aspect_relationship);
END_TYPE;
```

22.3.4 characterized_product_definition

The **characterized_product_definition** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

The **characterized_product_definition** type allows for the designation of a product_definition, or a product_definition_relationship, or a product_definition_relationship_relationship, or a product_definition_occurrence to which properties may be assigned.

Selection of a product_definition_relationship means that the property is applied to the related_product_definition attribute in the context of its relating_product_definition attribute. Selection of a product_definition_relationship_relationship means that the property is applied to the related attribute in the context of its relating attribute.

NOTE This enables properties of a given product that depend on the usages of this product to be described.

EXAMPLE The shape of a gasket depends upon whether or not it is an element of an assembly and, if it is an element of an assembly, the shape depends upon the assembly in which it participates.

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EXPRESS specification:

```
TYPE characterized_product_definition = SELECT
  (product_definition,
   product_definition_occurrence,
   product_definition_relationship,
   product_definition_relationship_relationship);
END_TYPE;
```

22.3.5 derived_property_select

The **derived_property_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

The **derived_property_select** type allows for the selection of an action_property, a property_definition, or a resource_property, or a dimensional_location, or a dimensional_size, or a geometric_tolerance.

EXPRESS specification:

```
TYPE derived_property_select = SELECT
  (property_definition,
   action_property,
   resource_property,
   dimensional_location,
   dimensional_size,
   geometric_tolerance);
END_TYPE;
```

22.3.6 shape_aspect_or_characterized_object

The **shape_aspect_or_characterized_object** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE shape_aspect_or_characterized_object = SELECT
  (shape_aspect,
   characterized_object);
END_TYPE;
```

22.3.7 shape_definition

The **shape_definition** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

The **shape_definition** type allows for the designation of a product_definition_shape, a shape_aspect, or a shape_aspect_relationship.

References to a shape_aspect_relationship are references to the relationship itself and not to one of the associated shape_aspect.

EXPRESS specification:

```
TYPE shape_definition = SELECT
  (product_definition_shape,
   shape_aspect,
   shape_aspect_relationship);
END_TYPE;
```


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22.3.8 internal_or_reflected_shape_aspect

The **internal_or_reflected_shape_aspect** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE internal_or_reflected_shape_aspect = SELECT
  (shape_aspect,
   identifier);
END_TYPE;
```

22.3.9 multi_or_next_assembly_usage_occurrence

The **multi_or_next_assembly_usage_occurrence** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE multi_or_next_assembly_usage_occurrence = SELECT
  (multi_level_reference_designator,
   next_assembly_usage_occurrence);
END_TYPE;
```

22.4 Product property definition entity definitions

22.4.1 characterized_object

A **characterized_object** is the identification of an item that has associated property information.

NOTE 1 A **characterized_object** is characterized by the properties which refer to it.

NOTE 2 The properties of a **characterized_object** may be used as an environmental condition under which the properties of a product are measured.

EXAMPLE If a product has a set of properties that are measured within a room or an atmosphere, the room or atmosphere may be described with instances of the entity data type **characterized_object**.

EXPRESS specification:

```
ENTITY characterized_object;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **characterized_object** is known.

description: the text that characterizes the **characterized_object**. The value of the attribute need not be specified.

22.4.2 characterized_object_relationship

A **characterized_object_relationship** is a relationship between two instances of the entity data type characterized_object and provides an identification and description of this relationship.

NOTE 1 The role of **characterized_object_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

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NOTE 2 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 3 This entity, together with the characterized_object entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY characterized_object_relationship;
  name : label;
  description : OPTIONAL text;
  relating_object : characterized_object;
  related_object : characterized_object;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **characterized_object_relationship** is known.

description: the text that characterizes the **characterized_object_relationship**. The value of the attribute need not be specified.

relating_object: one of the instances of characterized_object that is a part of the relationship.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_object: the other instance of characterized_object that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 5 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

22.4.3 component_path_shape_aspect 

A **component_path_shape_aspect** is a type of shape_aspect that identifies a portion of a shape of an assembly by referencing a shape_aspect that represents a portion of the shape of an occurrence of a part within this assembly. The referenced part and its shape_aspect might be defined locally or externally.

EXPRESS specification:

```
ENTITY component_path_shape_aspect
  SUBTYPE OF (shape_aspect);
  location : multi_or_next_assembly_usage_occurrence;
  component_shape_aspect : internal_or_reflected_shape_aspect;
UNIQUE
  UR1: location, component_shape_aspect;
END_ENTITY;
```

Attribute definitions:

location: the occurrence of a part by its next_assembly_usage_occurrence or multi_level_reference_designator.

component_shape_aspect: the locally or externally defined shape_aspect. In the case that it is externally defined, only the id of the externally defined shape_aspect is provided.

22.4.4 constituent_shape_aspect 

A **constituent_shape_aspect** is a type of shape_aspect that identifies a portion of a shape_aspect.

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EXPRESS specification:

```
ENTITY constituent_shape_aspect
  SUBTYPE OF (shape_aspect);
  parent : shape_aspect;
DERIVE
  SELF\shape_aspect.of_shape : product_definition_shape := get_root_shape_aspect(SELF);
END_ENTITY;
```

Attribute definitions:

parent: the shape_aspect of which this entity is an aspect.

of_shape: the product_definition_shape of which this entity is an aspect.

22.4.5 general_property

A **general_property** is an identification of a type of property.

NOTE 1 An annotated Express schema can use this entity to identify a property selected in a library of properties.

NOTE 2 If needed, a **general_property** may be a member of a group.

EXAMPLE 1 "Kinematic viscosity", defined in ISO 31, is an example of a **general_property**.

EXPRESS specification:

```
ENTITY general_property;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **general_property**.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 4 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

name: the label by which the **general_property** is known.

EXAMPLE 2 'Kinematic viscosity' is an example of name.

description: the text that characterizes the **general_property**. The value of the attribute need not be specified.

22.4.6 general_property_association

A **general_property_association** is an association of an instance of one of the types listed in derived_property_select with a general_property. The general_property serves as the reference definition for the considered property.

EXAMPLE In order to characterize the kinematic viscosity of a fluid, an instance of **general_property_association** would relate an instance of general_property that defines the concept of kinematic viscosity and an instance of property_definition that defines the kinematic characteristics of an instance of product_definition defining the fluid.

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EXPRESS specification:

```
ENTITY general_property_association;
  name : label;
  description : OPTIONAL text;
  base_definition : general_property;
  derived_definition : derived_property_select;
WHERE
  WR1: (SIZEOF(['SHAPE_DIMENSION_SCHEMA.DIMENSIONAL_LOCATION', 'SHAPE_DIMENSION_SCHEMA.
DIMENSIONAL_SIZE', 'SHAPE_TOLERANCE_SCHEMA.GEOMETRIC_TOLERANCE'] * TYPEOF(derived_
definition)) > 0) OR (SIZEOF(USEDIN(derived_definition, ('PRODUCT_PROPERTY_DEFINITION_
SCHEMA.' + 'GENERAL_PROPERTY_ASSOCIATION.') + 'DERIVED_DEFINITION')) = 1);
  WR2: (SIZEOF(['SHAPE_DIMENSION_SCHEMA.DIMENSIONAL_LOCATION', 'SHAPE_DIMENSION_SCHEMA.
DIMENSIONAL_SIZE', 'SHAPE_TOLERANCE_SCHEMA.GEOMETRIC_TOLERANCE'] * TYPEOF(derived_
definition)) > 0) OR (derived_definition.name = base_definition.name);
END_ENTITY;
```

Attribute definitions:

name: the label by which the **general_property_association** is known.

description: the text that characterizes the **general_property_association**. The value of the attribute need not be specified.

base_definition: the general_property that provides the definition of the considered characteristic.

derived_definition: the derived_property_select, for which the identification of the considered characteristic is provided by a general_property.

Formal propositions:

WR1: In the case that the **derived_definition** is of type property_definition, action_property or resource_property it shall be referred to by at most one **general_property_association**.

WR2: In the case that the **derived_definition** is of type property_definition, action_property or resource_property the attribute name of the **derived_definition** shall be the same as the attribute name of the **base_definition**.

22.4.7 general_property_relationship 

A **general_property_relationship** is a relationship between two instances of the entity data type general_property and provides an identification and description of this relationship.

NOTE 1 This entity, together with the general_property entity, is based on the relationship template that is described in annex E.3

EXPRESS specification:

```
ENTITY general_property_relationship;
  name : label;
  description : OPTIONAL text;
  relating_property : general_property;
  related_property : general_property;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **general_property_relationship** is known.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

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description: the text that characterizes the **general_property_relationship**. The value of the attribute need not be specified.

relating_property: one of the instances of general_property that is a part of the relationship.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_property: the other instance of general_property that is a part of the relationship. If one element of the relationship is dependent upon the other this attribute shall be the dependent one.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

22.4.8 product_definition_shape



A **product_definition_shape** is a type of property_definition. It identifies the shape of an object. A **product_definition_shape** identifies the shape of a characterized_object or of one of the types reachable as characterized_product_definition.

NOTE 1 A **product_definition_shape** need not be associated with any geometric representation.

NOTE 2 Early in the design of a product there may not be a specific idea about the shape of the product but there may be certain characteristics of the shape that are to be represented. Those product shape characteristics can be attached to the product shape using this entity.

EXAMPLE A geometric representation of shape is not needed to assert facts such as, "a shape must fit within a 5 centimetre cube".

EXPRESS specification:

```
ENTITY product_definition_shape
  SUBTYPE OF (property_definition);
UNIQUE
  UR1: SELF\property_definition.definition;
WHERE
  WR1: SIZEOF(['PRODUCT_PROPERTY_DEFINITION_SCHEMA.CHARACTERIZED_PRODUCT_DEFINITION',
'PRODUCT_PROPERTY_DEFINITION_SCHEMA.CHARACTERIZED_OBJECT'] * TYPEOF(SELF\property_
definition.definition)) > 0;
END_ENTITY;
```

Formal propositions:

WR1: The definition attribute shall not refer to a shape_definition.

22.4.9 property_definition



A **property_definition** is a property that characterizes a single object.

NOTE 1 The type of the characterized item is one of the entity types that may be selected directly or indirectly with characterized_definition.

NOTE 2 The role of **property_definition** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

EXPRESS specification:

```
ENTITY property_definition;
  name : label;
  description : OPTIONAL text;
  definition : characterized_definition;
DERIVE
```

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```
  id : identifier := get_id_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ID_ATTRIBUTE.IDENTIFIED_ITEM'))
<= 1;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **property_definition** is known.

description: the text that characterizes the **property_definition**. The value of the attribute need not be specified.

definition: the item whose property is identified.

id: the identifier that distinguishes the **property_definition**.

NOTE 3 This attribute is an upwardly compatible addition to **property_definition** as specified in ISO 10303-41:1994.

NOTE 4 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 5 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

Formal propositions:

WR1: Each **property_definition** shall be the identified_item in at most one id_attribute.

NOTE 6 The id_attribute data type is defined in support_resource_schema of this part of ISO 10303.

NOTE 7 A template for constraining the population of the entity data types defined in the support_resource_schema is described in annex E.

22.4.10 shape_aspect

A **shape_aspect** is an identified element of the shape of an object or is part of another **shape_aspect**.

NOTE 1 A **shape_aspect** that is part of another **shape_aspect** allows decomposition of a **shape_aspect**.

EXAMPLE 1 Consider the product_definition_shape of a bolt. One might distinguish, as an element of this shape, the concept of the threaded portion of its shank. This portion of the shape could be specified using a **shape_aspect** entity so that other properties, such as surface finish, may be associated with it.

EXPRESS specification:

```
ENTITY shape_aspect
  SUPERTYPE OF (ONEOF(shape_aspect_occurrence, component_path_shape_aspect) ANDOR
constituent_shape_aspect);
  name : label;
  description : OPTIONAL text;
  of_shape : product_definition_shape;
  product_definitional : LOGICAL;
DERIVE
  id : identifier := get_id_value(SELF);
UNIQUE
  UR1: id, of_shape;
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ID_ATTRIBUTE.IDENTIFIED_ITEM'))
<= 1;
```


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```
END_ENTITY;
```

Attribute definitions:

name: the label by which the **shape_aspect** is known.

description: the text that characterizes the **shape_aspect**. The value of the attribute need not be specified.

of_shape: the product_definition_shape of which this entity is an aspect.

EXAMPLE 2 If the identified aspect is the threaded portion of a bolt's shank; this attribute would be the product_definition_shape of the bolt.

product_definitional: an indication that the **shape_aspect** is on the physical boundary of the product_definition_shape. If the value of this attribute is TRUE, the **shape_aspect** being identified is on such a boundary. If the value is FALSE, the **shape_aspect** being identified is not on such a boundary. If the value is UNKNOWN, it is not known whether or not the **shape_aspect** being identified is on such a boundary.

EXAMPLE 3 If the identified **shape_aspect** is the threaded portion of a bolt's shank, the value of this attribute would be TRUE. If it is the centre-line, the value would be FALSE.

id: the identifier that distinguishes the **shape_aspect**.

NOTE 2 This attribute is an upwardly compatible addition to **shape_aspect** as specified in ISO 10303-41:1994.

NOTE 3 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 4 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

Formal propositions:

WR1: Each **shape_aspect** shall be the identified_item in at most one id_attribute.

NOTE 5 The id_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 6 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

22.4.11 shape_aspect_occurrence

A **shape_aspect_occurrence** is a type of shape_aspect that is defined by another shape_aspect or a characterized_object.

EXPRESS specification:

```
ENTITY shape_aspect_occurrence
  SUBTYPE OF (shape_aspect);
  definition : shape_aspect_or_characterized_object;
WHERE
  WR1: acyclic_shape_aspect_occurrence(SELF, definition);
END_ENTITY;
```

Attribute definitions:

definition: the shape_aspect or characterized_object that is defining this **shape_aspect_occurrence**.

Formal propositions:

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WR1: In the case that a **shape_aspect_occurrence** is defined by another **shape_aspect_occurrence**, the graph established by the attribute definition shall not be cyclic.

22.4.12 shape_aspect_relationship

A **shape_aspect_relationship** is a relationship between two instances of the entity data type shape_aspect and provides an identification and description of their relationship.

NOTE 1 The role of **shape_aspect_relationship** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

NOTE 2 If one shape_aspect is part of another, this entity could be used to associate the two shape_aspects.

NOTE 3 Each shape_aspect entity may have different properties.

NOTE 4 No actual physical relationship is established between related instances of the entity data type shape_aspect.

EXAMPLE 1 A **shape_aspect_relationship** might relate two instances of the entity data type shape_aspect whose representations are the equivalent surfaces of a mould and a moulded product. The shape of the mould is not spatially related to the moulded product.

NOTE 5 Relationships represented using this entity may be parent child relationship. Annotated EXPRESS schemas that use or specialize this entity specify whether or not the relationship is directed.

NOTE 6 This entity, together with the shape_aspect entity, is based on the relationship template that is described in annex E.3.

EXPRESS specification:

```
ENTITY shape_aspect_relationship;
  name : label;
  description : OPTIONAL text;
  relating_shape_aspect : shape_aspect;
  related_shape_aspect : shape_aspect;
DERIVE
  id : identifier := get_id_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'ID_ATTRIBUTE.IDENTIFIED_ITEM'))
<= 1;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **shape_aspect_relationship** is known.

description: the text that characterizes the **shape_aspect_relationship**. The value of the attribute need not be specified.

relating_shape_aspect: one of the instances of shape_aspect that is a part of the relationship.

NOTE 7 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

EXAMPLE 2 A shape_aspect with five faces would play the role of **relating_shape_aspect** in five instances of the entity data type **shape_aspect_relationship**: one per face.

related_shape_aspect: the other instance of shape_aspect that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

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NOTE 8 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

id: the identifier that distinguishes the **shape_aspect_relationship**.

NOTE 9 This attribute is an upwardly compatible addition to **shape_aspect_relationship** as specified in ISO 10303-41:1994.

NOTE 10 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

NOTE 11 The context in which id is used as a discriminating characteristic can be identified in an annotated Express schema that uses or specializes this entity, or by default, in an agreement of common understanding between partners sharing this information.

Formal propositions:

WR1: Each **shape_aspect_relationship** shall be the identified_item in at most one id_attribute.

NOTE 12 The id_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 13 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

22.5 Product property definition function definitions

22.5.1 acyclic_characterized_object_relationship

The **acyclic_characterized_object_relationship** function determines whether the graph of instances of the entity data type characterized_object that contains relation as one of its links contains a cycle. This function may be used to evaluate either a characterized_object_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the characterized_object_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_characterized_object_relationship(relation : characterized_object_
relationship;
                                                   relatives : SET [1:?] OF characterized_
object;
                                                   specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF characterized_object_relationship;
  END_LOCAL;
  IF relation.relating_object IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(ca <* bag_to_set(USEDIN(relation.relating_object, ('PRODUCT_PROPERTY_
DEFINITION_SCHEMA.' + 'CHARACTERIZED_OBJECT_RELATIONSHIP.') + 'RELATED_OBJECT')) |
specific_relation IN TYPEOF(ca));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_characterized_object_relationship(x[i], relatives + relation.relating_
object, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
```

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```
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate characterized_object_relationship to be checked.

relatives: (input) the set of instances of the entity data type characterized_object that the function is searching for in the relating_object parameter of the **relation** argument.

specific_relation: (input) the fully qualified entity name of a type of characterized_object_relationship entity.

22.5.2 acyclic_general_property_relationship

The **acyclic_general_property_relationship** function determines whether the graph of instances of the entity data type general_property that contains relation as one of its links contains a cycle. This function may be used to evaluate either a general_property_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the general_property_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_general_property_relationship(relation : general_property_relationship;
                                               relatives : SET [1:?] OF general_property;
                                               specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF general_property_relationship;
  END_LOCAL;
  IF relation.relating_property IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(genp <* bag_to_set(USEDIN(relation.relating_property, ('PRODUCT_PROPERTY_
DEFINITION_SCHEMA.' + 'GENERAL_PROPERTY_RELATIONSHIP.') + 'RELATED_PROPERTY')) | specific_
relation IN TYPEOF(genp));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_general_property_relationship(x[i], relatives + relation.relating_
property, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate general_property_relationship to be checked.

relatives: (input) the set of instances of the entity data type general_property that the function is searching for in the relating_property parameter of the **relation** argument.

specific_relation: (input) the fully qualified entity name of a type of general_property_relationship entity.

22.5.3 acyclic_shape_aspect_occurrence

The **acyclic_shape_aspect_occurrence** function determines whether the graph of instances of the entity data type shape_aspect_occurrence that are linked by the attribute definition contains a cycle.

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The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE The algorithm of the function is explained in annex E.2.

EXPRESS specification:

```
FUNCTION acyclic_shape_aspect_occurrence(sao : shape_aspect_occurrence;
                                         definition : shape_aspect_or_characterized_
object) : BOOLEAN;
  IF NOT ('PRODUCT_PROPERTY_DEFINITION_SCHEMA.SHAPE_ASPECT_OCCURRENCE' IN
TYPEOF(definition)) THEN
    RETURN (TRUE);
  END_IF;
  IF definition :=: sao THEN
    RETURN (FALSE);
  ELSE
    RETURN (acyclic_shape_aspect_occurrence(sao, definition\shape_aspect_occurrence.
definition));
  END_IF;
END_FUNCTION;
```

Argument definitions:

sao: (input) the candidate shape_aspect_occurrence to be checked.

definition: (input) the candidate shape_aspect_or_characterized_object to be checked.

22.5.4 acyclic_shape_aspect_relationship

The **acyclic_shape_aspect_relationship** function determines whether the graph of instances of the entity data type shape_aspect that contains relation as one of its links contains a cycle. This function may be used to evaluate either a shape_aspect_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the shape_aspect_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_shape_aspect_relationship(relation : shape_aspect_relationship;
                                           relatives : SET [1:?] OF shape_aspect;
                                           specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF shape_aspect_relationship;
  END_LOCAL;
  IF relation.relating_shape_aspect IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(sa <* bag_to_set(USEDIN(relation.relating_shape_aspect, ('PRODUCT_PROPERTY_
DEFINITION_SCHEMA.' + 'SHAPE_ASPECT_RELATIONSHIP.') + 'RELATED_SHAPE_ASPECT')) | specific_
relation IN TYPEOF(sa));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_shape_aspect_relationship(x[i], relatives + relation.relating_shape_
aspect, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

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relation: (input) the candidate shape_aspect_relationship to be checked.

relatives: (input) the set of instances of the entity data type shape_aspect that the function is searching for in the relating_shape_aspect parameter of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the shape_aspect_relationship entity.

22.5.5 get_shape_aspects

The **get_shape_aspects** function returns for any instance of characterized_definition the set of instances of shape_aspect that refer to the instance of characterized_definition through an instance of product_definition_shape.

NOTE This function is not used in this schema.

EXPRESS specification:

```
FUNCTION get_shape_aspects(c_def_instance : characterized_definition) : SET OF shape_
aspect;
  LOCAL
    pd_set : SET OF product_definition_shape := [];
    pdr_set : SET OF shape_aspect := [];
  END_LOCAL;
  pd_set := bag_to_set(QUERY(pd <* USEDIN(c_def_instance, 'PRODUCT_PROPERTY_DEFINITION_
SCHEMA.PROPERTY_DEFINITION.DEFINITION') | 'PRODUCT_PROPERTY_DEFINITION_SCHEMA.PRODUCT_
DEFINITION_SHAPE' IN TYPEOF(pd)));
  IF SIZEOF(pd_set) < 1 THEN
    RETURN (pdr_set);
  END_IF;
  REPEAT i := 1 TO HIINDEX(pd_set);
    pdr_set := pdr_set + bag_to_set(USEDIN(pd_set[i], 'PRODUCT_PROPERTY_DEFINITION_SCHEMA.
SHAPE_ASPECT.OF_SHAPE'));
  END_REPEAT;
  RETURN (pdr_set);
END_FUNCTION;
```

Argument definitions:

c_def_instance: (input) the candidate characterized_definition to be checked.

22.5.6 get_root_shape_aspect

The **get_root_shape_aspect** function returns the product_definition_shape of which the current shape_aspect is an identified aspect.

EXPRESS specification:

```
FUNCTION get_root_shape_aspect(current : shape_aspect) : product_definition_shape;
  REPEAT WHILE 'PRODUCT_DEFINITION_SCHEMA.CONSTITUENT_SHAPE_ASPECT' IN TYPEOF(current);
    current := current.parent;
  END_REPEAT;
  RETURN (current.of_shape);
END_FUNCTION;
```

Argument definitions:

current: (input) the shape_aspect that is the basis for the function.

```
END_SCHEMA; -- product_property_definition_schema
```


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23 Product property representation schema

23.1 General

The subject of the **product_property_representation_schema** is the representation of properties.

The requirements addressed by this schema are:

— the ability to associate a representation with a **property_definition** that it represents;

— the ability to identify a **representation** as representing a shape;

— the ability to identify relationships between the representations of shapes and to state that the identified relationships are the representations of zero, one or more instances of the entity data type **product_definition**.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **product_property_representation_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA product_property_representation_schema '{iso standard 10303 part(41) version(9)
object(1) product_property_representation_schema(20)}';

REFERENCE FROM basic_attribute_schema
  (description_attribute,
   description_attribute_select,
   get_description_value,
   get_name_value,
   name_attribute,
   name_attribute_select);
REFERENCE FROM material_property_definition_schema
  (property_definition_relationship);
REFERENCE FROM product_definition_schema
  (product_definition,
   product_definition_relationship);
REFERENCE FROM product_property_definition_schema
  (characterized_object,
   characterized_definition,
   general_property,
   product_definition_shape,
   property_definition,
   shape_aspect,
   shape_aspect_relationship);
REFERENCE FROM product_structure_schema
  (product_definition_specified_occurrence);
REFERENCE FROM representation_schema
  (representation,
   representation_item,
   representation_relationship,
   representation_reference,
   using_representations,
   mapped_item,
   representation_context,
   representation_map,
   list_representation_item,
   set_representation_item,
   get_representations_for_items);
REFERENCE FROM support_resource_schema
  (bag_to_set,
   label,
```

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```
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

basic_attribute_schema ISO 10303-41

:: ISO -

product_definition_schema ISO 10303-41

product_property_definition_schema ISO 10303-41

:: ISO -

:: ISO -

support_resource_schema ISO 10303-41

NOTE 2 See Annex D, Paragraph 21 for a graphical representation of this schema.

23.2 Fundamental concepts and assumptions

The following assumptions apply to the **product_property_representation_schema**:

— this International Standard will include various types of representations ranging from simple collections of geometry to more elaborate collections of representations of different kinds of properties;

— it is possible to have more than one representation of a single property;

— a single representation may be used to represent a property of zero, one, or many instances of the entity data type **product_definition**;

— any property of a **product_definition** is an identifiable concept independent of how, or even if, it is represented.

NOTE Representation schemas are defined in other parts of ISO 10303. This schema provides a structure within which these resources can be brought together to represent the properties of a product.

23.3 Product property representation type definitions

23.3.1 pprs_description_attribute_select

The **pprs_description_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

EXPRESS specification:

```
TYPE pprs_description_attribute_select = SELECT BASED_ON description_attribute_select WITH
  (context_dependent_shape_representation,
   property_definition_representation);
END_TYPE;
```

23.3.2 pprs_name_attribute_select

The **pprs_name_attribute_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

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EXPRESS specification:

```
TYPE pprs_name_attribute_select = SELECT BASED_ON name_attribute_select WITH
  (context_dependent_shape_representation,
   property_definition_representation);
END_TYPE;
```

23.3.3 chained_representation_link

The **chained_representation_link** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

The **chained_representation_link** type allows for the designation of a mapped_item, a representation_context, or a representation_relationship.

EXPRESS specification:

```
TYPE chained_representation_link = SELECT
  (mapped_item,
   representation_context,
   representation_relationship);
END_TYPE;
```

23.3.4 represented_definition

The **represented_definition** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

The **represented_definition** type allows for the designation of a general_property, a property_definition, a property_definition_relationship, a shape_aspect, or a shape_aspect_relationship.

EXPRESS specification:

```
TYPE represented_definition = SELECT
  (general_property,
   property_definition,
   property_definition_relationship,
   shape_aspect,
   shape_aspect_relationship);
END_TYPE;
```

23.3.5 item_identified_representation_usage_definition

The **item_identified_representation_usage_definition** type is an extensible list of alternate data types. It provides a mechanism to refer to instances of the data types included in the types that extend the **item_identified_representation_usage_definition** type.

NOTE This empty extensible select requires extension in a further schema to ensure that entities that refer to it have at least one valid instantiation.

EXPRESS specification:

```
TYPE item_identified_representation_usage_definition = EXTENSIBLE GENERIC_ENTITY SELECT
  (represented_definition);
END_TYPE;
```

23.3.6 item_identified_representation_usage_select

The **item_identified_representation_usage_select** type is a list of alternate data types. It provides a mechanism to refer to an instance of one of these data types.

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EXPRESS specification:

```
TYPE item_identified_representation_usage_select = SELECT
  (representation_item,
   list_representation_item,
   set_representation_item);
END_TYPE;
```

23.4 Product property representation entity definitions

23.4.1 chain_based_item_identified_representation_usage

A **chain_based_item_identified_representation_usage** is a type of item_identified_representation_usage that represents a chain of representations in a graph of representations, where the undirected links in the graph can be instances of entity data types representation_context, representation_relationship, or mapped_item. The directed links are derived from the nodes and undirected links and consist solely of instances of the entity data type representation_relationship. The attribute rep_2 of the derived instances is directed towards the root, while the attribute rep_1 is directed towards the leaf.

EXPRESS specification:

```
ENTITY chain_based_item_identified_representation_usage
  SUBTYPE OF (item_identified_representation_usage);
  nodes : LIST [2:?] OF representation;
  undirected_link : LIST [1:?] OF chained_representation_link;
DERIVE
  root : representation := nodes[1];
  SELF\item_identified_representation_usage.used_representation RENAMED leaf :
representation := nodes[HIINDEX(nodes)];
  directed_link : LIST [1:?] OF representation_relationship := get_directed_link(nodes,
undirected_link);
WHERE
  WR1: EXISTS(directed_link);
END_ENTITY;
```

Attribute definitions:

nodes: the list of representation in the chain.

undirected_link: the list of items that relate nodes in the chain.

NOTE 1 The items in the **undirected_link** are placed correctly relative to the nodes in the chain but are not guaranteed to have a consistent direction along the chain.

root: the node that is the initial terminus of the chain.

leaf: the node that is the final terminus of the chain.

directed_link: the list of representation_relationship that are derived from the **undirected_link** and that are consistently directed from the root to the leaf.

Formal propositions:

WR1: The directed_link attribute shall be valid for the given set of nodes and undirected_links.

NOTE 2 the get_directed_link function returns indeterminate if it cannot construct a valid list. The EXPRESS EXISTS function returns FALSE with an indeterminate input.

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23.4.2 characterized_chain_based_item_within_representation

A **characterized_chain_based_item_within_representation** is a type of characterized_item_within_representation that represents a chain of representations in a graph of representations, where the undirected links in the graph can be instances of entity data types representation_context, representation_relationship, or mapped_item. The directed links are derived from the nodes and undirected links and consist solely of instances of the entity data type representation_relationship. The attribute rep_2 of the derived instances is directed towards the root, while the attribute rep_1 is directed towards the leaf.

EXPRESS specification:

```
ENTITY characterized_chain_based_item_within_representation
  SUBTYPE OF (characterized_item_within_representation);
  nodes : LIST [2:?] OF representation;
  undirected_link : LIST [1:?] OF chained_representation_link;
DERIVE
  root : representation := nodes[1];
  SELF\characterized_item_within_representation.rep RENAMED leaf : representation :=
nodes[HIINDEX(nodes)];
  directed_link : LIST [1:?] OF representation_relationship := get_directed_link(nodes,
undirected_link);
WHERE
  WR1: EXISTS(directed_link);
END_ENTITY;
```

Attribute definitions:

nodes: the list of representation in the chain.

undirected_link: the list of items that relate nodes in the chain.

NOTE 1 The items in the **undirected_link** are placed correctly relative to the nodes in the chain but are not guaranteed to have a consistent direction along the chain.

root: the node that is the initial terminus of the chain.

leaf: the node that is the final terminus of the chain.

directed_link: the list of representation_relationship that are derived from the **undirected_link** and that are consistently directed from the root to the leaf.

Formal propositions:

WR1: The directed_link attribute shall be valid for the given set of nodes and undirected_links.

NOTE 2 the get_directed_link function returns indeterminate if it cannot construct a valid list. The EXPRESS EXISTS function returns FALSE with an indeterminate input.

23.4.3 characterized_item_within_representation

A **characterized_item_within_representation** is a type of characterized_object that provides the characterization of a representation_item within a specified representation.

EXPRESS specification:

```
ENTITY characterized_item_within_representation
  SUBTYPE OF (characterized_object);
  item : representation_item;
  rep : representation;
UNIQUE
  UR1: item, rep;
WHERE
```

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```
  WR1: rep IN using_representations(item);
END_ENTITY;
```

Attribute definitions:

item: the representation_item that is characterised.

rep: the representation in which the characterization of the representation_item applies.

Formal propositions:

WR1: The **item** shall be founded in the specified representation.

23.4.4 context_dependent_shape_representation

A **context_dependent_shape_representation** is the association of a shape_representation_relationship with a product_definition_shape. The product_definition_shape identifies the shape of a product_definition as it plays the role of the related_product_definition in a product_definition_relationship.

NOTE 1 The role of **context_dependent_shape_representation** can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

EXAMPLE 1 A gasket's shape depends upon whether the gasket is participating in an assembly relationship and, if it is a part of an assembly, the product or products with which it is assembled.

EXAMPLE 2 The position of a bolt's shape depends upon the way it is positioned in a given assembly.

EXPRESS specification:

```
ENTITY context_dependent_shape_representation;
  representation_relation : shape_representation_relationship;
  represented_product_relation : product_definition_shape;
DERIVE
  description : text := get_description_value(SELF);
  name : label := get_name_value(SELF);
WHERE
  WR1: 'PRODUCT_DEFINITION_SCHEMA.PRODUCT_DEFINITION_RELATIONSHIP' IN TYPEOF(SELF.
represented_product_relation.definition);
  WR2: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
  WR3: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'NAME_ATTRIBUTE.NAMED_ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

representation_relation: a shape_representation_relationship that is associated with the product_definition_shape.

represented_product_relation: a product_definition_shape that identifies the shape of the related product_definition in the context of a product_definition_relationship.

description: the text that characterizes the **context_dependent_shape_representation**.

NOTE 2 This attribute is an upwardly compatible addition to **context_dependent_shape_representation** as specified in ISO 10303-41:1994.

name: the label by which the **context_dependent_shape_representation** is known.

NOTE 3 This attribute is an upwardly compatible addition to **context_dependent_shape_representation** as specified in ISO 10303-41:1994.

Formal propositions:

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WR1: The **represented_product_relation** shall identify the shape of a product_definition_relationship.

WR2: Each **context_dependent_shape_representation** shall be the described_item in at most one description_attribute.

NOTE 4 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

WR3: Each **context_dependent_shape_representation** shall be the named_item in at most one name_attribute.

NOTE 5 The name_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 6 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

23.4.5 item_identified_representation_usage

An **item_identified_representation_usage** is an identification of a single or a set or a list of representation_item(s) within a representation as being the element(s) that describes a particular component or part of the property that is described by the representation. More than one representation. may be bound to a product data item by members of **item_identified_representation_usage** and more than one member of representation_item(s) may be bound to a product data item by members of **item_identified_representation_usage**. The same item of product data shall not be bound to more than one member of representation_item within a member of representation.

EXAMPLE 1 In an application protocol, an instance of representation describes the shape of a product. One element of the representation — a curve — represents the boundary of a **shape_aspect**, a hole, in the product. **Item_identification_representation_usage** is used to state that the curve referenced by 'identified_item' is a representation item for the **shape_aspect** referenced by 'definition' and that the whole representation is referenced by 'used_representation' to indicate the structural context for the curve. **Geometric_representation_context** plays no role in this example.

EXAMPLE 2 In an application protocol, an instance of representation describes the shape of a product. One element of the representation — a curve — represents the boundary of a **shape_aspect**, a hole, in the product. A member of **Item_identification_representation_usage** is used to state that the curve referenced by 'identified_item' is a representation item for the **shape_aspect#1** referenced by 'definition' and that the whole representation is referenced by 'used_representation' to indicate the structural context for the curve. A second member of **Item_identification_representation_usage** references a different curve in that same representation and references the same **shape_aspect#1**. The data population is invalid because only one item within a representation can be bound to a **shape_aspect**.

EXPRESS specification:

```
ENTITY item_identified_representation_usage;
  name : label;
  description : OPTIONAL text;
  definition : item_identified_representation_usage_definition;
  used_representation : representation;
  identified_item : item_identified_representation_usage_select;
UNIQUE
  UR1: used_representation, identified_item;
  ;
  UR2: used_representation, definition;
WHERE
  WR1: NOT (SIZEOF(identified_item) = 1) OR (used_representation IN using_
representations(identified_item));
  WR2: NOT (SIZEOF(identified_item) > 1) OR (used_representation IN get_representations_
for_items(identified_item));
END_ENTITY;
```

Attribute definitions:

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name: the label by which the **item_identified_representation_usage** is known.

description: the text that characterizes the **item_identified_representation_usage**. The value of the attribute need not be specified.

definition: the general_property, property_definition, property_definition_relationship, shape_aspect, or shape_aspect_relationship that is represented.

used_representation: the representation that describes the property or aspect of which the definition is a part or component, and that contains the **identified_item** (directly or indirectly) as one of its items.

identified_item: the representation_item or list_representation_item or set_representation_item that describes the identified property or aspect.

Formal propositions:

WR1: If the size of identified_item is one then the used_representation shall reference the item directly or indirectly.

WR2: If the size of identified_item is greater than one then the used_representation shall be one of the representations that reference the item directly or indirectly.

23.4.6 property_definition_representation 

A **property_definition_representation** is an association between a property_definition and a representation of the property.

EXPRESS specification:

```
ENTITY property_definition_representation;
  definition : represented_definition;
  used_representation : representation;
DERIVE
  description : text := get_description_value(SELF);
  name : label := get_name_value(SELF);
WHERE
  WR1: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'DESCRIPTION_ATTRIBUTE.DESCRIBED_
ITEM')) <= 1;
  WR2: SIZEOF(USEDIN(SELF, 'BASIC_ATTRIBUTE_SCHEMA.' + 'NAME_ATTRIBUTE.NAMED_ITEM')) <= 1;
END_ENTITY;
```

Attribute definitions:

definition: the identification of the general_property, property_definition, property_definition_relationship, shape_aspect, or shape_aspect_relationship that is represented.

used_representation: the representation of the property or aspect of the representation.

description: the text that characterizes the **property_definition_representation**.

NOTE 1 This attribute is an upwardly compatible addition to **property_definition_representation** as specified in ISO 10303-41:1994.

name: the label by which the **property_definition_representation** is known.

NOTE 2 This attribute is an upwardly compatible addition to **property_definition_representation** as specified in ISO 10303-41:1994.

Formal propositions:

WR1: Each **property_definition_representation** shall be the described_item in at most one description_attribute.

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NOTE 3 The description_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

WR2: Each **property_definition_representation** shall be the named_item in at most one name_attribute.

NOTE 4 The name_attribute data type is defined in basic_attribute_schema of this part of ISO 10303.

NOTE 5 A template for constraining the population of the entity data types defined in the basic_attribute_schema is described in annex E.

23.4.7 shape_definition_representation

A **shape_definition_representation** is a type of property_definition_representation. A **shape_definition_representation** is the association of a shape_representation with a product_definition_shape or with a property_definition characterizing a shape_definition.

EXPRESS specification:

```
ENTITY shape_definition_representation
  SUBTYPE OF (property_definition_representation);
  SELF\property_definition_representation.definition : property_definition;
  SELF\property_definition_representation.used_representation : shape_representation;
WHERE
  WR1: ('PRODUCT_PROPERTY_DEFINITION_SCHEMA.PRODUCT_DEFINITION_SHAPE' IN
TYPEOF(definition)) OR ('PRODUCT_PROPERTY_DEFINITION_SCHEMA.SHAPE_DEFINITION' IN
TYPEOF(definition\property_definition.definition));
END_ENTITY;
```

Attribute definitions:

definition: an inherited attribute whose value shall be of type property_definition

used_representation: an inherited attribute whose value shall be of type shape_representation.

Formal propositions:

WR1: The inherited definition attribute shall be a product_definition_shape or a property_definition whose definition attribute is a shape_definition.

23.4.8 shape_representation

A **shape_representation** is a type of representation that represents a shape.

EXPRESS specification:

```
ENTITY shape_representation
  SUBTYPE OF (representation);
END_ENTITY;
```

23.4.9 shape_representation_reference

A **shape_representation_reference** is a type of representation_reference in which the referenced data type is a shape_representation.

EXPRESS specification:

```
ENTITY shape_representation_reference
  SUBTYPE OF (representation_reference);
END_ENTITY;
```

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23.4.10 shape_representation_relationship

A **shape_representation_relationship** is a type of representation_relationship in which at least one of the instances of the entity data type representation is a shape_representation or shape_representation_reference.

EXAMPLE The representation of the shape of a bolt may be related to the representation of a position if the bolt is a part of an assembly.

EXPRESS specification:

```
ENTITY shape_representation_relationship
  SUBTYPE OF (representation_relationship);
WHERE
  WR1: SIZEOF(['PRODUCT_PROPERTY_REPRESENTATION_SCHEMA.SHAPE_REPRESENTATION',
'PRODUCT_PROPERTY_REPRESENTATION_SCHEMA.SHAPE_REPRESENTATION_REFERENCE'] * (TYPEOF(SELF
\representation_relationship.rep_1) + TYPEOF(SELF\representation_relationship.rep_2))) >=
1;
END_ENTITY;
```

Formal propositions:

WR1: At least one of the two representations in the **shape_representation_relationship** shall be a shape_representation or shape_representation_reference.

23.4.11 specified_occurrence_context_dependent_shape_representation

A **specified_occurrence_context_dependent_shape_representation** is a type of context_dependent_shape_representation that established a relation for a hierarchical product structure using product_definition_specified_occurrence.

EXPRESS specification:

```
ENTITY specified_occurrence_context_dependent_shape_representation
  SUBTYPE OF (context_dependent_shape_representation);
  sub_element : product_definition_specified_occurrence;
WHERE
  WR1: 'PRODUCT_DEFINITION_SCHEMA.PRODUCT_DEFINITION_OCCURRENCE' IN TYPEOF(SELF.
represented_product_relation.definition.related_product_definition);
  WR2: sub_element IN represented_product_relation.definition.related_product_definition.
descendant_occurrences;
END_ENTITY;
```

Attribute definitions:

sub_element: the product_definition_specified_occurrence for which the association established by the inherited attributes **representation_relation** and **represented_product_relation** applies.

Formal propositions:

WR1: The related_product_definition of the **represented_product_relation** shall be of type product_definition_occurrence.

WR2: The **sub_element** shall be within the descendant_occurrences of the product_definition_occurrence that is referenced by the related_product_definition of the **represented_product_relation**.

23.5 Product property representation function definitions

23.5.1 get_directed_link

The **get_directed_link** function returns for any list of representation and associated list of chained_representation_link the resolved list of representation_relationship when each member of

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undirected_link references the correct members of **nodes**. The **get_directed_link** function returns UNKNOWN when the input does not satisfy the correctness requirement.

EXPRESS specification:

```
FUNCTION get_directed_link(nodes : LIST OF representation;
                           undirected_link : LIST OF chained_representation_link) : LIST
OF representation_relationship;
  LOCAL
    directed_link : LIST OF representation_relationship := [];
  END_LOCAL;
  IF (SIZEOF(nodes) <> SIZEOF(undirected_link) + 1) OR (value_unique(nodes) = FALSE) THEN
    RETURN (?);
  END_IF;
  REPEAT i := 1 TO SIZEOF(undirected_link);
    CASE TRUE OF
    'REPRESENTATION_SCHEMA.REPRESENTATION_CONTEXT' IN TYPEOF(undirected_link[i]) :
      BEGIN
        IF (nodes[i]\representation.context_of_items :=: undirected_link[i]) AND (nodes[i
+ 1]\representation.context_of_items :=: undirected_link[i]) THEN
          INSERT(directed_link, representation_relationship('', '', nodes[i + 1],
nodes[i]), i - 1);
        ELSE
          RETURN (?);
        END_IF;
      END;
    'REPRESENTATION_SCHEMA.REPRESENTATION_RELATIONSHIP' IN TYPEOF(undirected_link[i]) :
      BEGIN
        IF (nodes[i] :=: undirected_link[i]\representation_relationship.rep_1) AND
(nodes[i + 1] :=: undirected_link[i]\representation_relationship.rep_2) OR (nodes[i] :=
: undirected_link[i]\representation_relationship.rep_2) AND (nodes[i + 1] :=: undirected_
link[i]\representation_relationship.rep_1) THEN
          INSERT(directed_link, representation_relationship('', '', nodes[i + 1],
nodes[i]), i - 1);
        ELSE
          RETURN (?);
        END_IF;
      END;
    'REPRESENTATION_SCHEMA.MAPPED_ITEM' IN TYPEOF(undirected_link[i]) :
      BEGIN
        IF (nodes[i] IN using_representations(undirected_link[i])) AND (nodes[i + 1] :=:
undirected_link[i]\mapped_item.mapping_source\representation_map.mapped_representation)
THEN
          INSERT(directed_link, representation_relationship('', '', nodes[i + 1],
nodes[i]), i - 1);
        ELSE
          RETURN (?);
        END_IF;
      END;
    OTHERWISE :
      RETURN (?);
    END_CASE;
  END_REPEAT;
  IF value_unique(directed_link) = FALSE THEN
    RETURN (?);
  END_IF;
  RETURN (directed_link);
END_FUNCTION;
```

Argument definitions:

nodes: (input) the list of instances of the entity data type representation to be used as a reference for constructing the directed_link.

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undirected_link: (input) the list of instances of the entity data type mapped_item, representation_context, or representation_relationship that the function is comparing against the reference to establish the directed_link.

23.5.2 relatives_of_product_definitions

The **relatives_of_product_definitions** function finds all of the instances of the product_definition entity data type that are related to one or more elements of the **definition_set** argument. Only those relationships that are established by the subtype of the product_definition_relationship entity given in the **relation_subtype** argument are considered by this function.

EXPRESS specification:

```
FUNCTION relatives_of_product_definitions(definition_set : SET OF product_definition;
                                          relation_subtype : STRING) : SET OF product_
definition;
  FUNCTION local_relatives_of_product_definitions(definition_set : SET OF product_
definition;
                                                  total_definitions : SET OF product_
definition;
                                                  relation_subtype : STRING) : SET OF
product_definition;
    LOCAL
      local_def : SET OF product_definition := [];
      local_pdr : SET OF product_definition_relationship := [];
      local_total : SET OF product_definition := [];
    END_LOCAL;
    REPEAT i := 1 TO HIINDEX(definition_set);
      local_pdr := local_pdr + bag_to_set(USEDIN(definition_set[i], relation_subtype + '.
RELATING_PRODUCT_DEFINITION'));
    END_REPEAT;
    REPEAT i := 1 TO HIINDEX(local_pdr);
      local_def := local_def + local_pdr[i].related_product_definition;
    END_REPEAT;
    IF SIZEOF(local_def) - SIZEOF(total_definitions) = 0 THEN
      RETURN (local_def);
    ELSE
      local_total := total_definitions + local_def;
      RETURN (local_def + local_relatives_of_product_definitions(local_def - total_
definitions, local_total, relation_subtype));
    END_IF;
  END_FUNCTION;
  RETURN (local_relatives_of_product_definitions(definition_set, definition_set, relation_
subtype));
END_FUNCTION;
```

Argument definitions:

definition_set: (input) the set of instances of the entity data type product_definition upon which operations are performed.

relation_subtype: the fully qualified name of an entity that is a subtype of the product_definition_relationship entity.

23.5.3 relatives_of_shape_representations

The **relatives_of_shape_representations** function finds all of the instances of the shape_representation entity data type that are related to one or more elements of the **shape_representation_set** argument.

NOTE This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the identification_assignment_relationship entity include rules that use this function.

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EXPRESS specification:

```
FUNCTION relatives_of_shape_representations(shape_representation_set : SET OF shape_
representation) : SET OF shape_representation;
  FUNCTION local_relatives_of_shape_representations(shape_representation_set : SET OF
shape_representation;
                                                    total_reps : SET OF shape_
representation) : SET OF shape_representation;
    LOCAL
      local_shape_rep : SET OF shape_representation := [];
      local_srr : SET OF shape_representation_relationship := [];
      local_total : SET OF shape_representation := [];
    END_LOCAL;
    REPEAT i := 1 TO HIINDEX(shape_representation_set);
      local_srr := local_srr + QUERY(rr <* bag_to_set(USEDIN(shape_representation_
set[i], 'REPRESENTATION_SCHEMA.REPRESENTATION_RELATIONSHIP.REP_1')) | 'PRODUCT_PROPERTY_
REPRESENTATION_SCHEMA.SHAPE_REPRESENTATION_RELATIONSHIP' IN TYPEOF(rr));
    END_REPEAT;
    REPEAT i := 1 TO HIINDEX(local_srr);
      IF 'PRODUCT_PROPERTY_REPRESENTATION_SCHEMA.' + 'SHAPE_REPRESENTATION_RELATIONSHIP'
IN TYPEOF(local_srr[i]) THEN
        local_shape_rep := local_shape_rep + local_srr[i].rep_2;
      END_IF;
    END_REPEAT;
    IF SIZEOF(local_shape_rep - total_reps) = 0 THEN
      RETURN (shape_representation_set);
    ELSE
      local_total := total_reps + local_shape_rep;
      RETURN (local_shape_rep + local_relatives_of_shape_representations(local_shape_rep
- total_reps, local_total));
    END_IF;
  END_FUNCTION;
  RETURN (local_relatives_of_shape_representations(shape_representation_set, shape_
representation_set));
END_FUNCTION;
```

Argument definitions:

shape_representation_set: (input) the set of instances of the entity data type shape_representation to be operated on.

23.5.4 get_property_definition_representations

The **get_property_definition_representations** function returns for any characterized_definition the set of property_definition_representation objects that refer to the characterized_definition through a property_definition.

EXPRESS specification:

```
FUNCTION get_property_definition_representations(c_def_instance : characterized_
definition) : SET OF property_definition_representation;
  LOCAL
    pd_set : SET OF property_definition := [];
    pdr_set : SET OF property_definition_representation := [];
  END_LOCAL;
  pd_set := bag_to_set(USEDIN(c_def_instance, 'PRODUCT_PROPERTY_DEFINITION_SCHEMA.
PROPERTY_DEFINITION.DEFINITION'));
  IF SIZEOF(pd_set) < 1 THEN
    RETURN (pdr_set);
  END_IF;
  REPEAT i := 1 TO HIINDEX(pd_set);
    pdr_set := pdr_set + bag_to_set(USEDIN(pd_set[i], 'PRODUCT_PROPERTY_REPRESENTATION_
SCHEMA.PROPERTY_DEFINITION_REPRESENTATION.DEFINITION'));
  END_REPEAT;
  RETURN (pdr_set);
```

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```
END_FUNCTION;
```

Argument definitions:

c_def_instance: (input) the candidate characterized_definition to be checked.

```
END_SCHEMA; -- product_property_representation_schema
```

24 Qualifications schema

24.1 General

The subject of the **qualifications_schema** is the identification of types of qualifications and the relationships between them. This schema provides for the definition of generic data relating to types of qualifications.

EXAMPLE The identification of types of qualifications and the definition of various kinds of relationship between types of qualifications are examples of generic data relating to types of qualification.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **qualifications_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA qualifications_schema '{iso standard 10303 part(41) version(6) object(1)
qualifications_schema(21)}';

REFERENCE FROM support_resource_schema
  (identifier,
   label,
   text,
   bag_to_set);
```

NOTE 1
The schemas referenced above are specified in the following parts:

support_resource_schema	ISO 10303-41

NOTE 2 See Annex D, Paragraph 22 for a graphical representation of this schema.

24.2 Fundamental concepts and assumptions

This schema provides resources for the identification, naming, and description of:

— qualifications;

— types of qualification;

— relationships between types of qualification.

24.3 Qualifications entity definitions

24.3.1 qualification 

A **qualification** is a particular instance of the formal identification of a capability or aptitude in a specific person or organization.

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EXAMPLE 1 John Smith's driving licence.

EXAMPLE 2 Mary Brown's doctorate entitled "Datamodelling Using EXPRESS2".

NOTE A relationship between a qualification_type object and one or more **qualification** objects is established by the declaration of a qualification_type_assignment subtype containing a single attribute that references a set of a SELECT type that contains an qualification.

EXPRESS specification:

```
ENTITY qualification;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **qualification**.

name: the label by which the **qualification** is known.

description: the text that characterizes the **qualification**. The value of this attribute need not be specified.

24.3.2 qualification_relationship

A **qualification_relationship** is a relationship between two instances of the qualification entity data type and provides for the identification, naming, and description of that relationship.

EXPRESS specification:

```
ENTITY qualification_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating_qualification : qualification;
  related_qualification : qualification;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **qualification_relationship**.

name: the label by which the **qualification_relationship** is known

description: the text that characterizes the **qualification_relationship**. The value of this attribute need not be specified.

relating_qualification: one of the instances of qualification that is a part of the relationship.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_qualification: the other instance of qualification that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 2 the meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

24.3.3 qualification_type

An **qualification_type** is the formal identification of a capability or aptitude.

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EXAMPLE 1 A driving licence.

EXAMPLE 2 A university degree.

NOTE A relationship between a **qualification_type** object and one or more qualification objects is established by the declaration of an qualification_type_assignment subtype containing a single attribute that references a set of a SELECT type that contains a qualification.

EXPRESS specification:

```
ENTITY qualification_type;
  id : identifier;
  name : label;
  description : OPTIONAL text;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **qualification_type**.

name: the label by which the **qualification_type** is known.

description: the text that characterizes the **qualification_type**. The value of this attribute need not be specified.

24.3.4 qualification_type_relationship 

A **qualification_type_relationship** is a relationship between two instances of the qualification_type entity data type and provides for the identification, naming, and description of that relationship.

EXPRESS specification:

```
ENTITY qualification_type_relationship;
  id : identifier;
  name : label;
  description : OPTIONAL text;
  relating_qualification_type : qualification_type;
  related_qualification_type : qualification_type;
END_ENTITY;
```

Attribute definitions:

id: the identifier that distinguishes the **qualification_type_relationship**.

name: the label by which the **qualification_type_relationship** is known.

description: the text that characterizes the **qualification_type_relationship**. The value of this attribute need not be specified.

relating_qualification_type: one of the instances of qualification_type that is a part of the relationship.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_qualification_type: the other instance of qualification_type that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

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24.4 Qualifications function definitions

24.4.1 acyclic_qualification_relationship

The **acyclic_qualification_relationship** function determines whether the graph of instances of the entity data type qualification that contains relation as one of its links contains a cycle. This function may be used to evaluate either an qualification_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the qualification_relationship entity include rules that use this function.

EXPRESS specification:

```
FUNCTION acyclic_qualification_relationship(relation : qualification_relationship;
                                            relatives : SET OF qualification;
                                            specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF qualification_relationship;
  END_LOCAL;
  IF relation.relating_qualification IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(qual <* bag_to_set(USEDIN(relation.relating_qualification, ('QUALIFICATIONS_
SCHEMA.' + 'QUALIFICATION_RELATIONSHIP.') + 'RELATED_QUALIFICATION')) | specific_relation
IN TYPEOF(qual));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_qualification_relationship(x[i], relatives + relation.relating_
qualification, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate qualification_relationship to be checked.

relatives: (input) the set of instances of the entity data type qualification that the function is searching for in the relating_qualification parameter of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the qualification_relationship entity.

24.4.2 acyclic_qualification_type_relationship

The **acyclic_qualification_type_relationship** function determines whether the graph of instances of the entity data type qualification_type that contains relation as one of its links contains a cycle. This function may be used to evaluate either a qualification_type_relationship or any of its subtypes.

The function returns TRUE if no cycle has been detected. Otherwise it returns FALSE.

NOTE 1 The algorithm of the function is explained in annex E.2.

NOTE 2 This function is not used in this schema. It is defined here because other ISO 10303 integrated resources and application protocols that use the qualification_type_relationship entity include rules that use this function.

EXPRESS specification:

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```
FUNCTION acyclic_qualification_type_relationship(relation : qualification_type_
relationship;
                                                 relatives : SET OF qualification_type;
                                                 specific_relation : STRING) : BOOLEAN;
  LOCAL
    x : SET OF qualification_type_relationship;
  END_LOCAL;
  IF relation.relating_qualification_type IN relatives THEN
    RETURN (FALSE);
  END_IF;
  x := QUERY(qultyp <* bag_to_set(USEDIN(relation.relating_qualification_type,
('QUALIFICATIONS_SCHEMA.' + 'QUALIFICATION_TYPE_RELATIONSHIP.') + 'RELATED_QUALIFICATION_
TYPE')) | specific_relation IN TYPEOF(qultyp));
  REPEAT i := 1 TO HIINDEX(x);
    IF NOT acyclic_qualification_type_relationship(x[i], relatives + relation.relating_
qualification_type, specific_relation) THEN
      RETURN (FALSE);
    END_IF;
  END_REPEAT;
  RETURN (TRUE);
END_FUNCTION;
```

Argument definitions:

relation: (input) the candidate qualification_type_relationship to be checked.

relatives: (input) the set of instances of the entity data type qualification_type that the function is searching for in the relating_qualification_type parameter of the **relation** argument.

specific_relation: (input) the fully qualified name of a subtype of the qualification_type_relationship entity.

```
END_SCHEMA; -- qualifications_schema
```

25 Security classification schema

25.1 General

The subject of the **security_classification_schema** is the definition of degrees of secrecy to be specified.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **security_classification_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA security_classification_schema '{iso standard 10303 part(41) version(3) object(1)
security_classification_schema(22)}';

REFERENCE FROM support_resource_schema
  (label,
   text);
```

NOTE 1
The schemas referenced above are specified in the following parts:

support_resource_schema	ISO 10303-41

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NOTE 2 See Annex D, Paragraph 23 for a graphical representation of this schema.

25.2 Fundamental concepts and assumptions

A security classification is the level of confidentiality that is required for the purpose of product data protection. Security classifications are assigned by some authoritative agency. Security classification information can be attached to any aspect of product data.

25.3 Security classification entity definitions

25.3.1 security_classification

A **security_classification** is the level of confidentiality that is required for the purpose of product data protection.

NOTE The role of can be defined in the annotated EXPRESS schemas that use or specialize this entity, or by default, in an agreement of common understanding between the partners sharing this information.

EXPRESS specification:

```
ENTITY security_classification;
  name : label;
  purpose : text;
  security_level : security_classification_level;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **security_classification** is known.

EXAMPLE 'National Security' is a security classification role. A security classification assignment role assigns the security classification role to a security_classification_assignment.

purpose: an informal description of the intent of the **security_classification**.

security_level: the security_classification_level that specifies the required degree of security.

25.3.2 security_classification_level

A **security_classification_level** is a category of security required for product data protection.

EXAMPLE 'Confidential', 'secret', and 'top secret' are examples of security_classification levels.

EXPRESS specification:

```
ENTITY security_classification_level;
  name : label;
END_ENTITY;
```

Attribute definitions:

name: the label by which the **security_classification_level** is known.

NOTE The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

```
END_SCHEMA; -- security_classification_schema
```

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26 Support resource schema

26.1 General

The subject of the **support_resource_schema** is a description of EXPRESS declarations that are shared by more than one of the ISO 10303 integrated resource schemas.

This clause defines the information requirements to which implementations shall conform using the EXPRESS language as defined in ISO 10303-11. The following EXPRESS declaration begins the **support_resource_schema** and identifies the necessary external references.

Short names of entities defined in this schema are described in Annex A. Unambiguous identification of this schema is defined in Annex B.

EXPRESS specification:

```
SCHEMA support_resource_schema '{iso standard 10303 part(41) version(7) object(1) support_
resource_schema(23)}';
```

NOTE 1 The schemas referenced above are specified in the following parts:

NOTE 2 See Annex D, Paragraph 24 for a graphical representation of this schema.

26.2 Fundamental concepts and assumptions

An identifier is a means that enables a processor to distinguish an information from other information of the same kind. A label is textual information assigned to an item of information in order to facilitate its access by a human being.

Product data items may have an identification. The formats used to define identifiers are quite various. This International Standard defines identifiers as strings. It does not specify the internal structure and content of these strings.

Product data items may have a label. The formats used to define labels are quite various. This International Standard defines identifiers as strings. It does not specify the internal structure and content of these strings.

26.3 Support resource type definitions

26.3.1 identifier 

An **identifier** is a string suitable for identifying some product data.

EXPRESS specification:

```
TYPE identifier = STRING;
END_TYPE;
```

26.3.2 label 

A **label** is an alphanumeric string that represents the human-interpretable name of something and shall have a natural-language meaning.

EXPRESS specification:

```
TYPE label = STRING;
END_TYPE;
```

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26.3.3 text

A **text** is an alphanumeric string intended to be read and understood by a human being. It is for information purposes only.

EXPRESS specification:

```
TYPE text = STRING;
END_TYPE;
```

26.4 Support resource function definitions

26.4.1 bag_to_set

The **bag_to_set** function converts BAGs into SETs.

EXPRESS specification:

```
FUNCTION bag_to_set(the_bag : BAG OF GENERIC) : SET OF GENERIC;
  LOCAL
    the_set : SET OF GENERIC := [];
  END_LOCAL;
  IF SIZEOF(the_bag) > 0 THEN
    REPEAT i := 1 TO HIINDEX(the_bag);
      the_set := the_set + the_bag[i];
    END_REPEAT;
  END_IF;
  RETURN (the_set);
END_FUNCTION;
```

Argument definitions:

the_bag: (input) the BAG that is to be converted into a SET.

26.4.2 type_check_function

The **type_check_function** function compares the names of the data types of the instance that is referred to by the parameter **the_type** with the set of entity names specified by **sub_names**.

Depending on the value of **criterion**, it returns TRUE if:

— case 0: the intersection of the two aggregates is not empty.

NOTE 1 If **sub_names** contains the entity names of all the subtypes of a particular entity data type, this case can be used to ensure that no instance of the supertype exists.

— case 1: the intersection of the two aggregates is empty.

NOTE 2 If **sub_names** contains the entity names of subtypes of a particular entity data type, this function can be used to ensure that no instance of these subtypes exists.

— case 2: the intersection of the two aggregates contains exactly one name.

NOTE 3 If **sub_names** contains the entity names of all the subtypes of a particular entity data type, this function can be used to ensure that no complex entity instance of subtypes of the entity exists and that only instances of subtypes of the entity exist.

NOTE 4 The concept of complex entity instance is defined in ISO 10303-11.

— case 3: the intersection of the two aggregates contains at most one name.

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NOTE 5 If **sub_names** contains the entity names of subtypes of a particular entity data type, this function can be used to ensure that no complex entity instance of these subtypes of the entity exists.

EXPRESS specification:

```
FUNCTION type_check_function(the_type : GENERIC;
                             sub_names : SET OF STRING;
                             criterion : INTEGER) : LOGICAL;
  IF NOT EXISTS(the_type) OR (SIZEOF(sub_names) = 0) THEN
    RETURN (UNKNOWN);
  ELSE
    CASE criterion OF
    0 :
      RETURN (SIZEOF(sub_names * TYPEOF(the_type)) > 0);
    1 :
      RETURN (SIZEOF(sub_names * TYPEOF(the_type)) = 0);
    2 :
      RETURN (SIZEOF(sub_names * TYPEOF(the_type)) = 1);
    3 :
      RETURN (SIZEOF(sub_names * TYPEOF(the_type)) <= 1);
    OTHERWISE :
      RETURN (UNKNOWN);
    END_CASE;
  END_IF;
END_FUNCTION;
```

Argument definitions:

the_type: (input) the instance whose data type is checked.

sub_names: (input) the set of names of the entity data types that are candidates for the type of **the_type**.

criterion: (input) the integer number value between 0 and 3.

```
END_SCHEMA; -- support_resource_schema
```

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Annex A
(normative)

Short names of entities

Requirements on the use of the short names are found in the implementation methods included in ISO 10303. EXPRESS entity names and the equivalent short names are available from:

http://standards.iso.org/iso/10303/tech/short_names/short-names.txt

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Annex B
(normative)

Information object registration

B.1 Document Identification

To provide for unambiguous identification of an information object in an open system, the object identifier

{ iso standard 10303 part(41) version(6) }

is assigned to this part of ISO 10303. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2 Schema identification

B.2.1 action-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(8) object(1) action_schema(1)}

is assigned to the action-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.2 application-context-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(8) object(1) application_context_schema(2)}

is assigned to the application-context-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.3 approval-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(7) object(1) approval_schema(3)}

is assigned to the approval-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.4 basic-attribute-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(9) object(1) basic_attribute_schema(4)}

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is assigned to the basic-attribute-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.5 certification-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(3) object(1) certification_schema(5)}

is assigned to the certification-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.6 contract-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(3) object(1) contract_schema(6)}

is assigned to the contract-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.7 date-time-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(7) object(1) date_time_schema(7)}

is assigned to the date-time-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.8 document-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(4) object(1) document_schema(8)}

is assigned to the document-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.9 effectivity-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(8) object(1) effectivity_schema(9)}

is assigned to the effectivity-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.10 experience-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(4) object(1) experience_schema(10)}

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is assigned to the experience-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.11 external-reference-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(7) object(1) external_reference_schema(11)}

is assigned to the external-reference-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.12 group-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(7) object(1) group_schema(12)}

is assigned to the group-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.13 language-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(1) object(1) location_schema(13)}

is assigned to the language-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.14 location-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(4) object(1) location_schema(14)}

is assigned to the location-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.15 management-resources-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(8) object(1) management_resources_schema(15)}

is assigned to the management-resources-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.16 measure-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(8) object(1) measure_schema(16)}

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is assigned to the measure-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.17 person-organization-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(8) object(1) person_organization_schema(17)}

is assigned to the person-organization-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.18 product-definition-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(9) object(1) product_definition_schema(18)}

is assigned to the product-definition-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.19 product-property-definition-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(9) object(1) product_property_definition_schema(19)}

is assigned to the product-property-definition-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.20 product-property-representation-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(9) object(1) product_property_representation_schema(20)}

is assigned to the product-property-representation-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.21 qualifications-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(6) object(1) qualifications_schema(21)}

is assigned to the qualifications-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.22 security-classification-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(3) object(1) security_classification_schema(22)}

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is assigned to the security-classification-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

B.2.23 support-resource-schema schema identification

To provide for unambiguous identification of the schema specifications given in this integrated resource in an open information system, the object identifiers are assigned as follows:

{iso standard 10303 part(41) version(7) object(1) support_resource_schema(23)}

is assigned to the support-resource-schema schema. The meaning of this value is defined in ISO/IEC 8824-1, and is described in ISO 10303-1.

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Annex C
(normative)

Computer interpretable listings

This annex references a listing of the EXPRESS entity names and corresponding short names as specified or referenced in this part of ISO 10303. It also provides a listing of each EXPRESS schema specified in this part of ISO 10303 without comments or other explanatory text. These listings are available in computer-interpretable form in Table C.1 and can be found at the following URLs:

Short names http://standards.iso.org/iso/10303/tech/short_names/short-names.txt

EXPRESS http://standards.iso.org/iso/10303/smrl/v8/tech/smrlv8.zip

Table C.1

HTML file	ASCII file
action_schema.htm0; / current-metanorma-collection/action_schema.htm	action_schema.exp0; / current-metanorma-collection/action_schema.exp
application_context_schema.htm0; / current-metanorma-collection/application_context_schema.htm	application_context_schema.exp0; / current-metanorma-collection/application_context_schema.exp
approval_schema.htm0; / current-metanorma-collection/approval_schema.htm	approval_schema.exp0; / current-metanorma-collection/approval_schema.exp
basic_attribute_schema.htm0; / current-metanorma-collection/basic_attribute_schema.htm	basic_attribute_schema.exp0; / current-metanorma-collection/basic_attribute_schema.exp
certification_schema.htm0; / current-metanorma-collection/certification_schema.htm	certification_schema.exp0; / current-metanorma-collection/certification_schema.exp
contract_schema.htm0; / current-metanorma-collection/contract_schema.htm	contract_schema.exp0; / current-metanorma-collection/contract_schema.exp
date_time_schema.htm0; / current-metanorma-collection/date_time_schema.htm	date_time_schema.exp0; / current-metanorma-collection/date_time_schema.exp
document_schema.htm0; / current-metanorma-collection/document_schema.htm	document_schema.exp0; / current-metanorma-collection/document_schema.exp
effectivity_schema.htm0; / current-metanorma-collection/effectivity_schema.htm	effectivity_schema.exp0; / current-metanorma-collection/effectivity_schema.exp
experience_schema.htm0; / current-metanorma-collection/experience_schema.htm	experience_schema.exp0; / current-metanorma-collection/experience_schema.exp
external_reference_schema.htm0; / current-metanorma-collection/external_reference_schema.htm	external_reference_schema.exp0; / current-metanorma-collection/external_reference_schema.exp
group_schema.htm0; / current-metanorma-collection/group_schema.htm	group_schema.exp0; / current-metanorma-collection/group_schema.exp
language_schema.htm0; / current-metanorma-collection/language_schema.htm	language_schema.exp0; / current-metanorma-collection/language_schema.exp
location_schema.htm0; / current-metanorma-collection/location_schema.htm	location_schema.exp0; / current-metanorma-collection/location_schema.exp
management_resources_schema.htm0; / current-metanorma-collection/management_resources_schema.htm	management_resources_schema.exp0; / current-metanorma-collection/management_resources_schema.exp
measure_schema.htm0; / current-metanorma-collection/measure_schema.htm	measure_schema.exp0; / current-metanorma-collection/measure_schema.exp
person_organization_schema.htm0; / current-metanorma-collection/person_organization_schema.htm	person_organization_schema.exp0; / current-metanorma-collection/person_organization_schema.exp

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(continued)

HTML file	ASCII file
product_definition_schema.htm0; / current-metanorma-collection/product_definition_schema.htm	product_definition_schema.exp0; / current-metanorma-collection/product_definition_schema.exp
product_property_definition_schema.htm0; / current-metanorma-collection/product_property_definition_schema.htm	product_property_definition_schema.exp0; / current-metanorma-collection/product_property_definition_schema.exp
product_property_representation_schema.htm0; / current-metanorma-collection/product_property_representation_schema.htm	product_property_representation_schema.exp0; / current-metanorma-collection/product_property_representation_schem
qualifications_schema.htm0; / current-metanorma-collection/qualifications_schema.htm	qualifications_schema.exp0; / current-metanorma-collection/qualifications_schema.exp
security_classification_schema.htm0; / current-metanorma-collection/security_classification_schema.htm	security_classification_schema.exp0; / current-metanorma-collection/security_classification_schema.exp
support_resource_schema.htm0; / current-metanorma-collection/support_resource_schema.htm	support_resource_schema.exp0; / current-metanorma-collection/support_resource_schema.exp
NOTE The information provided in computer-interpretable form at the above URLs is informative. The information that is contained in the body of this part of ISO 10303 is normative.	

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Annex D
(normative)

EXPRESS-G diagrams

The diagrams in this annex correspond to the EXPRESS schemas specified in this part of ISO 10303. The diagrams use the EXPRESS-G graphical notation for the EXPRESS language. EXPRESS-G is defined in ISO 10303-11.

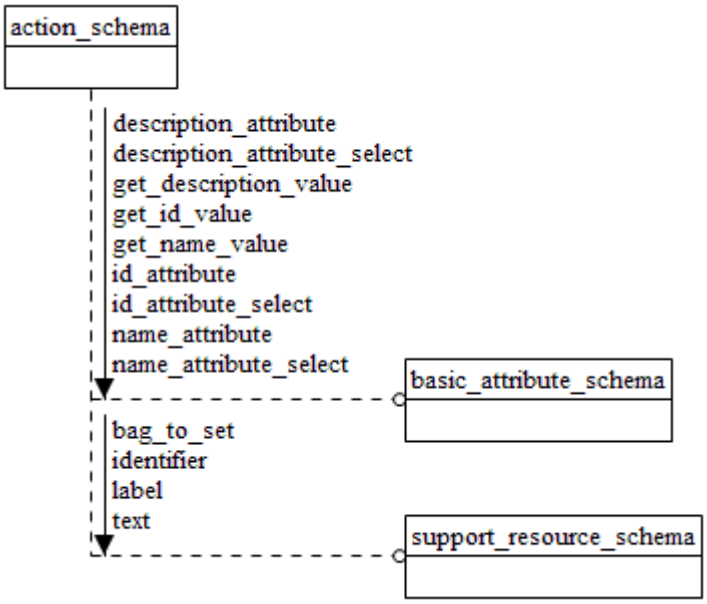


Figure D.1 — EXPRESS-G diagram of the action_schema (1 of 2)

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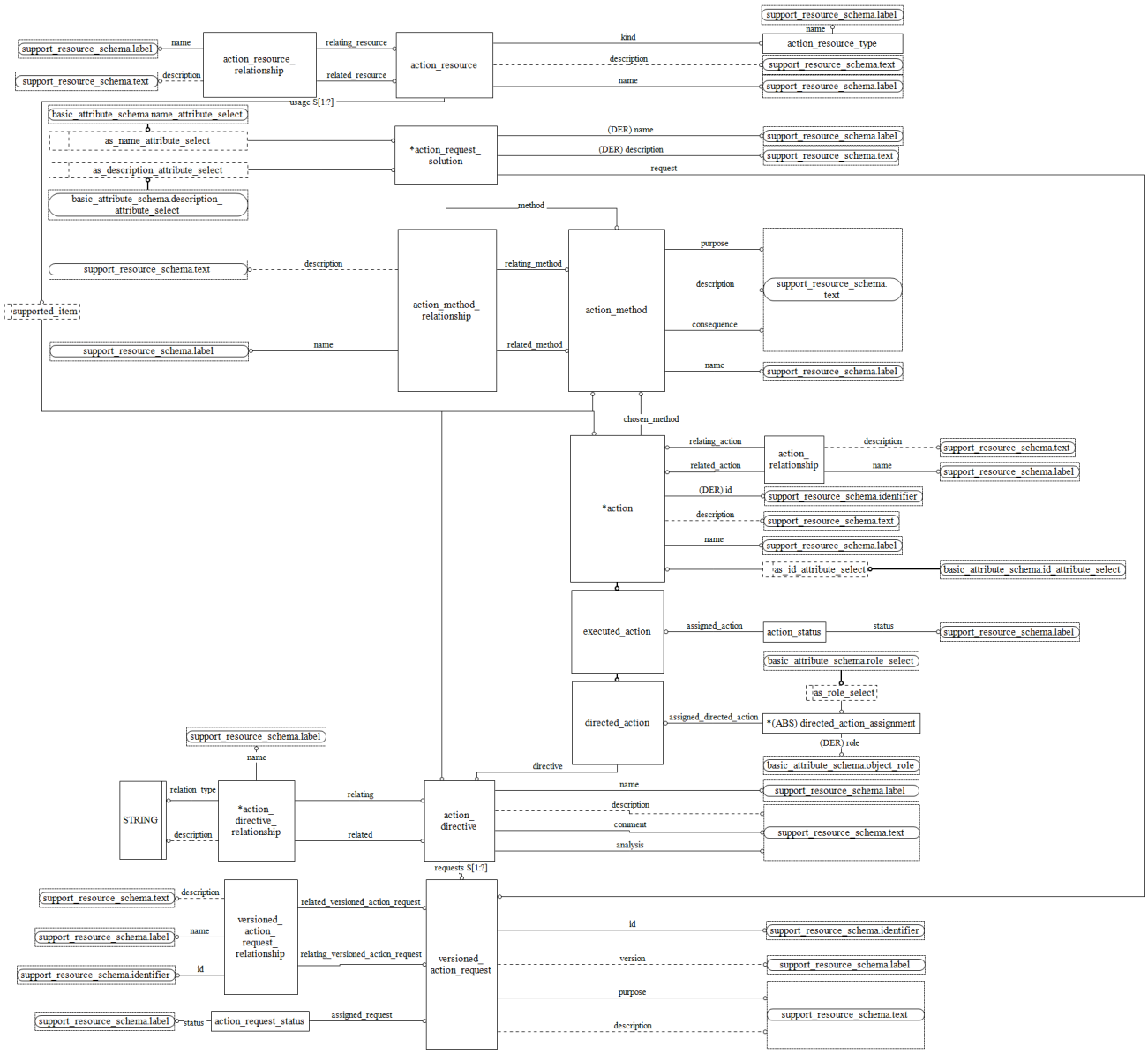


Figure D.2 — EXPRESS-G diagram of the action_schema (2 of 2)

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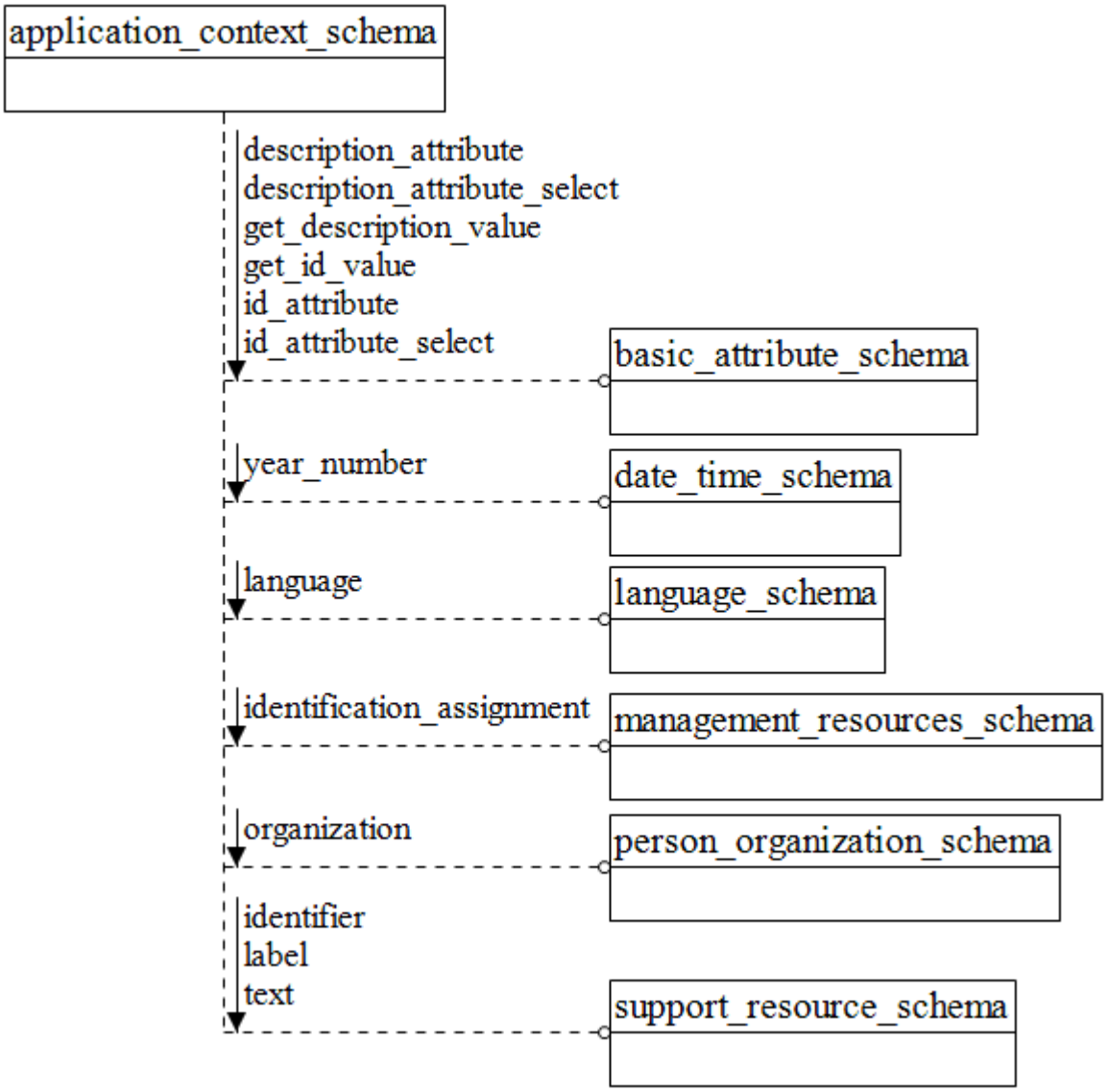


Figure D.3 — EXPRESS-G diagram of the application_context_schema (1 of 2)

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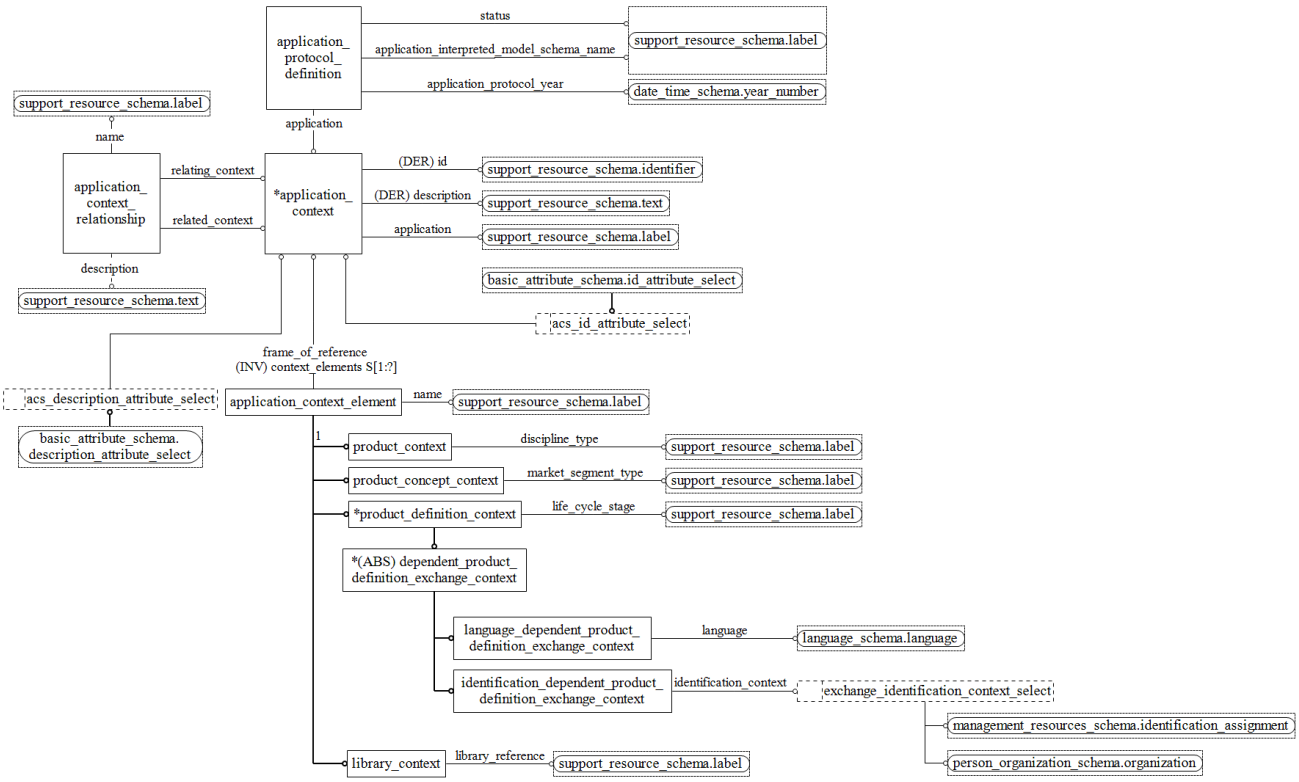


Figure D.4 — EXPRESS-G diagram of the application_context_schema (2 of 2)

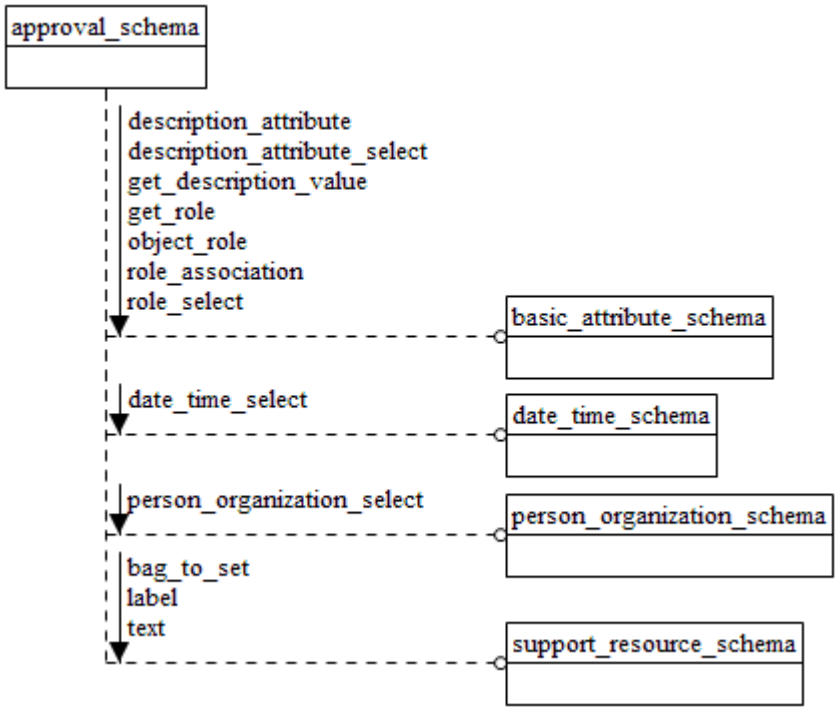


Figure D.5 — EXPRESS-G diagram of the approval_schema (1 of 2)

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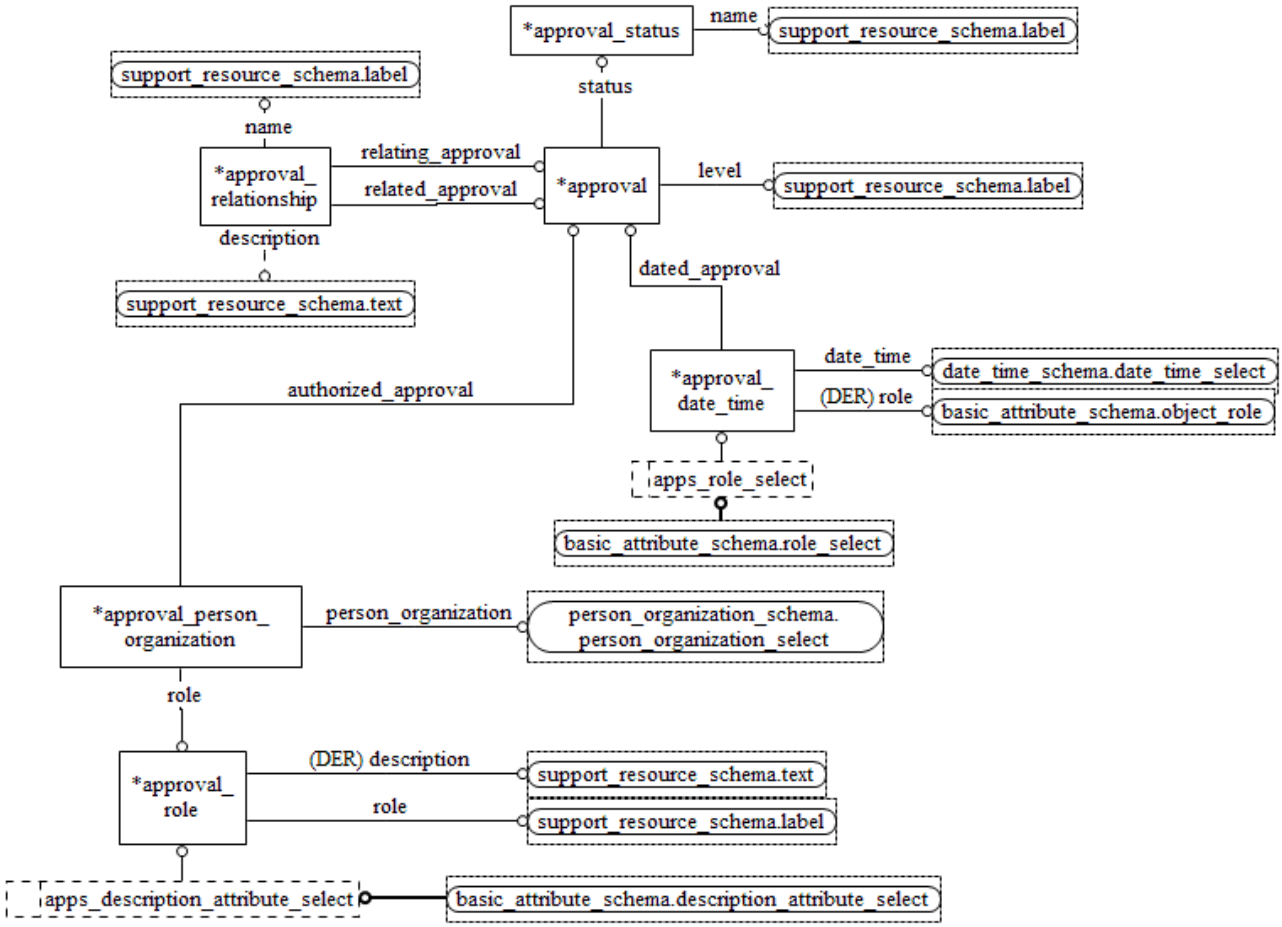


Figure D.6 — EXPRESS-G diagram of the approval_schema (2 of 2)

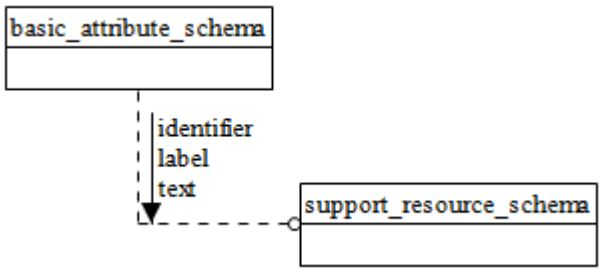


Figure D.7 — EXPRESS-G diagram of the basic_attribute_schema (1 of 5)

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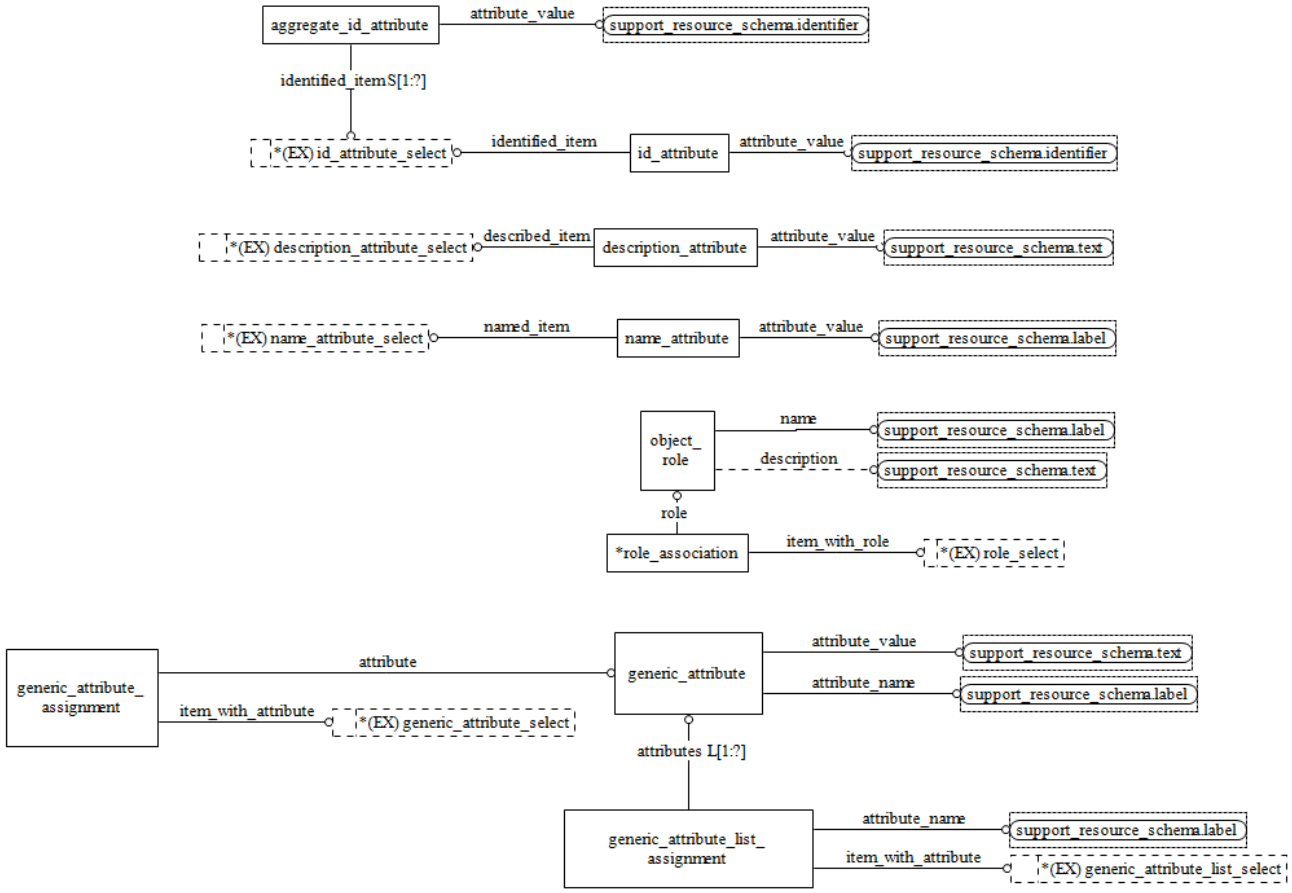


Figure D.8 — EXPRESS-G diagram of the basic_attribute_schema (2 of 5)

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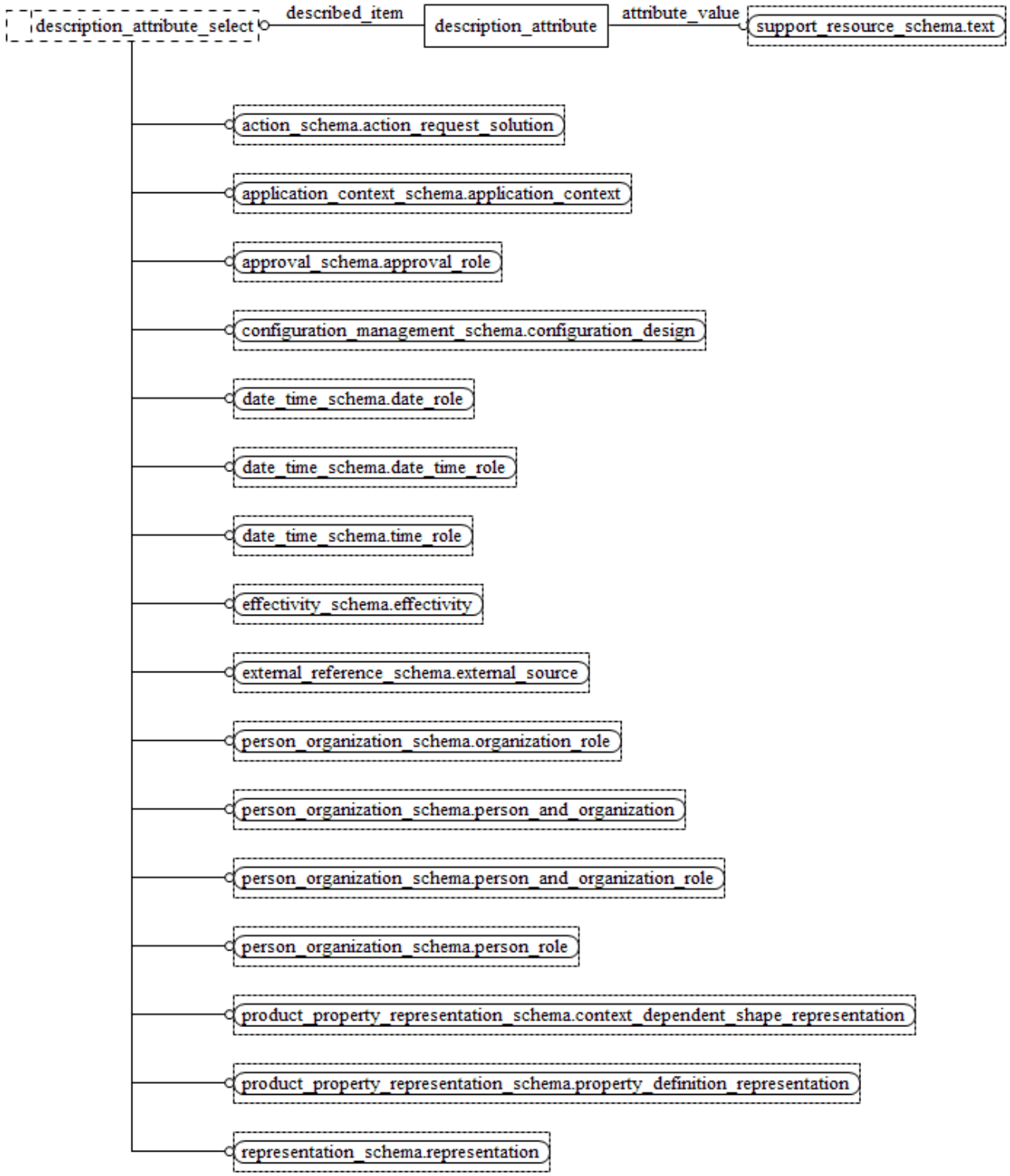


Figure D.9 — EXPRESS-G diagram of the basic_attribute_schema (3 of 5)

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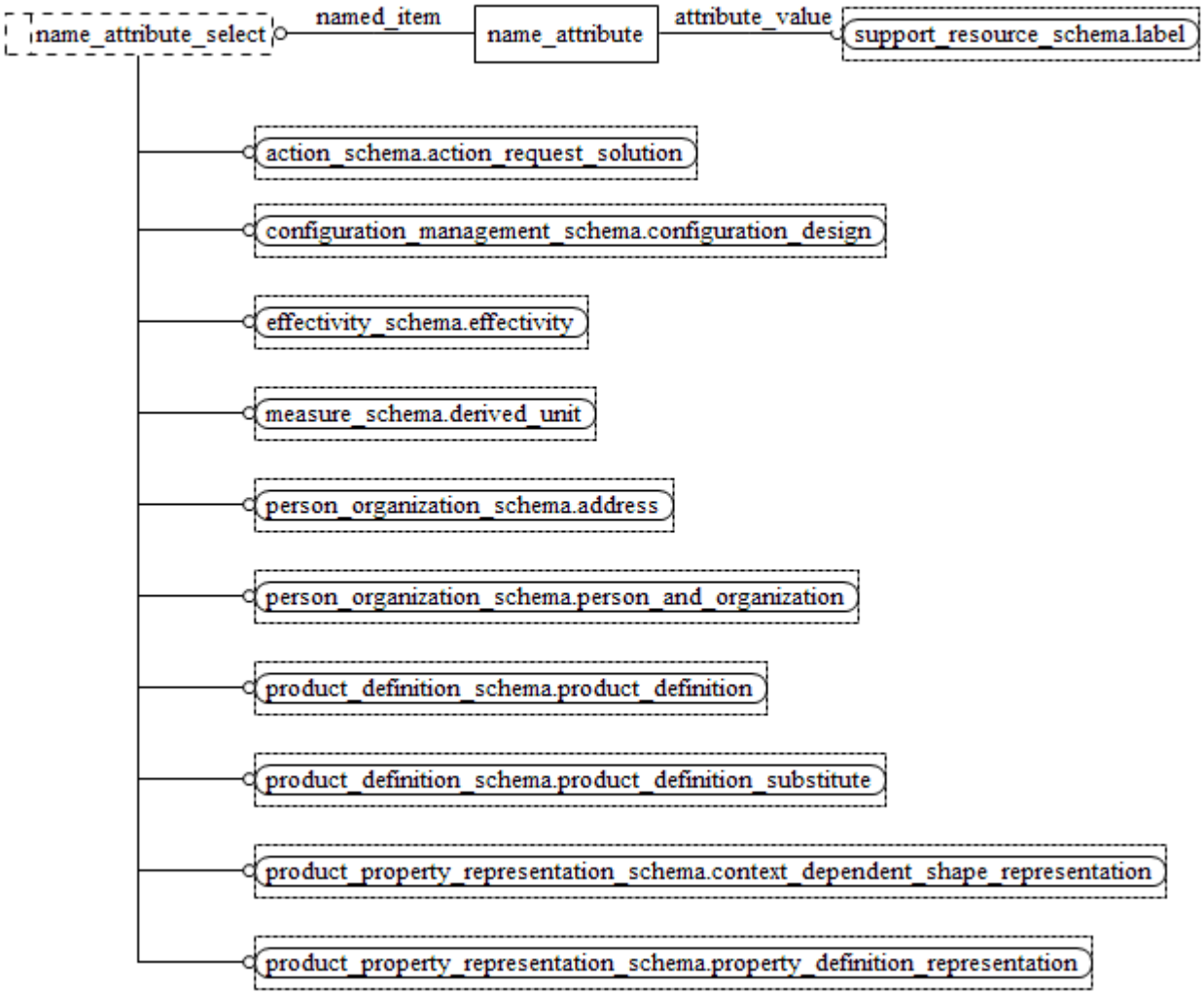


Figure D.10 — EXPRESS-G diagram of the basic_attribute_schema (4 of 5)

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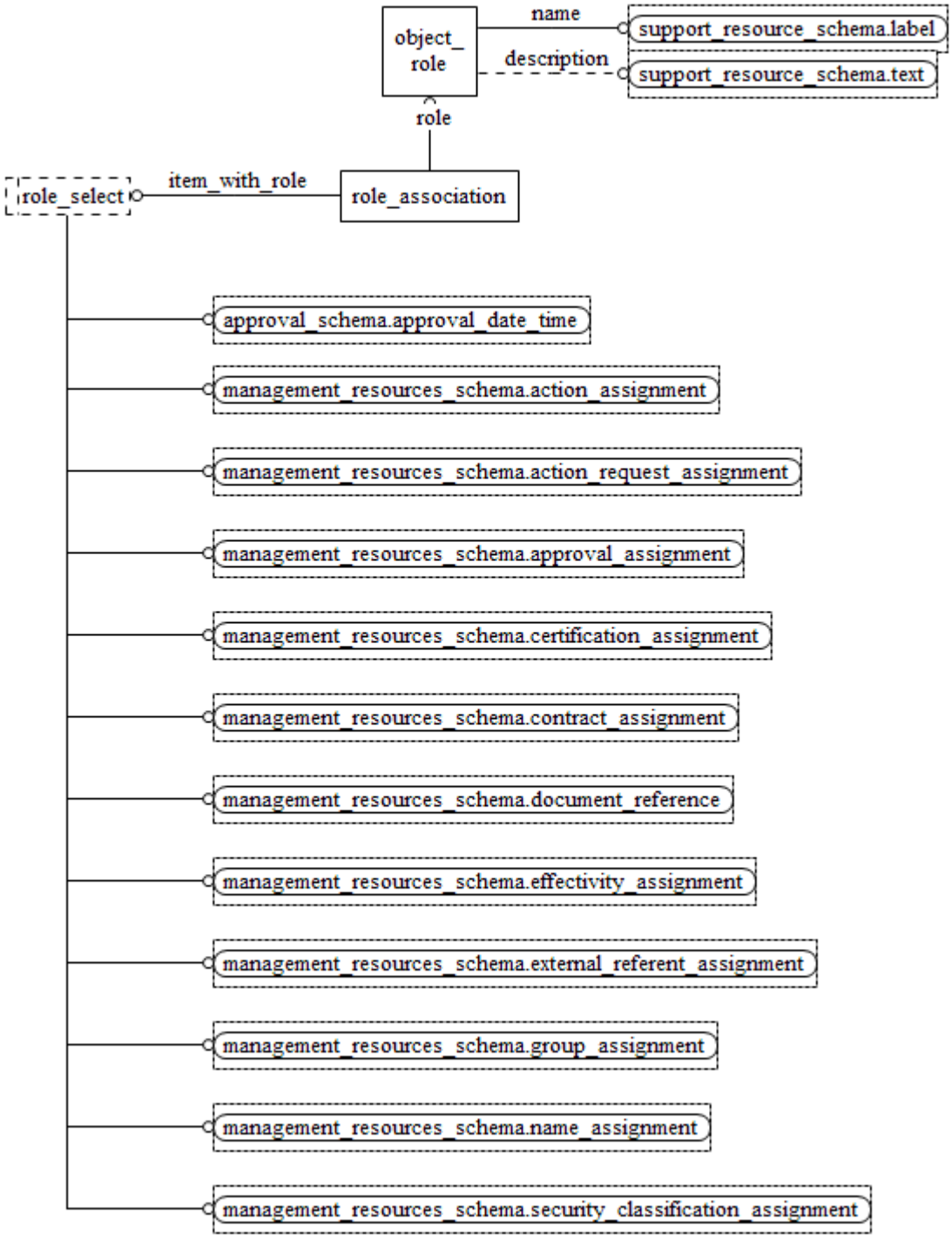


Figure D.11 — EXPRESS-G diagram of the basic_attribute_schema (5 of 5)

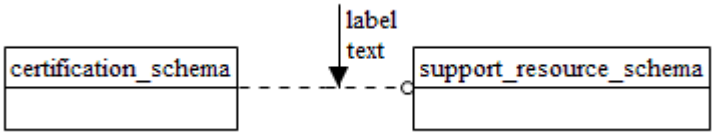


Figure D.12 — EXPRESS-G diagram of the certification_schema (1 of 2)

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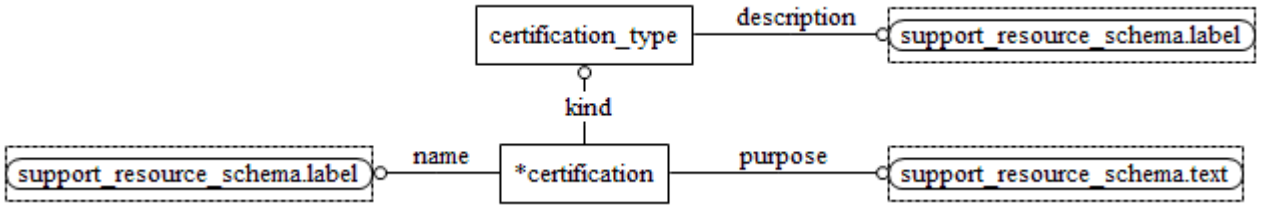


Figure D.13 — EXPRESS-G diagram of the certification_schema (2 of 2)

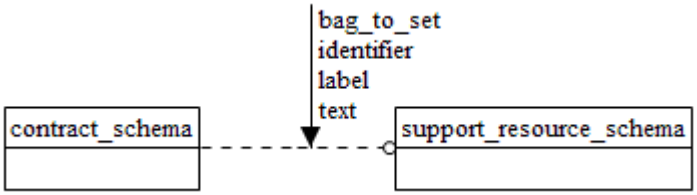


Figure D.14 — EXPRESS-G diagram of the contract_schema (1 of 2)

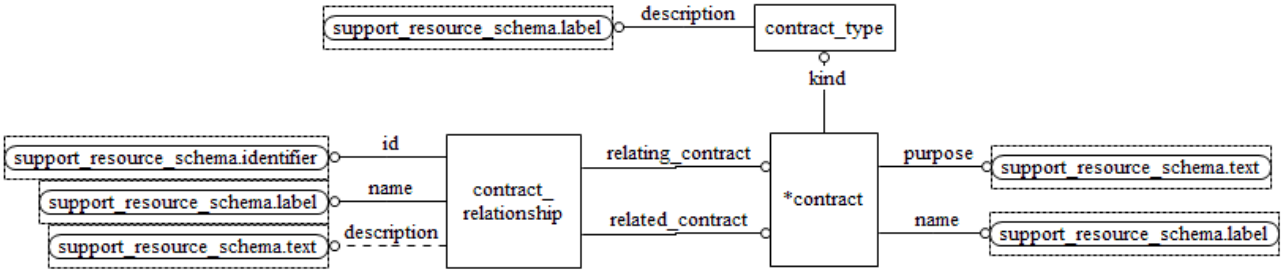


Figure D.15 — EXPRESS-G diagram of the contract_schema (2 of 2)

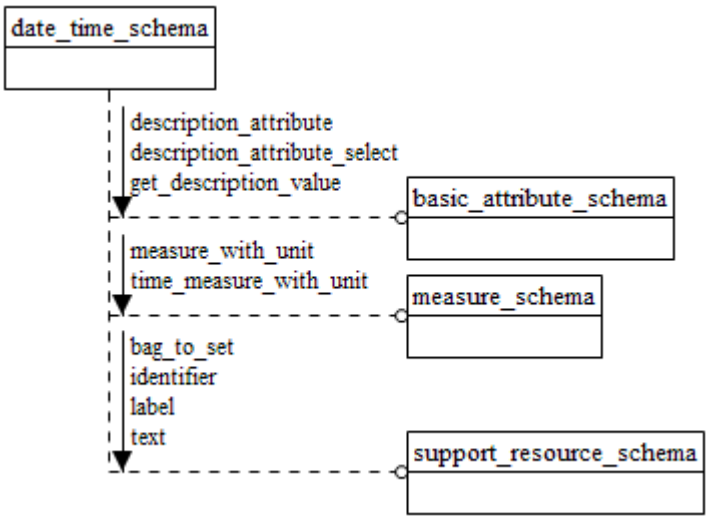


Figure D.16 — EXPRESS-G diagram of the date_time_schema (1 of 3)

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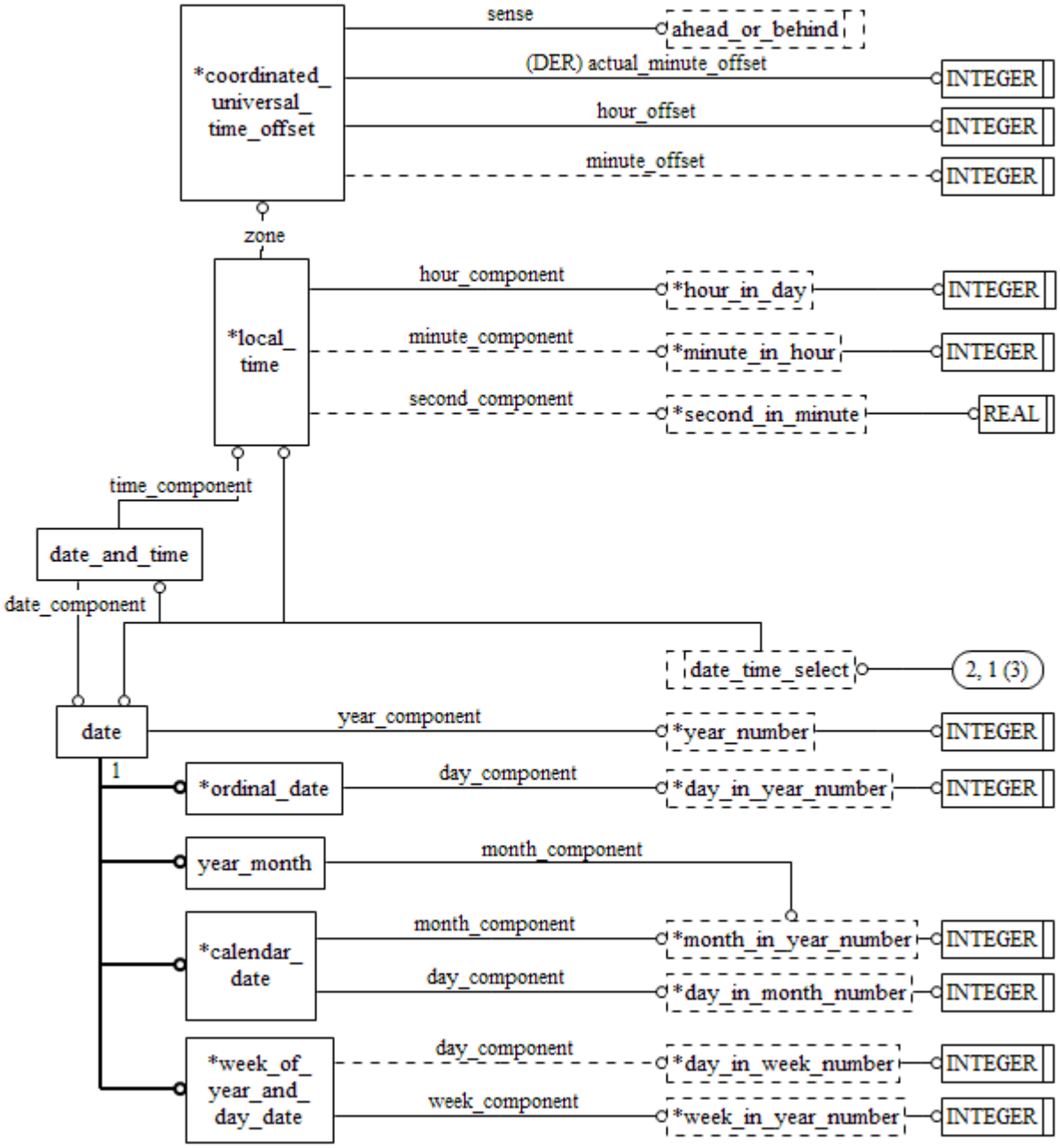


Figure D.17 — EXPRESS-G diagram of the date_time_schema (2 of 3)

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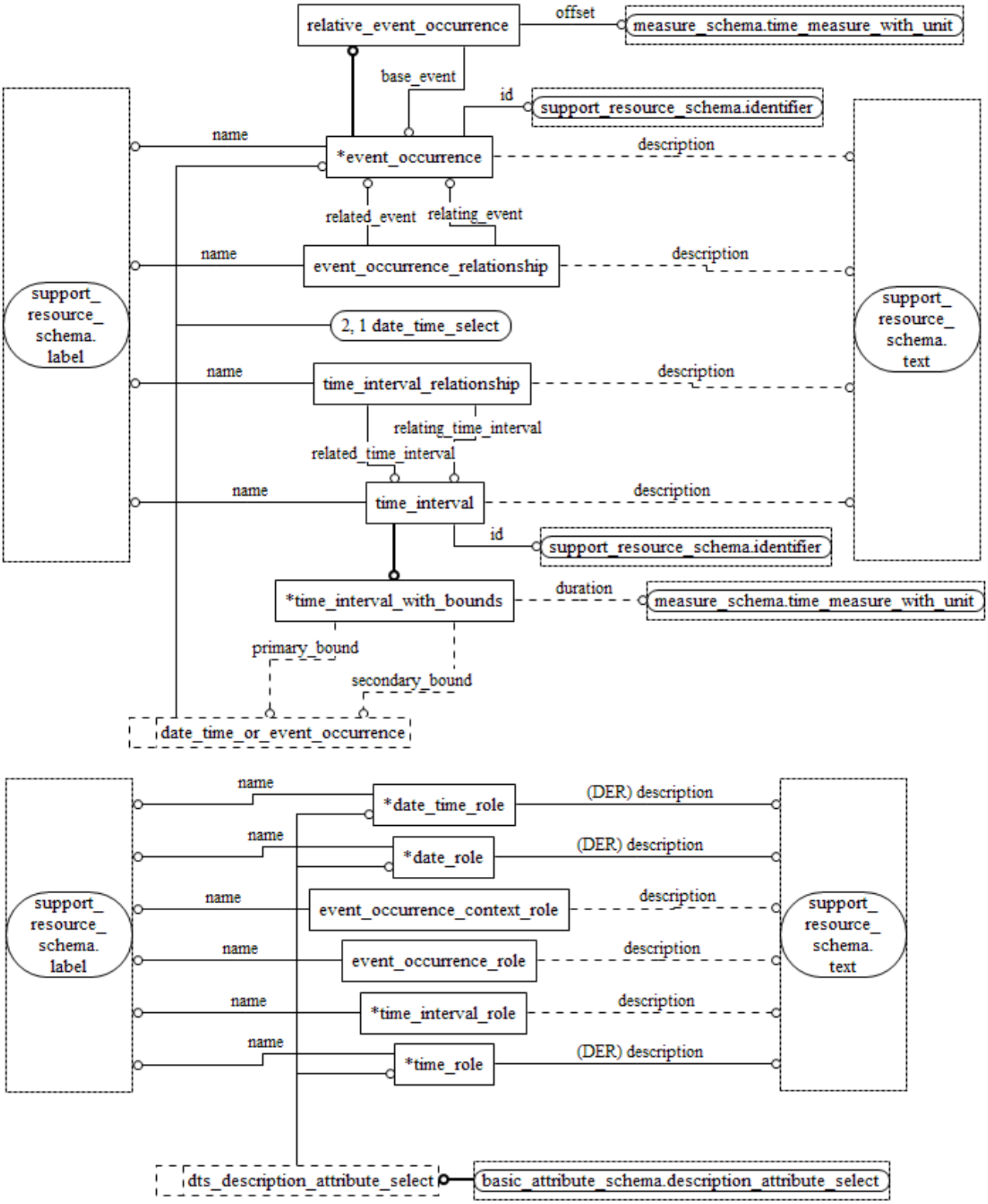


Figure D.18 — EXPRESS-G diagram of the date_time_schema (3 of 3)

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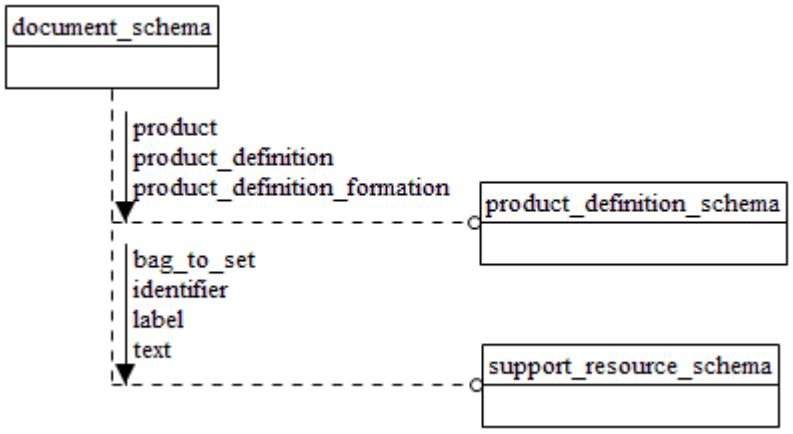


Figure D.19 — EXPRESS-G diagram of the document_schema (1 of 2)

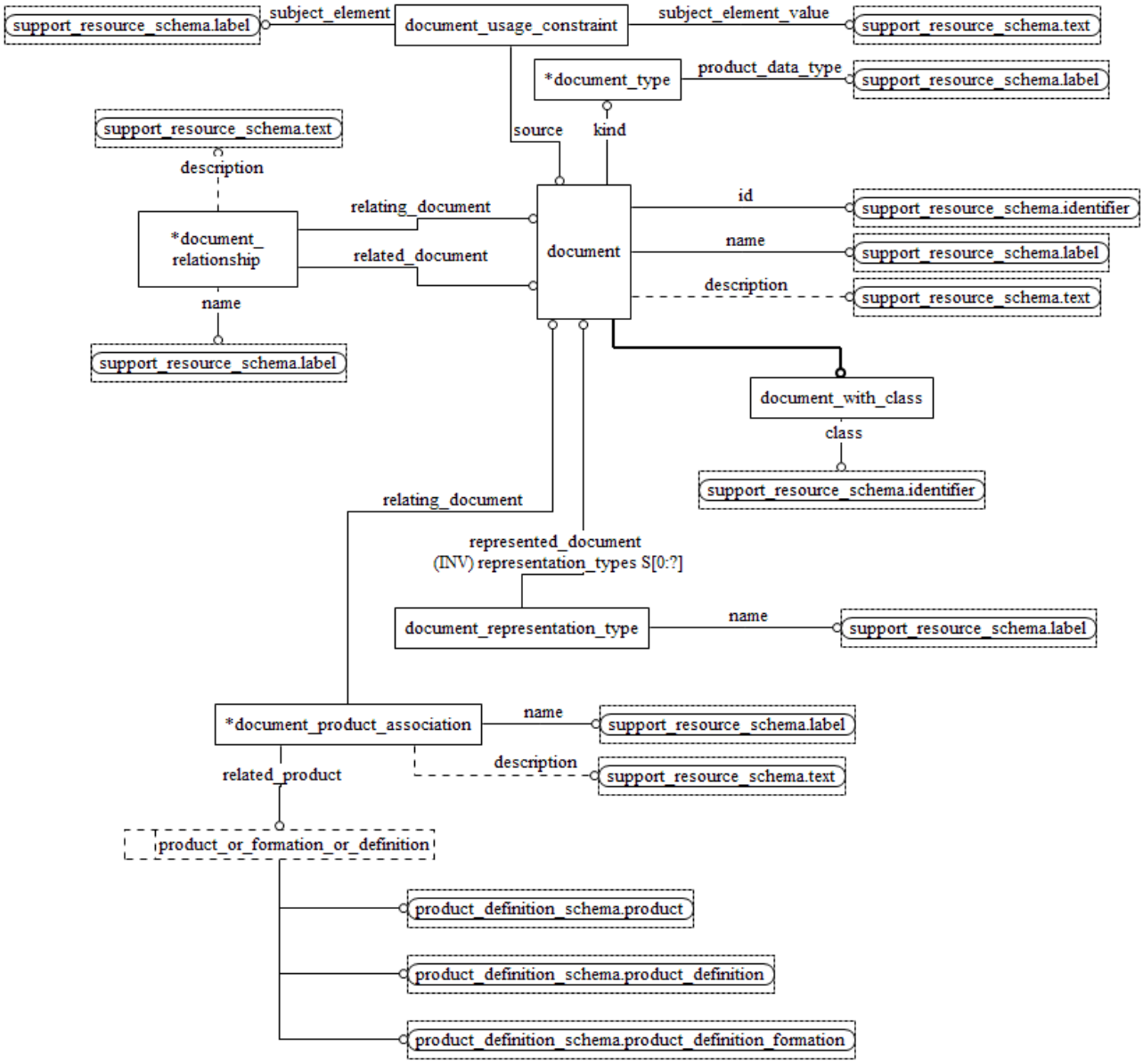


Figure D.20 — EXPRESS-G diagram of the document_schema (2 of 2)

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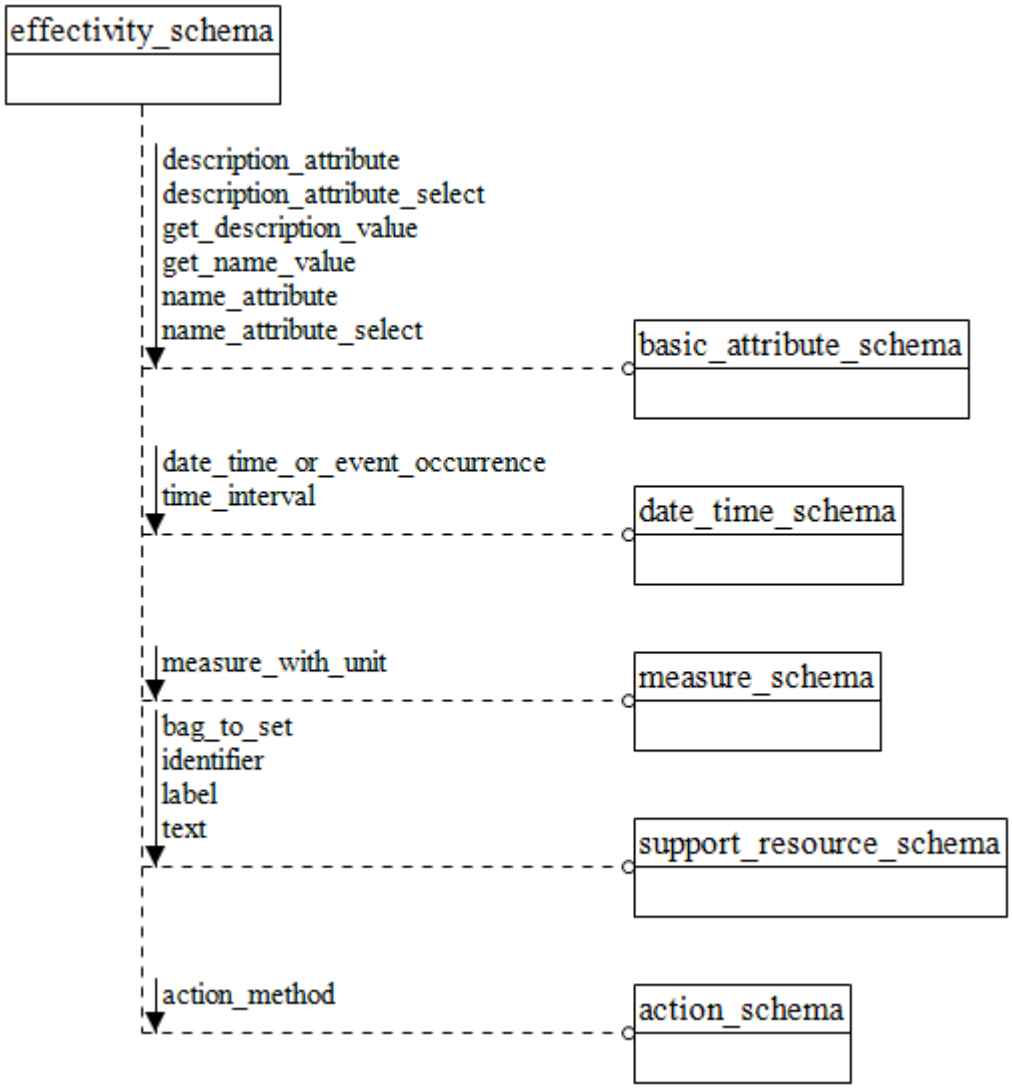


Figure D.21 — EXPRESS-G diagram of the effectivity_schema (1 of 2)

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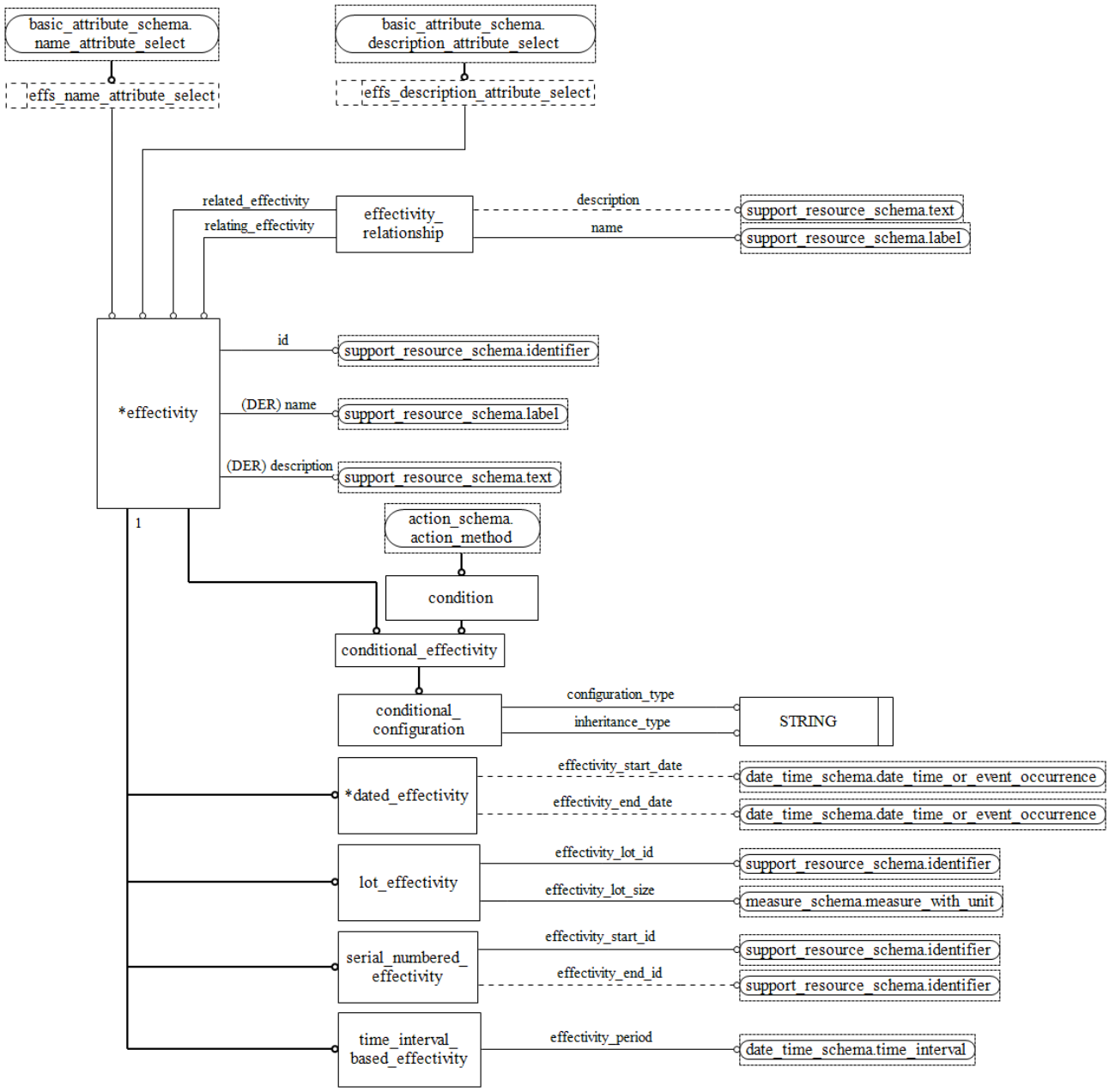


Figure D.22 — EXPRESS-G diagram of the effectivity_schema (2 of 2)

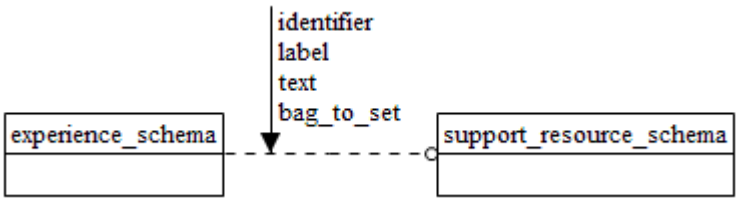


Figure D.23 — EXPRESS-G diagram of the experience_schema (1 of 2)

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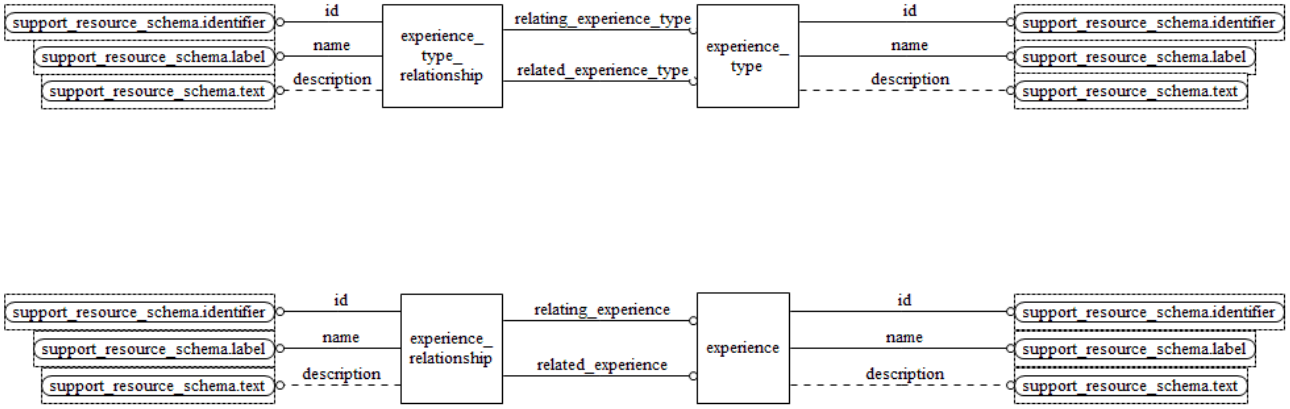


Figure D.24 — EXPRESS-G diagram of the experience_schema (2 of 2)

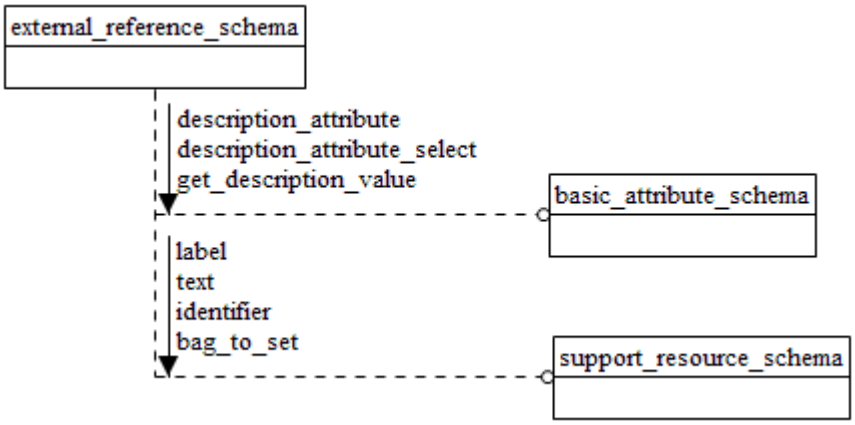


Figure D.25 — EXPRESS-G diagram of the external_reference_schema (1 of 2)

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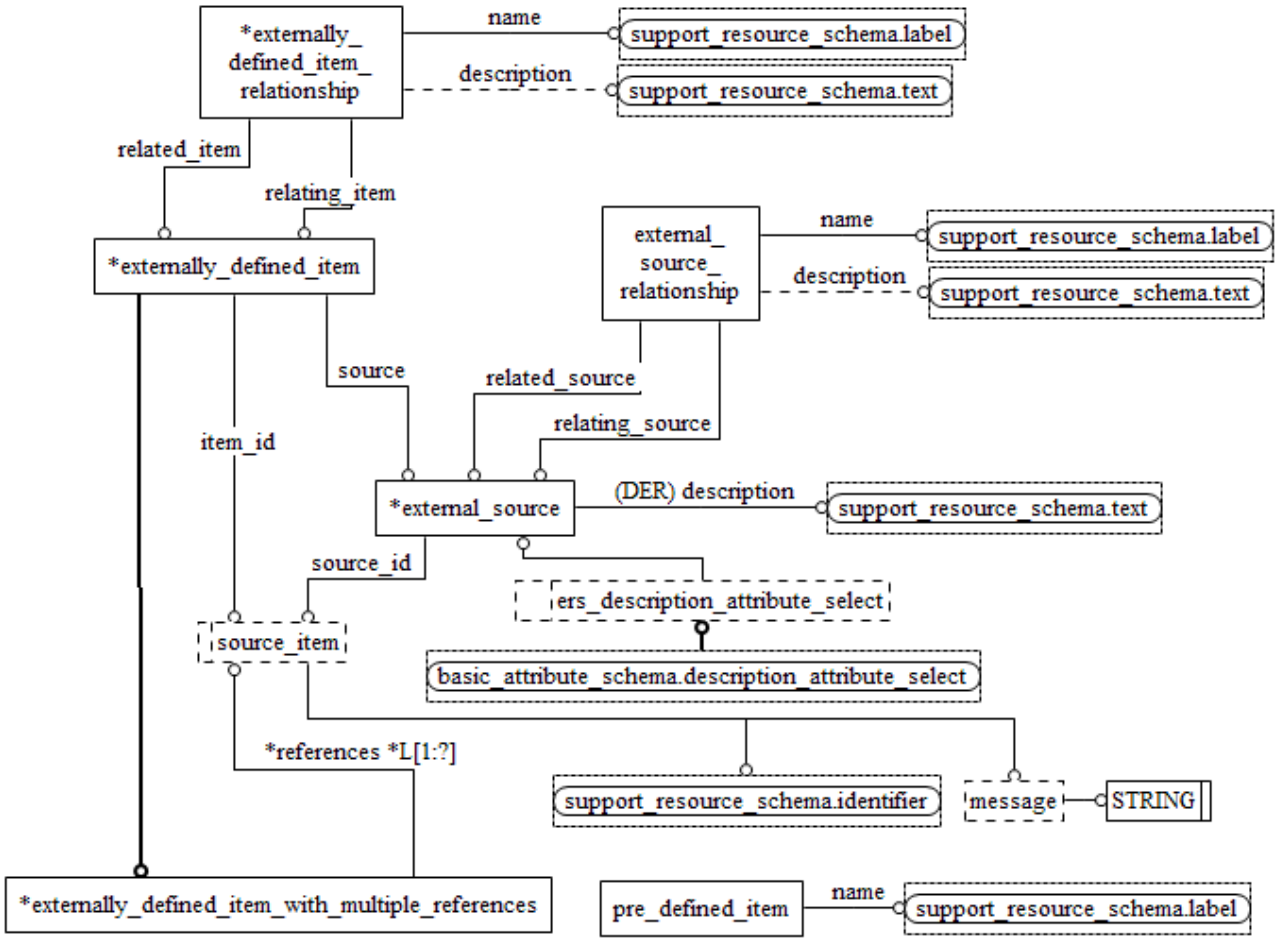


Figure D.26 — EXPRESS-G diagram of the external_reference_schema (2 of 2)

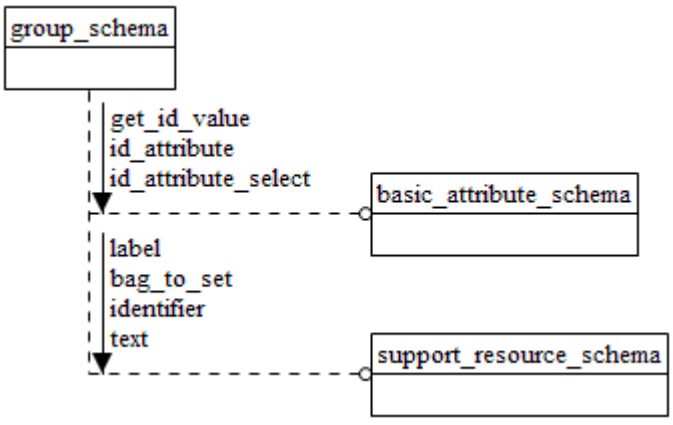


Figure D.27 — EXPRESS-G diagram of the group_schema (1 of 2)

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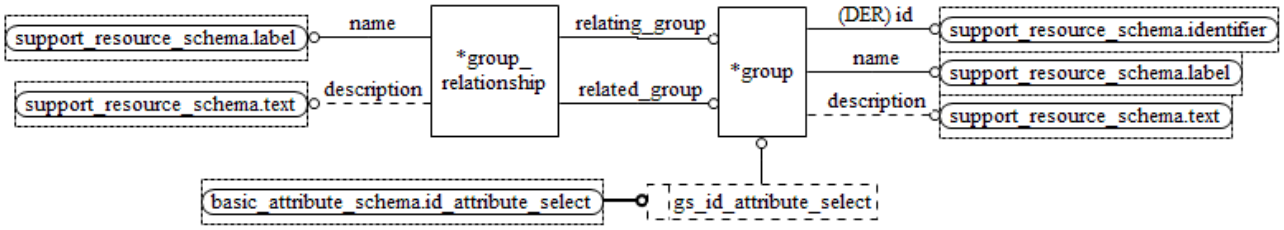


Figure D.28 — EXPRESS-G diagram of the group_schema (2 of 2)

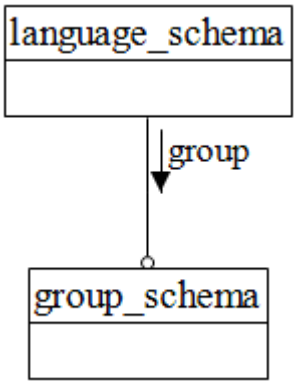


Figure D.29 — EXPRESS-G diagram of the language_schema (1 of 2)

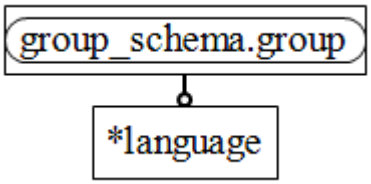


Figure D.30 — EXPRESS-G diagram of the language_schema (2 of 2)

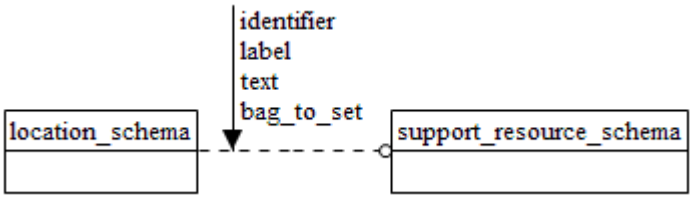


Figure D.31 — EXPRESS-G diagram of the location_schema (1 of 2)

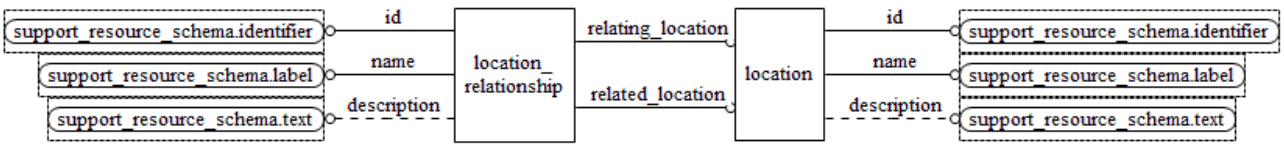


Figure D.32 — EXPRESS-G diagram of the location_schema (2 of 2)

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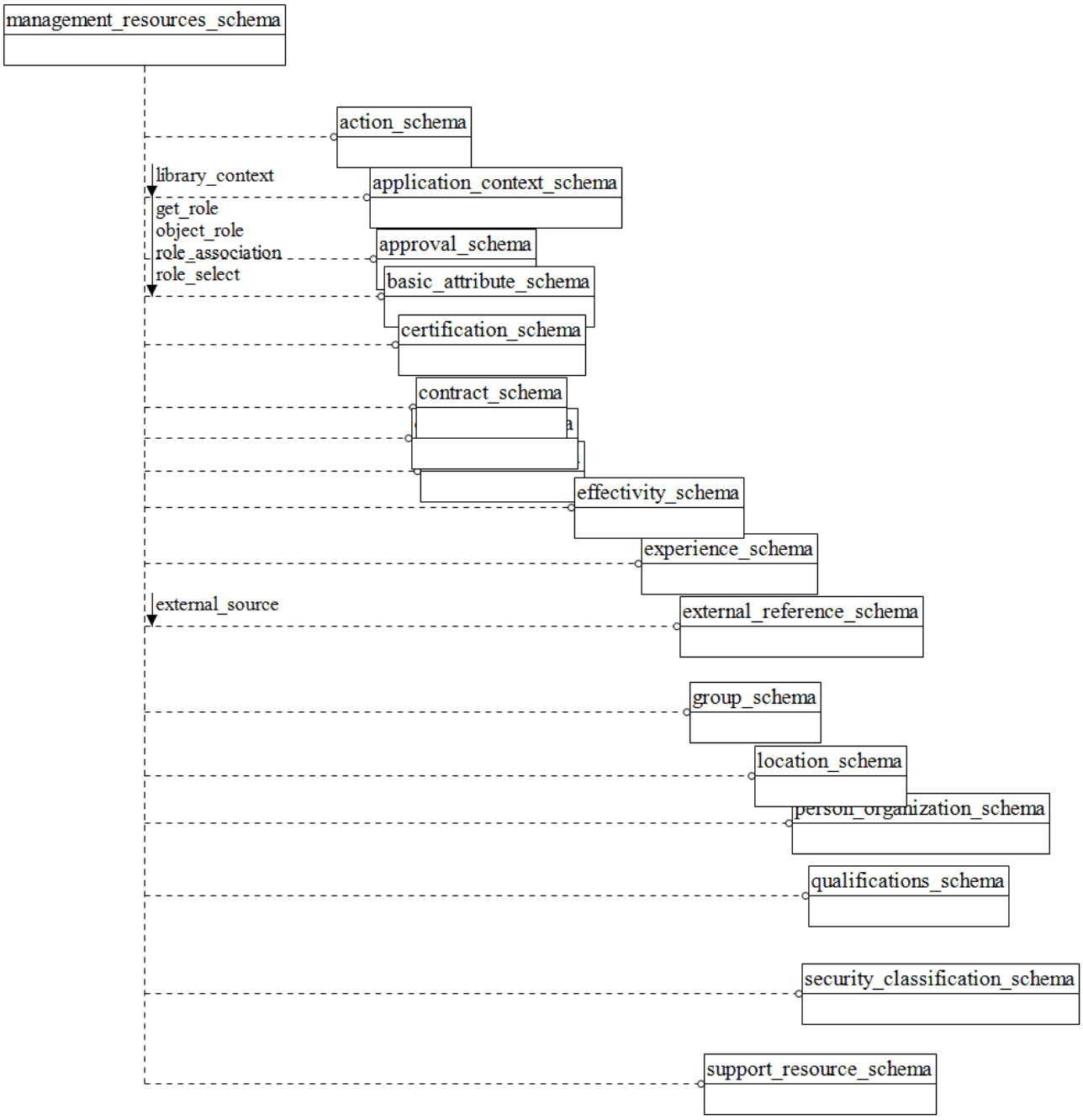


Figure D.33 — EXPRESS-G diagram of the management_resources_schema (1 of 9)

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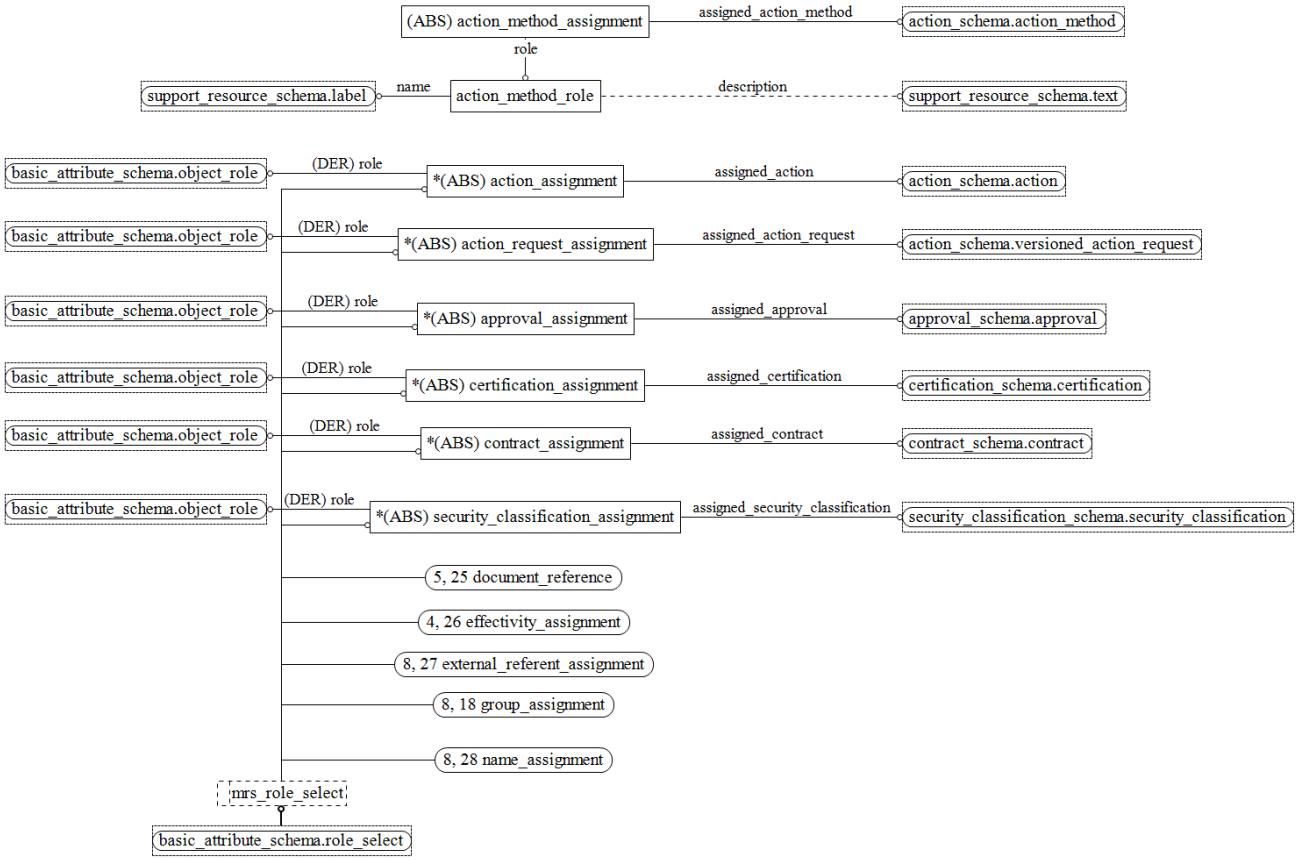


Figure D.34 — EXPRESS-G diagram of the management_resources_schema (2 of 9)

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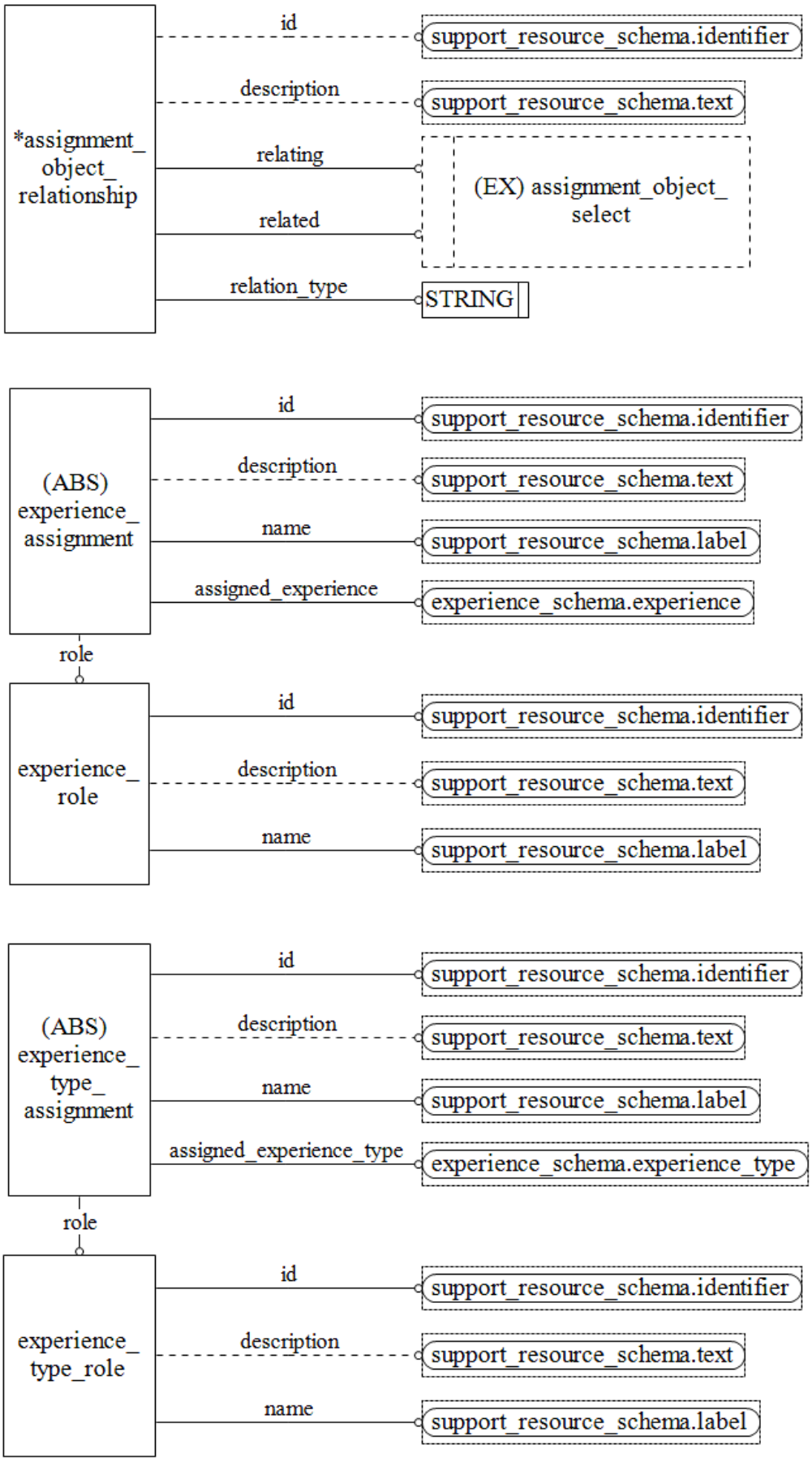


Figure D.35 — EXPRESS-G diagram of the management_resources_schema (3 of 9)

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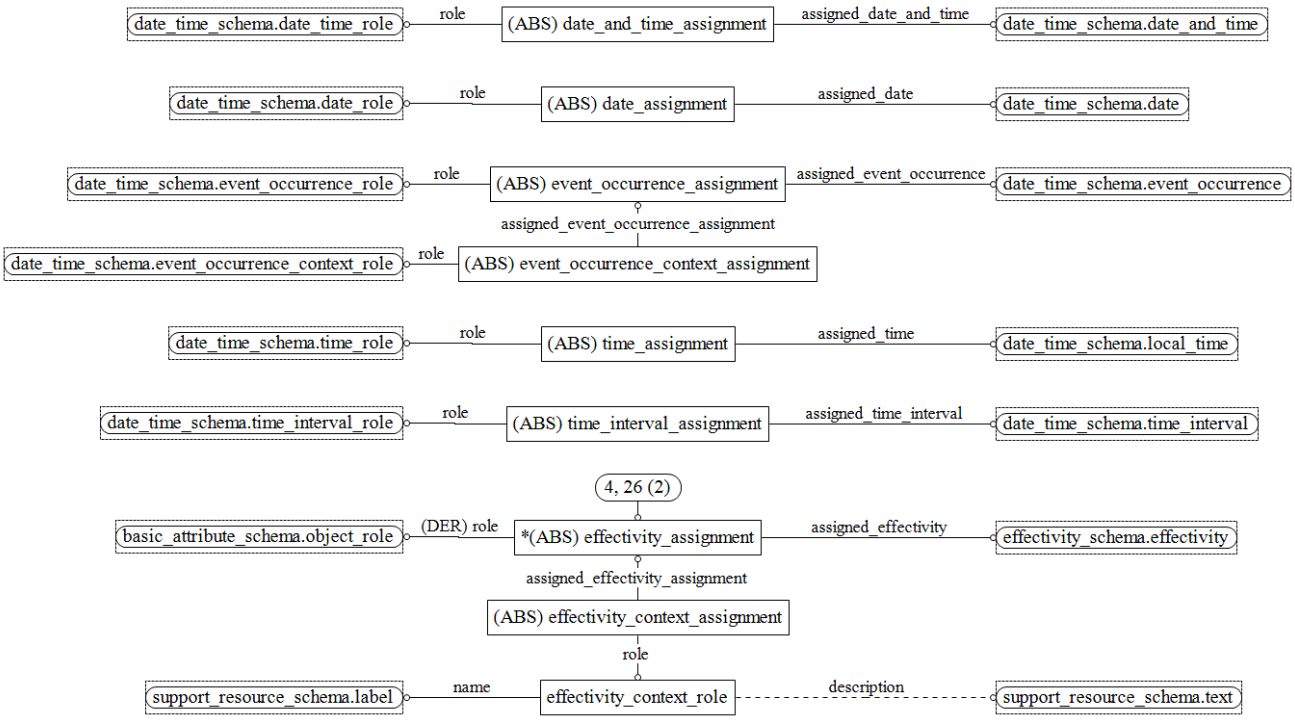


Figure D.36 — EXPRESS-G diagram of the management_resources_schema (4 of 9)

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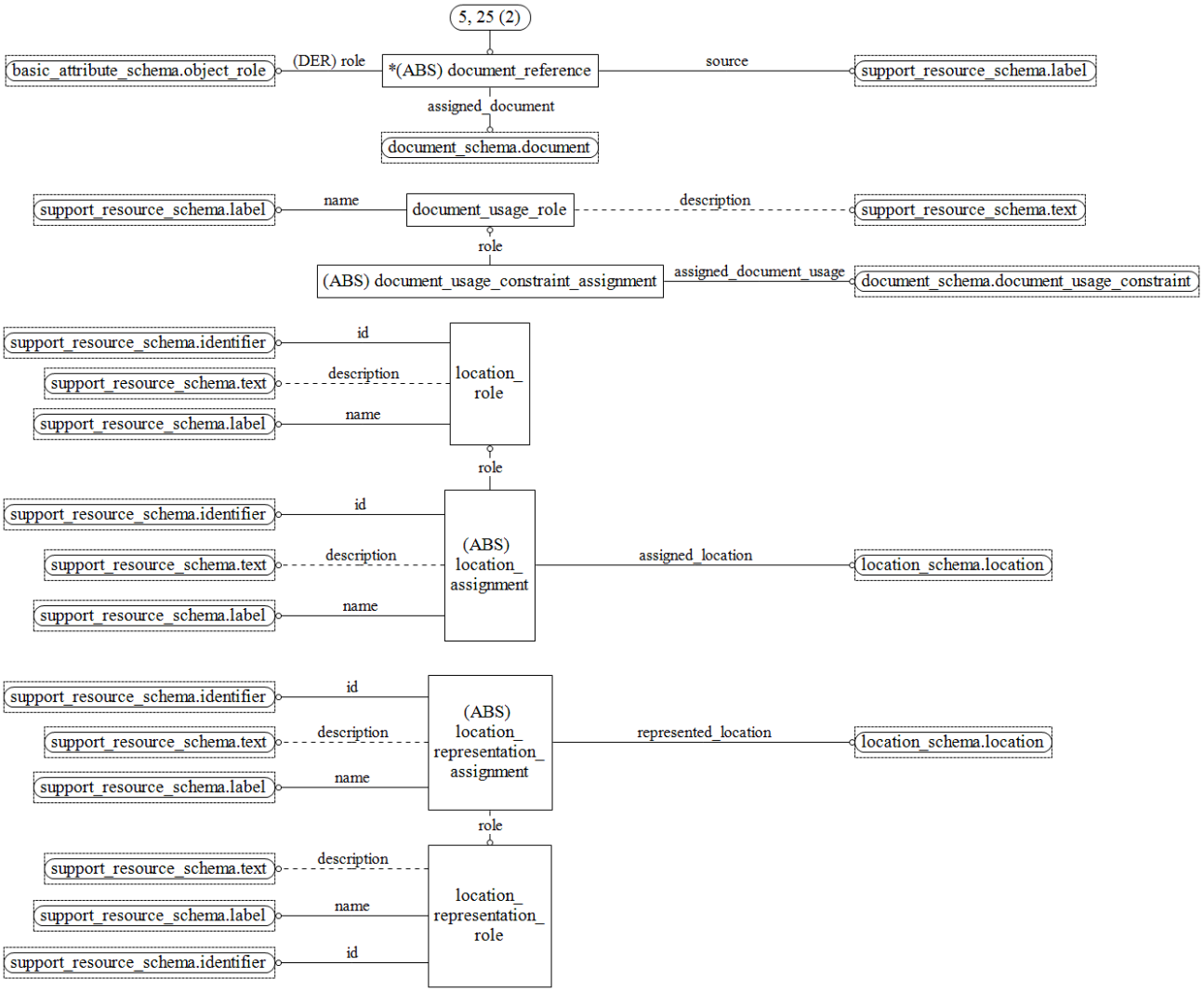


Figure D.37 — EXPRESS-G diagram of the management_resources_schema (5 of 9)

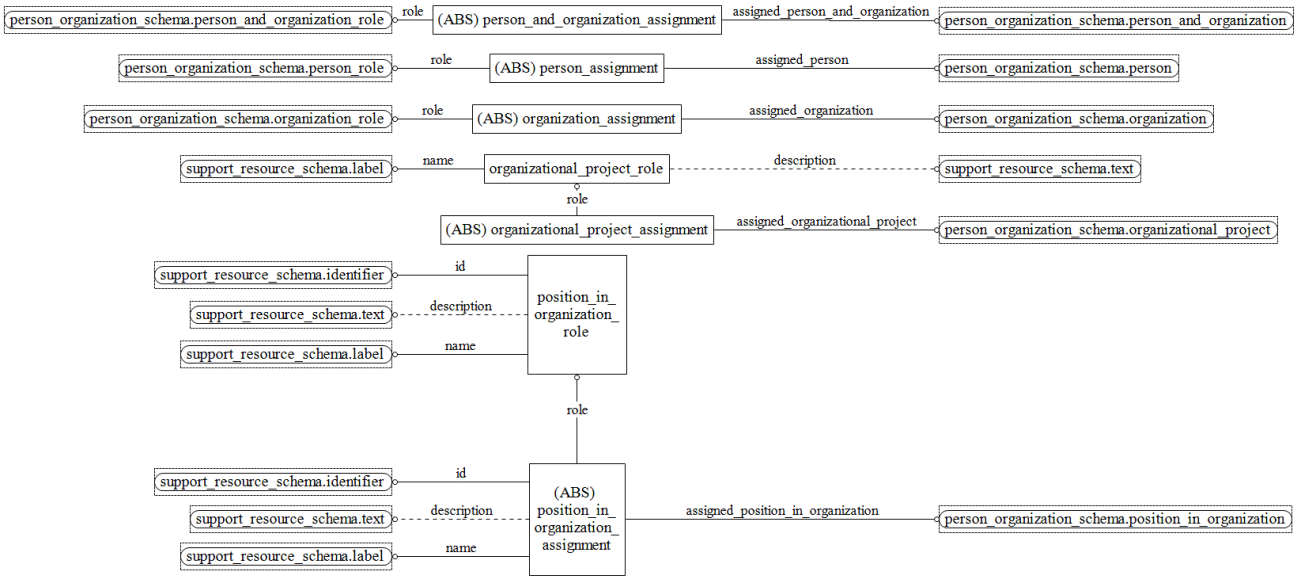


Figure D.38 — EXPRESS-G diagram of the management_resources_schema (6 of 9)

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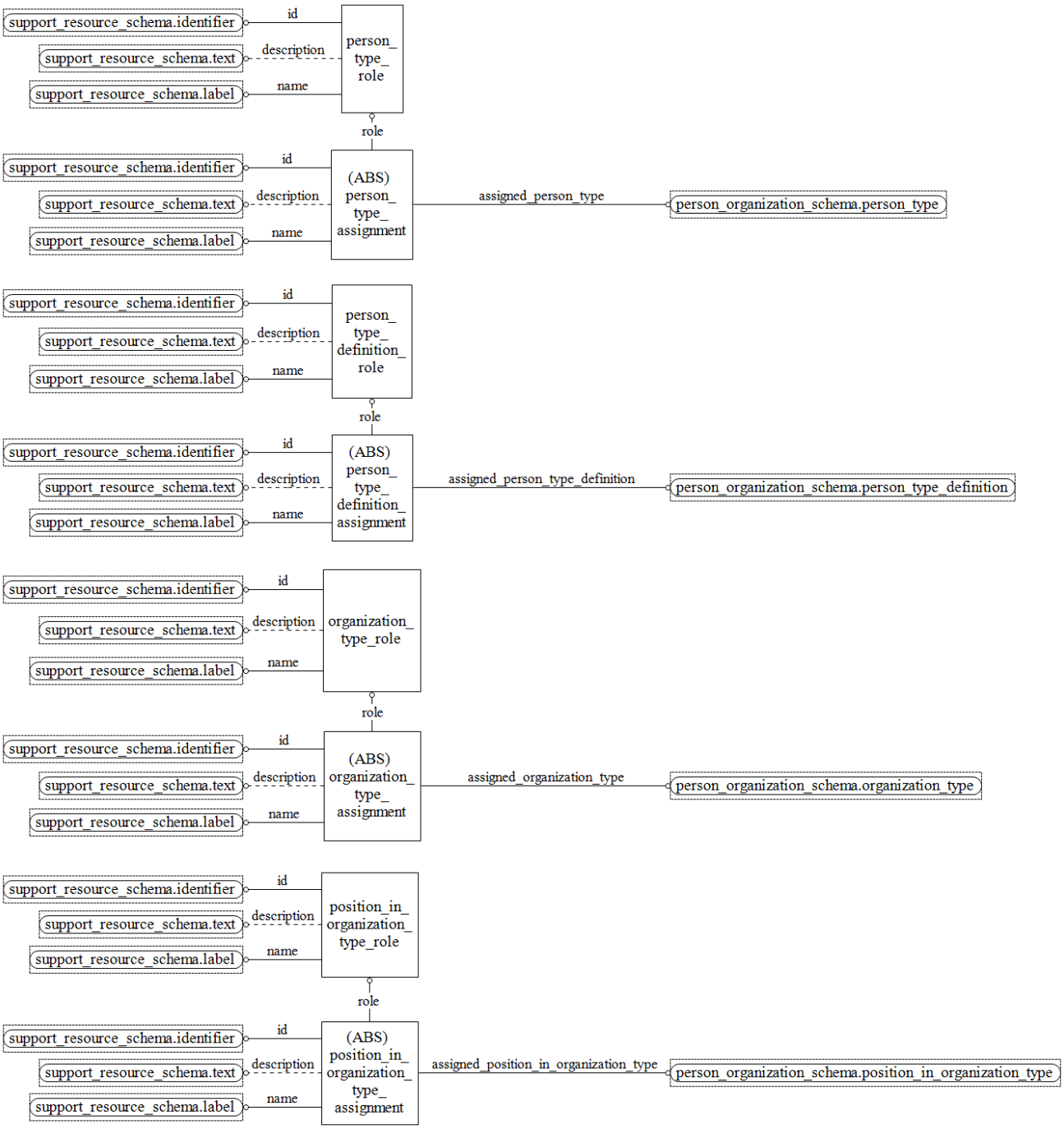


Figure D.39 — EXPRESS-G diagram of the management_resources_schema (7 of 9)

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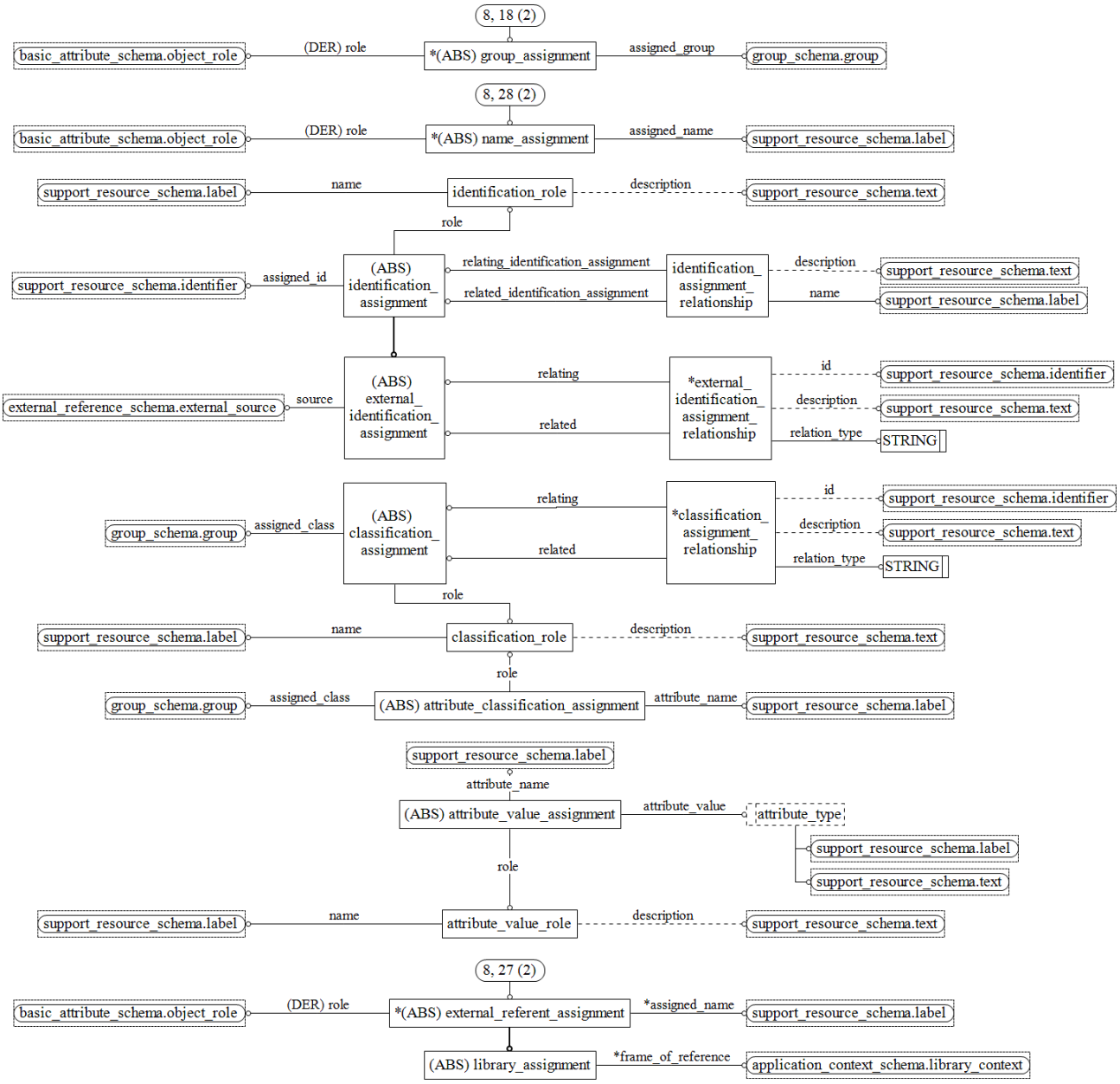


Figure D.40 — EXPRESS-G diagram of the management_resources_schema (8 of 9)

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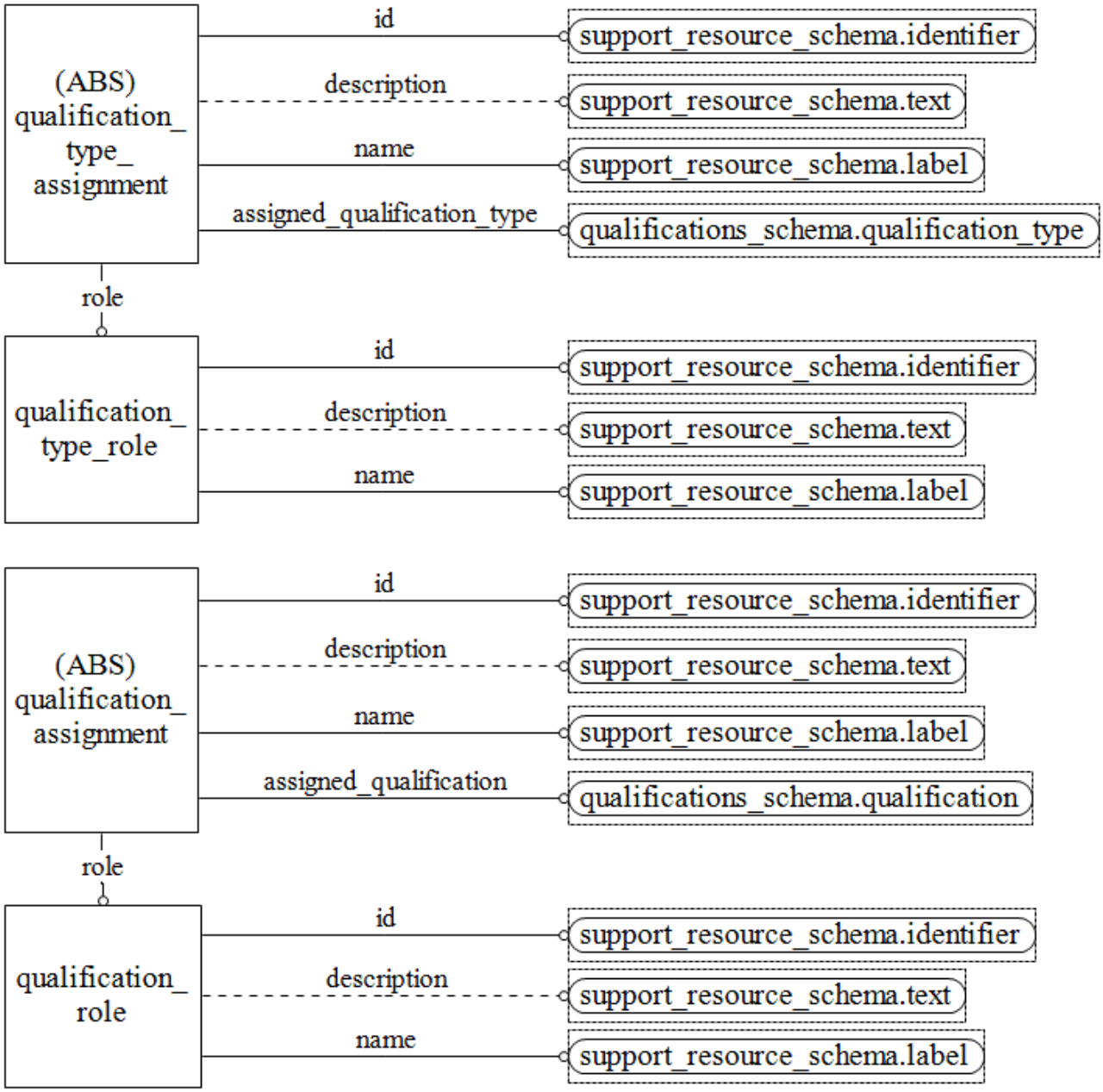


Figure D.41 — EXPRESS-G diagram of the management_resources_schema (9 of 9)

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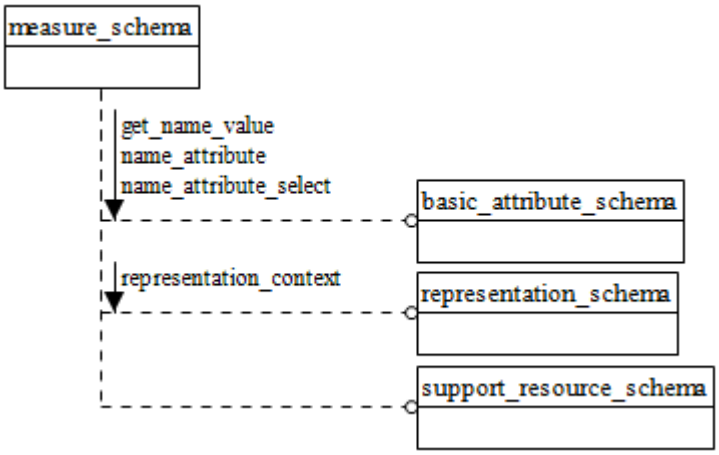


Figure D.42 — EXPRESS-G diagram of the measure_schema (1 of 5)

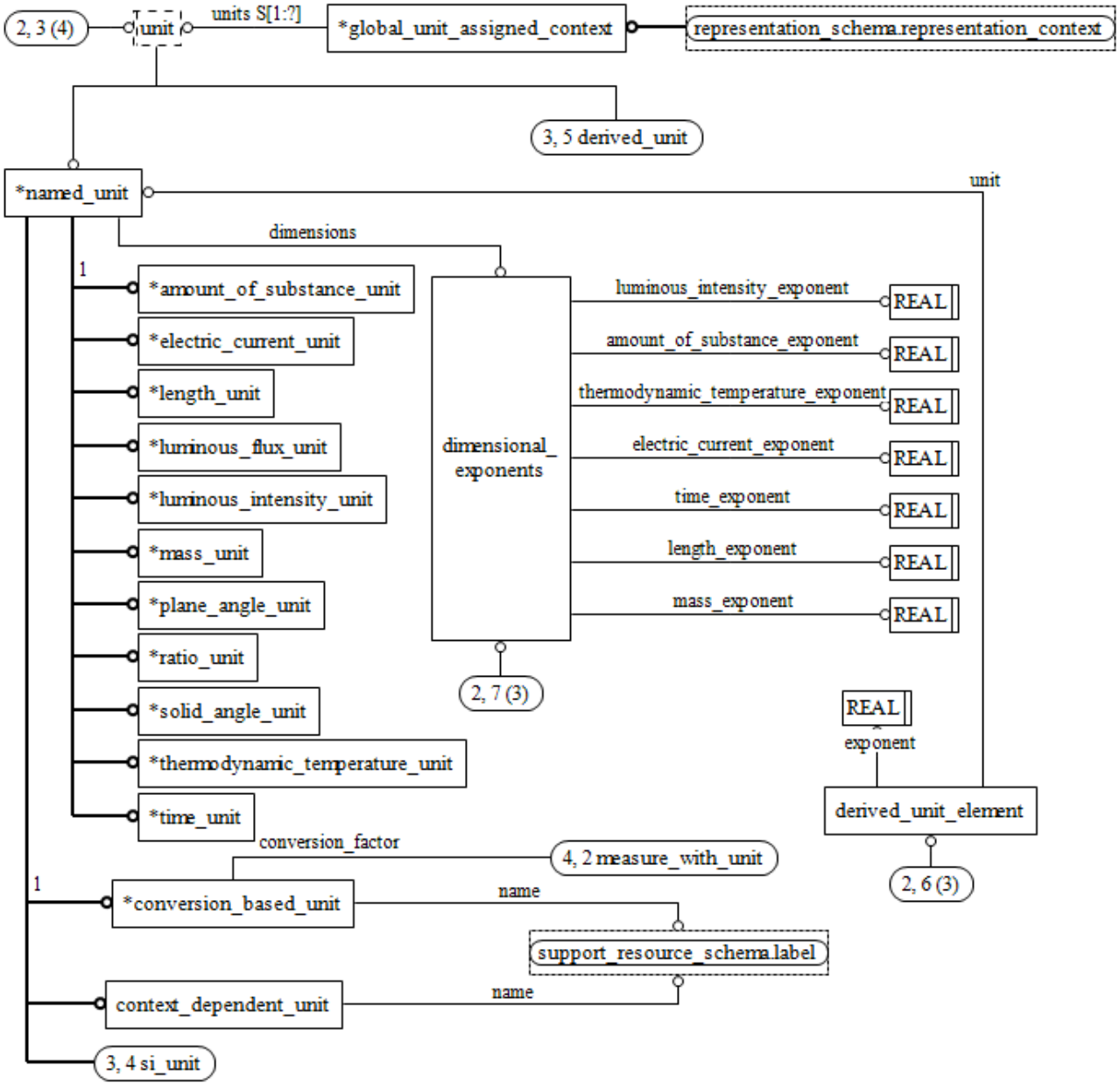


Figure D.43 — EXPRESS-G diagram of the measure_schema (2 of 5)

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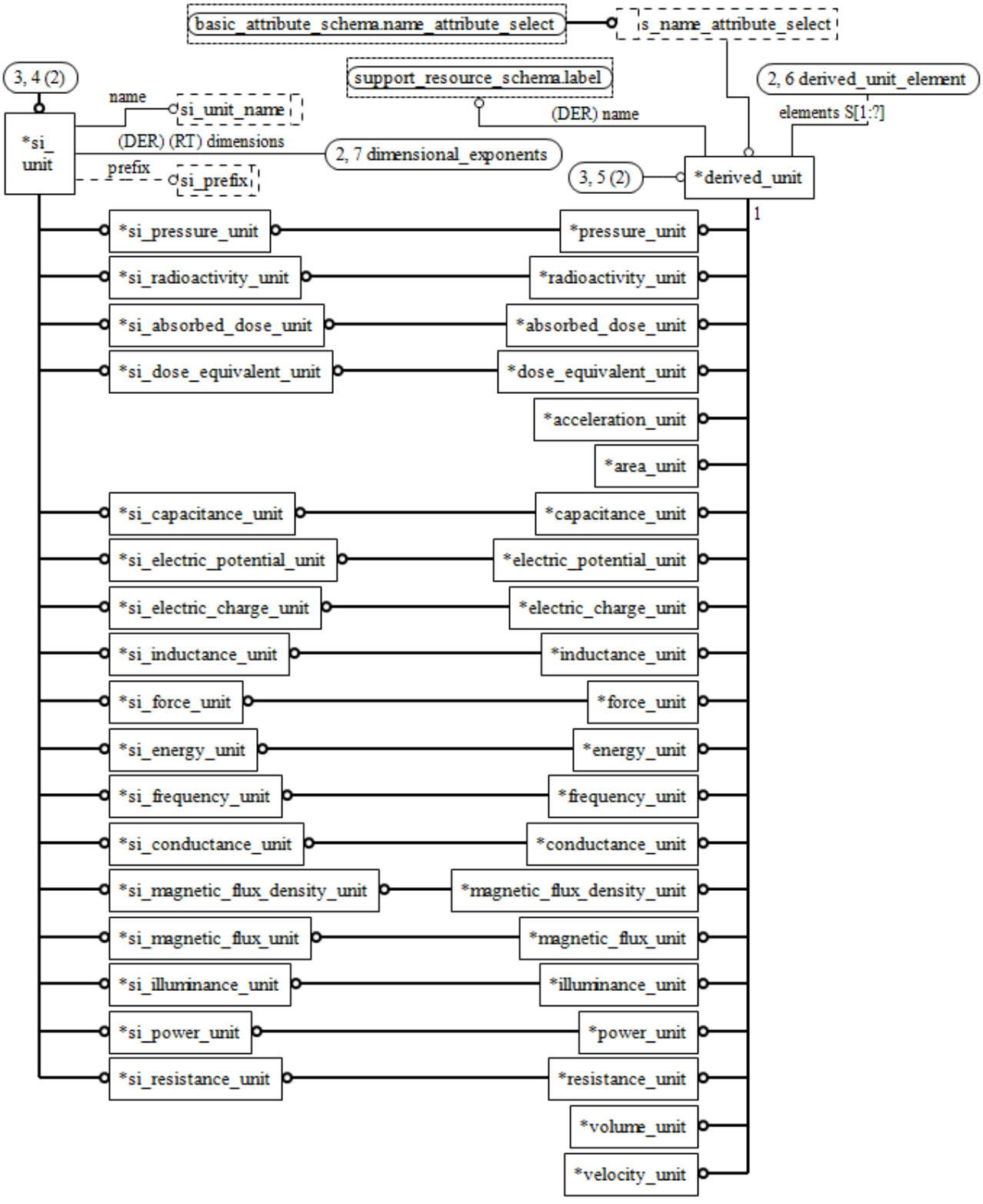


Figure D.44 — EXPRESS-G diagram of the measure_schema (3 of 5)

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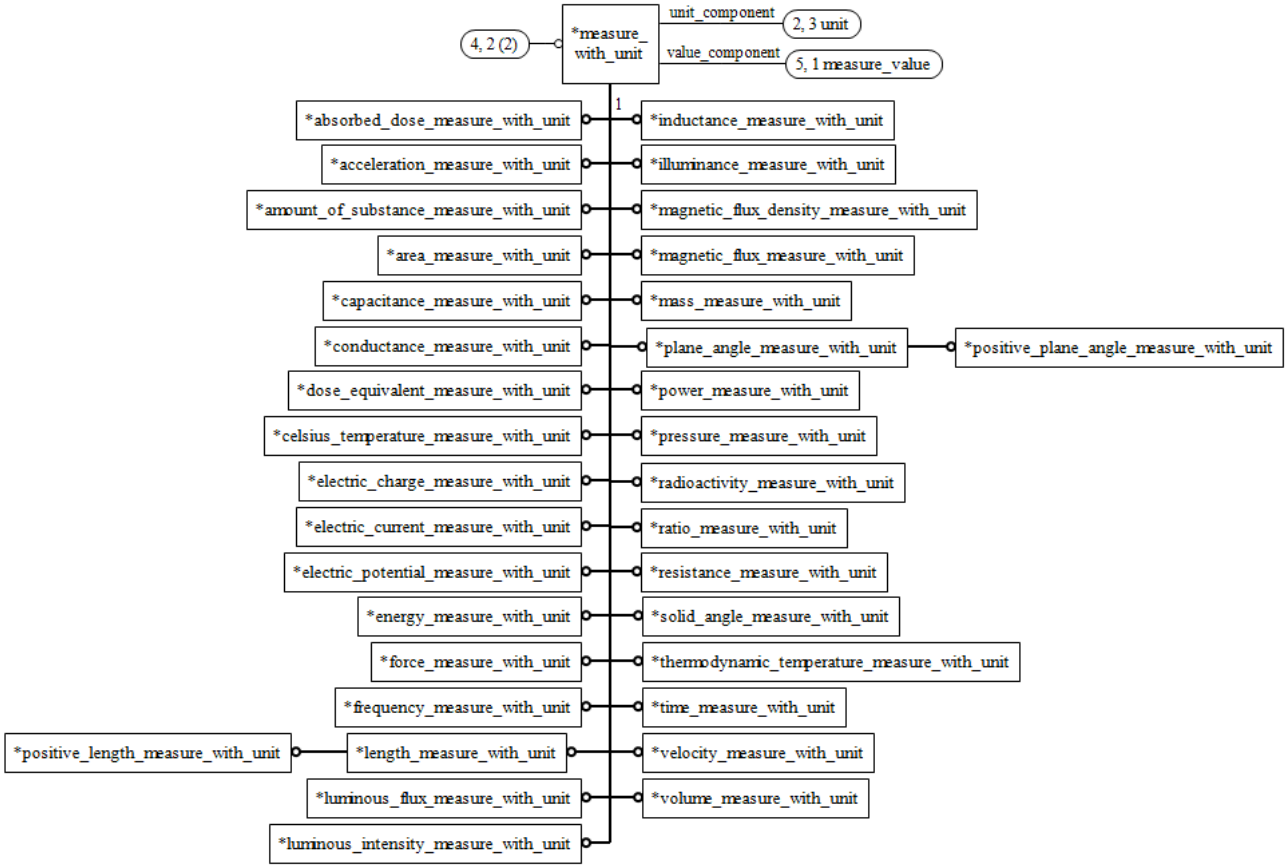


Figure D.45 — EXPRESS-G diagram of the measure_schema (4 of 5)

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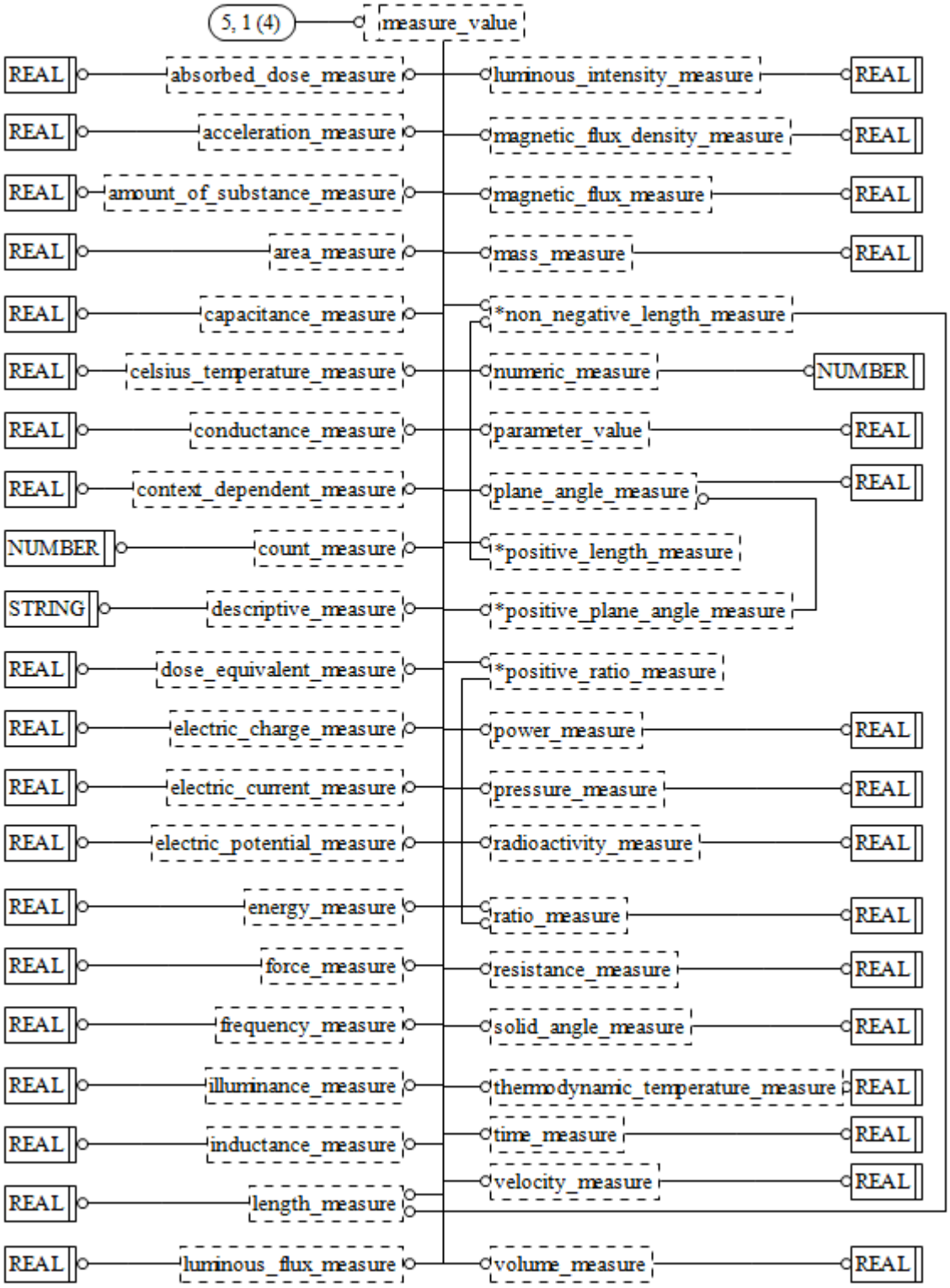


Figure D.46 — EXPRESS-G diagram of the measure_schema (5 of 5)

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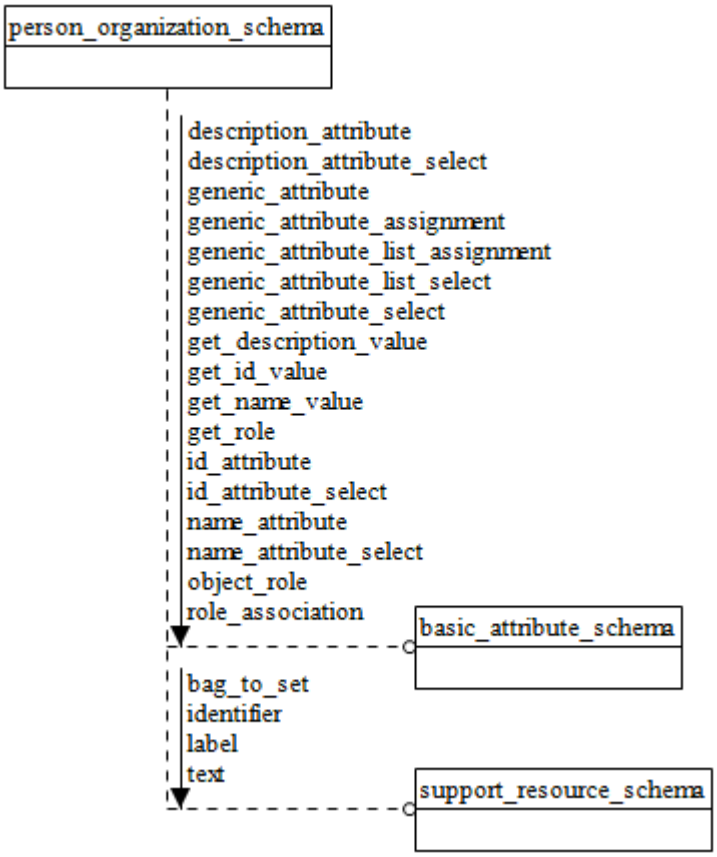


Figure D.47 — EXPRESS-G diagram of the person_organization_schema (1 of 4)

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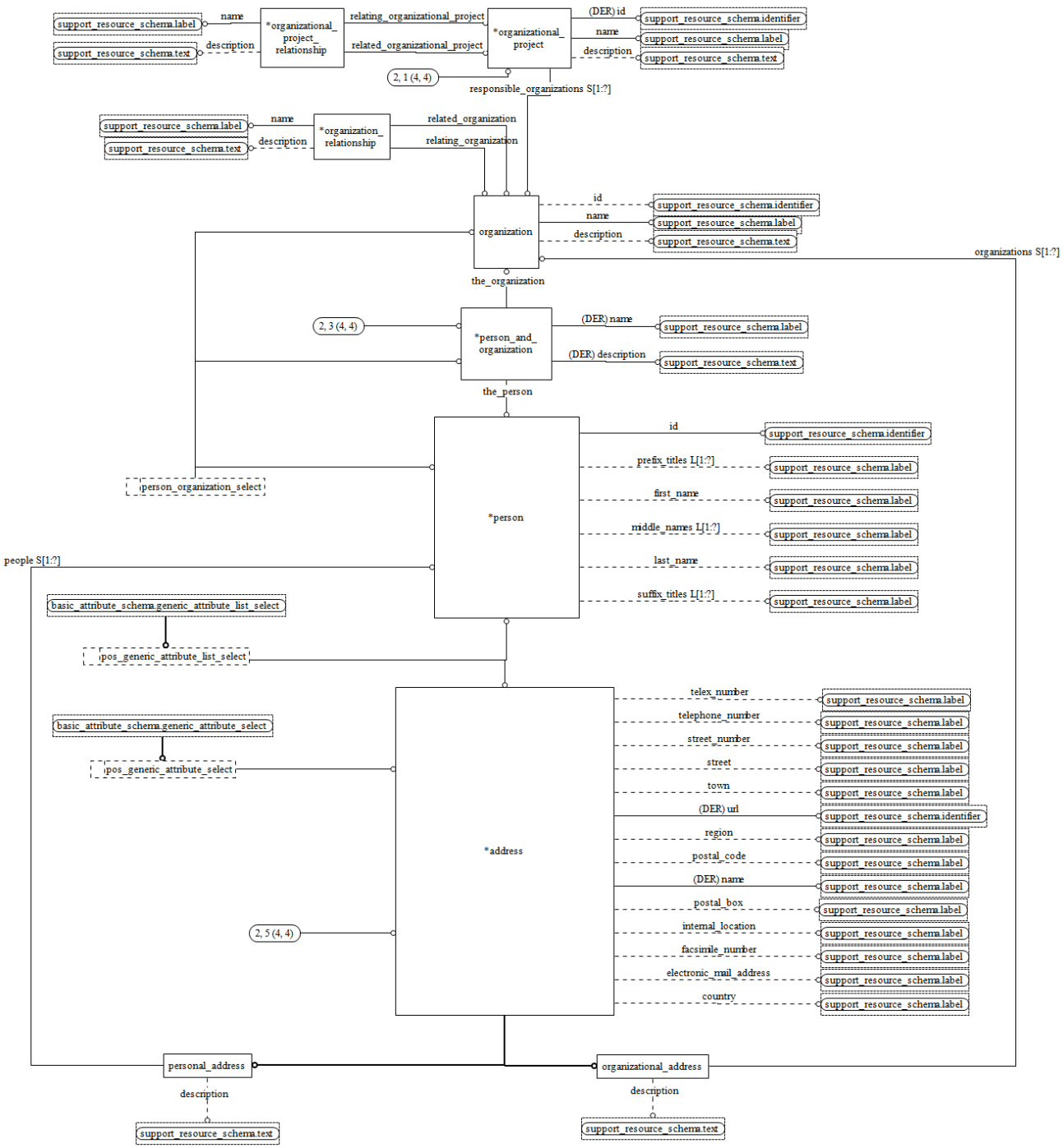


Figure D.48 — EXPRESS-G diagram of the person_organization_schema (2 of 4)

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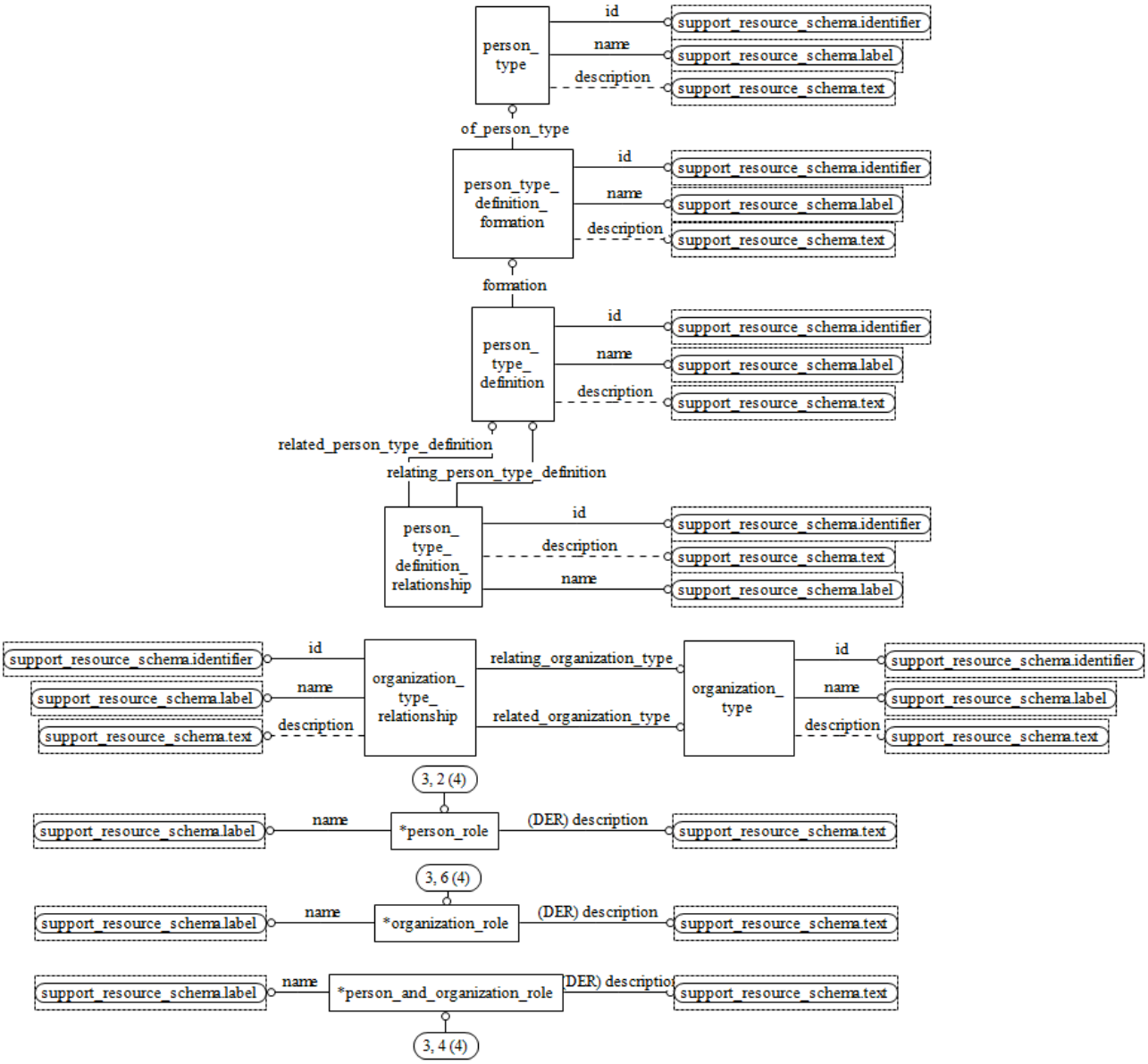


Figure D.49 — EXPRESS-G diagram of the person_organization_schema (3 of 4)

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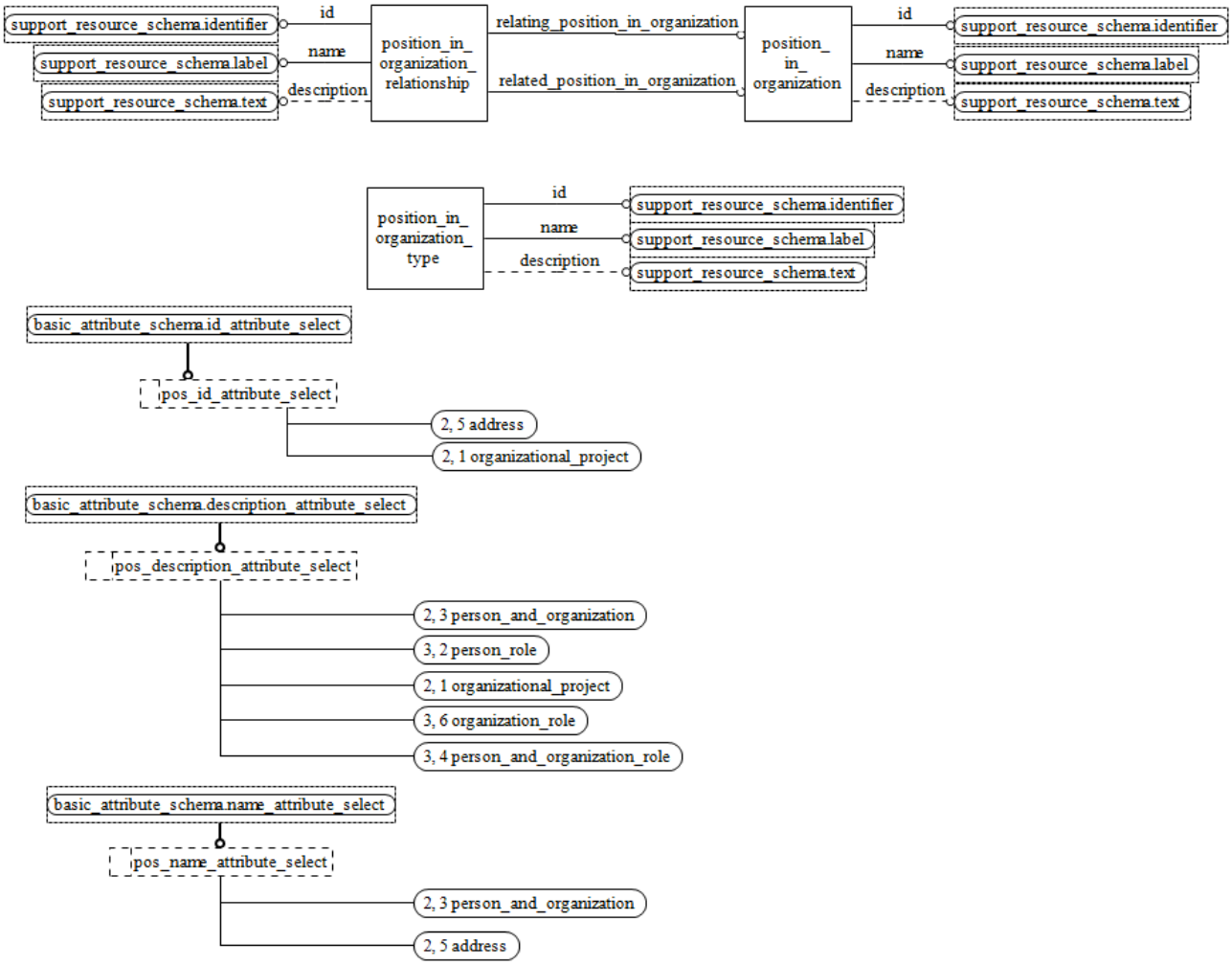


Figure D.50 — EXPRESS-G diagram of the person_organization_schema (4 of 4)

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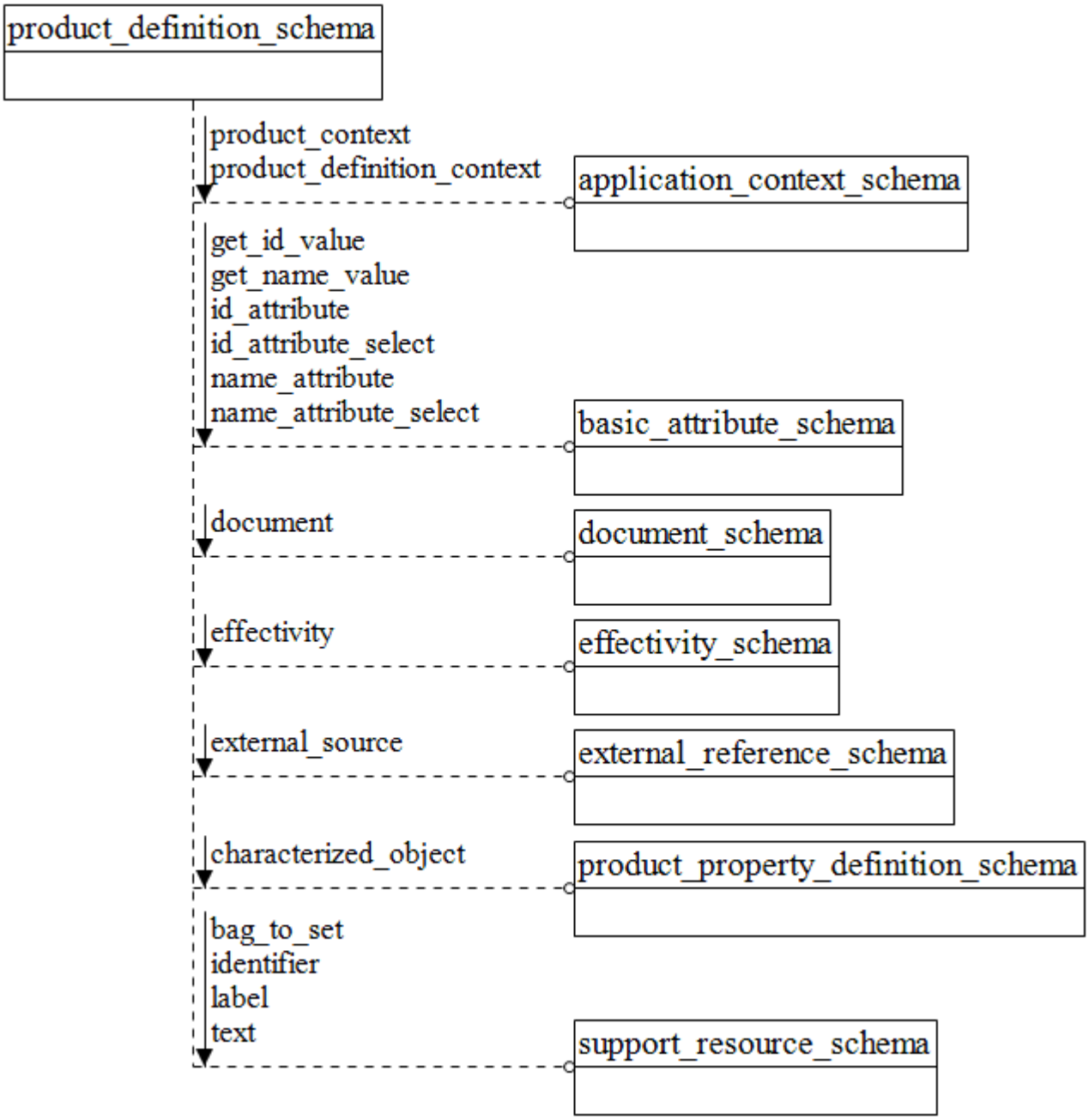


Figure D.51 — EXPRESS-G diagram of the product_definition_schema (1 of 3)

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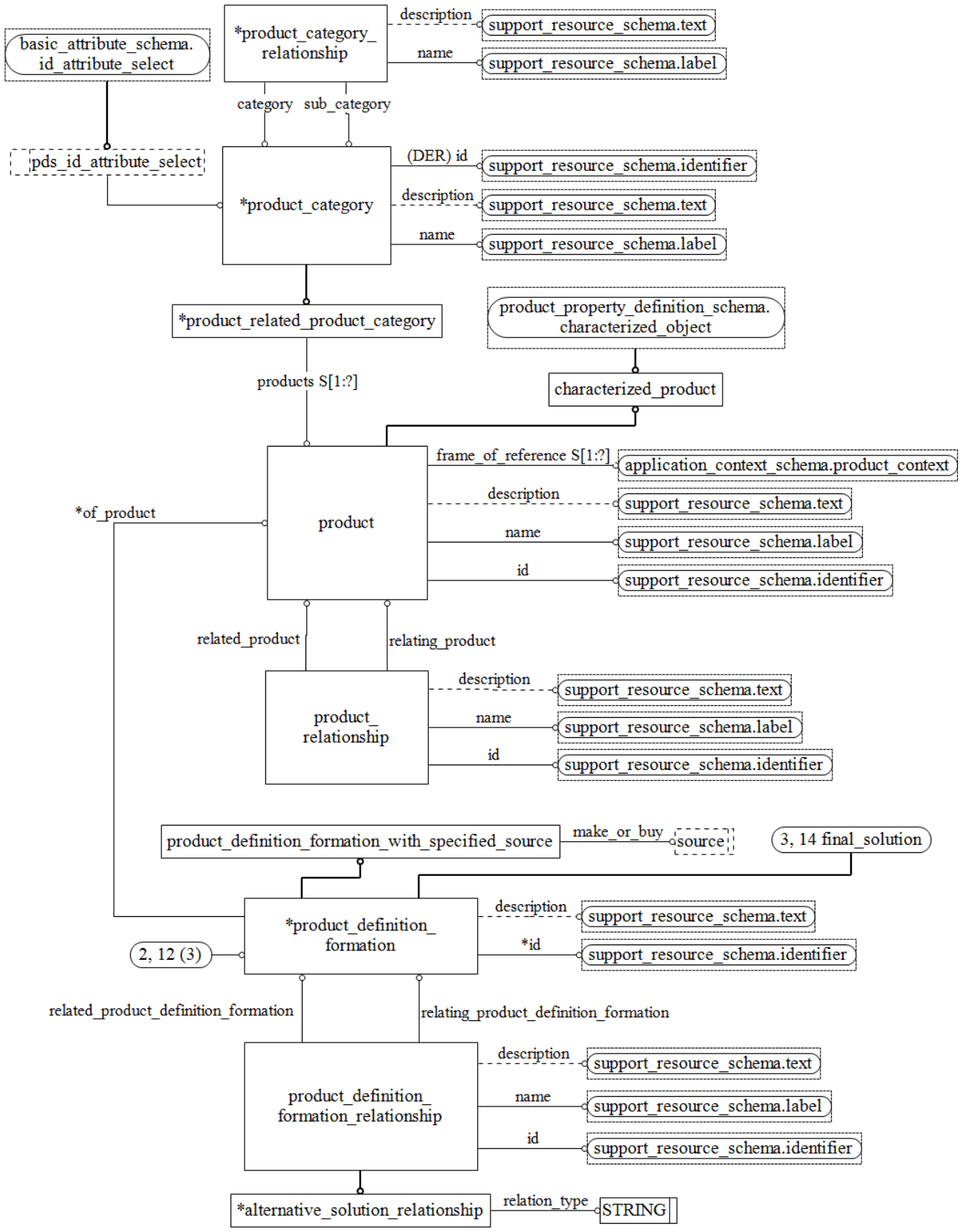


Figure D.52 — EXPRESS-G diagram of the product_definition_schema (2 of 3)

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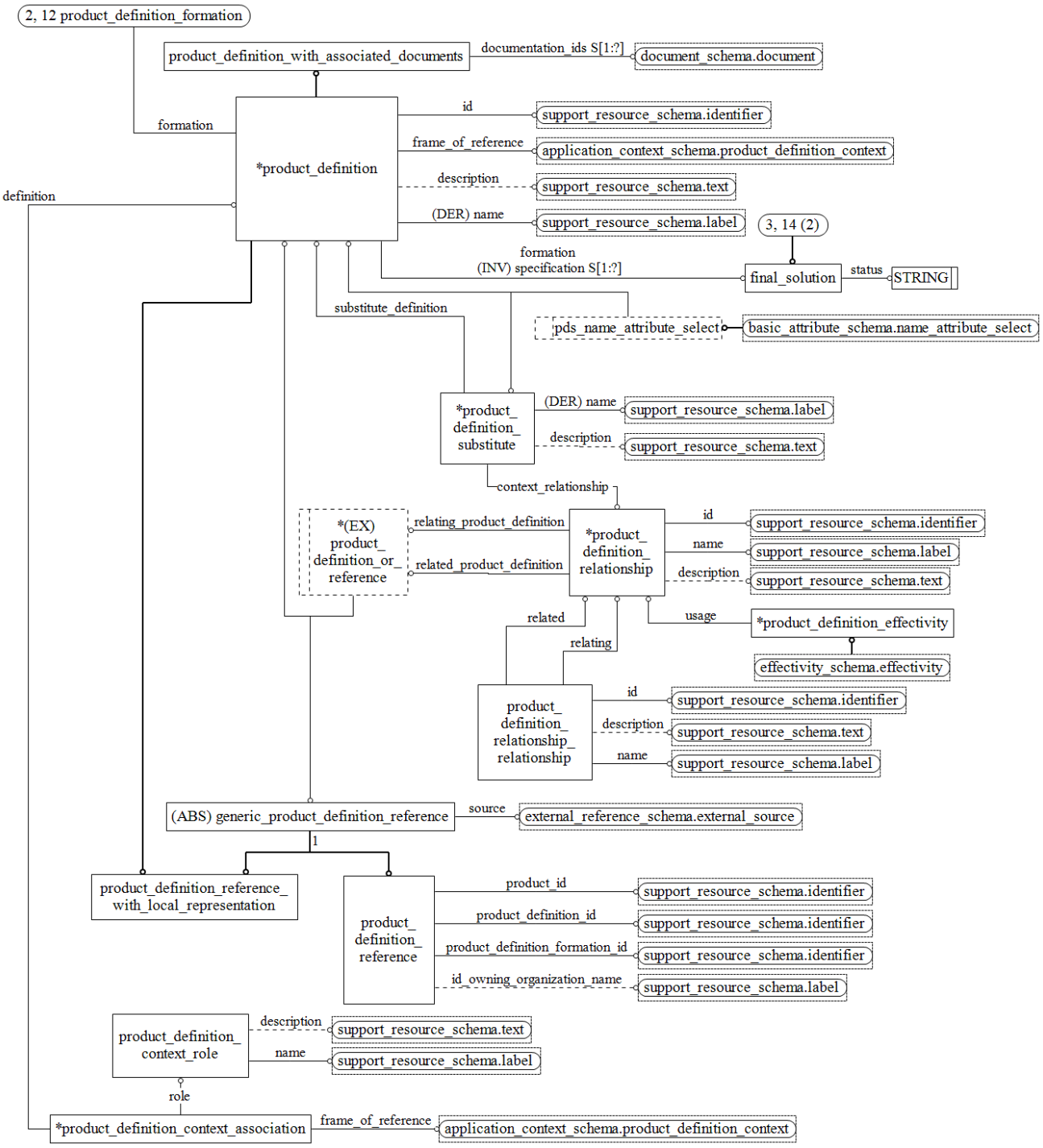


Figure D.53 — EXPRESS-G diagram of the product_definition_schema (3 of 3)

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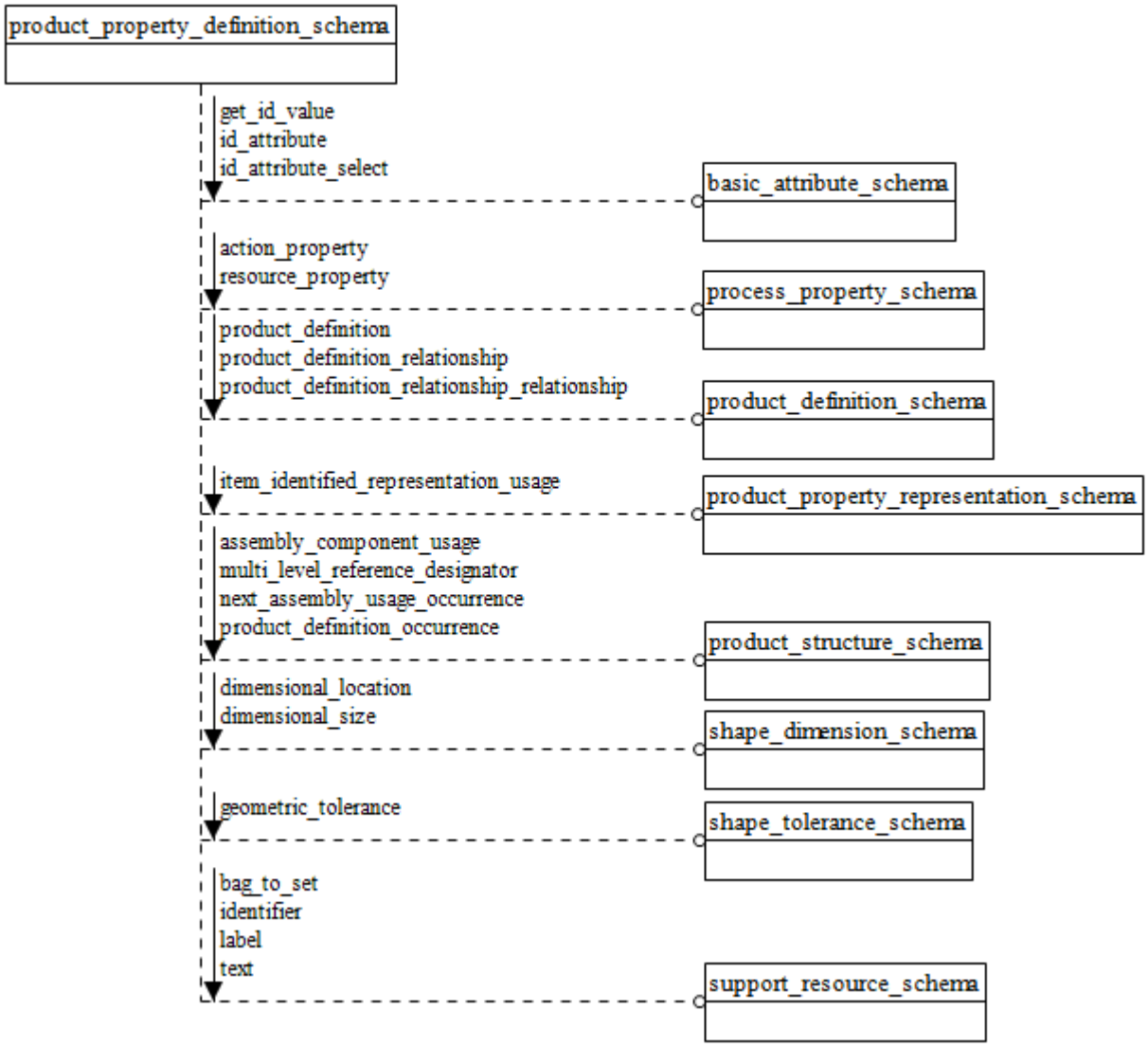


Figure D.54 — EXPRESS-G diagram of the product_property_definition_schema (1 of 2)

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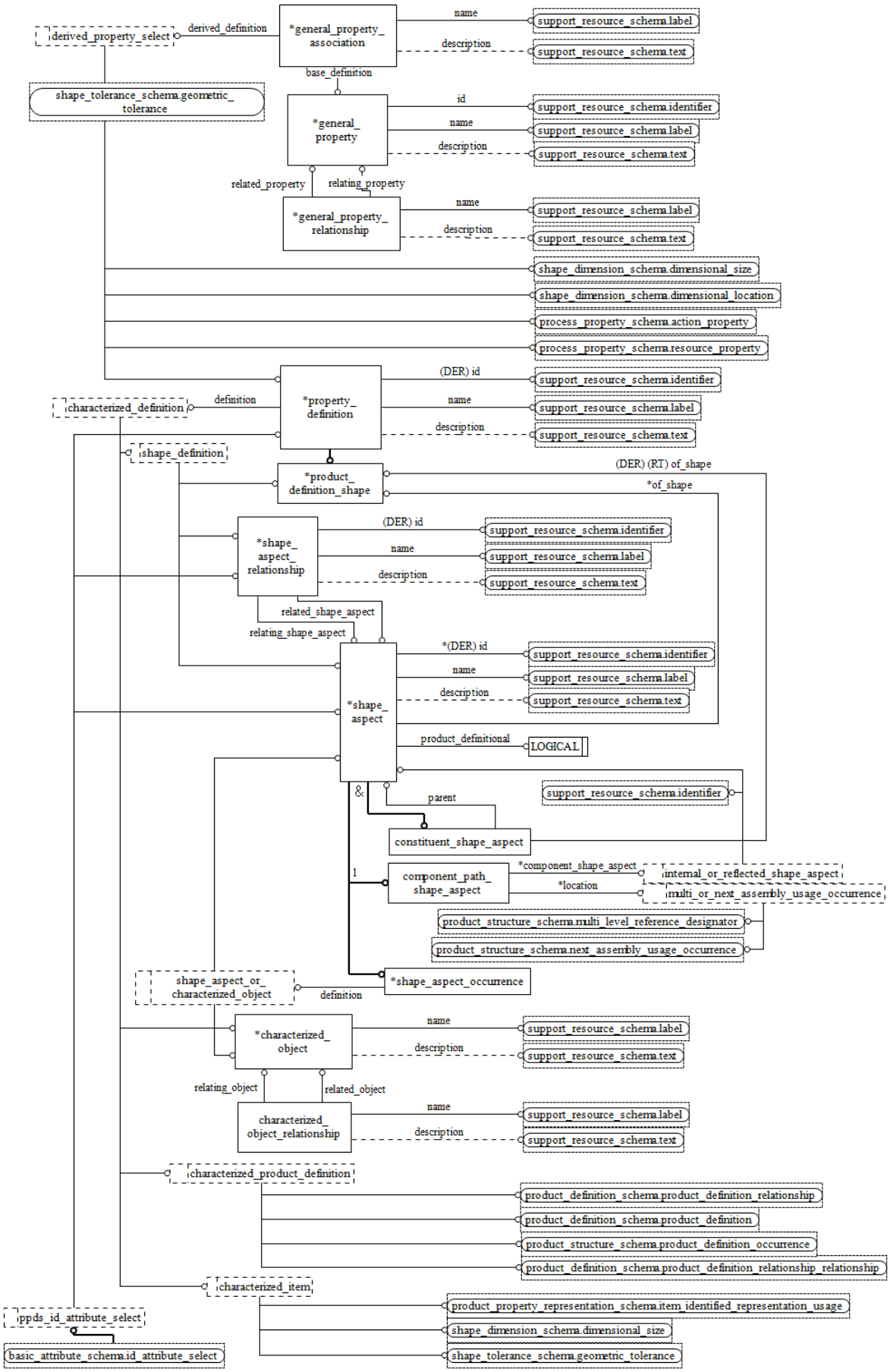


Figure D.55 — EXPRESS-G diagram of the product_property_definition_schema (2 of 2)

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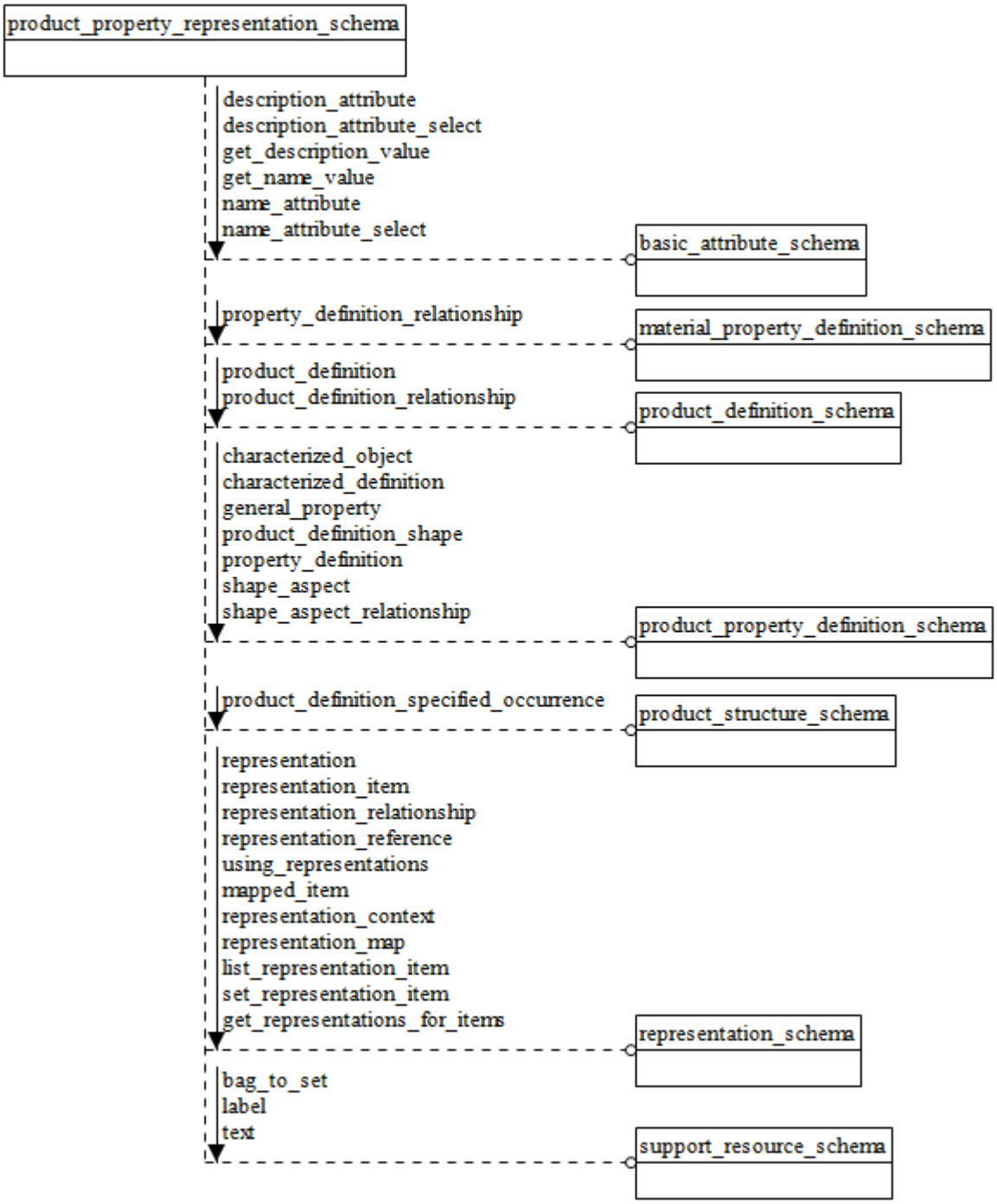


Figure D.56 — EXPRESS-G diagram of the product_property_representation_schema (1 of 2)

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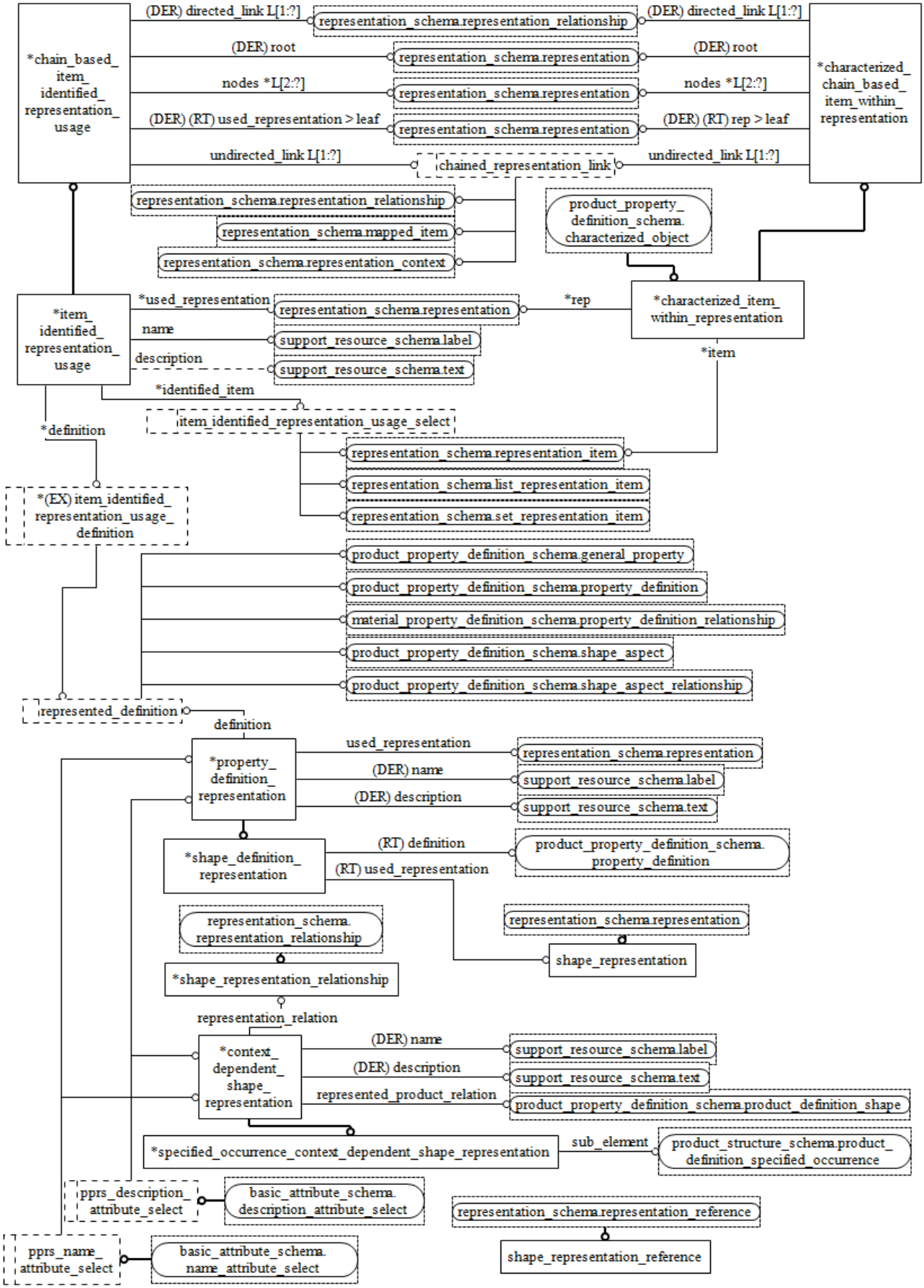


Figure D.57 — EXPRESS-G diagram of the product_property_representation_schema (2 of 2)

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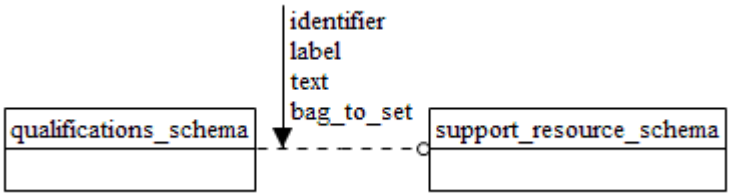


Figure D.58 — EXPRESS-G diagram of the qualifications_schema (1 of 2)

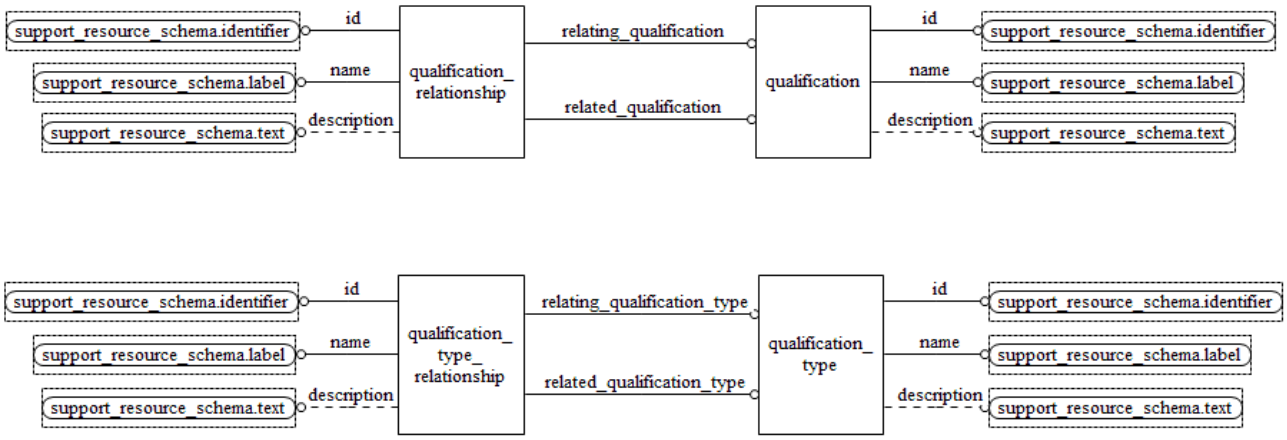


Figure D.59 — EXPRESS-G diagram of the qualifications_schema (2 of 2)

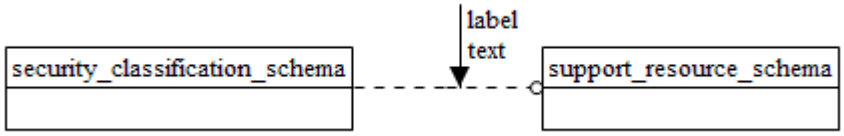


Figure D.60 — EXPRESS-G diagram of the security_classification_schema (1 of 2)

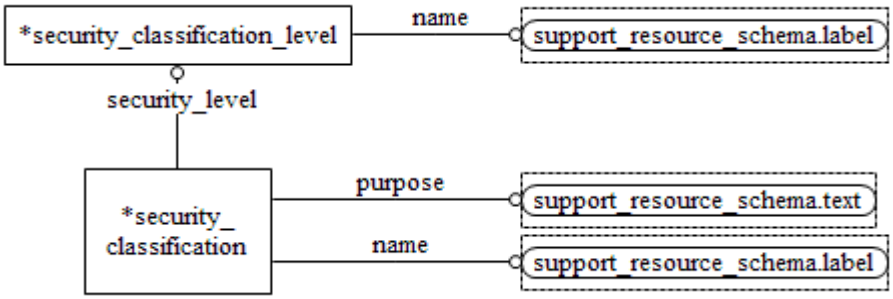


Figure D.61 — EXPRESS-G diagram of the security_classification_schema (2 of 2)

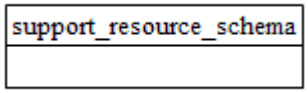


Figure D.62 — EXPRESS-G diagram of the support_resource_schema (1 of 2)

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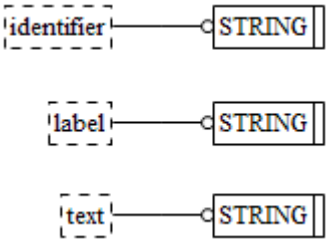


Figure D.63 — EXPRESS-G diagram of the support_resource_schema (2 of 2)

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Annex E
(informative)

Technical discussion

E.1 Generic product description resource structure

The generic product description resource is made up of schemas that reflect the different kinds of data that may be associated with a product.

The **application_context_schema** allows for the description of the conditions under which a given instance of the generic product description resource was defined.

The **product_definition_schema** supports the description of data that relates to a particular product.

The **product_property_definition_schema** supports the description of the properties of this product, for example, its shape and material.

The **product_property_representation_schema** specifies the ways in which a property of a product may be represented.

The relationships between these schemas are shown in Figure 1.

— Application context: a **product_definition** is defined in at least one application context and may be used in zero, one, or many application contexts. A single application context may be used to define zero, one, or more **product_definition**s.

— Product property definition: a **product_definition** may have one or more property definitions associated with it;

— Property representation: each property definition may be represented in one or more ways. The only property that is expanded in this edition of this part is shape.

E.2 Function template for cycle detection

E.2.1 acyclic_object_relationship

The template described in this clause is used in various ISO 10303 integrated resources. It is a function that detects cyclic definitions.

The **acyclic_object_relationship** function determines whether the graph of instances of the entity data type object that contains relation as one of its links contains a cycle. This function may be used to evaluate either a **object_relationship** or any of its subtypes.

Let Z be a set of instances of the entity data type **object**. The initial content of Z is the content of the set provided by the parameter **relatives**. Let R be the instance of object referred to by the attribute **relation.relating_object**. The functions looks for the instances of **specific_relation** that refer to R through their attribute **object_relationship.related_object**. It adds to Z the set of instances of object that are referred to by these instances of **specific_relation** through their attribute **object_relationship.relating_object**.

The function then recursively applies the same search for each new element of Z until each branch of the graph has been fully explored or until a cycle has been detected. The function detects a cycle and returns

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FALSE if, at any stage of the search, an instance proposed for addition to Z already exists in Z. Otherwise, it returns TRUE.

NOTE 1 Calls to functions based on this template should be done as follows.

WR1: `acyclic_object_relation (SELF, [SELF.related_object], '….' );`

EXPRESS specification:

```
FUNCTION acyclic_object_relationship
 (relation     : object_relationship;
  relatives      : SET [1:?] OF object;
  specific_relation : STRING) : BOOLEAN;

  LOCAL
 x       : SET OF object_relationship;
  END_LOCAL;

  IF relation.relating_object IN relatives THEN
 RETURN (FALSE);
  END_IF;       -- IN is based in instance equality

  x := QUERY (oor &lt;* bag_to_set (USEDIN
      (relation.relating_object,
       'OBJECT_SCHEMA.' +
       'OBJECT_RELATIONSHIP.' +
       'RELATED_OBJECT')) |
        specific_relation IN TYPEOF (oor));
  REPEAT I := 1 TO HIINDEX(x);       -- pre-checked loop
    IF NOT acyclic_object_relationship
        (x[i],
         relatives + relation.relating_object,
         specific_relation) THEN
      RETURN(FALSE);
    END_IF;
  END_REPEAT;

  RETURN(TRUE);
END_FUNCTION;
```

Attribute definitions

relation — (input) the candidate object_relationship to be checked.

NOTE 2 The instance of the **object_relationship** entity specified in the relation parameter of **acyclic_object_relationship** function is either an **object_relationship** entity or one of its subtypes.

relatives — (input) the set of instances of the entity data type object that defines the domain for the detection of cycles.

specific_relation — (input) the fully qualified entity name of the entity data type object_relationship or of one of its subtypes.

E.3 Relationship template

E.3.1 object_relationship

The template described in this clause is used in various ISO 10303 integrated resources. It supports the description of graphs of like objects.

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An **object_relationship** relates two objects with a description of their relationship.

EXPRESS specification:

```
ENTITY object_relationship;
  name    : label;
  description    : text;
  relating_object : object;
  related_object  : object;
END_ENTITY;
```

Attribute definitions

name — the **label** by which the **object_relationship** is known.

description — the **text** that characterizes the **object_relationship**.

relating_object — one of the instances of **object** that is a part of the relationship.

NOTE 1 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

related_object — the other instance of **object** that is a part of the relationship. If one element of the relationship is dependent upon the other, this attribute shall be the dependent one.

NOTE 2 The meaning of this attribute can be defined in the annotated EXPRESS schemas that use or specialize this entity, or in an agreement of common understanding between the partners sharing this information.

E.3.2 Constraining entity instances of the basic_attribute_schema

In order to constrain the population of instances of the **basic_attribute_schema**, the following rule may be used in EXPRESS schemas that use or specialize the constructs of this part of ISO 10303. The rule **attribute_x_not_allowed** enforces the restriction that the attribute x of any instance of the entity data type **entity_type** not be present.

NOTE Such a rule can only be specified in cases in which the referenced attribute is optional.

EXPRESS specification:

```
RULE  attribute_x_not_allowed FOR (entity_type) ;
WHERE
   wr1: SIZEOF (QUERY (ent &lt;* entity_type | EXISTS (ent.x))) = 0;
END_RULE;
```

E.4 Use of the generic management resource constructs

The EXPRESS elements specified in the **management_resource_schema** are used to associate management type data with product data in specific application interpreted models. This clause describes the EXPRESS mechanism that is used to make these associations.

The ABSTRACT SUPERTYPE statement from the EXPRESS language is used in the **management_resource_schema** to define template structures that are used to associate management type data with product data in an application interpreted model.

The **approval_assignment** entity from the **management_resource_schema** defines a template structure that is used to associate approvals with product data.

EXPRESS specification:

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```
ENTITY approval_assignment
  ABSTRACT SUPERTYPE;
   assigned_approval : approval;
  DERIVE
     role            : object_role := get_role (SELF);
WHERE
   WR1 : SIZEOF (USEDIN (SELF, 'BASIC_ATTRIBUTE_SCHEMA.' +
      'ROLE_ASSOCIATION.ITEM_WITH_ROLE')) &lt;= 1;
END_ENTITY; -- approval_assignment
```

These template structures are used in application interpreted models in which the management type data is to be associated with product data, in the following way:

— a SELECT type, and the necessary EXPRESS USE and REFERENCE statements, that defines the EXPRESS elements that are to have a particular kind of management type data associated with them is specified;

— an entity, and the necessary EXPRESS USE and REFERENCE statements, that is a SUBTYPE of the required template structure and that has an attribute that is of the type of the SELECT type is specified.

— If approvals are assigned to drawing sheet revisions and drawing revisions, the following SELECT type would be defined:

EXPRESS specification:

```
TYPE approval_item = SELECT
     (drawing_sheet_revision,
      drawing_revision );
END_TYPE;
```

The following **applied_approval_assignment** entity could be used to associate **approval** with drawing sheet revisions and drawing revisions:

EXPRESS specification:

```
ENTITY applied_approval_assignment
    SUBTYPE OF (approval_assignment);
    item : approval_item;
END_ENTITY;
```

NOTE In order to facilitate interoperability among application protocols, subtypes of the entity data type **xxx_assignment** are conventionally named **applied_xxx_assignment** in the application interpreted models.

In the **management_resource_schema** each abstract **xxx_assignment** entity has a **role** attribute. This attribute allows the characterization of the resource construct **xxx** with respect to the product data to which it will be assigned.

For example, in particular instances of the entity data type **applied_contract_assignment**, the derived attribute **role** can refer to instances of the entity data type **object_role** with name 'does not apply', or 'is the binding agreement for the production of', in order to characterize the role of a contract with respect to a particular product.

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Annex F
(informative)

Examples

F.1 Use of the product_definition_schema

This clause provides two examples of the use of the entity data type **product_definition** to characterize a version of a product.

The first example considers the case in which a version of the product is defined in three distinct views, each associated with a particular life cycle stage: 'specification stage', 'design stage', and 'manufacturing design stage'.

In this situation, there would be:

— three instances of the entity data type **product_definition_context**: one per life cycle stage;

— three instances of the entity data type **product_definition**. The **frame_of_reference** attribute of each instance of **product_definition** would refer to an instance of **product_definition_context**;

— an instance of the entity data type **product_definition_formation**, referred to by the three above mentioned instances of **product_definition**;

— an instance of the entity data type product, referred to by the instance of **product_definition_formation**.

The second example considers the case where a version of the product is defined in a primary view and where it is later considered that this definition is also relevant for the other life cycle stages.

In this situation, there would be:

— three instances of the entity data type **product_definition_context**, one per life cycle stage;

— one instance of the entity data type **product_definition**. The attribute **frame_of_reference** of this instance of **product_definition** would refer to an instance of **product_definition_context** associated with the primary context;

— two instances of the entity data type **product_definition_context_association** relating the other instances of **product_definition_context** with the instance of **product_definition**;

— an instance of the entity data type **product_definition_formation**, referred to by the instance of **product_definition**;

— an instance of the entity data type **product**, referred to by the instance of **product_definition_formation**.

F.2 Document as product

This part of ISO 100303 provides resources that support the association of a reference to an external document with any product data. This association uses the entity data types **document_reference** and **document**.

This part of ISO 10303 also provides resources that support the representation of a document in the case it is managed within the information system as a particular case of product.

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This clause explains how to use these resources to represent the information that characterize the document. It specifies how to identify the document, how to identify its versions and how to characterize its definitions.

NOTE Considering a document as a kind of product is a decision that may depend on the application domain and on the emphasis in this domain that is made on the documents versus that made on other kinds of parts. In mechanical products, documents such as operator's manuals are usually listed in the Bill of Materials of the products to be manufactured and delivered. In such a case, these documents may be tracked in the information systems as particular products.

The interpretation of 'Document as Product' uses basic product master identification for the fundamental requirements of document identification, versioning, and definition.

The following entity data types should therefore be used:

— **product**;

— **product_definition_formation**;

— **product_related_product_category**;

— **product_definition**.

F.2.1 Identification of a document

The identification information, common to all the historical versions of a document, managed as a product, should be conveyed in an instance of the entity data type **product**. The document identifier should be conveyed in the attribute **product.id**.

In order to characterize the fact that the instance of **product** actually identifies a document, this instance should be referred to by an instance of the entity data type **product_related_product_category** whose attribute name is assigned the value 'document'.

NOTE Further classifications of the document can be conveyed using appropriate instances of the entity data types **product_category** and **product_category_relationship**.

F.2.2 Identification of a version of a document

The identification information of a version of the document should be conveyed in an instance of the entity data type **product_definition_formation**.

F.2.3 Identification of a definition of document

The identification information of a definition view of a version of the document should be conveyed in an instance of the entity data type **product_definition**.

If the definition is digital, the attribute **product_definition.frame_of_reference** should refer to an instance of the entity data type **product_definition_context** whose attribute name has the value 'digital document definition'.

If the definition is non-digital, the attribute **product_definition.frame_of_reference** should refer to an instance of the entity data type **product_definition_context** whose attribute name has the value 'physical document definition'.

EXAMPLE An example of where a definition might be non-digital might be a case where the physical aspects of a book are relevant.

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F.2.4 Assembly structure of a document

If a document is considered as a kind of product and if its composition needs to be represented, the resources provided in ISO 10303-44 to represent assembly structures should be used.

For example, the decomposition of an encyclopedia in volumes can be represented using instances of the entity data type **next_assembly_usage_occurrence**.

NOTE The entity data type **next_assembly_usage_occurrence** can also be used to represent the fact that a document is a constituent of any other kind of product. For example, the driver's manual of a car may be identified as a constituent of the car using an instance of **next_assembly_usage_occurrence** that relates two instances of the entity data type **product_definition**, identifying definitions of the driver's manual and of the car respectively.

F.2.5 Association of documentation to other data

If a document is considered as a kind of product and if the information it contains is used to document a product or an activity, the association of the definition, of the version or of the identification of the document with the product or activity data should only be dealt with using document specific resources, that is using the entity data types **document_reference**, **document_usage_constraint_assignment** or **product_definition_with_associated_documents**.

NOTE Consequently, in such a case, the association of document data with product data should not be dealt with using the entity data types **product_definition_relationship**, **product_definition_formation_relationship** or **product_relationship**.

F.2.6 Enabling use of document specific resources

In order to enable the use of the document specific resources, as listed above, for a document considered as a kind of product, one needs to relate an instance of the entity data type **document** with an instance corresponding either to the identification of the document, to a version of the document or to a definition of the document. This association can be done using the entity data type **document_product_association**. For example, in order to state that an instance of the entity data type **document** identifies the same collection of information as a version of the corresponding product, an instance of **document_product_association** will relate the instance of document and the corresponding instance of **product_definition_formation**. The attribute document_product_association.name will have the value 'equivalence'.

F.2.7 Properties of a document

In this part of ISO 10303, properties assigned to product data are conveyed using instances of the entity data type **property_definition**.

If the properties are independent from the realization of the document, the instances of the entity data type **property_definition** can refer, through the attribute **property_definition.definition**, to an instance of **product_definition**.

If properties that depend on the actual realization of the document are considered, a common subtype of **characterized_object** and of **document** should be created.

For example, an annotated EXPRESS schema may define the entity data type **document_file** as below, in order to allow the characterization of the size of files. The file size will then be assigned to the documents, using appropriate instances of the entity data types **property_definition**, **property_definition_representation** and **representation**.

EXPRESS specification:

```
ENTITY document_file
SUBTYPE OF (document, characterized_object);
```

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```
END_ENTITY;
```

F.3 Use of the measure_schema

The following examples show how the resources provided within this schema are intended to be used to represent measures and units.

F.3.1 Use of derived_unit subtypes

F.3.1.1 SI base unit definitions

To support the discussion the following instance data is provided for SI base unit definitions.

Base unit reference instances:

```
#4  =(LENGTH_UNIT()              NAMED_UNIT(*)          SI_UNIT($,.METRE.));
#14 =(MASS_UNIT()                NAMED_UNIT(*)     SI_UNIT(.KILO.,.GRAM.));
#24 =(NAMED_UNIT(*)              SI_UNIT($, .SECOND.)          TIME_UNIT());
#26 =(ELECTRIC_CURRENT_UNIT()    NAMED_UNIT(*)       SI_UNIT($, .AMPERE.));
#426=(NAMED_UNIT(*) SI_UNIT($, .KELVIN.) THERMODYNAMIC_TEMPERATURE_UNIT());
#427=(AMOUNT_OF_SUBSTANCE_UNIT() NAMED_UNIT(*)          SI_UNIT($, .MOLE.));
#428=(LUMINOUS_INTENSITY_UNIT()  NAMED_UNIT(*)      SI_UNIT($, .CANDELA.));
```

NOTE #14 instance is created to support definition of SI derived units and is the formal definition that the kilogram is the SI unit of mass.

F.3.1.2 SI derived units

SI derived unit exchange should use the **derived_unit** and **unit_elements** referencing either a SI base unit or other SI derived units rather than relying directly on **dimensional_exponents**.

NOTE This part of ISO 10303 addresses the kilogram issue by isolating the application of .KILO. used to define a unit from the application of .KILO. used as a prefix in a measure value.

F.3.1.2.1 SI derived unit using user defined names

The list of entries in **si_unit_name** in this part of ISO 10303 is not exhaustive. In the case that the name of a derived unit is not included in **si_unit_name** then an instance of **derived_unit** (that is not also an instance of **si_unit**) should be populated. In that case, the **derived_unit.name** attribute should be populated to identify the unit.

F.3.1.2.2 SI derived unit using predefined names

Some SI derived unit names are included in **si_unit_name**. An instance of the specific multiple inheritance subtype of **derived_unit** and **si_unit** should be populated when it is desired to exchange these types of SI units:

— **absorbed_dose_unit**,

— **radioactivity_unit**,

— **capacitance_unit**,

— **dose_equivalent_unit**,

— **electric_charge_unit**,

— **conductance_unit**,

— **electric_potential_unit**,

— **energy_unit**,

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— **magnetic_flux_density_unit**,

— **force_unit**,

— **frequency_unit**,

— **illuminance_unit**,

— **inductance_unit**,

— **magnetic_flux_unit**,

— **power_unit**,

— **pressure_unit**,

— **resistance_unit**.

F.3.1.3 Derived units whose system of units is unspecified

The following derived units have unspecified system of units in this part of ISO 10303:

— acceleration,

— area,

— velocity,

— volume.

NOTE This part of ISO 10303 specifies the fact that the unit is a **derived_unit** and the **dimensional_exponents** values for the unit.

If all **derived_unit_element** instances reference SI units, then the derived unit is an SI derived unit.

For the case that other units are exchanged, such as English Imperial, each of the **derived_unit_element** instances referenced by the **derived_unit** should be in the same system of units.

F.3.1.4 Detailed examples of measure unit definitions

F.3.1.4.1 Definition of SI units

Definition of "newton":

A Newton (N) is $kg{\cdot}m{\cdot}s^{-2}$. Part41 provides the ability to explicitly state that force is derived, that the newton is a SI derived unit with a name.

This part of ISO 10303 requires that a data population include kilogram as the mass unit when newton is defined so as to make the mathematical properties of the data set consistent with SI system of units.

NOTE If the **mass_unit** prefix is not provided, even though the newton is declared to be the unit, the numerical instance data declares the unit to be the dyne.

Recommended approach:

```
/* establish system of units */
#5=DERIVED_UNIT_ELEMENT(#4,1.0);
#15=DERIVED_UNIT_ELEMENT(#14,1.0);
#25=DERIVED_UNIT_ELEMENT(#24,-2.0);
/* establish newton as SI force unit */
#4161100=SI_FORCE_UNIT((#5,#15,#25),*,$,.NEWTON.);
```

Definition of "pascal":

A Pascal (Pa) is $1N{\cdot}m^{-2}$.

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```
/* establish the multiplication by m^{-2}. */
#550005=DERIVED_UNIT_ELEMENT(#4,-2.0);
/* establish a reference to a newton already defined */
#5500025=DERIVED_UNIT_ELEMENT(#4161100,1.0);
#4161200=SI_PRESSURE_UNIT((#550005,#5500025),*,$,.PASCAL.);
```

Definition of "joule":

A Joule (J) is $1 N \cdot m$.

```
/* establish the multiplication by m. */
#650005=DERIVED_UNIT_ELEMENT(#4,1.0);
/* establish a reference to a newton already defined */
#6500025=DERIVED_UNIT_ELEMENT(#4161100,1.0);
#4161300=SI_ENERGY_UNIT((#650005,#6500025),*,$,.JOULE.);
```

Definition of "watt":

A Watt (W) is $1 J \cdot s^{-1}$.

```
/* establish the division by sec. */
#750005=DERIVED_UNIT_ELEMENT(#4,-1.0);
/* establish a reference to a joule already defined */
#7500025=DERIVED_UNIT_ELEMENT(#4161300,1.0);
#4161400=SI_POWER_UNIT((#750005,#7500025),*,$,.WATT.);
```

Definition of "coulomb":

A Coulomb (C) is $1 A \cdot s$.

```
/* establish system of units */
#8500015=DERIVED_UNIT_ELEMENT(#24,1.0);
#8500025=DERIVED_UNIT_ELEMENT(#26,1.0);
/* establish coulomb as SI electric charge unit */
#1001=SI_ELECTRIC_CHARGE_UNIT((#8500015,#8500025),*,$,.COULOMB.);
```

Definition of "volt":

A Volt (V) is $1 W \cdot A^{-1}$.

```
/* establish the division by amp. */
#950005=DERIVED_UNIT_ELEMENT(#26,-1.0);
/* establish a reference to a watt already defined */
#9500025=DERIVED_UNIT_ELEMENT(#4161400,1.0);
#1002=SI_ELECTRIC_POTENTIAL_UNIT((#950005,#9500025),*,$,.VOLT.);
```

Definition of "farad":

A Farad (F) is $1 C \cdot V^{-1}$.

```
/* establish a reference to a coulomb already defined. */
#860005=DERIVED_UNIT_ELEMENT(#1001,1.0);
/* establish a reference to a volt already defined */
#8600025=DERIVED_UNIT_ELEMENT(#1002,-1.0);
#4161500=SI_CAPACITANCE_UNIT((#860005,#8600025),*,$,.FARAD.);
```

Definition of "ohm":

An Ohm (Ω) is $1 V \cdot A^{-1}$.

```
/* establish a reference to a volt already defined. */
#870005=DERIVED_UNIT_ELEMENT(#1001,1.0);
/* establish a reference to amp*/
#8700025=DERIVED_UNIT_ELEMENT(#26,-1.0);
#10099=SI_RESISTANCE_UNIT((#870005,#8700025),*,$,.OHM.);
```

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Definition of “siemens”:

A Siemens (S) is $1\Omega^{-1}$.

```
/* establish a reference to an ohm already defined. */
#880005=DERIVED_UNIT_ELEMENT(#10099,-1.0);
#100=SI_CONDUCTANCE_UNIT((#880005),*,$,.SIEMENS.);
```

Definition of “weber”:

A Weber (Wb) is $1V{\cdot}s$.

```
/* establish a reference to a volt already defined. */
#890005=DERIVED_UNIT_ELEMENT(#1001,1.0);
/* establish a reference to second*/
#8900025=DERIVED_UNIT_ELEMENT(#24,1.0);
#10023=SI_MAGNETIC_FLUX_UNIT((#890005,#8900025),*,$,.WEBER.);
```

Definition of “tesla”:

A Tesla (T) is $1Wb{\cdot}m^{-2}$.

```
/* establish a reference to a weber already defined. */
#900005=DERIVED_UNIT_ELEMENT(#10023,1.0);
/* establish a reference to metre*/
#9000025=DERIVED_UNIT_ELEMENT(#4,-2.0);
#4161600=SI_MAGNETIC_FLUX_DENSITY_UNIT((#900005,#9000025),*,$,.TESLA.);
```

Definition of “henry”:

A Henry (H) is $1Wb{\cdot}A^{-1}$.

```
/* establish a reference to a weber already defined. */
#910005=DERIVED_UNIT_ELEMENT(#10023,1.0);
/* establish a reference to ampere*/
#9100025=DERIVED_UNIT_ELEMENT(#26,-1.0);
#47000=SI_INDUCTANCE_UNIT((#910005,#9100025),*,$,.HENRY.);
```

F.3.1.4.2 Non SI unit definitions

Well-known are the definition of length, area and volume. Examples included are square millimeters, cubic millimeters, inches, square inches and cubic inches.

Definition of Square Millimeter:

```
#613=DERIVED_UNIT_ELEMENT(#4,2.0);
#614=AREA_UNIT((#613));
#615=NAME_ATTRIBUTE('SQUARE MILLIMETRE',#614);
```

NOTE: In this example #614 defines an area unit of square millimeter. **Area_unit** is an instance of **derived_unit** but is not an instance of **si_unit** so **derived_unit.name** is populated.

Definition of Cubic Millimeter:

```
#610=DERIVED_UNIT_ELEMENT(#4,3.0);
#611=VOLUME_UNIT((#610));
#612=NAME_ATTRIBUTE('CUBIC MILLIMETRE',#611);
```

NOTE: In this example #611 defines a volume unit of cubic millimeter. **Volume_unit** is an instance of **derived_unit** but is not an instance of **si_unit** so **derived_unit.name** is populated.

Definition of Inch:

```
#71 =DIMENSIONAL_EXPONENTS(1.,0.,0.,0.,0.,0.,0.) ;--length
#2944=(LENGTH_UNIT() NAMED_UNIT(*) SI_UNIT(.MILLI.,.METRE.));
```

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```
/* Because the unit is an si_unit, the dimensional exponents for #2944 are not exchanged
but are calculated based on the enumeration values. */
#2945=LENGTH_MEASURE_WITH_UNIT(LENGTH_MEASURE(2.54E1),#2944);
#2946=(CONVERSION_BASED_UNIT('INCH',#2945) LENGTH_UNIT() NAMED_UNIT(#71));
```

NOTE 1 There is a (new) local rule on **conversion_based_unit** requiring the **dimensional_exponents** of a **conversion_based_unit** to be equal to the **dimensional_exponents** of the **unit_component attribute** of the **conversion_factor**. This example satisfies that rule because record #2946 references #71 directly and **dimensional_exponents** are derived by the receiver (if need be) for record #2944.

Definition of Square Inch:

```
#6130=DERIVED_UNIT_ELEMENT(#2946,2.0);
#6140=AREA_UNIT((#6130));
#6150=NAME_ATTRIBUTE('SQUARE INCH',#6140);
```

NOTE 2 In this example #6140 defines an area unit of square inch. **Area_unit** is an instance of **derived_unit** but is not an instance of **si_unit** so **derived_unit.name** is populated.

Definition of Cubic Inch:

```
#6100=DERIVED_UNIT_ELEMENT(#2946,3.0);
#6110=VOLUME_UNIT((#6100));
#6120=NAME_ATTRIBUTE('CUBIC INCH',#6110);
```

NOTE 3 In this example #6110 actually defines a volume unit of cubic inch. **Volume_unit** is an instance of **derived_unit** but is not an instance of **si_unit** so **derived_unit.name** is populated.

Definition of "pound force":

Recommended approach using existing newton declaration (#4161100):

```
/* conversion from newton to 'pound force' where 'pounds force is defined per the English
Gravitational System.  */
#111=DIMENSIONAL_EXPONENTS(1.,1.,-2.,0.,0.,0.,0.) ;--force
#101=FORCE_MEASURE_WITH_UNIT(FORCE_MEASURE(4.4482216152605),# 4161100);
#103=(CONVERSION_BASED_UNIT('pound force',#101) FORCE_UNIT() NAMED_UNIT(#111));
```

F.3.1.5 Example application of unit definitions to measure value

```
#714=AREA_MEASURE_WITH_UNIT(AREA_MEASURE(150.0),#614);
```

In this example #714 describes the surface area of a small cube with 5mm sides.

```
#711=VOLUME_MEASURE_WITH_UNIT(VOLUME_MEASURE(125.0),#611);
```

In this example #711 describes the volume of a small cube with 5mm sides.

```
#7140=AREA_MEASURE_WITH_UNIT(AREA_MEASURE(150.0),#6140);
```

In this example #7140 describes the surface area of a small cube with 5in sides.

```
#7110=VOLUME_MEASURE_WITH_UNIT(VOLUME_MEASURE(125.0),#6110);
```

In this example #7110 describes the volume of a small cube with 5in sides.

```
#8000=FORCE_MEASURE_WITH_UNIT(FORCE_MEASURE(250.2),#103);
```

In this example #8000 describes the force of 250.2 pounds force produced by an engine.

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F.3.2 Currency conversion

Consider the case in which an amount of money of 3.50 French Francs is to be expressed. In order to represent the unit French Franc, an instance of **context_dependent_unit** will be created. Its name attribute will have the value 'French Franc' and its dimensions attribute will refer to an instance of **dimensional_exponents** having the following attribute values:

— **length_exponent** = 0.0;

— **mass_exponent** = 0.0;

— **time_exponent** = 0.0;

— **electric_current_exponent** = 0.0;

— **thermodynamic_temperature_exponent** = 0.0;

— **amount_of_substance_exponent** = 0.0;

— **luminous_intensity_exponent** = 0.0.

Then, an instance of **measure_with_unit** will be created. Its **unit_component** attribute will refer to the instance depicted above and its **value_component** attribute will be of type **count_measure** and will have the value of 3.50.

NOTE A Euro may be specified relative to another currency using **conversion_based_unit**.

F.3.3 Context dependent unit

In describing a parts list, consider the case in which three occurrences of a given item are to be expressed. In order to represent the unit 'occurrence of an item', an instance of **context_dependent_unit** will be created. Its name attribute will have the value 'parts' and its dimensions attribute will refer to an instance of **dimensional_exponents** having the following attribute values:

— **length_exponent** = 0.0;

— **mass_exponent** = 0.0;

— **time_exponent** = 0.0;

— **electric_current_exponent** = 0.0;

— **thermodynamic_temperature_exponent** = 0.0;

— **amount_of_substance_exponent** = 0.0;

— **luminous_intensity_exponent** = 0.0.

Then, an instance of **measure_with_unit** will be created. Its **unit_component** will refer to the instance depicted above and its **value_component** attribute will be of type **count_measure** and will have the value of 3.

F.3.4 Unit conversion based on an algebraic expression

This example presents how to represent a unit that is converted from another unit, using any algebraic expression. In the following, the unit that is converted is designated as the reference unit.

NOTE 1 The case of a conversion expression that is a multiplication by a real coefficient can be dealt with using **conversion_based_unit**.

In order to provide the capability to represent, in an annotated EXPRESS schema, a unit that is defined by a conversion algebraic expression, this conversion expression is represented using resources defined in

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ISO 13584-20. The principle of the representation is to associate to each unit involved in the conversion expression the concept of variable as defined in ISO 13584-20 and to describe the algebraic expression that relate these variables.

For example, if the reference unit is a **named_unit**, the following subtype will be created:

EXPRESS specification:

```
ENTITY named_unit_variable
SUBTYPE OF (named_unit, variable_semantics);
INVERSE
      associated_variable_environment: environment FOR semantics;
END_ENTITY;
```

For the unit defined using the algebraic expression, the following subtype will be created:

EXPRESS specification:

```
ENTITY expression_conversion_based_unit
SUBTYPE OF (named_unit, variable_semantics);
INVERSE
      associated_variable_environment: environment FOR semantics;
END_ENTITY;
```

The following instances present an instantiation scheme in which the unit Degree Fahrenheit is defined with respect to the unit Degree Celsius.

NOTE 2 The mathematical formula relating both units is: $T_f = 1{,}8 x T_c + 32$

NOTE 3 The instances are expressed using the notation of ISO 10303-21 [2].

EXPRESS specification:

```
/* definition of the unit Degree Celsius */

#70 = (NAMED_UNIT(*) NAMED_UNIT_VARIABLE() SI_UNIT($, .DEGREE_CELSIUS. )
THERMODYNAMIC_TEMPERATURE_UNIT() VARIABLE_SEMANTICS());

/* Because the unit is an si_unit, the dimensional exponents for #70 are not exchanged
but are calculated based on the enumeration values. */

#71 = DIMENSIONAL_EXPONENTS(0.,0.,0.,0.,1.,0.,0.) ;

/* definition of the variable corresponding to T<sub>c</sub> and association with the
unit Degree Celsius */

#97 = REAL_NUMERIC_VARIABLE ();
#98 = ENVIRONMENT (#97, #70);

/* Representation of the expression '1.8 * T<sub>c</sub> + 32' */

#1005 = MULT_EXPRESSION ((#1006, #97));
#1006 = REAL_LITERAL (1.8);
#1007 = REAL_LITERAL (32);
#1008 = PLUS_EXPRESSION((#1005, #1007));

/* definition of the unit Degree Fahrenheit */

#170 = (EXPRESSION_CONVERSION_BASED_UNIT() NAMED_UNIT(#71)
THERMODYNAMIC_TEMPERATURE_UNIT() VARIABLE_SEMANTICS()) ;

/* definition of the variable corresponding to T<sub>f</sub> and association with the
unit Degree Fahrenheit */

#197 = REAL_NUMERIC_VARIABLE ();
```

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```
#198 = ENVIRONMENT (#197, #170);

/* expression to specify that T<sub>f</sub> is equal to the plus_expression */

#1010 = COMPARISON_EQUAL ((#197, #1008));
```

NOTE 4 If needed, the name of the unit Degree Fahrenheit can be represented using the entity **name_assignment**.

F.3.5 Use of area_unit and volume_unit

The following excerpt illustrates how an area unit and/or a volume unit can be constructed from the data types defined in clause **Measure**. Entity instance #611 is an example of a volume unit named 'cubic millimetre', and entity instance #614 is an area unit named 'square millimetre'.

NOTE 1 The instances are expressed using the notation of ISO 10303-21 [2].

```
#4=(LENGTH_UNIT() NAMED_UNIT(*) SI_UNIT(.MILLI.,.METRE.));
#610=DERIVED_UNIT_ELEMENT(#4,3.0);
#611=VOLUME_UNIT((#610));
#612=NAME_ATTRIBUTE('CUBIC MILLIMETRE',#611);
#613=DERIVED_UNIT_ELEMENT(#4,2.0);
#614=AREA_UNIT((#613));
#615=NAME_ATTRIBUTE('SQUARE MILLIMETRE',#614);
#711=VOLUME_MEASURE_WITH_UNIT(VOLUME_MEASURE(125.0),#611);
#714=AREA_MEASURE_WITH_UNIT(AREA_MEASURE(150.0),#614);
```

NOTE 2 In this example #611 actually defines a volume unit and #614 defines an area unit, these are used by #711 and #714 respectively to describe the volume and surface area of a small cube with 5mm sides.

F.3.6 Use of global_unit_assigned_context

The following example set of instances defines the context of a 3D geometric representation (#604), in which plane angles are expressed in radians, solid angles are expressed in steradians and lengths are expressed in millimetres.

NOTE The instances are expressed using the notation of ISO 10303-21 [2].

EXPRESS specification:

```
#3=(NAMED_UNIT(*)SI_UNIT($,.STERADIAN.)SOLID_ANGLE_UNIT());
#4=(LENGTH_UNIT()NAMED_UNIT(*)SI_UNIT(.MILLI.,.METRE.));
#5=(NAMED_UNIT(*)PLANE_ANGLE_UNIT()SI_UNIT($,.RADIAN.));

#604=(GEOMETRIC_REPRESENTATION_CONTEXT(3)
      GLOBAL_UNIT_ASSIGNED_CONTEXT( (#4,#5,#3))REPRESENTATION_CONTEXT('l-bracket','3D'));
```

F.4 Use of the person_organization_schema

F.4.1 Address of a person in an organization

In many cases, it will be desirable to associate and address with a person in the context of an organization. This address will usually be different from the home address of the person and may be different from the address of the organization itself. Such a requirement may be accomplished by a complex entity instance of a subtype whose supertypes are both **personal_address** and **organizational_address**.

An annotated EXPRESS schema that uses this part of ISO 10303 can create the following subtype to address this requirement.

EXPRESS specification:

```
ENTITY persons_in_organization_address
```

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```
SUBTYPE OF (personal_address, organizational_address)
WHERE
WR1:  SIZEOF(SELF\organizational_address.organizations)=1;
WR2:  SIZEOF(QUERY(pers &lt;* SELF\personal_address.people |
      SIZEOF(pers_in_org &lt;*
      USEDIN(pers,
 'PERSON_ORGANIZATION_SCHEMA.PERSON_AND_ORGANIZATION.THE_PERSON')
      | pers_in_org.the_organization IN
      SELF\organizational_address.organizations) =0)) =0;
END_ENTITY;
```

Formal propositions

WR1 The address of only one organization shall be considered;

WR2 The set of persons that are given an **address** and that are not related to the organization through an instance of **person_and_organization**, shall be empty.

F.4.2 Use of person_assignment

In some cases, it may be desirable to have more than one person associated with product data. For instance, one person may be the originator of the product data, and that data may subsequently be assigned to another person. In such cases, multiple instances of **person_assignment** may created, and the related **person_role** instances may be used to identify the originator and the successor.

In the following example, it is assumed that an annotated EXPRESS schema that uses this part of ISO 10303 has created a subtype of **person_assignment** called **applied_person_assignment**.

EXPRESS specification:

```
TYPE product_data_select = SELECT (applied_person_assignment, product,
...);
END_TYPE;

ENTITY applied_person_assignment
SUBTYPE OF person_assignment;
  assigned_items: SET[1:?] of product_data_select;
END_ENTITY;
```

The following set of instances record that Jim Smith was originally responsible for some product data; he was later replaced by Tom Jones.

NOTE The instances are expressed using the notation of ISO 10303-21 [2].

EXPRESS specification:

```
#1 = PERSON( '276 32 0402', 'Smith', 'Jim', '$' , '$', '$');
#2 = PERSON( '347 30 2476', 'Jones', 'Tom', '$' , '$', '$');
#10 = PERSON_ROLE( 'owner/originator', '$');
#11 = PERSON_ROLE( 'successor', 'takes over from Jim Smith');
#41 = (...some instance of product data);
#42 = (...another instance of product data);
#61 = APPLIED_PERSON_ASSIGNMENT( #1, #10, ( #41, # 42,...));
#62 = APPLIED_PERSON_ASSIGNMENT( #2, #11, (#1));
```

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